# K3 SURFACES NOTES

RAY SHANG

Abstract. Potpourri of supplementary notes to *Lectures on K3 surfaces* by Professor Huybrechts [Huy].

Contents

## 1. Smooth Quartic and Poincare Residue

A smooth quartic $X \subset \mathbb{P}^3$ is a K3 surface. In other words, it is a complete non-singular variety (separated, geometrically integral scheme of finite type over k) of dimension two, such that

$$\omega_{X/k} \cong \mathcal{O}_X \text{ and } H^1(X, \mathcal{O}_X) = 0.$$

The notion of a Poincare residue appears when one asks for an explicit trivializing section of its canonical line bundle. The Poincare residue generalizes the notion of residues in single variable complex analysis to several complex variables, and its origins may be found in Poincare's study of period integrals of a certain form [Poi87].

Suppose $Y \subset X$ is a smooth hypersurface in a complex manifold $X$ of dimension $n$. Let $\alpha \in H^0(\omega_X \otimes \mathcal{O}_X(Y))$ be a meromorphic section on $X$ with at most simple poles along $Y$. Locally, one can write $\alpha$ as $h\frac{dz_1}{z_1} \wedge dz_2 \wedge \cdots \wedge dz_n$, with $z_1$ defining $Y$. Then

$$Res_Y(\alpha) = (hdz_2 \wedge \cdots \wedge dz_n)|_Y.$$

– This map is well-defined. To see this, suppose on $U_i$ we can write $\alpha$ as $h\frac{dz_1}{z_1} \wedge dz_2 \wedge \cdots \wedge dz_n$, and on $U_j$ we can write $\alpha$ as $g\frac{dx_1}{x_1} \wedge dx_2 \wedge \cdots \wedge dx_n$. Let $\phi(x_1, \cdots, x_n) =$

$(\phi_1, \cdots, \phi_n)$ be an isomorphism on overlaps starting from the coordinates of $U_j$ to $U_i$. Then note that $\phi_1 = ux_1$ for some nowhere vanishing $u$. Then

$$Res(\phi^* h\frac{dz_1}{z_1} \wedge dz_2 \wedge \cdots \wedge dz_n) = Res(g\frac{dx_1}{x_1} \wedge dx_2 \wedge \cdots \wedge dx_n)$$

$$= Res(h \circ \phi\frac{d\phi_1}{\phi_1} \wedge d(\phi_2) \wedge \cdots \wedge d(\phi_n)) = Res(h \circ \phi(\frac{dx_1}{x_1} + \frac{du_1}{u_1}) \wedge d(\phi_2) \wedge \cdots \wedge d(\phi_n))$$

$$= Res(h \circ \phi\frac{dx_1}{x_1} \wedge d(\phi_2) \wedge \cdots \wedge d(\phi_n)) = h \circ \phi d(\phi_2) \wedge \cdots \wedge d(\phi_n)|_Y = \phi^* h dz_2 \wedge \cdots \wedge dz_n$$

$$= \phi^* Res(h\frac{dz_1}{z_1} \wedge dz_2 \wedge \cdots \wedge dz_n).$$

Thus, the Poincare residue gives a well-defined map

$$H^0(\omega_X \otimes \mathcal{O}_X(Y)) \to H^0(\omega_Y).$$

– Analyzing

$$0 \to \mathcal{I}_Y/\mathcal{I}_Y^2 \to \Omega_X|_Y \to \Omega_Y \to 0$$

to get the adjunction formula

$$(\omega_X \otimes \mathcal{O}_X(Y))|_Y \cong \omega_Y,$$

we see that the isomorphism is exactly given by Poincare residues.

– Another way to understand Poincare residue, which makes the anaology with residues from single variabel complex analysis clearer. Suppose we have a compact complex manifold $X$ of dimension $n$, and a smooth effective divisor $D$. Let $T_D \subset X \setminus D$ denote a tubular neighborhood of $D$.

Suppose we have a global holomorphic $n$-form with only a pole around $D$. This is a global section $\omega \in H^0(\omega_X \otimes \mathcal{O}_X(D))$. Then for every $p$ there exists local coordinates $U_p$ that expresses $\omega$ locally as

$$g\frac{dz_1}{z_1} \wedge dz_2 \wedge \cdots \wedge dz_n,$$

where $z_1$ locally defines $D$ in $U_p$, and $g$ is a holomorphic function. Then the Residue is locally given as

$$Res(\omega) = gdz_2 \wedge \cdots \wedge dz_n|_D.$$

More formally, the Poincare residue is the map

$$Res : H^n(X \setminus D; \mathbb{C}) \to H^{n-1}(D; \mathbb{C})$$

where given $n-1$ cycle $\sigma$ in $H^{n-1}(D; \mathbb{C})$, we take a tubular neighborhood $T_\sigma$ which is an $n$ cycle lying in $X \setminus D$, and integrate

$$\int_{T_\sigma} \omega : H^{n-1}(D; \mathbb{C}) \to \mathbb{C}.$$

By Poincare duality, this linear functional corresponds to an element of $H^{n-1}(D; \mathbb{C})$.

Let us return to the example of a smooth hypersurface $X$ of $\mathbb{P}^n$ which is cut out by $f \in H^0(\mathcal{O}_{\mathbb{P}^n}(n+1))$. The claim is that the canonical line bundle $\omega_X$ has trivializing section given by the Poincare residue

$$Res(\frac{\sum(-1)^i x_i dz_0 \wedge \cdots \wedge \widehat{dx_i} \wedge \cdots \wedge dz_n}{f}).$$

On the distinguished affine open $D_+(z_i)$, this expression means

$$Res(\frac{\sum(-1)^i dx_0 \wedge \cdots \wedge \widehat{dx_i} \wedge \cdots \wedge dx_n}{\frac{f}{x_0^{n+1}}}).$$

We claim that, first of all, these sections glue together to an actual global section of $H^0(\omega_{\mathbb{P}^n} \otimes \mathcal{O}_{\mathbb{P}^n}(X))$. Since the coordinate change on overlaps from $D_+(z_j) \to D_+(z_i)$ is

$$[y_0 : \cdots : y_{j-1} : 1 : y_j : \cdots : y_n] \to [\frac{y_0}{y_i} : \cdots : \frac{y_{i-1}}{y_i} : 1 : \frac{y_{i+1}}{y_i} : \cdots : \frac{y_{j-1}}{y_i} : \frac{1}{y_i} : \frac{y_{j+1}}{y_i} : \cdots : \frac{y_n}{y_i}],$$

then the pullback of the form on $D_+(z_i)$ is

$$(-1)^i \frac{1}{f/x_i^{n+1}} d(\frac{Y_0}{Y_i}) \wedge \cdots \wedge d(\frac{Y_{i-1}}{Y_i}) \wedge d(\frac{Y_{i+1}}{Y_i}) \wedge \cdots \wedge d(\frac{1}{Y_i}) \wedge \cdots \wedge d(\frac{Y_n}{Y_i})$$

$$= (-1)^i \frac{1}{f/x_i^{n+1}} (\frac{dY_0}{Y_i} - \frac{Y_0}{Y_i^2} dY_i) \wedge \cdots \wedge \frac{-1}{Y_i^2} dY_i \wedge \cdots \wedge (\frac{dY_n}{Y_i} - \frac{Y_n}{Y_i^2} dY_i) = (-1)^j \frac{1}{f/x_j^{n+1}} dy_0 \wedge \cdots \wedge \widehat{dy_j} \wedge \cdots \wedge dy_n.$$

Thus, $\alpha \in H^0(\omega_{\mathbb{P}^n} \otimes \mathcal{O}_{\mathbb{P}^n}(X))$ and $Res(\alpha) \in H^0(\omega_X)$. Furthermore, $Res(\alpha)$ is nowhere vanishing. Locally on $D_+(z_0) \cap X$, it is of the form

$$\frac{1}{f} dx_1 \wedge \cdots \wedge dx_n.$$

Note that since $f$ defines a smooth hypersurface on $D_+(z_0)$, 0 is a regular value. Then, for example, we can consider the open subset of $D_+(z_0)$ for which $\frac{\partial f}{\partial x_k} \neq 0$. We can change the local coordinates so that $\frac{\partial f}{\partial u_k} \neq 0$ but all the other partials are zero. Then we can write

$$(-1)^{k-1} \frac{1}{\frac{\partial f}{\partial x_k}} \frac{df}{f} dx_1 \wedge \cdots \wedge \widehat{dx_k} \wedge \cdots \wedge dx_n,$$

so that the residue is

$$(-1)^{k-1} \frac{1}{\frac{\partial f}{\partial x_k}} dx_1 \wedge \cdots \wedge \widehat{dx_k} \wedge \cdots \wedge d_n,$$

so this is nowhere vanishing.

## 2. Complete intersections and Koszul complexes

**Proposition 2.1.** *Let $F_1, \cdots, F_n$ be homogeneous polynomials of degree $d_i$, such that*

$$X = Z(F_1) \cap \cdots \cap Z(F_n)$$

*is a complete intersection (in other words, the homogeneous ideal of $X$ is generated exactly by the $n$ elements $F_1, \cdots, F_n$). Then $X$ is a K3 surface if and only if $\sum d_i = n + 3$.*

Another way to think of a complete intersection is that the polynomials $F_i$ define a map

$$F : \mathbb{C}^{n+3} \setminus \{0\} \to \mathbb{C}^n$$

where $(z_0, \cdots, z_{n+2}) \mapsto (F_1, \cdots, F_n)$, such that $0 \in \mathbb{C}^n$ is a regular value of the map $F$. Verifying proposition 2.1 requires verifying that

$$\omega_X \cong \mathcal{O}_X \text{ and } H^1(X, \mathcal{O}_X) = 0.$$

Let's focus on the part about having trivial canonical bundle. The first approach will be inductive, and the second approach will be an excuse to talk a little about Koszul resolutions.

**The first approach** is a very natural thing to try. We'd like to consider the short exact sequences

$$0 \to \mathcal{O}_{Z(F_1)}(-d_1) \to \Omega_{\mathbb{P}^{n+2}}|_{Z(F_1)} \to \Omega_{Z(F_1)} \to 0$$
$$0 \to \mathcal{O}_{Z(F_1)\cap Z(F_2)}(-d_2) \to \Omega_{Z(F_1)}|_{Z(F_1)\cap Z(F_2)} \to \Omega_{Z(F_1)\cap Z(F_2)} \to 0$$
$$\vdots$$
$$0 \to \mathcal{O}_{Z(F_1)\cap\cdots\cap Z(F_n)}(-d_n) \to \Omega_{Z(F_1)\cap\cdots\cap Z(F_{n-1})}|_{Z(F_1)\cap\cdots\cap Z(F_n)} \to \Omega_{Z(F_1)\cap\cdots\cap Z(F_n)} \to 0$$

and inductively use the adjunction formula to show that $\omega_{Z(F_1)\cap\cdots\cap Z(F_n)} \cong \mathcal{O}_{Z(F_1)\cap\cdots\cap Z(F_n)}$ if and only if $\sum d_i = n + 3$. There is a slight problem with this argument, however, namely that while $Z(F_1) \cap \cdots \cap Z(F_n)$ is smooth, the other $Z(F_1) \cap \cdots \cap Z(F_i)$ are not necessarily smooth. Failure of smoothness will mean the many of the short exact sequences above could fail to be left-exact. But note that the singular locus for any $X_i = Z(F_1) \cap \cdots \cap Z(F_i)$ will be a closed subset of $X_i$. So here is a remedy: let $S = \bigcup_{i=1}^{n} Sing(X_i)$. Then consider the chain

$$(X_1 \setminus S) \supset (X_2 \setminus S) \supset \cdots \supset (X_{n-1} \setminus S) \supset (X_n \setminus S) = (X \setminus S).$$

Note $(X_i \setminus S) \cap Z(F_{i+1}) = (X_{i+1} \setminus S)$. But by analyzing Jacobians, we know $(X \setminus S) = X$. Then we can apply the induction for short exact sequences defined with respect to this chain.

**The second approach** involves the Koszul complex, which was originally used by Jean-Louis Koszul to define a cohomology theory for Lie algebras. Very useful general construction in homological algebra. It is related to notions of regular sequences, which is the algebraic analog of complete intersections, and in certain situations, it tells you about generators of a module, the relations between these relations, and so forth.

**Definition 2.2.** *Let $A$ be a commutative. Let $s : M \to A$ be a map of $A$-modules. Then the Koszul complex $K_s$ is:*

$$\cdots \to \bigwedge^r M \to \bigwedge^{r-1} M \to \cdots \to \bigwedge^2 M \to M \to^s A,$$

*where the map from $\bigwedge^k M \to \bigwedge^{k-1} M$ sends*

$$m_1 \wedge \cdots \wedge m_k \mapsto \sum_{i=1}^{k} (-1)^{i+1} s(m_i) \cdot m_1 \wedge \cdots \wedge \widehat{m_i} \wedge \cdots \wedge m_k.$$

In particular, the Koszul complex is a chain complex. In algebraic geometry, we may be given a ring $A$ and elements $f_1, \cdots, f_r$, and we will want to understand how the vanishing loci of $f_1, \cdots, f_r$ all intersect one another. We can study this by the Koszul complex $K(f)$ associated to $f = (f_1, \cdots, f_r)$.

**Definition 2.3.** *Let $A$ be a commutative ring. Let $f = (f_1, \cdots, f_r)$, where $f_i \in A$. Then we define $K(f)$ to be the Koszul complex associated to*

$$s : A^{\oplus r} \to^{\cdot (f_1, \cdots, f_r)} A$$

*where $s(g_1, \cdots, g_r) \mapsto \sum f_i g_i$. The Koszul complex is then the chain complex*

$$0 \to \bigwedge^r A^{\oplus r} \to \cdots \to \bigwedge^2 A^{\oplus r} \to A^{\oplus r} \to^{\cdot (f_1, \cdots, f_r)} A.$$

For a regular sequence $(f_1, \cdots, f_r)$ in $A$, the Koszul complex

$$0 \to \bigwedge^r A^{\oplus r} \to \cdots \to A^{\oplus r} \to A \to \frac{A}{\langle f_1, \cdots, f_r \rangle} \to 0$$

is exact [Wei, Corollary 4.5.5] [Eis95].

**Proposition 2.4** (Zero locus of a section of a vector bundle)**.** *Let $E$ be a vector bundle of rank $r$ on a variety $X$, and let $\sigma$ be a section of $E$.*

(1) *Regarding $\sigma$ as a map $E^* \to \mathcal{O}_X$, there is a Koszul complex*

$$0 \to \bigwedge^r E^* \to \cdots \to \bigwedge^2 E^* \to E^* \to \mathcal{O}_X$$

(2) *If the zero locus $Z$ of $\sigma$ has codimension $r$, the "expected value," and every local ring $\mathcal{O}_{X,x}$ has the property that any sequence of elements $x_1, \cdots, x_r$ generating a codimension $r$ ideal is a regular sequence, then the Koszul complex is a resolution of the sheaf $\mathcal{O}_Z$ by vector bundles.*

This is [Eis95, Exercise 17.20]. Let's unpack the map behind the Koszul complex associated to $\sigma \in H^0(X, E)$. Let's work over $\mathrm{Spec}(A) \subset X$. Since $\sigma$ is a global section, it is equivalent to the map $\mathcal{O}_X \to E$, where over $\mathrm{Spec}(A)$, the map $\mathcal{O}_X(\mathrm{Spec}(A)) \to E(\mathrm{Spec}(A))$ sends $1 \mapsto \sigma_A$. Dualizing, we obtain $E^* \to \mathcal{O}_X$, where over $\mathrm{Spec}(A)$, the map looks like

$$\mathrm{Hom}_A(E(\mathrm{Spec}(A)), A) \to \mathrm{Hom}_A(A, A),$$

where $\phi \in \mathrm{Hom}_A(E(\mathrm{Spec}(A)), A) \mapsto \phi(\sigma_a)$. The map $\bigwedge^2 E^*$ over $\mathrm{Spec}(A)$ looks like

$$\phi_1 \wedge \phi_2 \mapsto -\phi_1(\sigma_a) \cdot \phi_2 + \phi_2(\sigma_a) \cdot \phi_1.$$

Let us now apply this to the question we began with. We want to show that, if $X$ is a complete intersection, then $\omega_X \cong \mathcal{O}_X$ if and only if $\sum_{i=1}^n d_i = n+3$. Note by the adjunction formula, $\omega_X \cong \mathcal{O}_X(n+3) \otimes \bigwedge^n (\mathcal{I}_X/\mathcal{I}_X^2)^*$.

Consider $E = \bigoplus_{i=1}^n \mathcal{O}_{\mathbb{P}^{n+2}}(d_i)$. Then $F = (F_1, \cdots, F_n) \in H^0(E)$. Then we have Koszul resolution

$$0 \to \bigwedge^n E^* \to \cdots \to \bigwedge^2 E^* \to E^* \to \mathcal{O}_{\mathbb{P}^{n+2}}.$$

Note $E^* \cong \bigoplus_{i=1}^n \mathcal{O}_{\mathbb{P}^{n+2}}(-d_i)$. Over for example $D_+(z_0)$, the local section of $E^*$ look like

$$\bigoplus_{i=1}^n \frac{1}{Z_0^{d_i}} \mathbb{C}[\frac{z_1}{z_0}, \cdots, \frac{z_{n+2}}{z_0}].$$

The map $E^* \to \mathcal{O}_{\mathbb{P}^{n+2}}$ over $D_+(z_0)$ is given exactly by $\cdot(F_1, \cdots, F_n)$. The image of the map over $D_+(z_0)$ is $\langle \frac{F_1}{Z_0^{d_1}}, \cdots, \frac{F_n}{Z_0^{d_n}} \rangle$. The second part of proposition 2.4 implies we have an exact sequence

$$0 \to \bigwedge^n E^* \to \cdots \to \bigwedge^2 E^* \to E^* \to \mathcal{O}_{\mathbb{P}^{n+2}} \to \iota_* \mathcal{O}_X \to 0,$$

which factors as

$$0 \to \bigwedge^n E^* \to \cdots \to \bigwedge^2 E^* \to E^* \to \mathcal{I}_X \to 0.$$

Pulling back this exact sequence over $X$, the map $\bigwedge^2 E^*|_X \to E^*|_X$, which over $D_+(z_0)$ becomes a zero map. Then

$$E^*|_X \cong \mathcal{I}_X/\mathcal{I}_X^2 \implies \bigwedge^n (\mathcal{I}_X/\mathcal{I}_X^2)^* \cong \mathcal{O}_X(\sum d_i).$$

Thus, we have $\omega_X \cong \mathcal{O}_X \iff \sum d_i = n+3$. Furthermore, if this holds, we also have $H^1(X, \mathcal{O}_X)$. To see why, note sheaf cohomology on

$$0 \to \mathcal{I}_X \to \mathcal{O}_{\mathbb{P}^{n+2}} \to \iota^* \mathcal{O}_X \to 0$$

implies $H^1(X, \mathcal{O}_X) \cong H^2(\mathbb{P}^{n+2}, \mathcal{I}_X)$. By Serre duality, this is isomorphic to $H^n(\mathbb{P}^{n+2}, \mathcal{I}_X^* \otimes \mathcal{O}_{\mathbb{P}^{n+2}}(-n-3))$. Note dualizing the Koszul resolution for $\mathcal{I}_X$ and twisting yields

$$0 \to \mathcal{I}_X^* \otimes \mathcal{O}_{\mathbb{P}^{n+2}}(-n-3) \to E(-n-3) \to (\bigwedge^2 E)(-n-3) \to \cdots \to (\bigwedge^n E)(-n-3) \to 0 \to \cdots$$

which is a resolution by acyclic sheaves. We see that $(\bigwedge^n E)(-n-3) \to 0 \to 0$ immediately implies that $H^n(\mathbb{P}^{n+2}, \mathcal{I}_X^* \otimes \mathcal{O}_{\mathbb{P}^{n+2}}(-n-3)) = 0$, so $H^1(X, \mathcal{O}_X) = 0$.

**Corollary 2.5.** *The only unique cases of K3 surfaces which are complete intersections are*

$$d_1 = 4, (d_1, d_2) = (2,3), \text{ and } (d_1, d_2, d_3) = (2,2,2).$$

*This yields examples of K3 surfaces of degree four, six, and eight.*

*Proof.* By proposition 2.1, we must have $\sum_{i=1}^{n} d_i = n+3$. When one of the $d_i$ is 1, we are just cutting down by a hyperplane, so we are looking for unordered tuples of $d_i$, none of which are 1. For $n=1$, we just have $d_1 = 4$. When $n=2$, we have $(d_1, d_2) = (2,3)$. When $n=3$, we have $(d_1, d_2, d_3) = (2,2,2)$. Any other $n$ will yield tuples which include a 1. □

## 3. Koszul complexes and regular sequences

Regular sequences are the algebraic analogs of the geometric phenomena of complete intersections. Generally, (co)homology of Koszul complexes detect the length of regular sequences, and under certain hypotheses, can detect the regularity of given elements. Throughout this section, assume $M$ is a finitely generated $R$-module, where $R$ is a Noetherian ring.

**Definition 3.1.** *A regular sequence $x_1, \cdots, x_n \in R$ is an $M$-sequence if*

(1) $(x_1, \cdots, x_n)M \neq M$, *and*
(2) $x_i$ *is not a zerodivisor of* $\frac{M}{(x_1,\cdots,x_{i-1})M}$.

Given an $R$-module $N$ and an element $x \in N$, the **Koszul complex** of $x$ is:

$$K(x) : 0 \to R \to N \to \bigwedge N \to \bigwedge^2 N \to \bigwedge^3 N \to \cdots$$

where $\bigwedge^k N \to \bigwedge^{k+1} N$ is given by $x \wedge -$, and where we identify $\bigwedge^0 N \cong R$. In the context of regular sequences, the Koszul complexes we'll care about are for $N \cong R^{\oplus n}$. Specifically, given $x = (x_1, \cdots, x_n) \in R^n$, the Koszul complex

$$K(x) = K(x_1, \cdots, x_n) : 0 \to R \to R^n \to \bigwedge^2 R^n \to \cdots \to \bigwedge^{n-1} R^n \to \bigwedge^n R^n \to 0.$$

In general, given $x_1, \cdots, x_n \in R$, the Koszul homology of $M \otimes_R K(x_1, \cdots, x_n)$ detects the length of $M$-sequences contained in the ideal $(x_1, \cdots, x_n)$.

**Theorem 3.2.** *If for all $j < r$ we have*

$$H^j(M \otimes K(x_1, \cdots, x_n)) = 0,$$

*and $H^r(M \otimes K(x_1, \cdots, x_n)) \neq 0$, then every maximal $M$-sequence in $(x_1, \cdots, x_n) \subset R$ has length $r$.*

**Corollary 3.3.** *If $x_1, \cdots, x_n$ is an $M$-sequence, then considering the complex*

$$0 \to M \otimes_R R \to M \otimes_R R^n \to M \otimes_R \bigwedge^2 R^n \to \cdots \to M \otimes_R \bigwedge^n R^n \to 0,$$

*we have $H^n(M \otimes K(x_1, \cdots, x_n)) \cong M/(x_1, \cdots, x_n)M$ and*

$$H^i(M \otimes K(x_1, \cdots, x_n)) = 0, \text{ for all } i < n.$$

In particular, corollary 3.3 implies that when $x_1, \cdots, x_n$ is an $M$-sequence, the Koszul complex $M \otimes K(x_1, \cdots, x_n)$ provides a resolution for $M/(x_1, \cdots, x_n)M$. When $R$ is Noetherian local, we get an even stronger result.

**Theorem 3.4.** *Let $(R, m)$ be a local Noetherian ring and $x_1, \cdots, x_n \in m$. If for some $k$ we have $H^k(M \otimes K(x_1, \cdots, x_n)) = 0$, then*

$$H^j(M \otimes K(x_1, \cdots, x_n)) = 0, \forall j \leq k.$$

*In particular, $H^{n-1}(M \otimes K(x_1, \cdots, x_n)) = 0$ implies $x_1, \cdots, x_n$ is an $M$-sequence.*

A corollary of this is that when $(R, m)$ is local Noetherian, the order of the regular sequences does not matter, since we can construct an isomorphism between their Koszul complexes inducing an isomorphism on cohomology. Order does matter when $(R, m)$ is not local.

**Example 3.5.** *Let $R = k[x, y, z]/(x-1)z$ and let $f_1 = (x-1)y$ and $f_2 = x$. Then $f_2, f_1$ is an $R$-sequence, but $f_1, f_2$ is not a $R$-sequence.*

I don't find the proofs of theorem 3.2 and theorem 3.4 in [Eis95, section 17] to be particularly enlightening, but the homological techniques used seem good to know, so let me say a few things about them here.

**Tensor product of complexes.** Given two chain complexes

$$F : \cdots \longrightarrow F_{i-1} \longrightarrow F_i \longrightarrow F_{i+1} \longrightarrow \cdots$$

$$G : \cdots \longrightarrow G_{i-1} \longrightarrow G_i \longrightarrow G_{i+1} \longrightarrow \cdots$$

we define the tensor product chain complex to be $F \otimes G$ such that

$$(F \otimes G)_n = \bigoplus_{i+j=n} F_i \otimes G_i$$

and the differential $(F \otimes G)_n \longrightarrow (F \otimes G)_{n+1}$ is given on $F_i \otimes G_j$ to be the map $d(\alpha \otimes b) \mapsto d(\alpha) \otimes \beta + (-1)^i \alpha \otimes d(\beta)$.

**Lemma 3.6.** *Let $x, y \in R$. Then*

$$K(x, y) \cong K(x) \otimes K(y).$$

**Shifts**. Given a chain complex $G$, we can shift it by $i$ to obtain chain complex $G[i]$, such that $G[i]_n = G_{n+i}$. The differential for $G[i]_n$ is the differential given for $G_{i+n}$, but you multiply by $(-1)^i$. If we let $\mathcal{R}$ denote the complex $0 \to R \to 0$, then note that $G[n] \cong \mathcal{R}[n] \otimes G$.

**Mapping cones**. Let $y \in R$, and let $M_y$ be the complex $0 \longrightarrow R \longrightarrow^{\cdot y} R \longrightarrow 0$. Then there is a short exact sequence of chain complexes

$$0 \longrightarrow \mathcal{R}[-1] \longrightarrow \mathcal{M}_y \longrightarrow \mathcal{R} \longrightarrow 0,$$

and tensoring by a chain complex $G$, we obtain a short exact sequence of chain complexes

$$0 \to G[-1] \to \mathcal{M}_y \otimes G \to G \to 0,$$

which induces a long exact sequence in cohomology

$$H^i(G) \to^{\cdot y} H^{i+1}G[-1] \to H^{i+1}\mathcal{M}_y \otimes G \to H^{i+1}G \to^{\cdot y} H^{i+2}G[-1] \to \cdots,$$

where we note the boundary map $H^i(G) \to^{\cdot y} H^{i+1}G[-1] \cong H^iG$ is multiplication by $y$. The complex $\mathcal{M}_y \otimes G$ is called the mapping cone of $G$ with respect to multiplication by $y$. As a chain complex, it looks like:

$$\begin{tikzcd}
\cdots \arrow[r] & G_{n-1} \arrow[r, "d_{n-1}"] \arrow[dr, "\cdot y"'] & G_n \arrow[r, "d_n"] \arrow[dr, "\cdot y"] & G_{n+1} \arrow[r] & \cdots \\
& \bigoplus & \bigoplus & \bigoplus & \\
\cdots \arrow[r] & G_{n-2} \arrow[r, "-d_{n-2}"'] & G_{n-1} \arrow[r, "-d_{n-1}"'] & G_n \arrow[r] & \cdots
\end{tikzcd}$$

Altogether, the techniques of tensor products of chain complexes, shifts, and mapping cones plus their induced long exact sequence in cohomology provide the main technology for proving theorems 3.2 and 3.4.

**Self-duality for Koszul complexes.** Let $x = (x_1, \cdots, x_n) \in R^n$. Then we have the Koszul complex

$$0 \to R \to^{\cdot x} R^n \to \bigwedge^2 R^n \to \cdots \to \bigwedge^{n-1} R^n \to \bigwedge^n R^n \to 0.$$

Dualizing this, we obtain a complex

$$0 \to (\bigwedge^n R^n)^* \to (\bigwedge^{n-1} R^n)^* \to (\bigwedge^{n-2} R^n)^* \to \cdots \to (R^n)^* \to (R)^* \to 0$$

and identifying $(\bigwedge^{n-k} R^n)^* \cong \bigwedge^{n-k}(R^n)^*$, we obtain a complex

$$0 \to \bigwedge^n (R^n)^* \to \bigwedge^{n-1} (R^n)^* \to \bigwedge^{n-2} (R^n)^* \to \cdots \to (R^n)^* \to R^* \to 0.$$

Amazingly, these complexes are all isomorphic. By choice of an orientation $f : \bigwedge^n R^n \cong R$, one can construct maps $\bigwedge^k R^n \cong (\bigwedge^{n-k} R^n)^* \cong \bigwedge^{n-k}(R^n)^*$ which commute with the differentials. In other words, these Koszul complexes are self dual.

In geometric situations, we will frequently prefer the dual Koszul complex. To be clear, given $x_1, \cdots, x_n \in R$, this complex is of the form

$$K'(x_1, \cdots, x_n) : 0 \to \bigwedge^n R^n \to \cdots \to R^n \to^{\phi} R \to 0$$

where $\bigwedge^k R^n \to \bigwedge^{k-1} R^n$ is the map

$$\alpha_1 \wedge \cdots \wedge \alpha_k \mapsto \sum_{j=1}^{k} (-1)^{j+1} \phi(\alpha_j) \alpha_1 \wedge \cdots \wedge \widehat{\alpha_j} \wedge \cdots \wedge \alpha_k.$$

If $(R, m)$ is local and $x_1, \cdots, x_n \in m$, then self-duality and theorem 3.4 imply that if $H_1(K'(x_1, \cdots, x_n)) = 0$ then $x_1, \cdots, x_n$ is regular $R$-sequence.

**Some useful lemmas for doing geometry.**

**Lemma 3.7.** *Koszul homology can be related to the Tor functor in the following way. Suppose that $S \to R$ is a ring homomorphism, and suppose $y_1, \cdots, y_r \in S$ have images $x_1, \cdots, x_r \in R$. If $y_1, \cdots, y_r$ is a regular sequence in $S$, then*

$$H_i(K'(x_1, \cdots, x_r) \otimes M) = Tor_i^S(\frac{S}{y_1, \cdots, y_r}, M).$$

*Proof.* Let $M$ be an $R$-module. Note that $M$ also admits the structure of an $S$-module by the ring homomorphism $S \to R$. Consider the Koszul complex

$$K'(y_1, \cdots, y_r) : 0 \to \bigwedge^r S^r \to \cdots \to S^r \to S \to 0.$$

Note that since $y_1, \cdots, y_s$ is a regular sequence, then $K'(y_1, \cdots, y_r)$, by duality and corollary of theorem 3.2, provides a resolution for $S/y_1, \cdots, y_r$. Then tensoring by $M$ as an $S$-module, we see that

$$H_i(K'(y_1, \cdots, y_r) \otimes_S M) = Tor_i^S(\frac{S}{y_1, \cdots, y_r}, M).$$

But note that

$$\bigwedge^k S^r \otimes_S M \cong \bigwedge^k S^r \otimes_S R \otimes_R M.$$

We have $K'(y_1, \cdots, y_r) \otimes_S R \cong K'(x_1, \cdots, x_r)$, thus we see that

$$H_i(K'(x_1, \cdots, x_r) \otimes M) \cong Tor_i^S(\frac{S}{y_1, \cdots, y_r}, M).$$

□

In particular, given $x_1, \cdots, x_r \in R$ and $R$-module $M$, we can always compute the Koszul homology via a Tor functor by setting $S = R[Z_1, \cdots, Z_r]$, and considering the ring homomorphism $S \to R$ where $Z_i \mapsto x_i$. In particular, lemma 3.7 implies

$$H_i(K'(x_1, \cdots, x_r) \otimes M) = Tor_i^{R[Z_1, \cdots, Z_r]}(R, M).$$

**Lemma 3.8.** *Let $E$ be a locally free sheaf of rank $r$ on a variety $X$. Let $\sigma$ be a global section of $E$.*

(1) *Regarding $\sigma$ as map $F := E^* \to \mathcal{O}_X$, there is a Koszul complex*

$$0 \to \bigwedge^r F \to \cdots \to \bigwedge^i F \to \cdots \to F \to \mathcal{O}_X \to 0.$$

(2) *Suppose the zero locus $Z$ of $\sigma$ has codimension $r$. Suppose every local ring $\mathcal{O}_{X,x}$ has the property that any sequence of elements $x_1, \cdots, x_r$ generating a codimension $r$ ideal is a regular sequence ($X$ is locally Cohen-Macaulay). Then $K$ is a resolution of $\mathcal{O}_Z$ by vector bundles.*

*Proof.* (1) On $\mathrm{Spec}(A)$ of $X$, we have $E(\mathrm{Spec}(A)) \cong A^{\oplus r}$. The global section $\sigma$ of $E$ gives us elements $\sigma_a = (\sigma_{a,1}, \cdots, \sigma_{a,r}) \in A^{\oplus r}$. Then we have the Koszul complex

$$K(\sigma_{a,1}, \cdots, \sigma_{a,r}) : 0 \to A \to A^{\oplus r} \to \cdots \to \bigwedge^r A^{\oplus r} \to 0.$$

Functoriality of Koszul complexes implies that we can glue these Koszul complexes together to form a complex

$$0 \to \mathcal{O}_X \to E \to \bigwedge^2 E \to \cdots \to \bigwedge^r E \to 0.$$

Then dualizing, we obtain a Koszul complex

$$0 \to \bigwedge^r F \to \cdots \to F \to \mathcal{O}_X \to 0.$$

(2) Pick any point $p \in X$. We can localize the complex to

$$0 \to \bigwedge^r F_p \to \cdots \to F_p \to \mathcal{O}_{X,p} \to 0.$$

If $p \notin Z$ then the $r$ elements of $\mathcal{O}_{X,p}$ that the Koszul complex is defined with respect to are units, so the homology is all zero by [Eis95, proposition 17.14], so the above short exact sequence is exact. Otherwise, if $p \in Z$, then the $r$ elements of $\mathcal{O}_{X,p}$ locally cut out $Z$ around $p$, and thus they form a regular sequence. By theorem 3.4, this implies that

$$0 \to \bigwedge^r F_p \to \cdots \to F_p \to \mathcal{O}_{X,p} \to \mathcal{O}_{Z,p} \to 0$$

is exact. Altogether, this implies we have a resolution

$$0 \to \bigwedge^r F \to \cdots \to F \to \mathcal{O}_X \to \iota_* \mathcal{O}_Z \to 0.$$

□

## 4. Basic intersection theory of curves on algebraic surfaces and Riemann Roch for surfaces

To study K3 surfaces, we'll want to set up a basic intersection theory of curves. Let us recall some basic facts about curves.

**Definition 4.1.** *A curve $X$ is a 1 dimensional projective $k$-scheme, all of whose closed points have depth 1, i.e. all its local rings are Cohen-Macaulay. Assume $k$ is algebraically closed.*

If $D \subset X$ is an effective Cartier divisor on $X$, let $f_p$ be a local equation for $D$ at $p \in X$. For all but a finite set of points, say $p_1, \cdots, p_n$, we have $f_p$ is invertible. We define the **degree** of $D$ to be

$$\deg(D) = \sum_{i=1}^{n} \dim_k \frac{\mathcal{O}_{X,p_i}}{f_i}.$$

Note the structure sheaf $\mathcal{O}_D$ is just $\frac{\mathcal{O}_{X,p_i}}{f_i}$ at the points $p_i$, and 0 elsewhere. Thus,

$$\deg(D) = \sum_{i=1}^{n} \dim_k \frac{\mathcal{O}_{X,p_i}}{f_i} = H^0(X, \iota_* \mathcal{O}_D).$$

When $X$ is nonsingular and $(t_i)$ is the maximal ideal at $p_i$, then $f_{p_i} = ut_i^{e_i}$. Then $D$ is simply the Weil divisor $D = \sum_{i=1}^n e_i p_i$, and $\deg D = \sum_{i=1}^n e_i$.

The degree depends only on the divisor class of $D$, and not $D$ itself. The quickest way to see this is to consider the ideal sheaf sequence $0 \to \mathcal{O}_X(-D) \to \mathcal{O}_X \to \mathcal{O}_D \to 0$, which implies

$$\deg D = \chi(\mathcal{O}_X) - \chi(\mathcal{O}_X(-D)).$$

This also lets us extend the definition of degree to arbitrary invertible sheaves $\mathcal{L} \in \mathrm{Pic}(X)$:

$$\deg \mathcal{L} = \chi(\mathcal{O}_X) - \chi(\mathcal{L}^{-1}).$$

Riemann's half of the Riemann-Roch theorem tells you that

**Lemma 4.2.** *Riemann's half of the Riemann-Roch theorem tells you that*

(1) $\deg(L \otimes M) = \deg L + \deg M$, *so that*
(2) $\deg L^{-1} = \chi(\mathcal{O}_X) - \chi(L) = -\deg L$

$$\implies \chi(L) = \deg L + \chi(\mathcal{O}_X) = \deg L + 1 - p_a(X),$$

(3) *and we have a degree homomorphism*

$$\mathrm{Pic}(X) \to^{\deg} \mathbb{Z}.$$

Now fix a nonsingular projective surface $S$ over an algebraically closed field $k$. On $S$, we have divisors (Weil or Cartier, it makes no difference), and the group of divisor classes $\mathrm{Pic}(S)$. Among divisors, the effective divisors will be referred to simply as curves: these are now 1-dimensional closed subschemes, but they are not necessarily reduced or irreducible.

**Definition 4.3.** *Suppose $D_1, D_2$ are curves (effective Weil divisors) on $S$ such that*

$$\dim(Supp(D_1) \cap Supp(D_2)) = 0.$$

*In other words, $D_1$ and $D_2$ are divisors do not share any irreducible components.*

*Let $\{p_1, \cdots, p_n\} = Supp(D_1) \cap Supp(D_2)$. Let $f_{1,i}$ and $f_{2,i}$ be local equations for $D_1$ and $D_2$ around $p_i$. Then define*

$$(D_1 \cdot D_2) = \sum_{i=1}^{n} \dim_k \frac{\mathcal{O}_{S,p_i}}{\langle f_{1,i}, f_{2,i} \rangle}.$$

The number $\dim_k \frac{\mathcal{O}_{S,p_i}}{\langle f_{1,i}, f_{2,i}\rangle}$ is the local intersection multiplicity of $D_1$ and $D_2$ at $p_i$. We say $D_1$ and $D_2$ intersect transversely at $p_i$ if they share no irreducible components at $p_i$, and their local equations $f_{1,i}$ and $f_{2,i}$ at $p_i$ span the maximal ideal of $\mathcal{O}_{S,p_i}$. Note $m_{S,p_i}/m^2_{s,P_i}$ is the Zariski tangent space of $S$, so we see that transverse intersection equivalently means that the tangent lines of $D_1$ and $D_2$ at $p_i$ span $T_{p_i}S$.

Another note to make: if $\eta$ is the generic point of irreducible curve $C \subset S$, then $\mathcal{O}_{S,\eta}$ is a discrete valuation ring. So one can think of local equation $f$ for $C$ as an element of $\mathcal{O}_{S,p}$ which inside of $\mathcal{O}_{S,\eta}$ it has exponent 1, but is equivalent to identity in $\mathcal{O}_{S,\eta'}$ for all generic points $\eta'$ of other irreducible curves.

**Remark 4.4.** *Suppose $D_1$ and $D_2$ are effective Weil divisors which share irreducible components. If one can find $D_1' \sim D_1$ and $D_2' \sim D_2$ such that $D_1'$ and $D_2'$ share no irreducible components, then one can compute $(D_1.D_2)$ in terms of the local intersection multiplicities of $D_1'$ and $D_2'$.*

*For example, suppose we have degree $a$ and degree $b$ curve in $\mathbb{P}^2$. Then these are linearly equivalent to $aH$ and $bH$. But $H \sim H'$ where the lines intersect transversely, so the intersection of a degree $a$ adn degree $b$ curve is $(aH.bH') = ab$. This is the celebrated Bezout's theorem.*

We can express the intersection pairing cohomologically as the following.

**Proposition 4.5.** *Suppose the hypotheses of definition 4.3 are satisfied. Then*

$$(D_1 \cdot D_2) = \chi(\mathcal{O}_S) - \chi(-D_1) - \chi(-D_2) + \chi(-D_1 - D_2).$$

*Proof.* Note that the structure sheaf of $\mathcal{O}_{D_1 \cap D_2}$ is exactly $\frac{\mathcal{O}_{S,p_i}}{\langle f_{1,i}, f_{2,i}\rangle}$ over $p_i$, and 0 elsewhere. Thus, $h^0(\mathcal{O}_{D_1\cap D_2}) = (D_1 \cdot D_2) = \chi(\mathcal{O}_{D_1\cap D_2})$. Note we have the Koszul complex

$$0 \to \mathcal{O}_S(-D_1 - D_2) \to \mathcal{O}_S(-D_1) \oplus \mathcal{O}_S(-D-2) \to \mathcal{O}_S \to 0.$$

Since over the points $p \in S$ which are not contained in $D_1 \cap D_2$ the functions defining $D_1$ and $D_2$ around $p$ are units, thus the localization of the above short exact sequence is exact at $p$. However, if we localize at $p_i$, since $S$ is nonsingular it is locally Cohen-Macaulay. Thus, the $f_{1,i}, f_{2,i}$ are regular sequences in $\mathcal{O}_{S,p_i}$. Altogether, this implies we have a resolution

$$0 \to \mathcal{O}_S(-D_1 - D_2) \to \mathcal{O}_S(-D_1) \oplus \mathcal{O}_S(-D-2) \to \mathcal{O}_S \to \iota_*\mathcal{O}_{D_1\cap D_2} \to 0.$$

Additivity of the Euler characteristic implies the claim. □

Proposition 4.5 implies that we can extend our intersection formula to line bundles more generally.

**Definition 4.6.** [1] *Let $L_1, L_2 \in \mathrm{Pic}(S)$. Then define*

$$(L_1 \cdot L_2) = \chi(\mathcal{O}_S) - \chi(L_1^{-1}) - \chi(L_2^{-1}) + \chi(L_1^{-1} \otimes L_2^{-1}).$$

[1] This is also the coefficient of $n_1 n_2$ in the polynomial $\chi(X, L_1^{n_1} \otimes L_2^{n_2})$.

**Proposition 4.7.** *The intersection product is a symmetric bilinear form.*

*Proof.* The intersection product is clearly symmetric. Let's show it is bilinear. First suppose $D$ is a curve on $S$. Then

$$(D \cdot L) = \chi(\mathcal{O}_S) - \chi(-D) - \chi(L^{-1}) + \chi(-D \otimes L^{-1}).$$

Note $\chi(-D) + \chi(\mathcal{O}_D) = \chi(\mathcal{O}_S)$ and $\chi(-D \otimes L^{-1}) + \chi(\mathcal{L}^{-1}|_D) = \chi(\mathcal{L}^{-1})$. Then

$$(D \cdot L) = \chi(\mathcal{O}_D) - \chi(\mathcal{L}^{-1}|_D) = \deg_D(L|_D).$$

This implies that if $L_1$ has a global section, then $(L_1 \cdot L_2)$ is bilinear in $L_2$. Now suppose $L_1$ does not have a global section. Note $S$ is projective, so let $\mathcal{O}_S(1)$ be a very ample line bundle on $S$. By Serre's theorem, we can twist $L_1$ by a high enough power of $\mathcal{O}_S(1)$ so that $L_1$ admits a global section. Consider the expression

$$(L_1 \cdot L_2) + (L_1' \cdot L_2) - (L_1 \otimes L_1' \cdot L_2).$$

This expression is symmetric in the variables $L_1, L_1', L_2$. When $L_2$ admits a global section, this expression becomes zero. Thus, when $L_1$ or $L_1'$ admits a global section, this expression also becomes zero. If we let $L_1'$ be $\mathcal{O}_S(n)$, this implies that

$$(L_1 \cdot L_2) = (L_1(n) \cdot L_2) - (\mathcal{O}_S(n) \cdot L_2).$$

Both $L_1(n)$ and $\mathcal{O}_S(n)$ admit sections, so even if $L_1$ does not admit a section, the product is linear in the $L_2$ variable. By symmetry, altogether this implies bilinearity of the intersection product. □

In particular, since $(\mathcal{O}_S \cdot L) = 0$ for all $L$, we have $(L_1 \cdot L_2) = -(L_1^{-1} \cdot L_2)$.

**Lemma 4.8.** *Let $L$ be very ample. Let $D$ be a curve (effective divisor). Then*

$$(\mathcal{O}_S(D) \cdot L) > 0.$$

*Proof.* Let $|L| : S \to \mathbb{P}^n$ provide an embedding for $S$. Then we can pick a hyperplane $H$ of $\mathbb{P}^n$ which intersects $|L|(p)$ for a closed point $p \in Supp(D)$, but does not intersect $|L|(Supp(D))$ generically. Then $|L|^*(H)$ will be a curve $C'$ on $S$, and

$$(\mathcal{O}_S(D) \cdot L) = (C \cdot C') = \sum_{i=1}^{n} \frac{\mathcal{O}_{S,p}}{f_p, f_p'} > 0.$$

□

Note that in lemma 4.8, $L$ can be replaced by an ample divisor.

**Theorem 4.9** (Riemann-Roch for Surfaces)**.** *Let $L \in \mathrm{Pic}(S)$. Then*

$$\chi(L) = \frac{1}{2}(L \cdot L \otimes \omega_S^{-1}) + \chi(\mathcal{O}_S).$$

*Proof.* Let us simply the expression. We would like to show

$$2\chi(L) - 2\chi(\mathcal{O}_S) = (L \cdot L \otimes \omega_S^{-1})$$

$$\implies 2\chi(L) - 2\chi(\mathcal{O}_S) = -(L^{-1} \cdot L \otimes \omega_S^{-1})$$

$$\implies 2\chi(L) - 2\chi(\mathcal{O}_S) = -\chi(\mathcal{O}_S) + \chi(L) + \chi(L^{-1} \otimes \omega_S) - \chi(\omega_S).$$

$$\implies \chi(L) - \chi(\mathcal{O}_S) = \chi(L^{-1} \otimes \omega_S) - \chi(\omega_S).$$

If $L \cong \mathcal{O}_S(-D)$, then this becomes

$$\chi(\mathcal{O}_S(-D)) - \chi(\mathcal{O}_S) - \chi(\mathcal{O}_S(D) \otimes \omega_S) + \chi(\omega_S)$$

$$= -\chi(\mathcal{O}_D) - \chi((\mathcal{O}_S(D) \otimes \omega_S)|_D) = -\chi(\mathcal{O}_D) - \chi(\omega_D) = 0,$$

where the last two equalities are by the adjunction formula and by Serre duality. Thus, the Riemann Roch for surfaces holds for when $L^{-1}$ is the line bundle associated to a curve. We now show it holds generally. First, let

$$\Phi(L) = \chi(L) - \chi(\mathcal{O}_S) - \chi(L^{-1} \otimes \omega_S) + \chi(\omega_S).$$

Then we claim that

$$\Phi(L \otimes M) = \Phi(L) + \Phi(M).$$

This is a simple calculation that just requires expanding out the terms. Fix a very ample line bundle $\mathcal{O}_S(1)$. Then note by Serre's theorem that for large enough $n$, the line bundle $L^{-1}(n)$ has a nontrivial global section. Then

$$\Phi(L(-n)) = \Phi(L) + \Phi(\mathcal{O}_S(-n)),$$

and both $\Phi(L(-n)) = \Phi(\mathcal{O}_S(-n)) = 0$ by what we showed earlier. Thus, $\Phi(L) = 0$.

□

We are a priori more interested in intersection theory. But the great thing about Riemann-Roch is that it tells us to care about sheaf cohomology. Now we can use cohomological techniques to study the intersection theory on surfaces. The really important numerical characters of an invertible sheaf are

$$(L \cdot \mathcal{O}_S(1)), (L \cdot L), \text{ and } (L \cdot \omega_S).$$

## 5. Algebraic Hodge Index Theorem, Picard/Neron-Severi group of algebraic K3 surface

Let $\mathrm{Pic}^\tau(S) = \{L | (L.L') = 0, \forall L' \in \mathrm{Pic}(S)\}$ denote the subgroup of numerically trivial line bundles on $S$. Then the intersection pairing defines a non-degenerate bilinear pairing

$$Num(S) \times Num(S) \to \mathbb{Z}$$

where $Num(S) = \mathrm{Pic}(S)/\mathrm{Pic}^\tau(S)$.[2] Analyzing some consequences of the Riemann-Roch theorem for surfaces (4.9) will lead us to deduce the algebraic Hodge index theorem, which tells us something about how the intersection pairing behaves on $Num(S)$.

Let $D$ be a divisor such that $(D.D) > 0$. Then theorem 4.9 says that

$$\chi(nD) = h^0(nD) - h^1(nD) + h^2(nD) \geq \frac{1}{2}n^2(D.D) - \frac{1}{2}n(D.K) + \chi(\mathcal{O}_S)$$

$$\implies h^0(nD) + h^0(K - nD) \geq \frac{1}{2}n^2(D.D) - \frac{1}{2}n(D.K) + \chi(\mathcal{O}_S).$$

Let's analyze what happens when $n$ goes off to infinity.

(1) Suppose $n \to +\infty$.
   (a) Suppose $h^0(nD) \to \infty$. Then for any ample divisor $H$, $(nD.H) > 0$ for $n$ sufficiently large. Then $(D.H) > 0$.
   (b) Suppose $h^0(K - nD) \to \infty$. Then for any ample divisor $H$, $(K - nD.H) > 0$ for $n$ sufficiently large, so $(K.H) > n(D.H)$.

(2) Suppose $n \to -\infty$.
   (a) Suppose $h^0(nD) \to \infty$. Then for any ample divisor $H$, $(nD.H) > 0$ for $n$ sufficiently negative. Then $(D.H) < 0$.
   (b) Suppose $h^0(K - nD) \to \infty$. Then for any ample divisor $H$, $(K - nD.H) > 0$ for $n$ sufficiently negative, so $(K.H) > n(D.H)$.

It appears at first that the only situations we can have are 1(a) + 2(b), 1(b) + 2(a), and 1(b) + 2(b). But upon further inspection, we cannot have situation 1(b)+2(b). This is because for 1(b)+2(b) to hold, we must have $(D.H) = 0$ for every ample $H$. But if $(D.D) > 0$, then $D + mH$ is ample for $m$ sufficiently large. Then $(D.D + mH) = (D.D) > 0$, which is a contradiction. Thus, we must have:

**Proposition 5.1.** *Suppose $D$ is a divisor on $S$, such that $(D.D) > 0$. Then either*

- *$nD$ is effective for $n >> 0$ and $(D.H) > 0$ for all ample $H$*
- *$-nD$ is effective for $n >> 0$ and $(D.H) < 0$ for all ample $H$.*

*In particular, if $(D.D) > 0$ and $(D.H) > 0$ for some ample $H$, then we know $nD$ is effective. Analagous claim for $(D.H) < 0$.*

---

[2]A nontrivial theorem is that $Num(S)$ is finitely generated. We denote its rank as $\rho(S)$. In facts, its easy to see that over characteristic 0 it has no torsion, so it is isomorphic to $\mathbb{Z}^{\rho(S)}$.

**Lemma 5.2.** *Let $H$ be an ample divisor on $S$. For a divisor $D$ on $S$, suppose $(H.D) = 0$. Then $(D.D) \leq 0$. Moreover, $(D.D) = 0$ if and only if $D$ is numerically trivial.*

*Proof.* If $(D.D) > 0$, then either $nD$ or $-nD$ is effective, contradicting $(H.D) = 0$. Thus, we must have $(D.D) \leq 0$.

If $D$ is numerically trivial then $(D.D) = 0$ trivially.

Now suppose $(D.D) = 0$ and $(H.D) = 0$. Assume $D$ is not numerically trivial. Then there would exist a curve $C$ such that $(D.C) \neq 0$. Let $E = (H.H)C - (H.C)H$. Then $(H.nD + E) = 0$. Then

$$(nD + E.nD + E) = n^2(D.D) + 2n(D.E) + (E.E) = 2n(D.E) + (E.E) \leq 0.$$

If $(D.C) > 0$, then $(D.E) = (H.H)(D.C) > 0$ so $2n(D.E) + (E.E) > 0$ for sufficiently large $n$. If $(D.C) < 0$, then $2n(D.E) + (E.E) > 0$ for sufficiently large $-n$, contradiction. Therefore, if $H.D = 0$ and $(D.D) = 0$, then $D$ must be numerically trivial. □

**Corollary 5.3.** *Let $H$ be an ample divisor on $S$. Then*

$$(H.D)^2 \geq (H.H)(D.D)$$

*for all divisors $D$.*

*Proof.* Note $(H.D - \frac{(H.D)}{(H.H)}H) = 0$, so

$$(D - \frac{(H.D)}{(H.H)}H.D - \frac{(H.D)}{(H.H)}H) \leq 0 \implies (H.H)(D.D) \leq (H.D)^2.$$

□

The conditions for both 5.2 and corollary 5.3 for $H$ to be ample can be replaced with $(H.H) > 0$.

**Theorem 5.4** (Algebraic Hodge Index Theorem)**.** *Let $D$ be a divisor on $S$ such that $(D.D) > 0$. Then the intersecting pairing acts on $Num(S)_{\mathbb{Q}} = \mathbb{Q}D \oplus \{\mathbb{Q}D\}^{\perp}$ with signature*

$$(1, \rho(S) - 1).$$

*In other words, $(D.D) > 0$ and $(L.L) < 0$ for $L \in \{\mathbb{Q}D\}^{\perp}$.*

*Proof.* Since $(D.D) > 0$, we have a decomposition

$$Num(S)_{\mathbb{Q}} = \mathbb{Q}H \oplus \{\mathbb{Q}H\}^{\perp}$$

simply by linear algebra. If $(L.D) = 0$, then by the generalize version of lemma 5.2, we have $(L.L) \leq 0$. But if $(L.L) = 0$, then $L$ is numerically trivial. Then $(L.L) < 0$. □

The Neron-Severi group of an algebraic surface $X$ is the quotient

$$NS(S) := \frac{\mathrm{Pic}(S)}{\mathrm{Pic}^0(S)},$$

the quotient by the subgroup of line bundles which are algebraically equivalent to zero. Clearly $\mathrm{Pic}^0(S) \hookrightarrow \mathrm{Pic}^\tau(S)$, so we have surjections

$$\mathrm{Pic}(S) \to NS(S) \to Num(S)$$

for any algebraic surface $S$.

**Theorem 5.5.** *Let $S$ be an algebraic K3 surface, i.e. a complete non-singular variety (separated, geometrically integral scheme of finite type over $k$) of dimension two with $\omega_X \cong \mathcal{O}_X$ and $H^1(X, \mathcal{O}_X) = 0$.*

*Then the surjections*

$$\mathrm{Pic}(S) \cong NS(S) \cong {}^3 Num(S)$$

*are isomorphisms. Furthermore, the intersection pairing on* $\mathrm{Pic}(S)$ *is non-degenerate, even, with signature* $(1, \rho(S) - 1)$.

*Proof.* Clearly the maps are surjections. It suffices to show that the surjection $\mathrm{Pic}(S) \to Num(S)$ is injective. Suppose $L \in \mathrm{Pic}(S)$ is such that $L$ is numerically zero. Thus, $(L.L') = 0$ for all line bundles $L' \in \mathrm{Pic}(S)$. Note by Riemann-Roch for surfaces we have

$$\chi(L) = h^0(L) - h^1(L) + h^2(L) = \frac{1}{2}(L.L - K) + \chi(\mathcal{O}_S) = \chi(\mathcal{O}_S).$$

Note $\chi(\mathcal{O}_S) = h^0(\mathcal{O}_S) - h^1(\mathcal{O}_S) + h^2(\mathcal{O}_S) = 2$. And by Serre duality, $h^2(L) = h^0(L^\vee)$. Then

$$h^0(L) - h^1(L) + h^0(L^\vee) = 2.$$

Note if $L$ is numerically zero, then so is $L^\vee$. But if they are numerically zero, then they cannot admit a section, since for some ample $H$, if for example $L$ admits a section, then $(L.H) > 0$. But this would imply $-h^1(L) = 2$, contradiction. Thus, $h^0(L)$ and $h^0(L^\vee)$ must admit sections, which implies that $L$ must be trivial. *Note that this implicitly depends on knowing the existence of an ample line bundle on an algebraic K3 surface.*

The fact that the pairing is non-degenerate and has signature $(1, \rho(S) - 1)$ follows from the Hodge index theorem.

To see that the pairing is even, note

$$\chi(L) = \frac{1}{2}(L.L - K) + 2 \implies 2\chi(L) - 2 = (L.L).$$

□

In particular, there is no torsion in $\mathrm{Pic}(S)$ for a K3 surface $S$. It is a free module. Although, one does not need the above isomorphism to show this. An easy way to see there is no torsion is the following. If $L \in \mathrm{Pic}(S)$ is such that $L^{\otimes n} \cong \mathcal{O}_S$ for some $n$, then note by Riemann-Roch we must have

$$\chi(L) = 2 \implies h^0(L) - h^1(L) + h^0(L^\vee) = 2.$$

[3] This isomorphism does not hold for general complex K3 surfaces.

Then at least one of $L$ or $L^\vee$ is effective. WLOG suppose $L$ admits a section $s$. Then $s^n$ is a section of $L^{\otimes n}$, and this forces $s$ to be an everywhere nonzero trivializing section of $L$, so $L \cong \mathcal{O}_S$.

Let me make a comment on the Hodge index theorem from the analytic point of view. Suppose $S$ is a compact complex surface with a positive line bundle. In particular, it is a Kahler surface and admits an embedding into projective space by Kodaira's embedding theorem. In particular, by Chow's theorem, $S$ is actually an algebraic surface.

The sheaf cohomology of the exponential exact sequence on $S$ yields

$$\cdots \to H^1(S; \mathcal{O}_S^\times) \to^{c_1} H^2(S; \mathbb{Z}) \to H^2(S; \mathcal{O}_S) \to \cdots,$$

where $c_1(\mathcal{L}) \in H^2(S; \mathbb{Z})$ sends a line bundle $\mathcal{L} \in H^1(S, \mathcal{O}_S^\times)$ to a positive (1,1) form. In particular, every element of $H^{1,1}(S) \cap H^2(S; \mathbb{Z})$ is the first Chern class of some line bundle by the Lefschetz theorem (Hodge conjecture). There is a natural

$$H^2(S; \mathbb{Z}) \times H^2(S; \mathbb{Z}) \to \mathbb{Z}$$

where

$$(\alpha, \beta) \mapsto \int_S \alpha \wedge \beta.$$

This extends naturally to the intersection pairing on $H^2(S; \mathbb{R})$.

**Theorem 5.6** (Analytic Hodge Index Theorem)**.** *Let $S$ be a compact Kahler surface. The intersection pairing*

$$H^2(S; \mathbb{R}) \times H^2(S; \mathbb{R}) \to \mathbb{R}, \text{ where } ([\alpha], [\beta]) \mapsto \int_S \alpha \wedge \beta$$

*has signature* $(2h^{2,0}(S) + 1, h^{1,1}(S) - 1)$.

In particular, $H^2(S; \mathbb{C}) \cong H^{2,0}(S) \oplus H^{1,1}(S) \oplus H^{0,2}(S)$ by the Hodge decomposition. Then the intersection pairing on the subspace

$$(H^{2,0}(S) \oplus H^{0,2}(S)) \cap H^2(X; \mathbb{R}) = \{\alpha^{2,0} + \alpha^{0,2} | \overline{\alpha^{0,2}} = \alpha^{2,0}\}$$

is positive-definite by Hodge-Riemann bilinear relations, contributing $2h^{2,0}(S)$ to the signature.

The rest of the signature comes from analyzing the form on $H^{1,1}(X)$. This has Lefschetz decomposition

$$H^{1,1} \cong H_0^{1,1} \oplus \mathbb{C}\omega,$$

The intersection pairing is positive definite on $\mathbb{C}\omega$, and is negative-definite on $H_0^{1,1}$. Something one can show is that the primitive cohomology $H_0^{1,1}$ is isomorphic to $\{\alpha \in H^{1,1} | \int_X \omega \wedge \alpha = 0\} \subseteq H^2(X; \mathbb{R})$. Thus, we see that there is a correspondence between $Num(S)_\mathbb{R}$ and $H^{1,1}$, a divisor $D$ such that $(D.D) > 0$ and a positive (1,1) form $\omega$, and $\{\mathbb{R}D\}^\perp$ and $H_0^{1,1}$.

**Remark 5.7.** *As we have just seen, for projective complete algebraic surfaces, the algebraic Hodge index theorem and the analytic Hodge index theorem (taking the analytification of the algebraic surface, or vice versa take the compact complex manifold with a positive line bundle and it will be algebraic by Chow's theorem) hold, and the two intersection pairings coincide.*

*In general, for compact complex surfaces, the analytic Hodge index theorem and the algebraic Hodge index theorem do not hold. For one, a compact complex surface is algebraic if and only if there exists a line bundle $L$ such that $(L.L) > 0$. So without projectivity, the algebraic Hodge index theorem does not hold.*

*However, for compact Kahler manifolds, the analytic Hodge index theorem still holds. In particular, for complex K3 surfaces which are not projective, the algebraic Hodge index theorem does not hold, but the analytic Hodge index theorem does hold, since all complex K3 surfaces happen to be Kahler (difficult result).*

## 6. Hirzebruch-Riemann-Roch formula and Hodge Diamond of K3 surface

The Hirzebruch-Riemann-Roch formula generalizes the Riemann-Roch theorem for curves and surfaces. It allows us to compute the Euler characteristic of any vector bundle and relates it to intersection-theoretic information on the variety.

**Theorem 6.1** (Hirzebruch-Riemann-Roch)**.** [4] *Let $X$ be a complete nonsingular variety. For any vector bundle $E$, we have*

$$\chi(E) = \int_X ch(E)td(T_X).$$

Let us explain what $\int_X ch(E)td(T_X)$ means; afterwards we do some computations. Let us fix $E$ to be a rank $r$ vector bundle. Let $A^*(X)$ denote the Chow ring of $X$. By the splitting principle, we can think of the Chern class of $E$ as

$$1 + c_1(E) + c_2(E) + \cdots = c(E) = \prod_{i=1}^{r}(1 + \alpha_j),$$

where $\alpha_j$ are called the Chern roots of $E$. Then the Chern character $ch(E)$ is defined to be

$$ch(E) = \sum_{i=1}^{r} e^{\alpha_i} = \sum_{i=1}^{r}\sum_{t=0}^{\infty}\frac{\alpha_i^t}{t!}.$$

Note that since $ch(E)$ is symmetric with respect to the Chern roots, it can be written as a polynomial in terms of the Chern classes $c_i(E)$, so $ch(E) \in A^*(X)$. More specifically,

$$ch(E) = \mathrm{rk}(E) + c_1(E) + \frac{1}{2}(c_1^2 - 2c_2) + \frac{1}{6}(c_1^3 - 3c_1c_2 + 3c_3) + \cdots$$

[4]Also holds for compact complex manifolds.

The Todd class $td(E)$ is defined to be

$$td(E) = \prod_{i=1}^{r} Q(\alpha_i)$$

where $Q(x) = \frac{x}{1-e^{-x}} = 1+\frac{1}{2}x+\sum_{i=1}^{\infty}\frac{B_{2i}}{(2i)!}x^{2i} = 1+\frac{x}{2}+\frac{x^2}{12}-\frac{x^4}{720}+\cdots$ and $B_{2i}$ denotes the $(2i)$-th Bernoulli number. Since $td(E)$ is symmetric, it is also a polynomial in terms of the Chern classes $c_i(E)$, so $td(E) \in A^*(X)$. More specifically, one can work backwards to show:

$$td(E) = 1 + \frac{1}{2}c_1 + \frac{1}{12}(c_1^2 + c_2) + \frac{1}{24}c_1c_2 + \frac{1}{720}(-c_1^4 + 4c_1^2c_2 + 3c_2^2 + c_1c_3 - c_4) + \cdots .$$

Finally, the group homomorphism

$$\int_X : A^*(X) \to \mathbb{Z}$$

counts the multiplicity of 0-dimensional cycles, and evaluates all positive-dimensional cycles as zero.

Remark: the todd class can be computed from the Chern character and Chern classes.

**Proposition 6.2.** *The todd class $td(E)$ can be computed in terms of Chern characters. More specifically,*

$$\sum_{p=0}^{r}(-1)^p ch(\bigwedge^p E^*) = \frac{c_r(E)}{td(E)}.$$

*Proof.* Note that

$$\frac{c_r(E)}{td(E)} = \frac{\prod_{j=1}^r \alpha_j}{\prod_{j=1}^r \frac{\alpha_j}{1-e^{-\alpha_j}}} = \prod_{j=1}^{r}(1-e^{-\alpha_j}).$$

Furthermore, note that $c(E^*) = \prod_{j=1}^r(1-\alpha_j)$ and

$$c(\bigwedge^p E) = \prod_{1\le j_1<\cdots<j_p\le r}(1+\alpha_{j_1}+\cdots+\alpha_{j_p}),$$

which means that

$$\sum_{p=0}^{r}(-1)^p ch(\bigwedge^p E^*) = \sum_{o=0}^{r}(-1)^p \sum_{1\le j_1<\cdots<j_p\le r} e^{-\alpha_{j_1}-\cdots-\alpha_{j_p}} = \prod_{j=1}^{r}(1-e^{-\alpha_j}) = \frac{c_r(E)}{td(E)}.$$

∎

Let's use Hirzebruch-Riemann-Roch on the easiest examples that comes to mind.

**Example 6.3.** *Let $X = \mathbb{P}^n$. First, what is $td(T_{\mathbb{P}^n})$? Note the Euler exact sequence*

$$0 \to \mathcal{O}_{\mathbb{P}^n} \to \mathcal{O}_{\mathbb{P}^n}(1)^{\oplus n+1} \to T_{\mathbb{P}^n} \to 0$$

*implies*

$$c(T_{\mathbb{P}^n}) = (1+\xi)^{n+1},$$

*where $\xi \in A^*(\mathbb{P}^n) \cong \mathbb{Z}[\xi]/(\xi^{n+1})$ is the hyperplane class. Then*

$$td(T_{\mathbb{P}^n}) = (\frac{\xi}{1-e^{-\xi}})^{n+1}$$

*and*

$$ch(\mathcal{O}_{\mathbb{P}^n}(1)) = e^{\xi}.$$

*Suppose $n = 2$. Then, for example, we can verify that*

$$\chi(\mathcal{O}_{\mathbb{P}^2}(1)) = \int_{\mathbb{P}^2} e^{\xi}(\frac{\xi}{1-e^{-\xi}})^3 = \int_{\mathbb{P}^2}(1+\xi+\frac{\xi^2}{2})(1+\frac{\xi}{2}+\frac{\xi^2}{12})^3 = \int_{\mathbb{P}^2}(\frac{1}{2}+\frac{3}{2}+\frac{3}{4}+\frac{3}{12})\xi^2 = 3.$$

We now use the Hirzebruch-Riemann-Roch formula to determine the Hodge diamond of every K3 surface.

**Example 6.4.** *Let $S$ be a K3 surface. We know $h^{0,0} = H^0(\mathcal{O}_S) = 1$, and by Serre duality, $h^{2,2} = 1$. Since $\omega_S \cong \mathcal{O}_S$, we have $h^{0,2} = h^{2,0} = 1$. Also, since $H^1(\mathcal{O}_S) = 0$, we have $h^{0,1} = h^{1,0} = h^{2,1} = h^{1,2} = 0$ by Serre duality and symmetry. It remains to determine $h^{1,1}$. Note that*

$$\chi(T_S) = h^0(T_S) - h^1(T_S) + h^2(T_S),$$

*and $h^0(T_S) = h^2(\Omega_S) = 0$ and $h^2(T_S) = h^0(\Omega_S) = 0$ and $h^1(T_S) = h^1(\Omega_S)$ by Serre duality and the fact that $\omega_S \cong \mathcal{O}_S$. Thus, by Hirzebruch-Riemann-Roch, we have*

$$-h^{1,1} = \chi(T_S) = \int_S ch(T_S)td(T_S).$$

*Note that $\bigwedge^2 T_S \cong \mathcal{O}_S$. This implies that if $\alpha_1, \alpha_2$ are the Chern roots of $T_S$, then actually $\alpha_1 = -\alpha_2$. Then let $\alpha, -\alpha$ denote the Chern roots of $T_S$. Then we immediately see that $c_1(T_S) = 0$. Thus,*

$$ch(T_S) = 2 - c_2(T_S) \text{ and } td(T_S) = 1 + \frac{1}{12}c_2(T_S).$$

*Then*

$$\int_S ch(T_S)td(T_S) = \int_S (2 - c_2(T_S))(1 + \frac{1}{12}c_2(T_S)) = \frac{-5}{6}\int_S c_2(T_S).$$

*How do we determine $\int_S c_2(T_S)$? Hirzebruch-Riemann-Roch tells us that*

$$\chi(\mathcal{O}_S) = 2 = \int_S ch(\mathcal{O}_S)td(T_S) = \int_S td(T_S) = \frac{1}{12}\int_S c_2(T_S) \implies \int_S c_2(T_S) = 24.$$

*Then*

$$h^{1,1} = \frac{5}{6}(24) = 20.$$

*So the Hodge diamond of every K3 surface is*

$$\begin{array}{ccccc} & & 1 & & \\ & 0 & & 0 & \\ 1 & & 20 & & 1 \\ & 0 & & 0 & \\ & & 1 & & \end{array}$$

Implicitly, based on the tools that we used, we have only proved this Hodge diamond for algebraic K3 surfaces over $\mathbb{C}$. But it actually holds for non-projective complex K3 surfaces, and for algebraic K3 surfaces over arbitrary fields (although, the proof is much more difficult).

**Remark 6.5.** *Expanding out the relevant Chern characters and Todd classes in terms of Chern classes, with sufficient focus, it is relatively straightforward to verify that the Riemann-Roch theorems over curves and surfaces are special cases of the Hirzebruch-Riemann-Roch theorem.*

Note Hirzebruch-Riemann-Roch is a special case of the more general Grothendieck-Riemann-Roch theorem: let $X$ be a complete nonsingular variety. On $X$, denote $K^0(X)$ to be the free group generated by vector bundles on $X$, modulo the relation $[F] + [G] = [H]$ for vector bundles $F, G, H$ related by a short exact sequence:

$$0 \to F \to H \to G \to 0.$$

Note $K^0(X)$ has the structure of a ring, as we define $[F] \cdot [G] = [F \otimes G]$. Note this is well-defined since short exact sequences are preserved by tensor products by locally free sheaves.

Next, we define $K_0(X)$ to be the free group generated by coherent sheaves on $X$, modulo $[F] + [G] = [H]$ for coherent sheaves $F, G, H$ related by a short exact sequence:

$$0 \to F \to H \to G \to 0.$$

Note $K_0(X)$ has the structure of a $K^0(X)$-module, where the multiplication by $K^0(X)$ is well-defined because, again, tensoring by locally free sheaves preserves exactness.

There is a map of $K^0(X)$-modules $K^0(X) \to K_0(X)$, sending a vector bundle $\mathcal{E}$ to its equivalence class in $K_0(X)$. In general, this is not an isomorphism, but when $X$ is nonsingular it is because by a result of Serre, every coherent sheaf can be finitely resolved by locally free sheaves:

$$0 \to \mathcal{E}^n \to \cdots \to \mathcal{E}^0 \to \mathcal{F} \to 0,$$

so we can define an inverse $K_0(X) \to K^0(X)$ by sending $[F] \in K_0(X)$ to $\sum_{i=0}^n (-1)^i [\mathcal{E}^i] \in K^0(X)$.

Thus, given that $X$ is smooth, we simply denote $K_0(X)$ and $K^0(X)$, with their identification to each other, as $K(X)$. Using pullback of vector bundles, note $K(-)$ defines a contravariant functor

$$K(-) : Sch_k \to \text{CommutativeRings}.$$

**Proposition 6.6.** *Let $f : X \to Y$ be a flat morphism of nonsingular complete varieties. Let $A(-)_{\mathbb{Q}} := A(-) \otimes_{\mathbb{Z}} \mathbb{Q}$ denote the rational chow ring functor. Then there is a commutative diagram:*

$$\begin{tikzcd} K(X) \arrow[r, "ch(-)"] & A(X)_{\mathbb{Q}} \\ K(Y) \arrow[u, "f^*"] \arrow[r, "ch(-)"'] & A(Y)_{\mathbb{Q}} \arrow[u, "f^*"'] \end{tikzcd}$$

*since $ch(-)$ is comprised of Chern classes and these commute with flat pullback.*

Note that actually $ch(-)$ defines an isomorphism between $K(X) \cong A(X)_{\mathbb{Q}}$. So studying vector bundles on a nonsingular $X$ can say something about the non-torsion cycles. The content of the Grothendieck-Riemann-Roch theorem is that we also have a commutative diagram for proper pushforwards.

**Theorem 6.7** (Grothendieck-Riemann-Roch)**.** *Let $f : X \to Y$ be a proper map of nonsingular varieties with $X$ quasiprojective. Then there is a commutative diagram*

$$\begin{tikzcd} K(X) \arrow[r, "ch(-)td(T_X)"] \arrow[d, "f_*"'] & A(X)_{\mathbb{Q}} \arrow[d, "f_*"] \\ K(Y) \arrow[r, "ch(-)td(T_Y)"'] & A(Y)_{\mathbb{Q}} \end{tikzcd}$$

*where $f_* : K(X) \to K(Y)$ is defined to be*

$$f_*(\mathcal{E}) = \sum (-1)^k [R^k f_* \mathcal{E}].$$

When we let $Y = \text{Spec}(k)$, we see that we recover the Hirzebruch-Riemann-Roch theorem.

## 7. Serre's GAGA Principle, Chow's theorem, algebraic vs. complex K3 surfaces

A deep study of algebraic K3 surfaces inevitably involves the theory of non-projective complex K3 surfaces. For example, the "twistor space construction," which is used in proving the Global Torelli Theorem for K3 surfaces, involves non-projective complex K3 surfaces. Thus, the study of algebraic K3 surfaces demands not only algebraic techniques but also analytic techniques. Serre's famous GAGA paper [Ser56], which exposes the intimate relationship between algebraic and analytic geometry.

For any locally ringed space $(X, \mathcal{O}_X)$ such that, locally, $X$ is isomorphic as locally ringed spaces to a Zariski locally closed subset of $\mathbb{A}^n_{\mathbb{C}}$, there is an analytic space $(X^h, \mathcal{O}^h_X)$ which we call its analytification, with a natural map

$$(X^h, \mathcal{O}^h_X) \to (X, \mathcal{O}_X).$$

Note there is a natural map $\mathcal{O}_{X,p} \to \mathcal{O}^h_{X,p}$. In particular, if $m_p$ and $m^h_p$ denote their corresponding maximal ideals, which are the ideals of germs of algebraic and holomorphic functions, respectively, which vanish at $p$, and we take their respective completions with respect to $m_p$ and $m^h_p$, then

$$\widehat{\mathcal{O}_{X,p}} \cong \widehat{\mathcal{O}^h_{X,p}}.^5$$

Intuitively, this makes sense. The simplest model of this phenomenon is on $\mathbb{C}^n$. The collection of germs of polynomial functions around the origin is $\mathbb{C}[Z_0, \cdots, Z_n]_{(Z_0,\cdots,Z_n)}$. The collection of germs of holomorphic functions around the origin is $\mathbb{C}\{Z_0, \cdots, Z_n\}$, which is the ring of germs of convergent power series (so, infinite expressions are allowed if it converges on some neighborhood of the origin). The respective completions of both of these is the formal power series ring $\mathbb{C}[[Z_0, \cdots, Z_n]]$.[6]

Now, if $F$ is a $\mathcal{O}_X$-module, then we can obtain its analytification $F^h$ by first taking the inverse image sheaf along $X^h \to X$. More concretely, one takes the etale escape $\epsilon(F)$, which is the set $\bigsqcup_{p\in X} F_p$ endowed with the topology generated by open sets $\bigsqcup_{p\in U}[s]_p$, for every open set $U$ and section $s \in F(U)$. Then if we take the fiber product in the category of topological spaces

$$\iota : X^h \times_X \epsilon(F),$$

which inherits the subspace topology from $X^h \times \epsilon(F)$, we can take the sheaf of sections to obtain a sheaf $F'$ over $X^h$. (In general, this is exactly how you get the inverse image sheaf along a continuous map $Y \to X$). One sees that $F'_p = F_p$. So the difference between these sheaves is the topology of their corresponding etale espaces. Finally,

$$F^h := F' \otimes_{\mathcal{O}'_X} \mathcal{O}^h_X = \iota^*(F).$$

In partiular, note that if $U$ is a Zariski open of $X$, then we can injectively pull back any section of $F(U)$ to a section of $F'(U)$. Furthermore, $F' \to F^h$ is an injective morphism of sheaves over $X^h$, since the morphism on stalks is injective.

[5] More specifically, they are a "flat couple. Many of the properties in the GAGA paper boil down to commutative algebra that is shown by properties of flat couples.

[6] One can use this idea to say, for example, that at the origin, the plane nodal cubic "looks like" $XY = 0$ at the origin.

This analytification functor is an exact functor from the category of $\mathcal{O}_X$-modules to the category of $\mathcal{O}_{X^h}$-modules. In particular, we have the following GAGA theorems.

**Theorem 7.1** (GAGA 1). *Let $X$ be a closed projective $\mathbb{C}$-variety, and $F \in Coh(X)$. Then*

$$H^q(X, F) \cong H^q(X^h, F^h),$$

*such that the isomorphism is functorial with respect to long exact sequences of coherent sheaves.*

*Proof.* Note that on a Zariski open $U \subset X$, we have a natural map $F(U) \to F^h(U)$. If we pick a finite cover $\underline{U}$ comprised of Zariski opens $U_i$, then we obtain a map of Cech complexes

$$\check{C}^*(\underline{U}, F) \to \check{C}^*(\underline{U}, F^h).$$

Taking direct limits over covers of Zariski opens, we obtain a map on sheaf cohomologies

$$H^q(X, F) \to H^q(X^h, F^h).$$

Also, note that analytification commutes with pushforward (extension by zero), and the pushfoward of a coherent sheaf is coherent and the analytification of a coherent sheaf is coherent [Ser56]. Thus, it suffices to prove the statement over $\mathbb{P}^n_{\mathbb{C}}$. There are three steps:

(1) Inductively show it holds for all $\mathcal{O}_{\mathbb{P}^n}(k)$. If one shows it holds for all $\mathcal{O}_{\mathbb{P}^{n-1}}(k)$, then use twists of the SES

$$0 \to \mathcal{O}_{\mathbb{P}^n}(-1) \to \mathcal{O}_{\mathbb{P}^n} \to \mathcal{O}_H \to 0,$$

where $H$ is a hyperplane, to show the statemenet inductively. Use five lemma on the map between long exact sequences.

(2) Show that any coherent sheaf is surjected onto by some direct sum of $\mathcal{O}_{\mathbb{P}^n}(k)$'s, and note the kernel is also coherent. Use five lemma on the map between long exact sequences.

□

**Theorem 7.2** (GAGA 2). *Let $X$ be a closed projective $\mathbb{C}$-variety, and let $X^h$ denote its analytification. Then the category of coherent algebraic sheaves over $X$ is isomorphic to the category of coherent analytic sheaves over $X^h$.*

*Proof.* Here is a sketch of the steps.

(1) First show that morphisms in each category correspond to each other. If $F \to G$ is a morphism between algebraic coherent sheaves, then analytification induces $F^h \to G^h$. But on the other hand, we need to show that any analytic morphism $F^h \to G^h$ comes from an algebraic morphism. This is done by examining sheaf homs. Note

$$H^0(X, \mathrm{Hom}_{\mathcal{O}_X}(F, G)) \cong H^0(X^h, [\mathrm{Hom}_{\mathcal{O}_X}(F, G)]^h)$$

by theorem 7.1. Then we claim

$$[\mathrm{Hom}_{\mathcal{O}_X}(F,G)]^h \cong \mathcal{H}om_{\mathcal{O}_X^h}(F^h,G^h),$$

which can be verified by noting there is a natural morphism and the stalks are isomorphic by $(\mathcal{O}_{X,p},\mathcal{O}_{X,p}^h)$ being an exact couple. Then

$$H^0(X,\mathrm{Hom}_{\mathcal{O}_X}(F,G)) \cong H^0(X^h,\mathcal{H}om_{\mathcal{O}_X^h}(F^h,G^h)).$$

(2) Every analytic coherent sheaf comes is the analytification of an algebraic coherent sheaf. This comes from proving Serre's Thoerem B, that every analytic coherent sheaf, up to a sufficient twist, is generated by global sections. Applying this also to the kernel, one obtains an exact sequence

$$A \to B \to F \to 0,$$

where $A,B$ are direct sums of $\mathcal{O}_{X^h}(k)$'s. We know that $A,B$, and $A \to B$ come from the analytification of algebraic objects, implying that $F$ does as well.

(3) Uniqueness, up to isomorphism, follows from uniqueness of morphisms (step 1).

□

**Corollary 7.3** (Chow's Theorem)**.** *Let $Z$ be a closed analytic subspace of $\mathbb{P}^n_{\mathbb{C}}$. Then $Z$ is algebraic.*

*Proof.* The sheaf of ideals $I^h$ of $Z$ is coherent. Then there is a corresponding algebraic coherent sheaf $I$. One has that the analytification of the short exact sequence

$$0 \to I \to \mathcal{O}_{\mathbb{P}^n} \to Q \to 0$$

is

$$0 \to I^h \to \mathcal{O}_{\mathbb{P}^n}^h \to i_*\mathcal{O}_Z \to 0.$$

Then $Q^h \cong \iota_*\mathcal{O}_Z$, and they have the same support, namely $Z$. Since the support of $Q$ is Zariski-closed, this implies $Z$ is Zariski closed thus algebraic. □

**Definition 7.4.** *An algebraic K3 surface is a complete nonsingular variety of dimension two such that*

$$\omega_{X/k} \cong \mathcal{O}_X, H^1(X,\mathcal{O}_X) = 0.$$

**Definition 7.5.** *A complex K3 surface is a compact connected complex manifold of dimension two such that*

$$\omega_X \cong \mathcal{O}_X, H^1(X,\mathcal{O}_X) = 0.$$

From this, noting that $\Omega^h_{X/\mathbb{C}} \cong \Omega_{X^h}$, we see every complex algebraic K3 surface corresponds, via GAGA, to complex K3 surface which are projective. And Chow's thoerem says that all projective complex K3 surfaces come from an algebraic K3 surface over $\mathbb{C}$. But there are many (in fact, most) complex K3 surfaces which are not projective.[7]

When working with complex K3 surfaces, an important tool that we can utilize is the exponential exact sequence:

$$0 \to \underline{Z}_X \to \mathcal{O}_X \to^{\exp} \mathcal{O}_X^\times \to 0.$$

In particular, if one begins with an algebraic K3 surface, one can mix algebraic and analytic techniques through the identification of the $H^q(X^h, \mathcal{O}_X^h)$ terms in the cohomology LES via GAGA.

## 8. Coverings and Hodge Diamond of Cyclic Covering

There are often scenarios in which a K3 surface arises as some kind of ramified covering map. We discuss some basic covering theory over $\mathbb{C}$, following [BVP04, Chapter 1, Section 16], and afterwards do a short example.

Let $\pi : X \to Y$ be a finite surjective proper holomorphic map between connected complex manifolds. Let

$$S = \{x \in X \,|\, \operatorname{rk} d\pi_x < \dim X\}.$$

Then $\pi : (X \setminus S) \to (Y \setminus \pi(S)$ is a topological covering. For each $x \in X$, we have a well-defined notion of a local degree $e_x$, and when $e_x \geq 2$, we say $\pi$ is ramified at $x$. We have $x \in S \iff e_x \geq 2$. Note that pulling back differential forms gives a map

$$\pi^*\omega_Y \to \omega_X$$

which is the canonical section of the line bundle

$$\mathrm{Hom}_{\mathcal{O}_X}(\pi^*\omega_Y, \omega_X).$$

The zero divisor $R$ associated to this section is called the ramification divisor. Note we have that

$$\omega_X = \pi^*\omega_Y \otimes \mathcal{O}_X(R).$$

**Lemma 8.1.** *If $R = \sum r_j R_j$ where $R_j$ are the irreducible components, then $r_j = e_j - 1$ where $e_j$ is the local degree at any point $x \in R_j$ which is smooth on $R_{red}$ and for which $y = \pi(x)$ is smooth on $B_j = \pi(R_j)$.*

*Proof.* Work locally and pull back a top form on $Y$. □

This gives the Riemann-Hurwitz formula,

$$\omega_X \cong \pi^*\omega_Y \otimes \mathcal{O}_X\big(\sum (e_j - 1)R_j\big).$$

Restricting our manifolds to Riemann surfaces recovers the usual Riemann-Hurwitz formula.

[7]It is in fact true, however, that all complex K3 surfaces are Kahler. Thus, we can always employ tools such as the Hodge decomposition.

**Lemma 8.2.** *Let $\pi : X \to Y$ be a covering of degree $d$. Suppose $\mathcal{L}$ is a line bundle on $Y$ such that $\pi^*\mathcal{L} \cong \mathcal{O}_X$. Then $\mathcal{L}^{\otimes d} \cong \mathcal{O}_Y$.*

Now we turn to a very nice construction called a cyclic covering, which under some hypotheses produces a covering map branched along some specified smooth effective divisor. Let $Y$ be a connected complex manifold and $D$ a divisor on $Y$ which is either effective or zero. Suppose we have a line bundle $\mathcal{L}$ on $Y$ such that

$$\mathcal{O}_Y(D) \cong \mathcal{L}^{\otimes d}.$$

Let $s$ be a global section of $\mathcal{O}_Y(D)$ corresponding to $D$. Now let $L$ denote the actual complex manifold such that the sheaf of sections of

$$p : L \to Y$$

is the locally free sheaf $\mathcal{L}$ of rank one. Pulling back $\mathcal{L}$, we obtain $p^*\mathcal{L}$ over $L$. Let $t \in H^0(L, p^*\mathcal{L})$ denote the tautalogical section. In other words, if $U$ is an open subset of $Y$ which locally trivializes $L$, so that $p^{-1}(U) \cong U \times \mathbb{C}$, then $p^*\mathcal{L} \to L$ is locally trivialized, and looks like $(U \times \mathbb{C}) \times \mathbb{C} \to (U \times \mathbb{C})$. Locally here, $t$ is the section $(U \times \mathbb{C}) \to (U \times \mathbb{C}) \times \mathbb{C}$ which maps $(x, \lambda) \mapsto (x, \lambda) \times \lambda$.

Consider the global section $p^*s - t^d \in H^0(L, p^*\mathcal{O}_Y(D))$. If $D \neq 0$ and reduced, then $X$ is an irreducible normal analytic subspace of $L$, and $\pi : X \to Y$ is a degree $d$ ramified covering of $Y$ branched along $D$. It is called the $d$-cyclic covering of $Y$ branched along $D$, determined by $\mathcal{L}$. $X$ has at most singularities over singular points of $D$. Thus, $X$ is smooth when $D$ is reduced and smooth.

**Lemma 8.3.** *Let $\pi : X \to Y$ be the d-cyclic covering of $Y$ branched along a smooth divisor $D$ and determined by $\mathcal{L}$, where $\mathcal{L}^{\otimes d} \cong \mathcal{O}_Y(D)$. Let $Q$ denote the reduced divisor $\pi^{-1}(D) \subset X$.*

(1) $\pi^*\mathcal{L} \cong \mathcal{O}_X(Q)$.
(2) $\pi^*\mathcal{O}_Y(D) \cong dQ$.
(3) $\omega_X \cong \pi^*(\omega_Y \otimes \mathcal{L}^{\otimes(d-1)}$.

*Proof.* (1) The tautological section $t$ restricted to $X$ is a global section of $\pi^*\mathcal{L}$. The effective divisor associated to $t$ on $X$ is when $p^*(s) = 0$, which is $Q$.

(2) The restriction of the section $p^*(s)$ to $X$ is a section of $\pi^*\mathcal{O}_Y(D)$. The effective divisor associated to this section is when $p^*(s) = t^n = 0$. We see then that the effective divisor is exactly $dQ$.

(3) Apply Riemann Hurwitz 8.1.

□

**Lemma 8.4.** *Let $\pi : X \to Y$ be a d-cyclic covering given by $\mathcal{L} \in \mathrm{Pic}(Y)$, branched along a reduced smooth effective divisor $D$ such that $\mathcal{L}^{\otimes d} \cong \mathcal{O}_Y(D)$. Then*

$$\pi_* \mathcal{O}_X \cong \bigoplus_{j=0}^{d-1} \mathcal{L}^{-j}$$

*Proof.* Let's first think about what functions look like on the complex manifold $L$ associated to $\mathcal{L}$. We have the projection

$$p : L \to Y$$

and given $U \subset Y$ which locally trivializes $\mathcal{L}$, we have a chart $U \times \mathbb{C} \subset L$. Then a function on $U \times \mathbb{C}$ looks like

$$\sum_{k=0}^{\infty} f_k t^k, \text{ where } f_k \in \mathcal{O}_Y(U).$$

Now suppose we have an open subset $V \subset L$. What will the holomorphic functions on $V$ look like? They will be glued together by the holomorphic functions on charts covering $V$ which are compatible with respect to the transition functions. If for example the transition function $(U_1 \times U_2) \times \mathbb{C} \to (U_1 \times U_2) \times \mathbb{C}$ is given by multiplication by $\lambda(z)$, then a holomorphic function on $U_1 \times \mathbb{C}$

$$\sum_{k=0}^{\infty} f_k t^k, \text{ where } f_k \in \mathcal{O}_Y(U_1)$$

and a holomorphic function on $U_2 \times \mathbb{C}$

$$\sum_{k=0}^{\infty} g_k t^k, \text{ where } g_k \in \mathcal{O}_Y(U_2)$$

are compatible if

$$f_k = g_k \cdot \lambda^k$$

which implies that the data of the holomorphic functions on $L$ can be as a subsheaf (because we need convergence) of $\bigoplus_{j=0}^{\infty} \mathcal{L}^{-j}$. But note that the holomorphic functions on $X$ come from restrictions of the holomorphic functions on $L$, and $p^* s - t^d = 0$ on $X$, and recall $s \in H^0(Y, \mathcal{L}^{\otimes d})$. Altogether, this implies that

$$\pi_* \mathcal{O}_X \cong \bigoplus_{j=0}^{d-1} \mathcal{L}^{-j}.$$

□

Let us now compute the Hodge diamond of a degree d covering. Suppose we have a Kahler manifold $X$ and $Z \subset X$ a smooth hypersurface. Let $\mathcal{L}$ be a line bundle on $X$ such that $\mathcal{L}^{\otimes d} \cong \mathcal{O}_X(Z)$, and let

$$\pi : Y \to X$$

be the $d$-cyclic covering constructed from $\mathcal{L}$. Let $Z' := \pi^{-1}(Z)^{\mathrm{red}}$. Note $\pi$ defines an isomorphism from $Z'$ to $Z$.

First, note that

$$\pi^*\Omega_X^p(\log Z) \cong \Omega_Y^p(\log Z').$$

Then the projection formula implies that

$$\pi_*(\Omega_Y^p(\log Z')) \cong \pi_*(\mathcal{O}_Y \otimes \pi^*\Omega_X^p(\log Z)) \cong \pi_*\mathcal{O}_Y \otimes \Omega_X^p(\log Z)) \cong (\bigoplus_{j=0}^{d-1} \mathcal{L}^{-j}) \otimes \Omega_X^p(\log Z).$$

Since $\pi$ is finite, all the higher direct images vanish. This implies[8] the isomorphisms

$$H^q(Y, \Omega_Y^p(\log Z')) \cong H^q(X, \pi_*\Omega_Y^p(\log Z')) \cong \bigoplus_{j=0}^{d-1} H^q(X, \Omega_X^p(\log Z) \otimes \mathcal{L}^{-j}).$$

Now this doesn't quite give us the Hodge numbers of the cyclic covering. We'll need to relate these logarithmic forms to the non-logarithmic forms. In particular, consider the fact that we have the residue short exact sequences

$$0 \to \Omega_X^p \to \Omega_X^p(\log Z) \to \Omega_Z^{p-1} \to 0,$$
$$0 \to \Omega_Y^p \to \Omega_Y^p(\log Z') \to \Omega_{Z'}^{p-1} \to 0.$$

Adding $(\bigoplus_{j=1}^{d-1} \mathcal{L}^{-j}) \otimes \Omega_X^p(\log Z)$ to the first residue SES yields:

$$0 \to \Omega_X^p \oplus [(\bigoplus_{j=1}^{d-1} \mathcal{L}^{-j}) \otimes \Omega_X^p(\log Z)] \to \Omega_X^p(\log Z) \oplus [(\bigoplus_{j=1}^{d-1} \mathcal{L}^{-j}) \otimes \Omega_X^p(\log Z)] \to \Omega_Z^{p-1} \to 0. \tag{1}$$

Pushing forward the second residue SES yields:

$$0 \to \pi_*\Omega_Y^p \to \pi_*\Omega_Y^p(\log Z') \to \pi_*\Omega_{Z'}^{p-1} \to 0. \tag{2}$$

Looking at the long exact sequences induced by short exact sequences 2 and 1, we see that

$$H^q(X, \pi_*\Omega_Y^p) \cong H^q(X, \Omega_X^p \oplus [(\bigoplus_{j=1}^{d-1} \mathcal{L}^{-j}) \otimes \Omega_X^p(\log Z)]).$$

By finiteness of $\pi$, this implies

$$H^q(Y, \Omega_Y^p) \cong H^q(X, \pi_*\Omega_Y^p) \cong H^q(X, \Omega_X^p) \oplus \bigoplus_{j=1}^{d-1} H^q(X, \Omega_X^p(\log Z) \otimes \mathcal{L}^{-j})$$

**Proposition 8.5.** *Suppose $X$ is a complex manifold and $Z \subset X$ a smooth hypersurface. Let $\mathcal{L}$ be a line bundle on $X$ such that $\mathcal{L}^{\otimes d} \cong \mathcal{O}_X(Z)$. Let $Y \to X$ denote the cyclic covering obtained from $\mathcal{L}$. Then*

$$H^q(Y, \Omega_Y^p) \cong H^q(X, \Omega_X^p) \oplus \bigoplus_{j=1}^{d-1} H^q(X, \Omega_X^p(\log Z) \otimes \mathcal{L}^{-j}).$$

Let us now give an example of a complex K3 surface obtained through cyclic coverings.

[8]This argument follows from an analytic version of [Har, Exercises III.8.1 and III.8.2].

**Example 8.6.** *Let $C \subset \mathbb{P}^2_{\mathbb{C}}$ be a smooth sextic. In other words, it is a smooth effective divisor associated to a global section of $\mathcal{O}_{\mathbb{P}^2}(6)$. Then using the theory of cyclic coverings, we can construct a 2-cyclic covering $\pi : X \to \mathbb{P}^2$ branched along $C$ using the line bundle $\mathcal{O}_{\mathbb{P}^2}(3)$. In particular, $X$ embeds in the manifold $Tot(\mathcal{O}_{\mathbb{P}^2}(3))$ as a hypersurface.*

*By Riemann-Hurwitz, we have that*

$$\omega_X \cong \pi^*(\omega_{\mathbb{P}^2} \otimes \mathcal{O}_{\mathbb{P}^2}(3)) \cong \pi^*(\mathcal{O}_{\mathbb{P}^2}) \cong \mathcal{O}_X.$$

*Next, we'd like to show that*

$$H^1(X, \mathcal{O}_X) = 0.$$

*We have by proposition 8.5 that*

$$H^1(X, \mathcal{O}_X) \cong H^1(\mathbb{P}^2, \mathcal{O}_{\mathbb{P}^2}) \oplus H^1(\mathbb{P}^2, \mathcal{O}_{\mathbb{P}^2}(\log C)) \oplus H^1(\mathbb{P}^2, \mathcal{O}_{\mathbb{P}^2}(\log C) \otimes \mathcal{O}_{\mathbb{P}^2}(-3)).$$

*Noting that $\mathcal{O}_{\mathbb{P}^2}(\log C) \cong \mathcal{O}_{\mathbb{P}^2}(6)$ (locally, algebra of functions with at most a simple pole along $C$), so*

$$H^1(X, \mathcal{O}_X) \cong H^1(\mathbb{P}^2, \mathcal{O}_{\mathbb{P}^2}(6)) \oplus H^1(\mathbb{P}^2, \mathcal{O}_{\mathbb{P}^2}(3)) = 0.$$

*Thus, the covering $X$ we obtain is a K3 surface.*

## 9. Complex Kummer Surfaces

We now turn to constructing the **Kummer surfaces**, which we show are complex K3 surfaces. Start with a 2-dimensional complex torus $A \cong \mathbb{C}^2/\Lambda$. Note $A$ admits an involution $i$ of order 2 so that $\tau(z) = -z$. There are exactly 16 fixed points of $i$, which we label as $p_1, \cdots, p_{16}$. Letting $G = \langle 1, \tau \rangle$, the quotient manifold $A/G$ is again a compact complex manifold since $G$ is a discrete group. Blowing up the double points on $A/G$ yields the Kummer surface.

Alternatively, if we blow up $A$ at each of the $p_i$, we get a compact complex manifold $\widetilde{A}$ such that $\pi_2 : \widetilde{A} \to A$ is an isomorphism over $A \setminus \{p_1, \cdots, p_{16}\}$, and $\pi_2^{-1}(p_i)$ is a smooth rational curve $E_i$. Note that if we consider a small open neighborhood of $p_i$, the involution $\tau$ will restrict to this open neighborhood, and lines through $p_i$ will be sent to themselves. Thus, the involution $\tau$ extends to an involution $\widetilde{\tau} : \widetilde{A} \to \widetilde{A}$, so that $\widetilde{\tau}$ is identity on the exceptional divisors $E_i$. Letting the group $\widetilde{G} := \langle 1, \tilde{\tau} \rangle$, we can consider the quotient $\widetilde{A}/\widetilde{G}$ which is again a compact complex manifold. This is isomorphic to the Kummer surface.

$$\begin{tikzcd} \widetilde{A} \arrow[r, "\pi_2"] \arrow[d, "\pi_1"'] & A \arrow[d] \\ \widetilde{A}/\widetilde{G} \arrow[r] & A/G \end{tikzcd}$$

Now we claim that the Kummer surface is a complex K3 surface. Thus, we must show that

$$H^1(\widetilde{A}/\widetilde{G}, \mathcal{O}_{\widetilde{A}/\widetilde{G}}) = 0. \text{ and } \omega_{\widetilde{A}/\widetilde{G}} \cong \mathcal{O}_{\widetilde{A}/\widetilde{G}}$$

First, we show the former. Note that there is an injection

$$H^1(\widetilde{A}/\widetilde{G}, \mathcal{O}_{\widetilde{A}/\widetilde{G}}) \hookrightarrow H^1(\widetilde{A}, \mathcal{O}_{\widetilde{A}}).$$

A quick way to see this is to note that for any open $U \subset \widetilde{A}/\widetilde{G}$, there is an injection

$$\mathcal{O}_{\widetilde{A}/\widetilde{G}}(U) \hookrightarrow \mathcal{O}_{\widetilde{A}}(\pi_1^{-1}(U)).$$

Then if we pick an open cover $\underline{U} = \{U_i\}$ of $\widetilde{A}/\widetilde{G}$, which induces an open cover $\underline{U}' = \{\pi_1^{-1}(U_i)\}$ of $\widetilde{A}$, we have an injection on Cech cohomology

$$\check{H}^*(\underline{U}, \mathcal{O}_{\widetilde{A}/\widetilde{G}}) \hookrightarrow \check{H}^*(\underline{U}', \mathcal{O}_{\widetilde{A}}).$$

By Leray's thoerem/Cartan's lemma, since both structure sheaves are fine sheaves, they are acylic on the respective covers, and thus the Cech cohomologies can be identified with sheaf cohomologies.

We also have $H^1(\widetilde{A}, \mathcal{O}_{\widetilde{A}}) \cong H^1(A, \mathcal{O}_A)$. This can be seen by almost exactly the same argument as above. The isomorphism follows from the fact that for every $U \subset A$, we have an isomorphism between $\mathcal{O}_A(U)$ and $\mathcal{O}_{\widetilde{A}}(\pi_2^{-1}(U))$. If $U$ contains none of the $p_i$, then we're done, since $U$ is then isomorphic to $\pi_2^{-1}(U)$. Otherwise, the isomorphism holds because of Hartog's extension theorem. Then we have

$$H^1(\widetilde{A}/\widetilde{G}, \mathcal{O}_{\widetilde{A}/\widetilde{G}}) \hookrightarrow H^1(\widetilde{A}, \mathcal{O}_{\widetilde{A}}) \cong H^1(A, \mathcal{O}_A).$$

Note the image must be the $\tau$-invariant subspace of $H^1(A, \mathcal{O}_A)$. But note that since $A$ is Kahler (can take the inherited euclidean metric), we have Hodge decomposition, and in particular, we have $H^1(A, \mathcal{O}_A) \cong H^{0,1}(A) \subseteq H^2(A; \mathbb{C})$. In particular, $H^2(A; \mathbb{C}) \cong \mathbb{C}\langle dx_i \wedge dx_j \rangle_{1 \le i < j \le 4}$, where the $x_i$ are coordinate functions on the $\mathbb{C}^2$ considered as $\mathbb{R}^4$ whose quotient by $\Lambda$ is $A$. In particular, it becomes clear from this that the only $\tau$-invariant element of $H^2(A; \mathbb{C})$ is 0. Thus,

$$H^1(\widetilde{A}/\widetilde{G}, \mathcal{O}_{\widetilde{A}/\widetilde{G}}) = 0.$$

It remains to verify that the canonical bundle of $\widetilde{A}/\widetilde{G}$ is trivial. Note that

$$\omega_{\tilde{A}} \cong \pi_2^* \omega_A \otimes \mathcal{O}_{\tilde{A}}(E_1 + \cdots + E_{16}) \cong \mathcal{O}_{\tilde{A}}(E_1 + \cdots + E_{16}).$$

Furthermore by the Riemann-Hurwitz formula (8.1) we have

$$\omega_{\widetilde{A}} \cong \pi_1^* \omega_{\widetilde{A}/\tilde{\imath}} \otimes \mathcal{O}_{\tilde{A}}(E_1 + \cdots + E_{16}) \implies \pi_1^* \omega_{\widetilde{A}/\tilde{\imath}} \cong \mathcal{O}_{\widetilde{A}}.$$

Now we also claim that $\pi_1^* \mathcal{O}_{\tilde{A}} \cong \mathcal{O}_{\tilde{A}/\tilde{G}} \oplus \mathcal{L}^\vee$, where $\mathcal{L}^{\otimes 2} \cong \mathcal{O}_{\tilde{A}/\tilde{G}}(\overline{E_1} + \cdots + \overline{E_{16}})$. We want to use lemma 8.4. What we will show is that $\tilde{A} \to \tilde{A}/\tilde{G}$ can be identified with a degree 2 cyclic covering branched along $\overline{E_1} + \cdots + \overline{E_{16}}$.

Note that there is a section $s \in \mathcal{O}_{\tilde{A}/\tilde{G}}(\overline{E_1} + \cdots + \overline{E_{16}})$ which corresponds to the smooth effective divisor $\overline{E_1} + \cdots + \overline{E_{16}}$. For every point $p \in \tilde{A}/\tilde{G}$, there exists a neighborhood $U$

such that $\pi_1^{-1}(U)$ can be identified with the subset

$$\pi_1^{-1}(U) \cong \{(p,t) \in U \times \mathbb{C} | s(p) = t^2\} \subset U \times \mathbb{C}.$$

Now suppose we have neighborhoods $U_1$ and $U_2$ such that

$$\pi_1^{-1}(U_1) \cong \{(p,t) \in U_1 \times \mathbb{C} | s(p) = t^2\}, \text{ and } \pi_1^{-1}(U_2) \cong \{(p,t) \in U_2 \times \mathbb{C} | s(p) = t^2\}.$$

Shrinking if necessary, we can assume that $U_1$ and $U_2$ locally trivialize $\mathcal{O}_{\tilde{A}/\tilde{G}}(\overline{E_1} + \cdots + \overline{E_{16}})$. What is the transition function between these two charts? Well, if over $U_1 \cap U_2$ the transition function of $\mathcal{O}_{\tilde{A}/\tilde{G}}(\overline{E_1} + \cdots + \overline{E_{16}})$ is given by $\lambda$, then we have $(p,t) \mapsto (p,t')$, and $s(p) \mapsto \lambda(p)s(p)$, so $(t')^2 = \lambda t^2$. Our specific situation here allows us to pick appropriate square roots $t' = \sqrt{\lambda}$ with respect to our covering, so that the square roots are compatible. Then we see that $\tilde{A}$ can be identified with being embedded in the complex manifold $L$ associated to a square root $\mathcal{L}$ of $\mathcal{O}_{\tilde{A}/\tilde{G}}(\overline{E_1} + \cdots + \overline{E_{16}})$, and we see that $\tilde{A} \to \tilde{A}/\tilde{G}$ can be identified as a degree 2 cyclic cover ramified along $E_1 + \cdots + E_{16} \subset \tilde{A}$.

Finally, this implies by lemma 8.4 that $\pi_1^* \mathcal{O}_{\tilde{A}} \cong \mathcal{O}_{\tilde{A}/\tilde{G}} \oplus \mathcal{L}^\vee$. Then by the projection formula, we have that

$$(\pi_1)_*(\mathcal{O}_{\tilde{A}} \otimes \pi_1^* \omega_{\tilde{A}/\tilde{G}}) \cong (\pi_1)_*(\mathcal{O}_{\tilde{A}}) \otimes \omega_{\tilde{A}/\tilde{G}} \implies \mathcal{O}_{\tilde{A}/\tilde{G}} \oplus \mathcal{L}^\vee \cong (\mathcal{O}_{\tilde{A}/\tilde{G}} \oplus \mathcal{L}^\vee) \otimes \omega_{\tilde{A}/\tilde{G}}.$$

The Krull-Schmidt theorem implies that $\omega_{\tilde{A}/\tilde{G}}$ is isomorphic to either $\mathcal{O}_{\tilde{A}/\tilde{G}}$ or $\mathcal{L}^\vee$. But in the latter case, we know the pullback has a global section, while the pullback $\pi_1^*(\mathcal{L}^\vee) \cong \mathcal{O}_{\tilde{A}}(-E_1 - \cdots - E_{16})$ has no global section. So we must have $\omega_{\tilde{A}/\tilde{G}} \cong \mathcal{O}_{\tilde{A}/\tilde{G}}$.

Some other useful discussions:

- `https://math.stackexchange.com/questions/1241894/references-to-understand-k3-surface`
- `https://math.stackexchange.com/questions/4920890/canonical-divisor-of-kummer-surface` `rq=1`

**Proposition 9.1.** *The Kummer surface $\tilde{A}/\tilde{G}$ is projective if and only if $A$ is projective.*

*Proof.* Suppose $A$ is projective. Then since $G$ is a finite group, GIT implies that $A/G$ is projective. Then since $\tilde{A}/\tilde{G}$ is the blow up of $A/G$ with respect to finitely many points, it is also projective.

Now suppose $\tilde{A}/\tilde{G}$ is projective. Then looking at the map $\tilde{A} \to \tilde{A}/\tilde{G}$ implies $\tilde{A}$ is projective, and thus the blow-down $A$ of $\tilde{A}$ is projective by the "Castelnuovo contractibility theorem" [HG94, "Castelnuovo-Enriques Criterion" page 476] [Bea96, II.17]. The contracting map must factor through the blow-down by the universal property, and from this it is an immediate exercise to show that it coincides with the blow-down. Alternatively, contracting a curve with negative square on a surface preserves projectivity. `https://mathoverflow.net/questions/123375/contracting-a-curve-of-negative-self-intersection-on-a-s` □

**Corollary 9.2.** *Most Kummer surfaces are not projective. This is because most tori* $\mathbb{C}^2/\Lambda$ *are not projective by Riemann's Criterion* [HG94, Chapter 2].

**Remark 9.3.** *Any K3 surface that contains 16 disjoint smooth rational curves* $C_1, \cdots, C_{16} \subset X$ *with* $(\frac{1}{2})\sum[C_i] \in NS(X)$ *is in fact a Kummer surface. An outline of the argument is given in* [Huy, Page 298].

## 10. The K3 Lattice

For complex K3 surfaces $X$, their middle cohomology $H^2(X;\mathbb{Z})$ is very interesting and important (for example, because of the Torelli theorem for complex K3 surfaces). Equipping $H^2(X;\mathbb{Z})$ with the intersection pairing, it turns out that:

**Theorem 10.1.** *The integral cohomology* $H^2(X;\mathbb{Z})$ *of a complex K3 surface* $X$ *endowed with the intersection form is abstractly isomorphic to the lattice*

$$E_8(-1)^{\oplus 2} \oplus U^{\oplus 3}.$$

This description of $H^2(X;\mathbb{Z})$ is nice because it provides an abstract basis whose interaction with the intersection pairing is explicit. It also extremely useful. Enumerative geometry problems on K3 surfaces can reduce to lattice theory computations, and many important theorems such as the Torelli theorem for K3 surfaces invoke the K3 lattice. In this section, we discuss the absolute basics of lattice theory – enough to understand what $E_8(-1)^{\oplus 2} \oplus U^{\oplus 3}$ means. But first, let's see what we can say about $H^2(X;\mathbb{Z})$ without lattice theory: namely that it is free of rank 22.

Fix a complex K3 surface $X$ and consider its exponential exact sequence. The cohomology LES yields

$$\begin{tikzcd}
0 \arrow[r] & H^1(X;\mathbb{Z}) \arrow[r] & H^1(X,\mathcal{O}_X) \arrow[r] & H^1(X,\mathcal{O}_X^\times) \arrow[dll] \\
& H^2(X;\mathbb{Z}) \arrow[r] & H^2(X,\mathcal{O}_X) \arrow[r] & H^2(X,\mathcal{O}_X^\times) \arrow[dll] \\
& H^3(X;\mathbb{Z}) \arrow[r] & \cdots &
\end{tikzcd}$$

Note that we can start with 0 because $X$ is compact so global sections are exact. Now we know that $H^1(X,\mathcal{O}_X) = 0$, which implies that $H^1(X;\mathbb{Z}) = 0$. By Poincare duality, this implies $H_3(X;\mathbb{Z}) = 0$, which means that $H^3(X;\mathbb{Z}) = 0$ up to torsion. Now note that we have

$$0 \to H^1(X,\mathcal{O}_X^\times) \to H^2(X;\mathbb{Z}) \to H^2(X,\mathcal{O}_X) \to \cdots.$$

We have $H^2(X, \mathcal{O}_X) \cong H^2(X, \omega_X) \cong H^{2,2}(X) \cong H^{0,0}(X) \cong \mathbb{C}$, and the Picard group of K3 surfaces are torsion-free. This implies that $H^2(X;\mathbb{Z})$ is also torsion free. A topology theorem says that the torsion of $H^i(X;\mathbb{Z})$ can be identified with the torsion of $H^{\dim_{\mathbb{R}} X-i+1}(X;\mathbb{Z})$. Thus, $H^2(X;\mathbb{Z})$ being torsion free impies $H^3(X;\mathbb{Z})$ is torsion free, and thus $H^3(X;\mathbb{Z}) = 0$.

The Hirzebruch-Riemann-Roch computation remains for complex K3 surfaces, so that $c_2(X) = 24$. For a complex surface, this can be read as an equality for the topological Euler number $e(X) = c_2(X) = 24$, so that $\sum(-1)^i b_i(X) = 24$. Since we have just shown that $b_1(X) = b_3(X) = 0$, and we know $b_0(X) = b_4(X) = 1$, we have $b_2(X) = 22$. Thus, $H^2(X;\mathbb{Z})$ is free abelian of rank 22. Now we also have the cup product structure on $H^2(X;\mathbb{Z})$, which coincides with the intersection pairing $\langle\alpha, \beta\rangle = \int_X \alpha \wedge \beta \in \mathbb{Z}$. To further describe $H^2(X;\mathbb{Z})$ we need some lattice theory.

**Definition 10.2.** *A lattice $\Lambda$ is a free $\mathbb{Z}$-module with a nondegenerate symmetric bilinear form*

$$\langle . \rangle : \Lambda \times \Lambda \to \mathbb{Z}.$$

*We say a lattice is even if $\langle x.x\rangle \in 2\mathbb{Z}$ for every $x \in \Lambda$.*

Fix a lattice $\Lambda$. Since $\langle . \rangle$ is nondegenerate, it defines an injection

$$\Lambda \hookrightarrow \Lambda^* := \mathrm{Hom}_{\mathbb{Z}}(\Lambda, \mathbb{Z})$$

into the dual lattice, by the map $x \mapsto \langle x.-\rangle$. The cokernel is called the **discriminant group**

$$A_\Lambda := \Lambda^*/\Lambda.$$

Now consider $\Lambda \otimes_{\mathbb{Z}} \mathbb{Q}$ and the $\mathbb{Q}$-bilinear extension $\langle . \rangle_{\mathbb{Q}}$ of $\langle . \rangle$. Then the dual lattice can be identified as

$$\Lambda^* \cong \{\langle x.-\rangle_{\mathbb{Q}} | x \in \Lambda \otimes_{\mathbb{Z}} \mathbb{Q} \text{ and } \langle x.\Lambda\rangle_{\mathbb{Q}} \in \mathbb{Z}\},$$

and $\Lambda^*$ naturally inherits a $\mathbb{Q}$-bilinear pairing $\Lambda^* \times \Lambda^* \to \mathbb{Q}$ where

$$(\langle x.-\rangle_{\mathbb{Q}}, \langle y.-\rangle_{\mathbb{Q}}) = \langle x.y\rangle_{\mathbb{Q}}.$$

This descends to pairing on the discriminant group $A_\Lambda \times A_\Lambda \to \mathbb{Q}/\mathbb{Z}$. When $\Lambda$ is an even lattice, we have a pairing $A_\Lambda \times A_\Lambda \to \mathbb{Q}/2\mathbb{Z}$, and this yields the map

$$q_\Lambda : A_\Lambda \to \mathbb{Q}/2\mathbb{Z},$$

the data of which recovers the aforementioned pairing. The **discriminant form associated to $\Lambda$**

$$(A_\Lambda, q_\Lambda)$$

is one of the main data-pieces attached to every even lattice $\Lambda$. Here are some other basic definitions/facts/data-pieces for lattices:

- If $I_\Lambda$ denotes the intersection matrix associated to $\Lambda$, then $|A_\Lambda| = |\det I_\Lambda|$.

– $\Lambda$ is **unimodular** when $\Lambda \hookrightarrow \Lambda^*$ is actually an isomorphism. So $\Lambda$ is unimodular $\iff A_\Lambda$ trivial $\iff \det I_\Lambda = \pm 1$.
– The **signature** of $\Lambda$ is defined as: take $\Lambda \otimes_{\mathbb{Z}} \mathbb{R}$, and note $I_\Lambda$ as a real symmetric matrix can then be diagonalized. Then the signature is $(n_+, n_-)$, where $n_+$ is the count of positive eigenvalues and $n_-$ is the count of negative eigenvalues.
– **Positive or negative definite** if $n_- = 0$ or $n_+ = 0$, respectively.

**Definition 10.3.** *For two lattices $\Lambda_1, \Lambda_2$, the direct sum $\Lambda_1 \oplus \Lambda_2$ is defined to be the lattice with pairing*

$$\langle x_1 + x_2 . y_1 + y_2 \rangle_{\Lambda_1 \oplus \Lambda_2} = \langle x_1 . y_1 \rangle_{\Lambda_1} + \langle x_2 . y_2 \rangle_{\Lambda_2}.$$

One can show quite easily that $A_{\Lambda_1 \oplus \Lambda_2} \cong A_{\Lambda_1} \oplus A_{\Lambda_2}$, and for even lattices we have

$$(A_{\Lambda_1 \oplus \Lambda_2}, q_{\Lambda_1 \oplus \Lambda_2}) \cong (A_{\Lambda_1}, q_{\Lambda_1}) \oplus (A_{\Lambda_2}, q_{\Lambda_2}).$$

**Definition 10.4.** *A morphism between even lattices $\Lambda_1 \to \Lambda$ is map of $\mathbb{Z}$-modules which respects the quadratic forms.*

Suppose $\phi : \Lambda_1 \to \Lambda$ is a morphism of even lattices. Then there is an induced map $\Lambda^* = \mathrm{Hom}_{\mathbb{Z}}(\Lambda, \mathbb{Z}) \to \mathrm{Hom}_{\mathbb{Z}}(\Lambda_1, \mathbb{Z}) = \Lambda_1^*$, where $\langle x.- \rangle \mapsto \langle x.\phi(-) \rangle$. To be a morphism of lattices means that $\phi$ must induce a map which respects discriminant forms. In other words, $\phi^*$ must descend to a map of discriminant forms:

$$(A_\Lambda, q_\Lambda) \to (A_{\Lambda_1}, q_{\Lambda_1}).$$

In particular, if $x \in \Lambda$, then $\langle x.\phi(-) \rangle$ must be represented by an element $y \in \Lambda_1$. Furthermore, for $\langle x.- \rangle \in A_\Lambda$, if it maps to $\langle x.\phi(-) \rangle = \langle y.- \rangle$, then $(x.x)_{\Lambda \otimes_{\mathbb{Z}} \mathbb{Q}} = (y.y)_{\Lambda_1 \otimes_{\mathbb{Z}} \mathbb{Q}}$.

If $\Lambda \to \Lambda_1$ is an injection of even lattices, then we claim that $A_{\Lambda_1}$ injects into $A_\Lambda$. Suppose $\langle x. \rangle \in A_{\Lambda_1}$ such that $\langle x. \rangle \mapsto \langle x.\iota(-) \rangle = \langle y. \rangle = 0 \in A_\Lambda$. This implies that $y \in \Lambda$. Since the morphism respects the discriminant forms, it also respects the $\mathbb{Q}/\mathbb{Z}$-bilinear pairing recovered from the quadratic form. Then

$$(\langle x.- \rangle, \langle x'.- \rangle) = (\langle y.- \rangle, \langle y'.- \rangle) \in \mathbb{Z}, \forall x' \in \Lambda_1 \otimes_{\mathbb{Z}} \mathbb{Q}.$$

But this means $\langle x.x' \rangle \in \mathbb{Z}$ for all $x' \in \Lambda_1 \otimes_{\mathbb{Z}} \mathbb{Q}$. This is only possible if $x \in \Lambda_1$. This shows

$$(A_{\Lambda_1}, q_{\Lambda_1}) \hookrightarrow (A_\Lambda, q_\Lambda).$$

**Proposition 10.5.** *Let $\Lambda \hookrightarrow \Lambda_1$ be an injection of even lattices such that $[\Lambda_1 : \Lambda]$ is finite. Then*

$$|A_\Lambda| = |A_{\Lambda_1}| \cdot [\Lambda_1 : \Lambda]^2.$$

We now discuss the notion of orthogonal lattices. This notion will arise when we discuss, for example, the Nevron-Severi lattice and its orthogonal complement, the transcendental lattice.

– A **primitive embedding** $\Lambda_1 \hookrightarrow \Lambda$ is an injection whose cokernel is torsion-free.

  - For example, the orthogonal complement of any sublattice $\Lambda_1 \hookrightarrow \Lambda$ is a primitive sublattice intersecting $\Lambda_1$ trivially. The induced embedding $\Lambda_1 \oplus \Lambda_2 \hookrightarrow \Lambda$ is of finite index, and in general not primitive, not even for primitive $\Lambda_1$.
- Two even lattices $\Lambda_1, \Lambda_2$ are **orthogonal** if there exists a primitive embedding $\Lambda_1 \hookrightarrow \Lambda$ into an even unimodular lattice with $\Lambda_1^\perp \cong \Lambda_2$.

**Proposition 10.6.** [Huy, Chapter 14, Prop 0.2] *Let $\Lambda_1, \Lambda_2$ be two even lattices. Then $\Lambda_1, \Lambda_2$ are orthogonal if and only if*

$$(A_{\Lambda_1}, q_{\Lambda_1}) \cong (A_{\Lambda_2}, -q_{\Lambda_2}).$$

If $\Lambda_1, \Lambda_2$ are orthogonal, then

$$|A_{\Lambda_1}| = |A_{\Lambda_2}| = [\Lambda : \Lambda_1 \oplus \Lambda_2],$$

and the signs of the discriminants of $\Lambda_1, \Lambda_2$ differ by the discriminant of $\Lambda$.

**Example 10.7.** *The hyperbolic plane, denoted $U$, $H$, or $\prod_{1,1}$, is the lattice $\cong \mathbb{Z}^2$ with intersection matrix $\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$. It is even unimodular of discriminant 1. The hyperbolic plane is very special. For example, if $U \hookrightarrow \Lambda$ is an embedding of even lattices, then*

$$\Lambda = U \oplus U^\perp.$$

*The same holds for $U^{\oplus k}$.*

**Example 10.8.** *The $E_8$ lattice is isomorphic to $\mathbb{Z}^8$ and given by the intersection matrix*

$$\begin{pmatrix}
2 & -1 & & & & & & \\
-1 & 2 & -1 & & & & & \\
 & -1 & 2 & -1 & -1 & & & \\
 & & -1 & 2 & 0 & & & \\
 & & -1 & 0 & 2 & -1 & & \\
 & & & & -1 & 2 & -1 & \\
 & & & & & -1 & 2 & -1 \\
 & & & & & & -1 & 2
\end{pmatrix}.$$

*Even, unimodular, positive definite of discriminant 1.*

**Example 10.9.** *For any lattice $\Lambda$, the twisted lattice $\Lambda(m)$ is given by multiplying the intersection form $\langle . \rangle$ by $m$. So $\Lambda(m) \cong \Lambda$ as $\mathbb{Z}$-modules, but*

$$\langle . \rangle_{\Lambda(m)} := m \langle . \rangle_\Lambda.$$

*Note that there is a surjection*

$$\Lambda(m)^* \to \Lambda^*,$$

*given by* $\langle x.\rangle_{\Lambda(m)} \mapsto \langle mx.\rangle_{\Lambda}$*. Then we have a short exact sequence of finite abelian groups*

$$0 \to \Lambda/m\Lambda \to A_{\Lambda(m)} \to A_{\Lambda} \to 0,$$

*which implies* $|A_{\Lambda(m)}| = |A_{\Lambda}| \cdot m^{\operatorname{rk}\Lambda}$.

## 11. Hyperelliptic curves, canonical curves, projective normality

By curve, we mean a smooth complete irreducible curve (or a compact Riemann surface). A hyperelliptic curve $C$ is a curve which admits a degree 2 covering $C \to \mathbb{P}^1$. We prove the following:

**Proposition 11.1.** *Let the curve* $C$ *have genus* $g \geq 2$*. Then* $C$ *is hyperelliptic* $\iff$ *the complete linear system* $|\omega_C|$ *fails to provide an embedding*

$$C \to \mathbb{P}(H^0(C, \omega_C)^\vee).$$

*Proof.* First note that $|\omega_C|$ is base point free. To see this, assume FSOC that there was some base point $p \in C$. Then looking at the SES

$$0 \to \omega_C(-p) \to \omega_C \to \omega_C(k(p)) \to 0,$$

we would need $H^0(\omega_C) \to H^0(\omega_C(k(p))$ to fail to be surjective. This would imply $H^0(\omega_C(-p)) \cong H^0(\omega_C)$. But since $\dim H^0(\omega_C) \geq 2$, this implies that there is a meromorphic funcion with pole at $p$, and holomorphic on $\mathbb{C} \setminus \{p\}$. In particular, this meromorphic function gives a biholomorphic map $C \to \mathbb{P}^1$, which is a contradiction because $C$ has genus $\geq 2$. So the complete linear system $|\omega_C|$ always provides a regular map from $C$ to $\mathbb{P}(H^0(C, \omega_C)^\vee)$.

Now let us examine when $|\omega_C|$ provides an embedding. It is when, given $p$ and $q$ not necessarily distinct, there exist some global section vanishing at $p$ but not $q$. When $p = q$, what we really mean is that there is some global section whose zero locus contains $p$ but not $2p$. Then we get an embedding when, for all $p, q$, we have

$$h^0(\omega_C(-p)) > h^0(\omega_C(-p-q)).$$

Note by *RR* we have $h^0(\omega_C(-p)) - h^1(\omega_C(-p)) = g - 1 + 1 - g = 0 \implies h^0(\omega_C(-p)) = h^0(\mathcal{O}_C(p)) = 1$. So this would mean $h^0(\omega_C(-p-q)) = 0$. But we fail to have an embedding if and only if $\exists p, q$ such that $h^0(\omega_C(-p)) = h^0(\omega_C(-p-q)) = 1$.

Note by RR we have $h^0(\omega_C(-p-q)) - h^0(\mathcal{O}_C(p+q)) = g-2+1-g = -1 \implies h^0(\mathcal{O}_C(p+q)) = 2$. So we fail to have an embedding $\iff$ $\exists p, q$ such that $h^0(\mathcal{O}_C(p+q)) = 2 \iff$ there exists meromorphic function with pole at some $p, q \iff$ double cover $C \to \mathbb{P}^1$ □

All curves of genus $g = 2$ are hyperelliptic. For curves of genus $g \geq 3$, most are not hyperelliptic by dimension counting. Furthermore, one can prove that the hyperelliptic curves are exactly those curves which are given by a cyclic degree 2 covering of $\mathbb{P}^1$. The embedding of a non-hyperelliptic curve $C \hookrightarrow \mathbb{P}^{g-1}$ is called a canonical curve. Furthermore, suppose we have a genus $g$ and degree $2g-2$ nondegenerate curve in $\mathbb{P}^{g-1}$. Then it must be a canonical curve, since it will be embedded by the complete linear system of a degree $2g-2$ line bundle $\mathcal{L}$, and looking at $H^0(\omega_C \otimes \mathcal{L}^{-1})$, it is 1 if the divisors are the same and 0 if they differ. And $h^0(\mathcal{L}) - h^0(\omega_C \otimes \mathcal{L}^{-1}) = 2g-2+1-g = g-1 \implies h^0(\omega_C \otimes \mathcal{L}^{-1}) = 1$. [9]

**Theorem 11.2** (Max Noether's Theorem)**.** [10] *The canonical embeddings of non-hyperelliptic curves are projectively normal.*

Let's talk a little about the definition of projectively normal and some geometric consequences. If we have a curve $X \subset \mathbb{P}^n$, it has a homogeneous coordinate ring, $k[X_0, \cdots, X_n]/I$ where $I$ is the ideal of homogeneous functions vanishing on $X$. If one considers

$$0 \to \mathcal{I}_X(k) \to \mathcal{O}_{\mathbb{P}^n}(k) \to \mathcal{O}_X(k) \to 0,$$

then the LES is

$$0 \to H^0(\mathcal{I}_X(k)) \to H^0(\mathcal{O}_{\mathbb{P}^n}(k)) \to H^0(\mathcal{O}_X(k)) \to H^1(\mathcal{I}_X(k)) \to 0.$$

Then we say $X$ is **projectively normal** if

$$H^0(\mathcal{O}_{\mathbb{P}^n}(k)) \to H^0(\mathcal{O}_X(k))$$

is a surjection for every $k$. In other words, $X$ is projectively normal when

$$\bigoplus_{k=0}^{\infty} H^0(\mathcal{O}_X(k))$$

is the homogeneous coordinate ring, where the multiplication on sections is induced from tensor product. In particular, being projectively normal means that when we intersect a hypersurface $Z$ with $X$, then all the other effective divisors linearly equivalent to $Z \cap X$ come from a hypersurface intersecting $X$. Now we present two equivalent definitions.

**Definition 11.3.** *Fix a hyperplane $H \subset \mathbb{P}^n$. Then $X$ is projectively normal if and only if every hypersurface in $H$ which contains $H \cap X$ comes from a hypersurface of $\mathbb{P}^n$ containing $X$.*

**Definition 11.4.** *$X$ is projectively normal $\iff$ $H^1(\mathcal{I}_X(k)) = 0$ for all $k \geq 0$.*

Let's suppose $X$ is projectively normal. Then taking the SES

$$0 \to \mathcal{I}_X(k) \to \mathcal{O}_{\mathbb{P}^n}(k) \to \mathcal{O}_X(k) \to 0,$$

we see surjectivity of $H^0(\mathcal{O}_{\mathbb{P}^n}(k)) \to H^0(\mathcal{O}_X(k))$ and $H^1(\mathcal{O}_{\mathbb{P}^n}(k)) = 0$ implies $H^1(\mathcal{I}_X(k)) = 0$. Conversely, vanishing of the $H^1$ implies projectively normal. This shows definition 11.4.

Let's look at definition 11.4. We want to show that $X$ is projectively normal if there is a surjection from

$$H^0(\mathcal{I}_X(k)) = \ker H^0(\mathcal{O}_{\mathbb{P}^n}(k)) \to H^0(\mathcal{O}_X(k))$$

[9] For an arbitrary curve of genus $g > 2$ and $k \geq 2$, or $g = 2$ and $k \geq 3$, we have $\omega_C^k$ is very ample.

[10] Max Noether was a mathematican at the University of Gottingen and was the father of Emmy Noether.

to

$$H^0(\mathcal{I}^H_{X\cap H}(k)) = \ker H^0(\mathcal{O}_H(k)) \to H^0(\mathcal{O}_{C\cap H}(k)).$$

But we have the SES

$$0 \to \mathcal{I}_H(k-1) \to \mathcal{I}_X(k) \to \mathcal{I}^H_{X\cap H}(k) \to 0.$$

Then if $X$ is projectively normal, we obtain the surjection. And if have the surjection

$$H^0(\mathcal{I}_X(k)) \to H^0(\mathcal{I}^H_{X\cap H}(k))$$

then $H^1(\mathcal{I}_H(k-1)) \hookrightarrow H^1(\mathcal{I}_H(k))$ for every $k$, and letting $k \to \infty$ we get vanishing, which implies that all of these vanish, which implies projective normality.

**Remark 11.5.** *In the course of the proof, we see the following fact. Suppose we have a nondegenerate degree $d$ genus $g$ curve $C$ in $\mathbb{P}^r$ embedded linearly normally, so that $H^0(\mathcal{O}_{\mathbb{P}^n}(1)) = H^0(\mathcal{O}_C(1))$. This implies that $H^1(\mathcal{I}_C(1)) = 0$, which implies that: take a hyperplane $H$, intersect with $H \cap X$. If a quadric of $H$ contains $H \cap C$, which will be $d$ points, then the quadric contains all of $C$. Any $\binom{r+2}{2} - 1$ points in $H$ determine quadric. So this implies that any degree $d < \binom{r+2}{2}$ curve will be contained in a quadric of $\mathbb{P}^r$.*

## 12. Arithmetic and geometric genus of curves

Let a curve $C$ mean a 1-dimensional projective $k$-scheme. The curve may be nonsingular, reducible, or non-reduced. We define the arithmetic genus of $C$ to be $p_a(C) = 1 - \chi(\mathcal{O}_C)$. When $C$ is non-singular, $p_a(C)$ coincides exactly with the geometric genus $\dim H^0(\Omega_C)$, as

$$p_a(C) = 1 - \chi(\mathcal{O}_C) = 1 - h^0(\mathcal{O}_C) + h^1(\mathcal{O}_C) = h^0(\Omega_C)$$

by Serre duality. The arithmetic genus is the preferred invariant for singular curves since it remains constant in flat families.

If we assume that $C$ is an integral curve, then we can relate the geometric and arithmetic genus in the following way. First, we define the geometric genus of $C$ to be the geometric genus of its normalization. Let

$$v : \tilde{C} \to C$$

denote the normalization map, where the map is built out of $\mathrm{Spec}(\tilde{A}) \to \mathrm{Spec}(A)$, where $\mathrm{Spec}(A) \subset C$ and $\tilde{A}$ is the integral closure of $A$ in its field of fractions. Then we have the short exact sequence

$$0 \to \mathcal{O}_C \to v_*\mathcal{O}_{\tilde{C}} \to v_*\mathcal{O}_{\tilde{C}}/\mathcal{O}_C \to 0,$$

which means

$$\begin{aligned} \chi(v_*\mathcal{O}_{\tilde{C}}) &= \chi(\mathcal{O}_C) + \chi(v_*\mathcal{O}_{\tilde{C}}/\mathcal{O}_C) \\ \implies p_a(C) &= g(C) + h^0(v_*\mathcal{O}_{\tilde{C}}/\mathcal{O}_C). \end{aligned}$$

Note that $v_*\mathcal{O}_{\tilde{C}}/\mathcal{O}_C$ is a skyscraper sheaf supported at the singular points of $C$. In particular, $p_a(C) \geq g(C)$, and $p_a(C) = g(C) \iff C$ is smooth. Now we do two examples to show that the nodal cubic and cupsidal cubic curves over algebraically closed $k$ are both singular curves with arithmetic genus 1 and geometric genus 0.

**Example 12.1** (Arithmetic Genus of Nodal Cubic)**.** *The nodal cubic is the curve*

$$\operatorname{Spec}\frac{k[X,Y]}{(Y^2-X^2-X^3)}.$$

*First, let's compute its normalization. We claim that its normalization is* $\operatorname{Spec} k[T]$*. But what should the normalization map be? Let's play around.*

*First, we want to think of the nodal cubic being parameterized. So let* $Y = XT$*. Then*

$$X^2T^2 = X^2 + X^3 \implies (T^2-1)X^2 = X^3 \implies T^2 - 1 = X, T(T^2-1) = Y.$$

*Then we claim that*

$$\operatorname{Spec} k[T] \to \operatorname{Spec}\frac{k[X,Y]}{(Y^2-X^2-X^3)},$$

*where* $(X,Y) \mapsto (T^2-1, T(T^2-1))$*, is the normalization map. To verify this, let's show that* $k[T]$ *is the integral closure of the integral domain in its fraction field. Note* $T = \frac{Y}{X}$*. Indeed,* $T$ *is integral because*

$$T^2 - 1 - X = 0$$

*in the fraction field. Furthermore,* $k[T]$ *is a UFD therefore integrally closed. This shows that* $k[T]$ *is the desired integral closure.*

*Now let's compute the fiber of the normalization map over the singular point, which is* $(X,Y) \in \operatorname{Spec}\frac{k[X,Y]}{(Y^2-X^2-X^3)}$*. This is the pullback of the diagram*

$$\begin{tikzcd} & \operatorname{Spec} k[T] \arrow[d] \\ \operatorname{Spec}(k) \arrow[r] & \operatorname{Spec}\frac{k[X,Y]}{(Y^2-X^3-X^2)} \end{tikzcd}$$

*Then the fiber is* $\operatorname{Spec} k[T] \otimes_{\frac{k[X,Y]}{(Y^2-X^3-X^2)}} k$*. Note* $T^2 \otimes 1 = 1 \otimes 1$*. Then we see that the elements are generated by* $(a \otimes bT) \otimes 1$ *and the multiplication is such that we can identify the fiber with*

$$\operatorname{Spec} k \oplus k = \operatorname{Spec} k \sqcup \operatorname{Spec} k.$$

*In particular, if we look at which points of* $\operatorname{Spec} k[T]$ *map to the singular point, we see that they are the points corresponding to the maximal ideals* $(T-1)$ *and* $(T+1)$*. Then we are interested in calculating the length of the quotient of local rings*

$$\frac{k[X,Y]}{(Y^2-X^3-X^2)}|_{(X,Y)} \hookrightarrow k[T]|_{(T-1)} \oplus k[T]|_{(T+1)}.$$

*From this, we see that* $p_a(C) = g(\mathbb{A}^1) + 1 \implies p_a(C) = 1$*.*

**Example 12.2** (Arithmetic Genus of Cupsidal Cubic)**.** *The cupsidal cubic is described as*

$$\operatorname{Spec} \frac{k[X,Y]}{(Y^2 - X^3)}.$$

*We claim the normalization is* $\operatorname{Spec} k[T]$*. Letting* $Y = XT$*, we have*

$$X^2T^2 = X^3 \implies T^2 = X.$$

*Then we claim the normalization is*

$$\operatorname{Spec} k[T] \to \operatorname{Spec} \frac{k[X,Y]}{(Y^2 - X^3)}$$

*where* $(X,Y) \mapsto (T^2, T^3)$*. Indeed, we have* $T = \frac{Y}{X}$ *is integral over* $\frac{k[X,Y]}{(Y^2-X^3)}$ *since* $T - X = 0$ *in the fraction field.*

*Now we calculate the fiber of the normalization over the singular point* $(X,Y) \in \operatorname{Spec} \frac{k[X,Y]}{(Y^2-X^3)}$*. This is the pullback of the diagram*

$$\begin{tikzcd} & \operatorname{Spec} k[T] \arrow[d] \\ \operatorname{Spec}(k) \arrow[r] & \operatorname{Spec} \frac{k[X,Y]}{(Y^2-X^3)} \end{tikzcd}$$

*which is* $\operatorname{Spec} k[T] \times_{\frac{k[X,Y]}{(Y^2-X^3)}} \operatorname{Spec}(k)$*, and noting that* $T^2 \otimes 1 = 0$*, we see elements can be described as* $(a \otimes Tb) \otimes 1$*, and the multiplication implies that this should be identified with* $k[T]/T^2$*.*

*Finally, we compute the arithmetic genus. We first want to calculate the quotient of stalks:*

$$\frac{k[X,Y]}{(Y^2 - X^3)}|_{(X,Y)} \to \frac{k[T]}{(T^2)},$$

*and see that the image is just* $k$*. So the image is* $T \cdot \frac{k[T]}{(T^2)}$*, so by Nakayama's, the length of this module is equal to the dimension of*

$$\frac{T \cdot \frac{k[T]}{(T^2)}}{T \cdot (T \cdot \frac{k[T]}{(T^2)})} = T \cdot \frac{k[T]}{(T^2)} \cong k\{T\},$$

*so 1 dimension. Thus,*

$$p_a(C) = g(\mathbb{A}^1) + 1 = 1.$$

## 13. Basics of (-2) curves on a K3 surface

Let $S$ be an algebraic K3 surface. The theory of (-2) curves play an important role in the theory of K3 surface. Here we briefly discuss what a (-2) curve is and their relations to linear systems.

Suppose $C \subset S$ is a curve, i.e. a 1-dimensional closed subscheme of $S$. Recall that we define the arithmetic genus of $C$ to be $p_a(C) := 1 - \chi(\mathcal{O}_C)$. We can relate this invariant to the curve's self intersection number: the short exact sequence

$$0 \to \mathcal{O}_S(-C) \to \mathcal{O}_S \to \mathcal{O}_C \to 0$$

$$\implies \chi(\mathcal{O}_S) = \chi(\mathcal{O}_S(-C)) + \chi(\mathcal{O}_C)$$

$$\implies p_a(C) = 1 - \chi(\mathcal{O}_S) + \chi(\mathcal{O}_S(-C)).$$

Then by Riemann-Roch for line bundles on surfaces, and knowing that $\omega_S \cong \mathcal{O}_S$, we have

$$p_a(C) = 1 - \chi(\mathcal{O}_S) + \frac{1}{2}(C.C) + \chi(\mathcal{O}_S)$$

which implies:

**Proposition 13.1.** *Let $C$ be a curve (1-dimensional closed subscheme) of a K3 surface $S$. Then*

$$2p_a(C) - 2 = (C.C).$$

In particular, when $C$ is an integral curve, we have $p_a(C) = p_g(C) + \delta(C)$, which means $p_a(C) \geq 0$. This implies:

**Proposition 13.2.** *If $C \subset S$ is an integral curve, then*

$$(C.C) \geq -2.$$

What happens when $(C.C) = -2$ for an integral curve? This implies that the integral curve has arithemtic genus 0. But this implies that it has geometric genus 0 and, in particular, no singular points. This means that $C$ is a rational curve! More generally, we say:

**Definition 13.3.** *A (-2) curve $C$ on a K3 surface $S$ is an irreducible curve such that $(C.C) = -2$.*

In particular, being irreducible implies that $C = nC'$ for some integral $C'$. But $(C.C) = -2 \implies n^2(C'.C') = -2$, and since $-2$ is square free, we must have $n = 1$. Thus, a $(-2)$ curve is automatically integral, and thus a $(-2)$ curve is a rational curve.

Now we relate linear series to self intersections. Let $\mathcal{L} \in \mathrm{Pic}(S)$. Riemann-Roch tells us that

$$h^0(\mathcal{L}) - h^1(\mathcal{L}) + h^0(\mathcal{L}^\vee) = \frac{1}{2}(L.L) + 2.$$

Then we see that if $(L.L) \geq -2$, then either $h^0(\mathcal{L}) \neq 0$ or $h^0(\mathcal{L}^\vee) \neq 0$. And if $(L.L) \geq 0$, then either $\mathcal{L} \cong \mathcal{O}_S$, or either $h^0(\mathcal{L})) \geq 2$ or $h^0(\mathcal{L}^\vee) \geq 2$. Now here is something more interesting:

**Proposition 13.4.** *Suppose $\mathcal{L} \in \mathrm{Pic}(S)$ such that $h^0(\mathcal{L}) = 1$ and the $\mathcal{L}$ is nontrivial. Letting $C$ denote the effective curve corresponding to $\mathcal{L}$, for any curve $D \subset C$, we have*

$$(D.D) \leq -2,$$

*and if $D$ is integral then $D \cong \mathbb{P}^1$.*

*Proof.* Let $C = \sum n_i C_i$ where $C_i$ are integral curves. Then $D$ is comprised of some $\sum d_i C_i$ where $0 \leq d_i \leq n_i$. Then we see that $h^0(D) = 1$ since $h^0(\mathcal{L}) = 1$. Then Riemann Roch tells us that

$$1 - h^1(D) = \frac{1}{2}(D.D) + 2 \implies -2 - 2h^1(D) = (D.D) \implies (D.D) \leq -2.$$

Indeed, if $D$ is integral then $(D.D) \geq -2 \implies (D.D) = -2$, which implies $D$ is rational. □

As a corollary, we see that for any curve $C$ such that $h^0(\mathcal{O}_S(C)) = 1$, we have $C = \sum n_i C_i$, where $C_i \cong \mathbb{P}^1$. A natural way in which such a line bundle appears on a K3 surface is as the fixed part of any line bundle.

Suppose we have a line bundle $\mathcal{L}$ and assume $h^0(\mathcal{L}) > 0$. Then the base locus $bs|L|$ denotes the locus on which $|L| : S \to \mathbb{P}(H^0(\mathcal{L}^\vee))$ is undefined. If we let $F$ denote the line bundle associated to the 1-dimensional part of $bs|L|$, if nonempty, then we must have $h^0(F) = 1$. We call $F$ the fixed part of $L$.

**Corollary 13.5.** *The fixed part of any line bundle on a K3 surface $S$ is a linear combination of rational curves.*

Note the fixed part allows us to define $|\mathcal{L}(-F)|$, which provides an extension of the map $|\mathcal{L}|$, so that the extension is defined everywhere except at possibly 0 dimensional subschemes.

## 14. Linear systems on K3 surfaces

In this section we relate smooth irreducible curves to the geometry of K3 surfaces. In particular, linear systems on K3 surfaces, when coming from a smooth irreducible curve, are relatively accessible. For this entire section, let $X$ be an algebraic K3 surface and $C \subset X$ a smooth irreducible curve of genus $g$. Let $L := \mathcal{O}_X(C)$.

(1) Note $(L.L) = 2p_a(C) - 2 = 2g - 2$.
(2) Then Riemann-Roch says $\chi(L) = g + 1 \implies h^0(L) - h^1(L) = g + 1$. But in fact, looking at the SES $0 \to \mathcal{O}_X \to L \to L|_C \to 0$ we must have

$$h^0(L) = g + 1 \text{ and } h^1(L) = 0.$$

(3) In general, by RR we have $\chi(L^k) = \frac{1}{2}k^2(2g - 2) + 2 = k^2(g - 1) + 2$, and one can inductively show that

$$h^0(L^k) = k^2(g - 1) + 2 \text{ and } h^1(L^k) = 0, \forall k \geq 1.$$

(4) Note by the adjunction formula, we have $L|_C \cong \omega_C$. So using the SES $0 \to L^{k-1} \to L^k \to \omega_C^k \to 0$ we have
$$H^0(L^k) \twoheadrightarrow H^0(\omega_C^k) \text{ and } h^0(L^k) = h^0(L^{k-1}) + h^0(\omega_C^k).$$

**Proposition 14.1.** *Let $C$ be a smooth irreducible curve of genus $g$ on an algebraic K3 surface $X$. Let $L := \mathcal{O}_X(C)$. The complete linear system $|L|$ is base point free. In particular, we have a map*
$$|L| : X \to \mathbb{P}^g$$
*whose restriction to $C$ is the canonical map $|\omega_C| : C \to \mathbb{P}^{g-1}$.*

*Proof.* Why is $|L|$ base point free? Since $H^0(X,L) \twoheadrightarrow H^0(C,L|_C)$, it suffices to know that $L|_C$ is base point free. But indeed it is because $\deg L|_C = (L.L) = 2g-2$, and any line bundle of degree $2g-2$ on a genus $g$ curve is base point free.

If we let $s_0 \in H^0(X,L)$ correspond to $C$, then $|L|$ can be written as
$$p \mapsto [s_0(p) : \cdots : s_g(p)].$$
There is a short exact sequence
$$0 \to H^0(X,\mathcal{O}_X) \to H^0(X,L) \to H^0(C,L|_C) \to 0$$
and we see that the kernel is exactly $s_0$. So when $s_0 = 0$, we restrict ourselves to the map
$$p \mapsto [s_1(p) : \cdots : s_g(p)]$$
and these are a basis for $H^0(C,L|_C) = H^0(C,\omega_C)$. □

**Remark 14.2.** *If $g(C) = 2$, then $|L|$ restricts to a 2 to 1 map from $C \to \mathbb{P}^1$. By Bertini's theorem, a generic element of $|L|$ is smooth and irreducible, and since the obvious family*
$$\chi \subseteq X \times \mathbb{P}(H^0(X,L)) \to \mathbb{P}(H^0(X,L))$$
*is a flat family, the generic element of $|L|$ will be smooth irreducible of genus 2. Since any point in $\mathbb{P}^g$ is contained in a hyperplane, we see this implies that $|L|$ is generically a 2 to 1 covering of $\mathbb{P}^g$.*

**Remark 14.3.** *If $g(C) > 2$, and the general element of $|L|$ is non-hyperelliptic, then $|L|$ is birational onto its image.*

**Remark 14.4.** *When $g \geq 2, k \geq 3$ or $g \geq 3, k \geq 2$, we have*
$$|L^k| : X \to \mathbb{P}^{k^2(g-1)+1}$$
*is birational onto its image. This is because of a theorem that $\omega_C^k$ is very ample for when $g \geq 2, k \geq 3$ or $g \geq 3, k \geq 2$.*

When $C$ is non-hyperelliptic with $g(C) > 2$, this map satisfies an analog of projective normality.

**Proposition 14.5.** *Suppose $g(C) > 2$ and $C$ is non-hyperelliptic. Then*

$$H^0(\mathbb{P}^g, \mathcal{O}_{\mathbb{P}^g}(n)) \to H^0(X, L^n)$$

*is surjective for all $n \geq 0$.*

*Proof.* Note that this map fits into the composition

$$H^0(\mathbb{P}^g, \mathcal{O}_{\mathbb{P}^g}(n)) \cong \operatorname{Sym}^n H^0(X, L) \to H^0(X, L^n) \to H^0(C, \omega_C^n).$$

Because $g(C) > 2$ and $C$ is non-hyperellliptic, Max Noether's theorem (11.2) implies that this composition is surjective. Furthermore, note for all positive integers $k$ we have $L^k$ is a big and nef line bundle, since $(L.L) > 0$ and $(L.C) \geq 0$. Thus $H^1(X, L^{n-1}) = 0$ by Kodaira-Ramanujam vanishing theorem and so $H^0(X, L^n) \to H^0(C, \omega_C^n)$.

But note that, if $s$ is the section corresponding to $C$, then the kernel of $H^0(X, L^n) \twoheadrightarrow H^0(C, \omega_C^n)$ is exactly $s \cdot H^0(X, L^{n-1})$. Then we see that we can induct to show that the map from $\operatorname{Sym}^n H^0(X, L)$ surjects not only to the quotient but also the kernel of $H^0(X, L^n) \to H^0(C, \omega_C^n)$, thus surjects onto $H^0(X, L^n)$. □

Now we collect an assortment of other interesting facts about curves and line bundles on K3 surfaces.

**Theorem 14.6** (Kodaira-Ramanujam). [11] *Let $X$ be a smooth projective surface over a field $k$ of characteristic 0. If $L$ is big and nef, i.e. $(L.L) > 0$ and $(L.C) \geq 0$ for all curves $C \subset X$, then*

$$H^i(X, \omega_X \otimes L) = 0, \forall i > 0.$$

In particular, on a K3 surface $X$, if $C$ is a curve whose arithmetic genus is $\geq 2$, then $(C.C) > 0$ which means $H^i(X, \mathcal{O}_X(C)) = 0$ for all $i > 0$.

**Proposition 14.7.** *For an irreducible curve $C$ on an algebraic K3 surface $X$ in characteristic $\neq 2$, with $(C.C) > 0$, the linear system $|\mathcal{O}_X(C)|$ is base point free and its generic member is smooth.*

We saw that this held true when $C$ is smooth. But this says that it also holds when $C$ is not necessarily smooth or reduced. Furthermore, there are again two cases. When the general member of $|\mathcal{O}_X(C)|$ is hyperelliptic, the map is of degree two. And when the general member is non-hyperelliptic, the map is of degree one, i.e. birational to its image.

When a line bundle has trivial self-intersection, we have the following result.

**Proposition 14.8.** *If a non-trivial nef line bundle $L$ on an algebraic K3 surface $X$ satisfies $(L.L) = 0$, then $L$ is base point free. If the characteristic of the field $\neq 2, 3$ then there exists a smooth irreducible elliptic curve $E$ such that $mE \in |L|$ for some $m > 0$.*

[11] The theorem holds true for smooth projective varieties in higher dimension, and holds for K3 surfaces in positive characteristic.

**Corollary 14.9.** *In characteristic $\neq 2, 3$, every line bundle with $(L.L) = 0$ and $(L.C) \geq 0$ for all $C \cong \mathbb{P}^1$ is isomorphic to $\mathcal{O}_X(mE)$ for some integer $m$ and smooth elliptic curve $E$.*

**Example 14.10.** *Let $X \subset \mathbb{P}^3_{\mathbb{C}}$ denote the Fermat quartic, cut out by $V(x_0^4 + x_1^4 + x_2^4 + x_3^4)$. Let $\ell \subset X$ denote the line given in $\mathbb{P}^3_{\mathbb{C}}$ as the vanishing locus of $V(x_1 - \xi x_0, x_3 - \xi x_2)$, where $\xi^4 = -1$ is a primitive eighth root of unity.*

*Indeed, one sees that the line $\ell$ is described as*

$$\{[x_0 : \xi x_0 : x_2 : \xi x_2]\} \cong \mathbb{P}^1 \subset X.$$

*Now consider the line bundle $L = \mathcal{O}_X(1) \otimes \mathcal{O}_X(-\ell)$. Note $H^0(\mathcal{O}_{\mathbb{P}^3}(1))$ surjects onto $H^0(\mathcal{O}_X(1))$. So $|L|$ is comprised of all intersections of $X$ with a hyperplane which contains $\ell$.*

*Note that $(L.C) \geq 0$ for every curve $C$ since $L$ is effective. Furthermore, note $(L.L) = 0$. This is because we can take two hyperplanes which contain $\ell$, and they will intersect nowhere else except $\ell$, so in the linear system of $\mathcal{O}_X(1) \otimes \mathcal{O}_X(-\ell)$, their intersection on $X$ will be empty once we've deleted $\ell$.*

*Then corollary 14.9 predicts that $L \cong \mathcal{O}_X(mE)$ for some elliptic curve $E$. Indeed, note that $h^0(X, L) = 2$, since once we fix $\ell$, the space of planes in $\mathbb{P}^3$ which contain $\ell$ is 2-dimensional. These descend to $X$.*

*Then we have $|L| : X \to \mathbb{P}^1$, and we see that this is given by the residual plane cubic curves – the hyperplane intersecting $X$ will give a degree four plane curve, but we know it contains $\ell$, so we divide by $\ell$. Explicitly, one can write*

$$[x_0 : x_1 : x_2 : x_3] \mapsto [x_0^2 + \xi^2 x_1^2 : x_2^2 - \xi^2 x_3^2].$$

So at the beginning of this section we saw that a (-2) curve is automatically rational. Here, if a line bundle has the correct "arithmetic genus 0" quantity, namely the self intersection is 0, plus its nef, then we know that it is linearly equivalent to some multiple of an elliptic curve. In fact we have the following existence result:

**Proposition 14.11.** [12] *A K3 surface $X$ in characteristic $\neq 2, 3$ is elliptic (admits an elliptic fibration) if and only if there exists a line bundle $L$ on $X$ with $(L.L) = 0$.*

Other useful facts:

**Proposition 14.12.** *A complete linear system $|L|$ on a K3 surface $X$ has no base points outside its fixed part, i.e. $Bs|L| = F$.*

## 15. Blowing up curves on a surface, normal crossing divisors

Let $S$ be a complex surface. We make no assumption of compactness. In this section we discuss the phenomena of blowing up curves on a surface, normal crossing divisors which

[12] The idea is that given such an L, one can find a nef L' with 0 self intersection.

can be obtained with enough blow-up procedures, and we also discuss a classification of "simple" singularities of curves, in that order.

Let $C \subset S$ be a curve on $S$, so locally $C$ is given as the zero locus of some holomorphic function $f$. If we choose local coordinates $(u, v)$ of $S$ so that a point $p \in C$ is identified with the origin $(0, 0)$, then the function $f$ can be written as a power series in terms of the variables $u, v$:

$$f = \sum_{d \geq 0} f_d(u, v),$$

where $f_d(u, v)$ is a homogeneous polynomial of degree $d$.

**Definition 15.1.** *We say $f$ vanishes to order $n$ at $(0, 0)$ if*

$$f = \sum_{d \geq n} f_d(u, v),$$

*where $f_n(u, v) \neq 0$.*

Now consider the blow up $\tilde{S}$ with respect to $p \in S$. With respect to the local coordinates $u, v$ around $p$, the blow up $\tilde{S}$ can be described locally around the exceptional divisor $E$ as

$$\{y \times [z_0 : z_1] \mid u(y) z_1 = v(y) z_0\}.$$

Furthermore, we obtain two charts, $U_0$ and $U_1$ respectively, locally around $E$:

(1) in $U_0$, where $z_0 \neq 0$, the local coordinates can be taken to be $u$ and $\frac{z_1}{z_0}$,
(2) and in $U_1$, where $z_1 \neq 0$, the local coordinates can be taken to be $v$ and $\frac{z_0}{z_1}$.

Note in the first case, the exceptional divisor is given by $u = 0$ and in the second case by $v = 0$. With respect to the first case, the projection $\pi : \tilde{S} \to S$ is locally described by $(u, \frac{z_1}{z_0}) \mapsto (u, u\frac{z_1}{z_0})$, and with respect to second case, the projection is locally described by $(\frac{z_0}{z_1}, v) \mapsto (\frac{z_0}{z_1} v, v)$.

Now let's discuss the transformation of $C \subset S$ in the blow up $\tilde{S}$ with respect to $p \in C$. For example, in $U_0$, the pullback $\pi^* f$ is described as

$$\pi^* f(u, \frac{z_1}{z_0}) = \sum_{d \geq n} u^d f_d(1, \frac{z_1}{z_0}).$$

In particular, this local description tells us that we have

$$\pi^* C = nE + \overline{\pi^{-1}(C \setminus p)},$$

where $n$ is the order of vanishing of $C$ at $p$. It remains to understand $\overline{\pi^{-1}(C \setminus p)}$. This is the curve described in $U_0$ as

$$\frac{\pi^* f}{u^n} = \sum_{d \geq n} u^{d-n} f_d(1, \frac{z_1}{z_0}).$$

(1) The first thing to understand is that since $C$ is a curve passing through $p$, especially when $C$ is singular there may be multiple tangent lines to $C$ at $p$. These tangent lines are in bijection with the zeroes $[z_0 : z_1]$ of $f_n(z_0, z_1)$, where we are thinking

again of $p$ as the origin in the local coordinates $u, v$. This bijection follows from thinking about Taylor series and Euler's identity for homogeneous polynomials and their partials.

(2) Once we blow up $C$, each of these tangent lines give the unique points $0\times[z_0 : z_1]$ at which the *proper transform* $\overline{C} := \overline{\pi^{-1}(C \setminus p)}$ of $C$ intersects the exceptional divisor $E$.

On the other hand, we call $\pi^* C = nE + \overline{C}$ the *total transform* of $C$. An important thing about this blowing up process is that the multiplicity $n$ of $C$ does not increase with blowing up. More precisely, if $[a_1 : b_1], \cdots, [a_k : b_k]$, denote the tangent lines, thus are the zeroes of $f_n(z_0, z_1)$, of $f$, then let $C_1, \cdots, C_k$ denote the components of $C$ which correspond to the branches of $C$ which have $[a_i : b_i]$ as a tangent line. Then $C_i$ is locally given by the function $f_i$, and $f = \prod_{i=1}^k f_i$. If $n_i$ denotes the order of vanishing of $C_i$, then $\sum n_i = n$.

The proper transforms $\overline{C_1}, \cdots, \overline{C_k}$ will intersect $E$ at $0 \times [a_i : b_i]$, respectively. Now, for example, if $[a_i : b_i]$ is such that $a_i \neq 0$, then we can observe that on $U_0$, the proper transform $\overline{C_i}$ is given as the zero locus of

$$\frac{\pi^* f_i}{u^{n_i}} = \sum_{d \geq n_i} u^{d-n_i} f_{i,d}(1, \frac{z_1}{z_0}),$$

and we see that the smallest nonzero homogeneous part of this will have degree $\overline{n_i} \leq n_i$. Then we see that the new multiplicities will be $\sum \overline{n_i} \leq \sum n_i = n$. Observe that $\overline{n_i} = n_i$ when $f_{i,n_i}$ has a monomial term of the form $v^{n_i}$. The nonincreasing of multiplicity of vanishing and the separation of tangent lines suggests that possibly, under certain hypotheses, iterated blow ups can desingularize the curve $C$, if singularities are present. Indeed, this is the case.

**Proposition 15.2.** *Let $S$ be a complex surface, and let $C \subset S$ be a reduced curve. Then there is a nonsingular surface $X$ and proper map $\tau : X \to S$ obtained by a series of blow-ups (finitely many locally on $S$), such that the proper transform $\overline{C}$ of $C$ with respect to $\tau$ is a nonsingular curve on $X$.*

*Proof.* It suffices to consider just one singularity $p \in C \subset S$. Now if the vanishing of $C$ at $p$ has multiple tangent lines, or the multiplicity of vanishing goes down, then we're good. So it suffices to consider the case when the vanishing of $C$ has just one tangent line, and one blow up doesn't reduce the multiplicity of vanishing. By a change of coordinates, we can assume that this tangent line, in the local coordinates $u, v$ around $p$ which identify $p$ as the origin, is given by the line $[1 : 0]$. Then we can write $f$ as

$$\lambda v^n + \sum_{\ell + m \geq n+1} \lambda_{\ell m} u^\ell v^m.$$

Note that since $C$ is assumed to be reduced, there will be some $\ell$ such that $\lambda_{\ell 0} \neq 0$. Let $_0$ be the minimum such $\ell$. Now in $U_0$, the proper transform is given by the equation

$$\frac{\pi^* f}{u^n} = \lambda(\frac{z_1}{z_0})^n + \sum_{\ell+m\geq n+1} \lambda_{\ell m} u^{\ell+m-n}(\frac{z_1}{z_0})^m.$$

Now we are assuming that the multiplicity of vanishing of the proper transform $\overline{C}$ is $n$ again, and that it also has only one tangent line to the vanishing, otherwise we're done. These assumptions imply that $\ell + 2m - n \geq n+1$, so $\ell + 2m \geq 2n+1$, which means that the $\lambda_{\ell m} = 0$ for $2n \geq \ell + 2m \geq n+1$. Now note that as we blew up, the power of $u$ associated to $\lambda_{\ell 0}$ decreased $n$, namely it is $u^{\ell - n}$ now. Then if we keep blowing up, we see this process must terminate, by reducedness. □

**Corollary 15.3.** *If $S$ is a compact complex surface, then the desingularization of a reduced curve can happen with finitely many blow ups.*

Now suppose $C \subset S$ was not a reduced curve, but we allowed it to be very general: some analytic space. Then through blow ups the total transform can be described as a normal crossing divisor.

**Proposition 15.4.** *Let $\mathcal{I}_X \subset \mathcal{O}_S$ be a coherent sheaf of ideals. Then there is a non singular surface and a proper map*

$$\tau : Y \to S$$

*obtained by a succession of, locally finitely many, blow ups such that $\tau^*\mathcal{I}_X$ is the ideal sheaf of a normal crossing divisor in $Y$.*

*Proof.* It suffices to do this locally. Pick a point $p \in S$. Then $\mathcal{I}_p$ is given by $(f_1, \cdots, f_n)$. Here's the strategy:

(1) First show $(f_1, \cdots, f_n)$ can be generated by monomials.
(2) Then show that it can be generated by monomials of distinct degrees.
(3) Then show that one can continue to blow up to reduce the number of generators until one gets a principal ideal generated by a single monomial (hence giving a local normal crossing divisor).

The first step is easy let $f = \prod f_1 \cdots f_n$. Then we can can successively blow up to obtain $\tau_1 : Y_1 \to S$ so that at each point $\tau_1^{-1}(p)$, the total transform is locally generated by monomials.

To show that it can be generated by monomials of distinct degrees, suppose we have $u^{n_1}v^{d-n_1}, \cdots, u^{n_k}v^{d-n_k}$. Then if we blow up and pull back these monomials, we end for example, on $U_0$ with the functions

$$u^d(\frac{z_0}{z_1})^{d-n_1}, \cdots, u^d(\frac{z_0}{z_1})^{d-n_k},$$

so these monomials become divisible by just one of the monomials of that degree. So finally we consider when our ideal is generated by monomials of all different degrees

$$u^{d_1}(\frac{z_0}{z_1})^{d_1-n_1}, \cdots, u^{d_k}(\frac{z_0}{z_1})^{d_k-n_k}.$$

Then blowing up gives us a prinicpal ideal in the new chart, unless $n_1 > \cdots > n_k$. But in this case, blowing up will reduce the number of terms in such a way that the maximal degree will decrease. So this will terminate finitely.

□

## 16. Fibrations of K3 Surfaces; fibrations on Fermat quartic

Fibrations are important mathematical objects. One reason for being interested in them is given a fibration on $X$, knowledge of properties about the base and the general fiber can sometimes let you deduce properties about $X$ itself. In this section we will work with the following definition of fibration.

**Definition 16.1.** *Let $X$ and $Y$ be compact Kahler manifolds. Then $f : X \to Y$ is called a fibration if it is surjective and the general fiber is connected.*

Note that any surjective map $f : X \to Y$ is proper since $X$ is compact. By Stein factorization, the map $f$ factors as $X \to Y_{st} \to Y$, where the fibers of $X \to Y_{st}$ fibers are connected, and $Y_{st} \to Y$ is finite. Thus, up to finite morphisms, every surjective map is a fibration. The intuition here is that the connected components of the fibers of $f$ can first be individually collapsed to points of a spcae $Y_{st}$.

Often algebraic geometers will relax the definition of a fibration to allow $f$ to be dominant, and $X$ and $Y$ to be normal instead of smooth. However, our definition suffices for our intents and purposes.

**Proposition 16.2.** [Voi] *Let $f : X \to Y$ be a surjective holomorphic map between compact Kahler manifolds.*

- *The morphisms*

  $$H^i(Y;\mathbb{Q}) \to H^i(X;\mathbb{Q})$$

  *are injections of Hodge structures.*
- *If the smooth fibers $X_s$ are connected and $H^0(X_s, \Omega^i_{X_s}) = 0$ for $i > 0$, then*

  $$f^* : H^0(Y, \Omega^i_Y) \to H^0(X, \Omega^i_X)$$

  *is an isomorphism for $i \geq 0$.*

*Proof.* The idea is that the map $f^* : H^i(Y;\mathbb{R}) \to H^i(X;\mathbb{R})$ has a left inverse, which is a map $H^i(X;\mathbb{R}) \to H^i(Y;\mathbb{R})$ given by

$$\alpha \mapsto f_*(\omega^{\wedge \dim X - \dim Y} \wedge \alpha),$$

where $\omega$ is a Kahler form normalized to have volume 1 along the fibers, and $f_*$ is integration along fibers. A good reference for this technique is in Bott and Tu's book. The left inverse in real coefficients gives the injection in rational coefficients. The key idea for the second fact is to utilize the short exact sequence

$$0 \to f^*\Omega_{Y,s} \to \Omega_X|_{X_s} \to \Omega_{X_s} \to 0.$$

□

**Proposition 16.3.** [Voi] *Let $f : X \to Y$ be a fibration. Then*

$$K_X|_{X_s} \cong K_{X_s}.$$

*Proof.* Consider again the short exact sequence

$$0 \to f^*\Omega_{Y,s} \to \Omega_X|_{X_s} \to \Omega_{X_s} \to 0.$$

Noting that $f^*\Omega_{Y_s}$ is trivial, taking determinants yields

$$K_X|_{X_s} \cong K_{X_s} \otimes \bigwedge^{\dim Y} f^*\Omega_{Y_s} \cong K_{X_s},$$

as desired. □

Often fibrations $X \to Y$ are such that the fibers $X_s$ are $d$-dimensional subvarieties of some space $S$. Then we have the following:

**Proposition 16.4.** [Voi] *Consider*

$$\begin{tikzcd} X \arrow[r, "\phi"] \arrow[d, "f"'] & S \\ Y & \end{tikzcd}$$

*where $f$ is a fibration, $X_s \subset S$ is a $d$-dimensional subvariety, and $\phi$ is a dominant and generically finite map. Then for general $X_s$, we have*

$$K_{X_s} \cong K_S|_{X_s} + E,$$

*where $E$ is an effective divisor on $X_s$.*

*Proof.* By the previous proposition, we have $K_X|_{X_s} \cong K_{X_s}$. By Riemann-Hurwitz, we have

$$K_X \cong \phi^*K_S + R,$$

where $R$ is the ramification divisor. For general $X_s$, we have $X_s$ is not contained in $R$. Then

$$K_X|_{X_s} \cong K_{X_s} \cong \phi^*K_S|_{X_s} + R|_{X_s}.$$

□

Now that we have discussed some general facts about fibrations, we claim the following:

**Proposition 16.5.** *Every fibration of a complex K3 surface*[13] *over a curve is an elliptic fibration over* $\mathbb{P}^1$*.*

*Proof.* Let $X \to C$ be an elliptic fibration. First note that $H^{1,0}(C) \hookrightarrow H^{1,0}(X)$, and $H^{1,0}(X) = 0$. Thus, $H^{1,0}(C) = 0$, which implies that $C \cong \mathbb{P}^1$.

Second, note that $K_X|_{X_s} \cong K_{X_s} \implies K_{X_s} \cong \mathcal{O}_{X_s}$. Thus, one of the fibers will be a smooth compact complex curve $X_s$ with $H^{1,0}(X_s) = 1 \implies X_s$ is an elliptic curve. □

**Definition 16.6.** *An elliptic K3 surface is a K3 surface X together with a surjective morphism* $\pi : X \to \mathbb{P}^1$ *such that the geometric generic fibre is a smooth integral curve of genus one or, equivalently, if there exists a closed point* $t \in \mathbb{P}^1$ *such that* $X_t$ *is a smooth integral curve of genus one.*

Note by [Har, II.Prop.9.7], an elliptic fibration of a K3 surface is automatically flat. Thus, the arithmetic genus of all fibers is automatically 1.

Let's give an example of an elliptic fibration. Consider the Fermat quartic surface $F := Z(X_0^4 + X_1^2 + X_2^4 + X_3^4) \subset \mathbb{P}^3_{\mathbb{C}}$. There are exactly 48 lines on the Fermat quartic. These have explicit parameterizations of the form

$$[x_0 : \xi x_0 : x_2 : \xi x_2],$$

for different "positions" of $x_0$ and $x_2$, and $\xi$ can be substituted with any of the other primitive eighth roots of unity so that $\xi^4 = -1$. We have that each line $\ell$ defines an elliptic fibration

$$|\mathcal{O}_F(1) \otimes \mathcal{O}_F(-\ell)| : X \to \mathbb{P}^1.$$

So one takes the pencil of hyperplanes in $\mathbb{P}^3_{\mathbb{C}}$ which contain $\ell$, and use the complete linear system to define the elliptic fibration. More geometrically, this map can be described in the following way: take a generic line $L \subset \mathbb{P}^3$, so that it does not intersect $\ell$. Then pick a point $p \in F$. Then $p$ and $\ell$ will span a plane, and this will intersect $L$ at some point. The fiber over a point is the following: take a corresponding plane, then intersect it with the Fermat $F$. This will yield a degree 4 curve in the plane which will also include $\ell$. Divide out $\ell$ so you're left with a degree 3 plane curve: an elliptic curve (if it is smooth).

What happens to the other 47 lines? The claim is that another line $\ell'$ collapses to a point if and only if $\ell'$ intersects $\ell$. Otherwise, $\ell'$ maps isomorphically onto $\mathbb{P}^1$. Such $\ell'$ are called sections.

**Definition 16.7.** *Suppose* $\pi : X \to \mathbb{P}^1$ *is an elliptic fibration of a K3 surface. Then a section is a divisor* $D \subset X$ *such that* $\pi|_D : D \cong \mathbb{P}^1$*.*

[13] Recall every complex K3 surface is Kahler, by a result of Siu.

The sections of an elliptic fibration are in one to one correspond with the rational points of the generic fiber, as each section will intersect the generic fiber at a rational point, and the closure of each rational point of the generic fiber in $X$ gives a section.

If we further analyze the geometry of the elliptic fibration $|\mathcal{O}_F(1) \otimes \mathcal{O}_F(-\ell)|$, we find that there are —– come back here to determine the singular fibers; there are at least two singular fibers of type IV—-. but we also know this because the singular fibers of elliptic fibrations are classified:

**Theorem 16.8** (Kodaira)**.** *Let $\pi : X \to \mathbb{P}^1$ be an elliptic fibration of a K3 surface. All possible singular fibers are completely classified, and any such fibration has exactly 24 singular fibers if "counted properly." More specifically, all the different types of singular fibers $X_t$ have euler numbers $e(X_t)$, and we must have*

$$\sum_t e(X_t) = 24.$$

| | | | |
|---|---|---|---|
| $\mathrm{I}_0$ | smooth elliptic | $\widetilde{A}_0$ | $e = 0$ |
| $\mathrm{I}_1$ | rational curve with DP | $\widetilde{A}_0$ | $e = 1$ |
| II | rational curve with cusp | $\widetilde{A}_0$ | $e = 2$ |
| III | | $\widetilde{A}_1$ | $e = 3$ |
| $\mathrm{I}_2$ | | $\widetilde{A}_1$ | $e = 2$ |
| IV | | $\widetilde{A}_2$ (1, 1, 1) | $e = 4$ |
| $\mathrm{I}_{n\geq 3}$ | | $\widetilde{A}_{n-1}$ (1 ⋯ 1) | $e = n$ |
| $\mathrm{I}^*_{n\geq 0}$ | | $\widetilde{D}_{n+4}$ (1, 1, 2, 2 ⋯ 2, 2, 1, 1) | $e = n + 6$ |
| $\mathrm{II}^*$ | | $\widetilde{E}_8$ (2, 4, 6, 5, 4, 3, 2, 1; 3) | $e = 10$ |
| $\mathrm{III}^*$ | | $\widetilde{E}_7$ (1, 2, 3, 4, 3, 2, 1; 2) | $e = 9$ |
| $\mathrm{IV}^*$ | | $\widetilde{E}_6$ (1, 2, 3, 2, 1; 2, 1) | $e = 8$ |

In the case of the Fermat quartic, the singular fibers are of type IV and thus have Euler number 4. It's not too difficult to find six of them, and from the 24 count we know that they are the only singular fibers.

**Question 16.9.** *The line $\ell$ will intersect each smooth elliptic curve fiber at 3 points. A line $\ell' \subset F$ which does not intersect $\ell$ will be a section of the fibration, and thus will intersect each smooth elliptic curve fiber at a single point. Does there exist a line such that it intersects an inflection point of each smooth elliptic fiber such that it defines an origin such that the three points of intersection from $\ell$ are the 2-torsion points of the fiber?*

*If there does exist such a configuration, it is likely unique, in the sense that no other line could provide the origin. This would imply that lines on the Fermat quartic come in natural pairs.*

Note that given another distinct line $\ell'$, we can define an elliptic fibration

$$|\mathcal{O}_F(1) \otimes \mathcal{O}_F(-\ell')| : F \to \mathbb{P}^1.$$

However, such a fibration will be isomorphic to the fibration defined with respect to $\ell$ via a projective automorphism. One can ask if there are other elliptic fibrations on a Fermat quartic that do not come from a line. The answer is yes, and here is one example: one can represent the Fermat quartic as a Kummer surface. Let $\Lambda_1 = \langle 1, i \rangle$ and $\Lambda_2 = \langle 1, 2i \rangle$ be lattices, and consider the abelian surface $A = \mathbb{C}/\Lambda_1 \oplus \mathbb{C}/\Lambda_2$. The Fermat quartic can be identified with the Kummer surface associated to $A$.

In general, given two elliptic curves $E_1$ and $E_2$, we can define the Kummer surface $Kum(E_1 \times E_2)$ associated to $E_1 \times E_2$, which is the minimal resolution of the quotient $(E_1 \times E_2) \sim$ at the sixteen fixed points of the involution action on $E_1 \times E_2$. Then we have natural elliptic fibrations

$$Kum(E_1 \times E_2) \to E_1/\!\sim\, \cong \mathbb{P}^1 \text{ and } Kum(E_1 \times E_2) \to E_2/\!\sim\, \cong \mathbb{P}^1.$$

For such elliptic fibrations, there are exactly four singular fibers, of type $I_0^*$. Thus, these elliptic fibrations of the Fermat quartic[14] are not isomorphic to the ones we constructed from a line.

One of the amazing applications of elliptic fibrations is that if it admits a section, then sometimes we can determine the picard number of the K3 surface.

**Theorem 16.10** (Shioda-Tate formula). *Let $\pi : X \to \mathbb{P}^1$ be an elliptic fibration of a complex K3 surface $X$ which admits a section. Let $r_t$ denote the number of irreducible components[15] of singular fiber $X_t$. Then*

$$\rho(X) = 2 + \sum_t (r_t - 1) + \operatorname{rk} MW(X),$$

*where $MW(X)$ denotes the Mordell-Weil group of $X$.*

The Mordell-Weil group $MW(X)$ of $X$ is defined in the following way: recall that there is a bijective correspondence between sections of $\pi$ and rational points on the generic fiber $X_\eta$. Since $X_\eta$ is itself an elliptic curve over the function field $k(T)$ of $\mathbb{P}^1_k$, then the rational points admit a group structure, which we identify with $MW(X)$.

[14]Be careful; there are 3 non projectively isomorphic projective models for the Fermat quartic. While the model we took has 48 lines, the others may not.

[15]Sidenote: any section of an elliptic fibration must intersect each fiber at a unique point. Such a point must lie uniquely on one irreducible component and at a smooth point

**Proposition 16.11.** *Let $K = Kum(E_1 \times E_2)$ denote the Kummer surface associated to a product of elliptic curves. Then $\rho(X) = 18$ if $E_1, E_2$ are not isogeneous, $\rho(X) = 19$ if $E_1, E_2$ are isogeneous without CM, and $\rho(X) = 20$ if $E_1, E_2$ are isogeneous with CM.*

**Corollary 16.12.** *Note $\mathbb{C}/\langle 1, i\rangle$ and $\mathbb{C}/\langle 1, 2i\rangle$ are isogeneous with complex multiplication. Thus, the Fermat quartic surface has picard number 20.*

## 17. Some motivation for lattice theory via elliptic fibrations

I want to give some motivation for one way in which we study the geometry of K3 surfaces. At the end of the day, we are doing geometry, so we should eventually answer some geometric questions. But a lot of our techniques focus on, for example, studying $H^2(X, \mathbb{Z})$ of a K3 surface. First of all, one motivation for studying Hodge structures are the Torelli theorems.

**Theorem 17.1.** *Two smooth compact complex curves $C, C'$ are isomorphic if and only if there is an isomorphism $H^1(C, \mathbb{Z}) \cong H^1(C', \mathbb{Z})$ compatible with polarizations.*

**Theorem 17.2.** *Complex K3 surfaces $S, S'$ are isomorphic if and only if there is an isomorphism $H^2(S, \mathbb{Z}) \cong H^2(S', \mathbb{Z})$ respecting intersection pairing.*

But there other ways in which studying $H^2(S; \mathbb{Z})$, along with the intersection pairing, of a K3 surface can unravels its geometry. The perspective I want to take is that of elliptic fibrations. Consider the following problem: I would like to find all elliptic fibrations on a K3 surface, modulo automorphisms of the K3 surface.

Let $X$ be an algebraic K3 surface over an algebraically closed field $k$. It is well known that the Picard lattice $\mathrm{Pic}(X) = H^{1,1}(X) \cap H^2(X; \mathbb{Z})$ is a hyperbolic even integral lattice of signature $(1, \rho(X) - 1)$, and can be an arbitrary even hyperbolic lattice of rank $\rho(X) \leq 22$[Nik14].

We have that elliptic fibrations on $X$ are in one-to-one correspondence with primitive isotropic nef elements $c \in \mathrm{Pic}(X)$[PSS71]. Such elements are defined to be $c \in \mathrm{Pic}(X)$ such that:

- $c \neq 0$ but $(c.c) = 0$;
- $\frac{c}{n} \in \mathrm{Pic}(X)$ only for integers $n = \pm 1$;
- $(c.D) \geq 0$ for any effective divisor $D$ on $X$.

For such $c \in \mathrm{Pic}(X)$, the complete linear system $|c|$ is one dimensional without base points, and it gives an elliptic fibration $|c| : X \to \mathbb{P}^1$. This one to one correspondence gives us the first clue that elliptic fibrations on $X$ can be understood by studying the Picard lattice. More precisely, let $O^+(\mathrm{Pic}(X))$ denote the group of isometries of the Picard lattice along with the induced quadratic form. The Weyl group $W(\mathrm{Pic}(X))$ is the subgroup of $O^+(\mathrm{Pic}(X))$ generated by reflections by $(-2)$ elements in $\mathrm{Pic}(X)$. In other words, $W(\mathrm{Pic}(X))$ is generated by the isometries $R_\delta : x \mapsto x = (x.\delta)\delta$ where $\delta \in \mathrm{Pic}(X)$ is such that $(\delta.\delta) = -2$ (thus corresponds to a rational curve). Then "up to finite groups," there is an "isomorphism" between

$$\mathrm{Aut}(X) \cong \frac{O^+(\mathrm{Pic}(X))}{W(\mathrm{Pic}(X))}.$$

From this, we know that there is a one to one correspondence between

$$\frac{\{\text{ elliptic fibrations on X }\}}{Aut(X)} \longleftrightarrow \frac{\{\text{ primitive isotropic nef elements of } \mathrm{Pic}(X)\}}{W(\mathrm{Pic}(X))}.$$

It is a theorem that there are actually finitely many elliptic fibrations on an algebraic K3 surface, modulo the action of the automorphism group, in every characteristic.

## 18. Axial elliptic fibration of Fermat Quartic, Mordell-Weil computation

Let $X = Z(x_0^4 + x_1^4 + x_2^4 + x_3^4) \subset \mathbb{P}^3_{\mathbb{C}}$ be the Fermat quartic surface. Let

$$\ell = \{[x_0 : \xi x_0 : x_2 : \xi x_2]\} \subset X$$

be a line on $X$, where $\xi$ denotes the first eighth primitive root of unity such that $\xi^4 = -1$. Then the elliptic fibration $\phi_\ell : X \to \mathbb{P}^1$ is given by the linear system $|\mathcal{O}_X(1) \otimes \mathcal{O}_X(-\ell)|$. Visually, one way of describing this map $\phi_\ell$ is: fix a disjoint line $L$ and a point $p \in X \setminus \ell$, then $\ell$ and $p$ will span a plane $\Lambda$, which will intersect $L$ at a unique point. Then $\phi_\ell(p)$ is that unique point of $\Lambda \cap L$. The following is easy to see.

**Lemma 18.1.** *A line $\ell' \neq \ell$ on the Fermat quartic collapses to a point if and only if $\ell'$ intersects $\ell$. Otherwise, $\ell'$ is disjoint from $\ell$ and is a section.*

Here are all the 48 lines on the Fermat quartic and what their intersection with $\ell$ is. If a line intersects $\ell$ then I state the coordinate at which they intersect; if they do not intersect then I do not say anything. Here we begin with the 16 where the $x_0$ and $x_2$ coordinates parameterize the line:

(1) The line $\{[a : \xi a : b : \xi b]\}$ is exactly the line $\ell$.
(2) The line $\{[a : \xi a : b : \xi^3 b]\}$ intersects $\ell$ at $[1 : \xi : 0 : 0]$.
(3) The line $\{[a : \xi a : b : \xi^5 b]\}$ intersects $\ell$ at $[1 : \xi : 0 : 0]$.
(4) The line $\{[a : \xi a : b : \xi^7 b]\}$ intersects $\ell$ at $[1 : \xi : 0 : 0]$.
(5) The line $\{[a : \xi^3 a : b : \xi b]\}$ intersects $\ell$ at $[0 : 0 : 1 : \xi]$.
(6) The line $\{[a : \xi^3 a : b : \xi^3 b]\}$
(7) The line $\{[a : \xi^3 a : b : \xi^5 b]\}$
(8) The line $\{[a : \xi^3 a : b : \xi^7 b]\}$
(9) The line $\{[a : \xi^5 a : b : \xi b]\}$ intersects $\ell$ at $[0 : 0 : 1 : \xi]$.
(10) The line $\{[a : \xi^5 a : b : \xi^3 b]\}$
(11) The line $\{[a : \xi^5 a : b : \xi^5 b]\}$
(12) The line $\{[a : \xi^5 a : b : \xi^7 b]\}$
(13) The line $\{[a : \xi^7 a : b : \xi b]\}$ intersects $\ell$ at $[0 : 0 : 1 : \xi]$.

(14) The line $\{[a : \xi^7 a : b : \xi^3 b]\}$
(15) The line $\{[a : \xi^7 a : b : \xi^5 b]\}$
(16) The line $\{[a : \xi^7 a : b : \xi^7 b]\}$

In particular, we see that lines (2), (3), and (4) all intersect $\ell$ at $[1 : \xi : 0 : 0]$, and one can see that these three lines and $\ell$ are contained in the same hyperplane $Z(x_1 - \xi x_0)$ in $\mathbb{P}^3_{\mathbb{C}}$. This suggests that the lines (2), (3), and (4) should collapse to the same point under $\phi_\ell$, the fiber over which could still be type $IV$.

Similarly, we see that lines (5), (9), (13) intersect $\ell$ at the same point $[0 : 0 : 1 : \xi]$, and they are also contained in the same hyperplane $Z(x_3 - \xi x_2)$, so these lines should also colllapse to a point under $\phi_\ell$, suggesting that the fiber over which could still be type $IV$.

Now we list the next 16 lines, where the $x_0$ and $x_1$ coordinates parameterize the line.

(1) The line $\{[a : b : \xi a : \xi b]\}$ intersects $\ell$ at $[1 : \xi : \xi : \xi^2]$.
(2) The line $\{[a : b : \xi a : \xi^3 b]\}$
(3) The line $\{[a : b : \xi a : \xi^5 b]\}$
(4) The line $\{[a : b : \xi a : \xi^7 b]\}$
(5) The line $\{[a : b : \xi^3 a : \xi b]\}$
(6) The line $\{[a : b : \xi^3 a : \xi^3 b]\}$ intersects $\ell$ at $[1 : \xi : \xi^3 : \xi^4]$.
(7) The line $\{[a : b : \xi^3 a : \xi^5 b]\}$
(8) The line $\{[a : b : \xi^3 a : \xi^7 b]\}$
(9) The line $\{[a : b : \xi^5 a : \xi b]\}$
(10) The line $\{[a : b : \xi^5 a : \xi^3 b]\}$
(11) The line $\{[a : b : \xi^5 a : \xi^5 b]\}$ intersects $\ell$ at $[1 : \xi : \xi^5 : \xi^6]$.
(12) The line $\{[a : b : \xi^5 a : \xi^7 b]\}$
(13) The line $\{[a : b : \xi^7 a : \xi b]\}$
(14) The line $\{[a : b : \xi^7 a : \xi^3 b]\}$
(15) The line $\{[a : b : \xi^7 a : \xi^5 b]\}$
(16) The line $\{[a : b : \xi^7 a : \xi^7 b]\}$ intersects $\ell$ at $[1 : \xi : \xi^7 : 1]$.

So lines (1), (6), (11), (16) intersect $\ell$ all at distinct points. Furthermore, none of them intersect one another, so they must all lie on different planes, so they must map to different points under $\phi_\ell$ (because all these planes already intersect at $\ell$, but $L$ is disjoint from $\ell$).

The last 16 lines are such that the $x_0, x_1$ coordinates still parameterize the lines, but the $x_2, x_3$ coordinates get "swapped" relative to the previous 16 lines we described.

(1) The line $\{[a : b : \xi b : \xi a]\}$
(2) The line $\{[a : b : \xi^3 b : \xi a]\}$
(3) The line $\{[a : b : \xi^5 b : \xi a]\}$
(4) The line $\{[a : b : \xi^7 b : \xi a]\}$ intersects $\ell$ at $[1 : \xi : 1 : \xi]$.
(5) The line $\{[a : b : \xi b : \xi^3 a]\}$ intersects $\ell$ at $[1 : \xi : \xi^2 : \xi^3]$.
(6) The line $\{[a : b : \xi^3 b : \xi^3 a]\}$

(7) The line $\{[a : b : \xi^5 b : \xi^3 a]\}$
(8) The line $\{[a : b : \xi^7 b : \xi^3 a]\}$
(9) The line $\{[a : b : \xi b : \xi^5 a]\}$
(10) The line $\{[a : b : \xi^3 b : \xi^5 a]\}$ intersects $\ell$ at $[1 : \xi : \xi^4 : \xi^5]$.
(11) The line $\{[a : b : \xi^5 b : \xi^5 a]\}$
(12) The line $\{[a : b : \xi^7 b : \xi^5 a]\}$
(13) The line $\{[a : b : \xi b : \xi^7 a]\}$
(14) The line $\{[a : b : \xi^3 b : \xi^7 a]\}$
(15) The line $\{[a : b : \xi^5 b : \xi^7 a]\}$ intersects $\ell$ at $[1 : \xi : \xi^6 : \xi^7]$.
(16) The line $\{[a : b : \xi^7 b : \xi^7 a]\}$

Among these last 16 lines, only four lines, namely lines $(4), (5), (10)$ and $(15)$, intersect $\ell$. None of the four lines do not intersect each other pairwise, so they are not contained in the same planes. Therefore, they individually collapse to distinct points under $\phi_\ell$.

While lines $\{(2), (3), (4)\}$ and lines $\{(5), (9), (13)\}$ give two singular fibers of type IV, we claim the other 8 lines are each irreducible components of 8 distinct singular fibers which are reducible singular cubics. In particular, we claim that they are fibers of type $I_2$. One way to see this is to use the Shioda-Tate formula and see the singular fibers must be a line and a conic with two double points.

Another way is to do this explicitly: consider for example the line $\ell' = \{[a : b : \xi a : \xi b]\}$. Then $\ell'$ and $\ell$ span the plane

$$\Lambda = V(x_3 - \xi x_2 - \xi x_1 + \xi^2 x_0).$$

Then the singular fiber which contains $\ell'$ is contained in the intersection

$$\Lambda \cap X = V(x_0^4 + x_1^4 + x_2^4 + x_3^4, x_3 - \xi x_2 - \xi x_1 + \xi^2 x_0) = \ell \cup \ell' \cup C.$$

Then calculate the equation for $C$ and the intersection $\ell'$ with $C$.

**Proposition 18.2.** *The elliptic fibrations on the Fermat quartic surface defined with respect to a line $\ell$ on the Fermat quartic has 10 singular fibers. Two of them of type $IV$, and 8 of them of type $I_2$. The fibration restricted to $\ell$ is a 3 to 1 map, and the other 33 lines on the Fermat quartic surface are sections.*

Now suppose we take a nonsingular fiber of this elliptic fibration $\pi_\ell$, and call it $H - \ell$. Choose a line $\ell_0$ which does not intersect any of the 8 lines $\ell_{s_j}$, $1 \leq j \leq 8$, in the singular reducible fibers. Then we can choose lines $\ell_1^1, \ell_2^1$ in one singular fiber of type $IV$ and choose lines $\ell_1^2, \ell_2^2$ in the other singular fiber of type $IV$ such that $\ell_0$ does not intersect any of $\ell_k^j$. Then we have the orthogonal complement of the embedding

$$\langle \ell_0, H - \ell \rangle \oplus \langle \ell_1^1, \ell_2^1 \rangle \oplus \langle \ell_1^2, \ell_2^2 \rangle \oplus \bigoplus_{j=1}^{8} \langle \ell_{s_j} \rangle \hookrightarrow \operatorname{Pic}(X)$$

in the Picard lattice is the Mordell-Weil lattice. Note this gives an embedding of lattices

$$R = \begin{pmatrix} -2 & 1 \\ 1 & 0 \end{pmatrix} \oplus \begin{pmatrix} -2 & 1 \\ 1 & -2 \end{pmatrix}^{\oplus 2} \oplus (-2)^{\oplus 8} \hookrightarrow E_8(-1)^{\oplus 2} \oplus U \oplus (-8)^{\oplus 2}.$$

**Proposition 18.3.** *The rank of the Mordell-Weil lattice, modulo torsion, is 6* [Shi15]. *Thus, the Fermat quartic surface has Picard rank*

$$\rho(X) = 20,$$

*by the Shioda-Tate formula.*

**Proposition 18.4.** *The torsion of the Mordell-Weil lattice of $X$ is given by*

$$R_{sat}/R,$$

*where $R_{sat}$ is the smallest primitive sublattice of* $\mathrm{Pic}(X)$ *which contains $R$. In particular, the torsion of the Mordell-Weil lattice is 0 in this case* [Shi15]. *Thus, the 33 line sections of this line fibration have infinite order on the generic fiber.*

The general formula for the pairing on the Mordell-Weil lattice is explicitly given in [Shi92, Theorem 2.4].

## 19. Kuga-Stake Construction

## 20. TGT - Towards Global Torelli for Complex K3 Surfaces

The global Torelli theorem for complex K3 surfaces states the following.

**Theorem 20.1.** *Let $S, S'$ be two complex K3 surfaces. Then $S \cong S'$ if and only if there is an isomorphism $H^2(S;\mathbb{Z}) \cong H^2(S';\mathbb{Z})$ compatible with intersection pairings.*

In the subsequent subsections, I review the main ideas behind the global Torelli theorem. First, we need to discuss deformations of complex structures. We do this by first discussing almost complex structures and deformations of complex structures. Afterwards, we sketch the classical Riemann-Hilbert correspondence, before moving onto discuss the local Torelli theorem for complex K3 surfaces.

### 20.1. **TGT1: Almost Complex Structures.**

This review of almost complex structures follows the discussion in [Huy05].

**Definition 20.2.** *An almost complex manifold is a differentiable manifold $X$ together with a vector bundle endomorphism*

$$I : TX \to TX, \text{ with } I^2 = -Id.$$

The endomorphism $I$ is called the *almost complex structure* on the underlying real manifold. If an almost complex structure exists, then the real dimension of $X$ is even.

**Proposition 20.3.** *Any complex manifold $X$ admits a natural almost complex structure.*

*Proof.* Cover $X$ by holomorphic charts $U_i$. Note at each point $T_xU$, where we consider the real $2n$ tangent space at $x$, the almost complex structure on this vector space is given by $\frac{\partial}{\partial x_i} \mapsto \frac{\partial}{\partial y_i}$, and $\frac{\partial}{\partial y_i} \mapsto -\frac{\partial}{\partial x_i}$. Note that when we complexify $T_xU$, we get a decomposition

$$T_{x,\mathbb{C}}U = T_x^{1,0}U \oplus T_x^{0,1}U,$$

where $T_x^{1,0}U$ and $T_x^{0,1}U$ are the $i$ and $-i$ eigenspaces of $I$. Note that $T_x^{1,0}U$ will then have basis

$$\frac{\partial}{\partial z_i} = \frac{1}{2}(\frac{\partial}{\partial x_i} - iI\frac{\partial}{\partial x_i}) = \frac{1}{2}(\frac{\partial}{\partial x_i} - i\frac{\partial}{\partial y_i})$$

and $T_x^{0,1}$ will have basis

$$\frac{\partial}{\partial \overline{z_i}} = \frac{1}{2}(\frac{\partial}{\partial x_i} + i\mathfrak{I}\frac{\partial}{\partial x_i}) = \frac{1}{2}(\frac{\partial}{\partial x_i} + i\frac{\partial}{\partial y_i}).$$

We see in general, this $I$ extends to $I : TU_i \to TU_i$, and since the transition functions are holomorphic, it extends to an endomorphism $I : TX \to TX$ to the entire tangent bundle, so that $I^2 = -Id$. To see why, note that if we complexify everything, we have this map

$$I_i : TX|_{U_i} \cong U_i \times \{\frac{\partial}{\partial z_i}, \frac{\partial}{\partial \overline{z_i}}\} \to U_i \times \{\frac{\partial}{\partial z_i}, \frac{\partial}{\partial \overline{z_i}}\} \cong TX|_{U_i},$$

and the same goes for

$$I_j : TX|_{U_j} \cong U_j \times \{\frac{\partial}{\partial z_i}, \frac{\partial}{\partial \overline{z_i}}\} \to U_j \times \{\frac{\partial}{\partial z_i}, \frac{\partial}{\partial \overline{z_i}}\} \cong TX|_{U_j}.$$

Now the transition function between $U_i$ and $U_j$ on overlap $U_i \cap U_j$ gives the transition function for $TX$ over $U_i \cap U_j$. This transition will preserve the $i$-eigenspace and $-i$ eigenspace. Thus, we see that these endomorphisms $I_i$ and $I_j$ over $U_i \cap U_j$ commute, by the diagram

$$\begin{tikzcd}
\frac{\partial}{\partial z_i} \arrow[r, "I_i"] \arrow[d] & i\frac{\partial}{\partial z_i} \arrow[d] \\
\mathfrak{J}(\phi_{ij})\frac{\partial}{\partial z_i} \arrow[r, "I_j"'] & i\mathfrak{J}(\phi_{ij})\frac{\partial}{\partial z_i}
\end{tikzcd}$$

and the same goes for the $-i$ eigenspace. Then by linearity one easily sees that we have commutativity for the $\frac{\partial}{\partial x_i}$ and $\frac{\partial}{\partial y_i}$. So the complex structure of a complex manifold $X$ gives a natural almost complex structure on $TX$.

□

**Remark 20.4.** *Not every real manifold of even dimenion admits an almost complex structure. For example, the four-dimensional sphere.*

Now, given a real manifold with complex structure $(X, I)$, if we complexify the tangent bundle of $X$, then we obtain a decomposition

$$T_{\mathbb{C}}X = T^{1,0}(X) \oplus T^{0,1}(X),$$

where $T^{1,0}(X)$ is the vector bundle of $(i)$-eigenspaces of $I$ and $T^{0,1}(X)$ is the vector bundle of $(-i)$-eigenspaces of $I$. Dualizing and wedging provides a decomposition of differential forms:

$$\bigwedge^k (T_{\mathbb{C}}X)^* = \bigoplus_{p+q=k} \bigwedge^p (T^{1,0}X)^* \otimes \bigwedge^q (T^{0,1}X)^*,$$

where we denote the sheaf of sections of these vector bundles as $\mathcal{A}_X^k$ and $\mathcal{A}_X^{p,q}$. Extending the exterior derivative $\mathbb{C}$-linearly yields

$$d : \mathcal{A}_X^k \to \mathcal{A}_X^k.$$

It would be very nice if $d = \partial + \overline{\partial}$, where $\partial : \mathcal{A}_X^{p,q} \to \mathcal{A}^{p+1,q}$ and $\overline{\partial} : \mathcal{A}_X^{p,q} \to \mathcal{A}_X^{p,q+1}$. In the familiar situation where we know $X$ is a complex manifold, we know this to be true. But in general, given a real manifold with almost complex structure $(X, I)$, this need not be true.

**Proposition 20.5.** *Let $(X, I)$ be a real manifold with almost complex structure. The following are equivalent:*

(1) $d = \partial + \overline{\partial}$
(2) *For $\alpha \in \mathcal{A}^{1,0}(X)$, $\Pi^{0,2} d\alpha = 0$, where $\Pi^{0,2}$ denotes projection to the $(0, 2)$ component.*

*Proof.* (1) implying (2) is almost immediate. Given $\alpha \in A^{1,0}(X)$, we have $d\alpha \in \mathcal{A}^{2,0}(X) \oplus \mathcal{A}^{1,1}(X)$, so projection to $\mathcal{A}^{0,2}(X)$ is zero.

To see that (2) implies (1), note that $d = \partial + \overline{\partial}$ if and only if for every $\alpha \in \mathcal{A}^{p,q}(X)$, we have $d\alpha \in \mathcal{A}^{p+1,q}(X) \oplus \mathcal{A}^{p,q+1}(X)$. We can locally express $\alpha$ as

$$f\omega_{i_1} \wedge \cdots \wedge \omega_{i_p} \wedge \omega'_{j_1} \wedge \cdots \wedge \omega'_{j_q}.$$

By the Leibniz formula, $d\alpha$ is written in terms of $df \in \mathcal{A}^{1,0}(X) \oplus \mathcal{A}^{0,1}(X)$, and $d\omega_{i_t} \in \mathcal{A}^{2,0}(X) \oplus \mathcal{A}^{1,1}(X)$, and $d\omega'_{j_t} = \overline{d\overline{\omega'_{j_t}}} \in \overline{\mathcal{A}^{2,0}(X) \oplus \mathcal{A}^{1,1}(X)} = \mathcal{A}^{1,1}(X) \oplus \mathcal{A}^{0,2}(X)$. We see then that $d\alpha \in \mathcal{A}^{p+1,q}(X) \oplus \mathcal{A}^{p,q+1}(X)$ as desired. □

**Definition 20.6.** *An an integrable complex structure $I$ on a real manifold $X$ is an almost complex structure on $X$ such that proposiiton 20.5 holds.*

Thus, one way of understanding an integrable complex structures $I$ is that it is an almost complex structure that the exterior derivative behaves "nicely" with respect to it. Another perspective on integrable complex structures is through vector fields.

**Proposition 20.7.** *An almost complex structure $I$ is integrable if and only if the Lie bracket of vector fields preserves $T_X^{0,1}$, i.e. $[T_X^{0,1}, T_X^{0,1}] \subseteq T_X^{0,1}$.*

*Proof.* Here we will use the invariant formula for the exterior derivative. If $\alpha$ is a $k$-form and $v_1, \cdots, v_{k+1}$ are vector fields, then

$$d\alpha(v_1, \cdots, v_{k+1}) = \sum_{i=1}^{k+1} (-1)^{i+1} v_i \cdot \alpha(v_1, \cdots, \widehat{v_i}, \cdots, v_{k+1})$$

$$+\sum_{1\le i<j\le k+1}(-1)^{i+j}\alpha([v_i,v_j],v_1,\cdots,\widehat{v_i},\cdots,\widehat{v_j},\cdots,v_{k+1}).$$

In the case of 1-forms, we have the that $d\alpha(v_1,v_2)=v_1\alpha(v_2)-v_2\alpha(v_1)-\alpha([v_1,v_2])$. This can be readily verified:

- we can assume that $\alpha=fdg$ for some smooth functions $f,g$. Then the left hand side simplifies to

$$d(fdg)(v_1,v_2)=(df\wedge dg)(v_1,v_2)=v_1(f)v_2(g)-v_2(f)dg(v_1).$$

The right hand side simplifies to

$$\begin{aligned}&v_1(fdg(v_2))-v_2(fdg(v_1))-fdg([v_1,v_2])\\ &=v_1(f)dg(v_2)+fv_1(dg(v_2))-v_2(f)dg(v_1)-fv_2(dg(v_1))-f[v_1v_2-v_2v_1](g)\\ &=v_1(f)v_2(g)+fv_1v_2(g)-v_2(f)v_1(g)-fv_2v_1(g)-f[v_1v_2-v_2v_1](g)\\ &=v_1(f)v_2(g)-v_2(f)dg(v_1).\end{aligned}$$

Now suppose that $I$ is integrable. Let $X,Y$ be any two sections of $T_X^{0,1}$. Then for any $\omega\in\mathcal{A}_X^{1,0}$, we have

$$d\omega(X,Y)=X\omega(Y)-Y\omega(X)-\omega([X,Y]).$$

Now since $I$ is integrable, we have $d\omega\in\mathcal{A}_X^{2,0}\oplus\mathcal{A}_X^{1,1}$. Then $d\omega(X,Y)=0$, and $\omega(Y)=\omega(X)=0$ as well, so $\omega([X,Y])=0$ for all $\omega\in\mathcal{A}_X^{1,0}$, which implies that $[X,Y]$ is again a section of $T_X^{0,1}$.

Now suppose that $[T_X^{0,1},T_X^{0,1}]\subseteq T_X^{0,1}$. Let $X,Y$ be sections of $T^{0,1}$. Then for every $\omega\in\mathcal{A}_X^{1,0}$, we have $\omega([X,Y])=0$, so

$$d\omega(X,Y)=X\omega(Y)-Y\omega(X)=0.$$

This implies that $d\omega\in\mathcal{A}^{2,0}\oplus\mathcal{A}^{1,1}$. □

**Corollary 20.8.** *If $I$ is an integrable almost complex structure, then $\partial^2=\overline{\partial}^2=0$, and $\partial\overline{\partial}=-\overline{\partial}\partial$. Conversely, if $\overline{\partial}^2=0$, then $I$ is integrable.*

*Proof.* The first claims follow immediately from integrability implying $d=\partial+\overline{\partial}$, combined with bidegree considerations and $d^2=0$.

Now suppose $\overline{\partial}^2=0$. We'd like to show $I$ is integrable. It suffices to show $[T^{0,1},T^{0,1}]\subseteq T^{0,1}$. Let $X,Y$ be sections of $T^{0,1}$. Note that if $\alpha$ is a (0,1) form, then $d\alpha(X,Y)=\overline{\partial}\alpha(X,Y)$. Then again, using the formula

$$d\omega(X,Y)=X\omega(Y)-Y\omega(X)-\omega([X,Y]),$$

setting $\omega=\overline{\partial}f$, we find

$$\begin{aligned}0=\overline{\partial}^2f=d\overline{\partial}f(X,Y)&=X\overline{\partial}f(Y)-Y\overline{\partial}f(X)-\overline{\partial}f([X,Y])\\ &=Xdf(Y)-Ydf(X)-\overline{\partial}f([X,Y])\\ &=d^2f(X,Y)+df([X,Y])-\overline{\partial}f([X,Y])=\partial f([X,Y]).\end{aligned}$$

Thus, since $\partial f([X,Y]) = 0$ for all $f$, this implies $[X,Y]$ is a section of $T^{0,1}$. □

The following highly non-trivial theorem tells us that a real manifold that admits an integrable almost complex structure is secretly a complex manifold. (And it is an exercise that an almost complex structure is induced by at most one complex structure)

**Theorem 20.9** (Newlander-Nierenberg)**.** *Any integrable almost complex structure is induced by a complex structure.*

Now that we have discussed complex structures, let us discuss deformations of complex structures.

20.2. **TGT2: Deformations of Complex Structures.** This section follows [Sch05]. Suppose we have a holomorphic submersion $\pi : \mathfrak{X} \to B$ between compact complex submanifolds. Let $0 \in B$ be a distinguished point. By Ehressmann's thoerem, there exists a contractible neighborhood $\Delta$ around $0 \in B$ such that $\mathfrak{X}|_\Delta \to \Delta$ is diffeomorphic as a family to $\mathfrak{X} \times \Delta \to \Delta$. In particular, all other fibers $\mathfrak{X}_t$ have underlying $C^\infty$ structures diffeomorphic to $\mathfrak{X}_0$. In this regard, around $\mathfrak{X}_0$, we can consider the family $\mathfrak{X}|_\Delta \to \Delta$ as a deformation of the complex structure of $\mathfrak{X}_0$. A natural question that arises from this is: given a priori just the projection $\mathfrak{X}_0 \times \Delta \to \Delta$ and some kind of data of almost complex structures on each $\mathfrak{X}_0 \times \{t\}$, how can we tell that this is actually a *deformation of complex structures*? Let's make this more precise. First of all, what kind of data are we even talking about?

Let's first discuss how we can compare different complex structures, and more broadly speaking almost complex structures. Let's suppose that $M$ is a compact complex manifold. Then $M$ has a natural complex structure which induces the direct sum

$$(TM)_{\mathbb{C}} \cong T_M^{1,0} \oplus T_M^{0,1}.$$

Let $\pi_{1,0}$ and $\pi_{0,1}$ denote the obvious projections. If we have some other almost complex structure $I : TM \to TM$, then it induces a different decomposition

$$(TM)_{\mathbb{C}} \cong T_{M,t}^{1,0} \oplus T_{M,t}^{0,1}.$$

Note that the data of the subbundles $T_M^{0,1}$ and $T_{M,t}^{0,1}$ is equivalent to their respective associated almost complex structures. We say that $I$ is sufficiently close to the first chosen complex structure on $M$ if the restriction of the projection

$$\pi_{0,1} : T_{M,t}^{0,1} \subset (TM)_{\mathbb{C}} \to T_M^{0,1}$$

is an isomorphism. Then when $I$ is sufficiently close the first chosen complex structure, we can keep track of its "difference" by the following map:

$$\phi : T_M^{0,1} \to^{\pi_{0,1}^{-1}} T_{M,t}^{0,1} \hookrightarrow (TM)_{\mathbb{C}} \to^{\pi_{1,0}} T_M^{1,0}.$$

In particular, we have
$$(id + \phi(t))(T^{0,1}) = T^{0,1}_{M,t}.$$
Note $\phi$ can be compactly described as an element of $\mathcal{A}^{0,1}(T^{1,0}_M)$, and in local coordinates is given by
$$\sum_j h_{ij}(z) d\overline{z_i} \otimes \frac{\partial}{\partial z_j},$$
so that locally $T^{0,1}_{M,t} \subset (TM)_{\mathbb{C}}$ is described as the span of the vector fields
$$\frac{\partial}{\partial \overline{z_k}} + \sum_j h_{kj} \frac{\partial}{\partial z_j}.$$
Let us return to the situation of a holomorphic surjective submersion $\pi : \mathfrak{X} \to B$ between compact complex manifolds, where we distinguish a fiber $\mathfrak{X}_0$ over $0 \in B$. Again, by Ehressmann's thoerem, this family is locally isotrivial in its $C^\infty$ structure. But in fact, we can do a little better:

**Lemma 20.10.** *For a sufficiently small open neighborhood $0 \in \Delta \subset B$, there exists a transversely holomorphic trivialization:*
$$\begin{tikzcd}
\mathfrak{X}|_\Delta \arrow[rr, "{(\phi,\pi)}"'] \arrow[dr] & & \mathfrak{X}_0 \times \Delta \arrow[dl] \\
& \Delta &
\end{tikzcd}$$
*so that*

- *$(\phi, \pi)$ is a diffeomorphism.*
- *$\phi$ is identity when restricted to $\mathcal{X}_0$*
- *the fibers of $\phi$ are holomorphic submanifolds of $\mathfrak{X}$ transverse to $\mathfrak{X}_0$ and isomorphic as complex manifolds to $\Delta$.*

Now, we can compare the transported complex structure on $\mathfrak{X}_0 \times \Delta$ to the natural complex structure it inherits as a product of two complex manifolds. The difference will be capture by some element of
$$\phi(t) \in \mathcal{A}^{0,1}(\mathfrak{X}_0 \times \Delta, T^{1,0}_{\mathfrak{X}_0 \times \Delta}),$$
which provides a map from
$$\pi^*_{\mathfrak{X}_0} T^{0,1}_{\mathfrak{X}_0} \oplus \pi^*_\Delta T^{0,1}_\Delta \cong T^{0,1}_{\mathfrak{X}_0 \times \Delta} \to T^{1,0}_{\mathfrak{X}_0 \times \Delta} \cong \pi^*_{\mathfrak{X}_0} T^{1,0}_{\mathfrak{X}_0} \oplus \pi^*_\Delta T^{1,0}_\Delta.$$
However, note that because $\pi$ is holomorphic, the element $\phi(t)$ will actually take values in just $\pi^*_{\mathfrak{X}_0} T^{1,0}_{\mathcal{X}_0}$. Furthermore, since anti holomorphic vector fields on $\{p\} \times \Delta$ correspond to an anti holomorphic vector field on $\phi^{-1}(p)$, we find that $\phi(t)$ maps $0 \oplus \pi^*_\Delta T^{0,1}_\Delta$ to 0. Then the difference between the almost complex structure is really captured by a morphism
$$\pi^*_{\mathcal{X}_0} T^{0,1}_{\mathfrak{X}_0} \to \pi^*_{\mathcal{X}_0} T^{1,0}_{\mathfrak{X}_0},$$

and is thus an element of $\pi^*_{\mathfrak{X}_0}\mathcal{A}^{0,1}(T^{1,0}_{\mathfrak{X}_0})$. But this element is not just any element of $\pi^*_{\mathfrak{X}_0}\mathcal{A}^{0,1}(T^{1,0}_{\mathfrak{X}_0})$ – it corresponds to a deformation of complex structures.

So here's the question: suppose we have the projection $\mathfrak{X}_0 \times \Delta \to \Delta$, where $\mathfrak{X}_0$ is a compact complex manifold and $\Delta$ is some contractible complex manifold (we can take it to be an open around the origin in $\mathbb{C}^m$), and we have *just a smooth section* $\phi(t) \in \pi^*_{\mathfrak{X}_0}\mathcal{A}^{0,1}(T^{1,0}_{\mathfrak{X}_0})$.... how do we know this data corresponds to a deformation of *complex structures*?

**Theorem 20.11.** *A smooth section $\phi(t) \in \pi^*_{\mathfrak{X}_0}\mathcal{A}^{0,1}(T^{1,0}_{\mathfrak{X}_0})$ corresponds to a complex structure on $\mathfrak{X}_0 \times \Delta$ if and only if*

$$\overline{\partial}_{\mathfrak{X}_0}\phi(t) + \frac{1}{2}[\phi(t), \phi(t)] = 0$$

*and $\phi(t)$ is holomorphic in $t$.*

*Proof.* We use the Newlander-Nirenberg criterion for an almost complex structure to be integrable. We need to show that

$$[T^{0,1}_{\mathfrak{X}_0\times\Delta,t}, T^{0,1}_{\mathfrak{X}_0\times\Delta,t}] \subseteq T^{0,1}_{\mathfrak{X}_0\times\Delta,t}.$$

It suffices to work locally. Suppose $z_1, \cdots, z_n$ are $t_1, \cdots, t_m$ are local coordinates for $\mathfrak{X}_0 \times \Delta$, and $\phi(t)$ is locally described as

$$\sum h_{ij}(z,t) d\overline{z_i} \otimes \frac{\partial}{\partial z_j}.$$

We have that $T^{0,1}_{\mathfrak{X}_0\times\Delta,t}$ is spanned by the vector fields

$$\frac{\partial}{\partial \overline{z_k}} + \sum_j h_{kj}(z,t)\frac{\partial}{\partial z_j} \text{ for } 1 \le k \le n$$

and

$$\frac{\partial}{\partial \overline{t_\ell}} \text{ for } 1 \le \ell \le m.$$

Note that

$$[\frac{\partial}{\partial \overline{t_\ell}}, \frac{\partial}{\partial \overline{z_k}} + \sum_j h_{kj}(z,t)\frac{\partial}{\partial z_j}] = \sum_j \frac{\partial h_{kj}}{\partial \overline{t_\ell}}\frac{\partial}{\partial z_j} = 0 \implies \frac{\partial h_{kj}}{\partial \overline{t_\ell}} = 0, \forall k, j, \ell.$$

So we see that in order to have a complex structure, we need $\phi(t)$ to be holomorphic in $t$.

Next, consider the conditions that fall out of

$$[\frac{\partial}{\partial \overline{z_k}} + \sum_j h_{kj}(z,t)\frac{\partial}{\partial z_j}, \frac{\partial}{\partial \overline{z_r}} + \sum_j h_{rj}(z,t)\frac{\partial}{\partial z_j}] \in T^{0,1}_{\mathfrak{X}_0\times\Delta,t}.$$

Expanding this out, we see that this holds if and only if $\overline{\partial}_{\mathfrak{X}_0}\phi(t) + \frac{1}{2}[\phi(t), \phi(t)] = 0$ holds. $\square$

The equation $\overline{\partial}_{\mathfrak{X}_0}\phi(t) + \frac{1}{2}[\phi(t), \phi(t)] = 0$ is called the Maurer-Cartan equation. Now, theorem 20.11 seems to just be talking about a complex structure on $\mathfrak{X}_0 \times \Delta$. But actually, the Maurer-Cartan equation is the necessary and sufficient condition for the almost complex structure on $\mathfrak{X}_0$ at time $t$ to be a complex structure, and the second necessary and sufficient condition of theorem 20.11 guarantees that the family is an analytic family. If we have the Maurer-Cartan equation but not the second condition, then we just have a $C^\infty$ deformation of complex manifolds. Together, these two conditions are necessary and sufficient for our data of $\mathfrak{X}_0 \times \Delta \to \Delta$ and $\phi(t)$ to correspond to an analytic family of compact complex manifolds.

Before we end this section, we should mention how the Kodaira-Spencer map can be explicitly related to the data of $\mathfrak{X}_0 \times \Delta \to \Delta$ and $\phi(t) \in \phi(t) \in \pi^*_{\mathfrak{X}_0}\mathcal{A}^{0,1}(T^{1,0}_{\mathfrak{X}_0})$ satisfying theorem 20.11. Recall that if we have an analytic family $\pi : \mathfrak{X} \to B$, then we have the short exact sequence

$$0 \to T_{\mathfrak{X}_0} \to T_{\mathfrak{X}}|_{\mathfrak{X}_0} \to (\pi^* T_B)|_{\mathfrak{X}_0} \to 0,$$

(note we're talking about holomorphic tangent bundles here) where $(\pi^*T_B)|_{\mathfrak{X}_0} \cong T_0B \otimes_{\mathbb{C}} \mathcal{O}_{\mathfrak{X}_0}$. This induces a map

$$T_0B \to H^1(T_{\mathfrak{X}_0}),$$

which is the Kodaira-Spencer map. Recall that $H^1(T_{\mathfrak{X}_0})$ is in one to one correspondence with infinitesimal deformations $\mathfrak{X} \to \operatorname{Spec} \frac{\mathbb{C}[\epsilon]}{\epsilon^2}$. Intuitively, the Kodaira-Spencer map lifts a vector $v \in T_0B$ to a family of normal vectors which infinitesimally deform $\mathfrak{X}_0$ in the direction of $v$.

When we have the setup of $\mathfrak{X}_0 \times \Delta \to \Delta$ and $\phi(t) \in \pi^*_{\mathfrak{X}_0}\mathcal{A}^{0,1}(T^{1,0}_{\mathfrak{X}_0})$ satisfying theorem 20.11, then the Kodaira-Spencer map is quite explicit. We can associate to every holomorphic tangent vector $v \in T^{1,0}_0\Delta$ an infinitesimal deformation $v \cdot \phi(t) \in H^{0,1}(\mathfrak{X}_0, T^{1,0}_{\mathfrak{X}_0})$, where if we locally write $v = \sum_{i=1}^m \lambda_i \frac{\partial}{\partial t_i}$, then $v \cdot \phi(t)$ is the directional derivative of $\phi(t)$ in the direction of $v$ at $t = 0$:

$$\sum_{j,k,\ell} \lambda_\ell \frac{\partial h_{jk}}{\partial t_\ell}(z, 0) d\overline{z_j} \otimes \frac{\partial}{\partial z_k} \in H^{0,1}(\mathfrak{X}_0, T^{1,0}_{\mathfrak{X}_0}).$$

Back in the day, Kodaira and Spencer wanted to find a family for a given $\mathfrak{X}_0$ which witnessed every possible infinitesimal deformation of $\mathfrak{X}_0$. They conjectured it, but only could prove it when Nirenberg joined them:

**Theorem 20.12** (Kodaira-Spencer-Nirenberg)**.** [16] *Let $\mathfrak{X}_0$ be a compact complex manifold such that $H^2(\mathfrak{X}_0, T^{1,0}_{\mathfrak{X}_0}) = 0$. Then there is a small open ball $\Delta \subseteq H^1(\mathfrak{X}_0, T^{1,0}_{\mathfrak{X}_0})$ centered at 0*

[16] This condition on second cohomology of the holomorphic tangent bundle is not strictly necessary. There are examples of compact complex manifolds which also have this family which witnesses all their infiniteismal deformations, with their $H^2$ nonvanishing. A famous example is that of all Calabi-Yau manifolds, which is the famous Tian-Todorov theorem.

*and a family*

$$\pi : \mathfrak{X}_0 \times \Delta \to \Delta$$

*along with an element* $\phi(t) \in \pi^*_{\mathfrak{X}_0}\mathcal{A}^{0,1}(T^{1,0}_{\mathfrak{X}_0})$ *satisfying theorem 20.11, so that the Kodaira-Spencer map*

$$T^{1,0}_0\Delta \cong H^1(\mathfrak{X}_0, T^{1,0}_{\mathfrak{X}_0}) \to H^1(\mathfrak{X}_0, T^{1,0}_{\mathfrak{X}_0})$$

*is the identity map.*

In particular, the family $\mathfrak{X}_0 \times (\Delta \cap \mathbb{C}\cdot v) \to (\Delta \cap \mathbb{C}\cdot v)$ over the line $\Delta \cap \mathbb{C}\cdot v$ provides a family which witnesses an infinitesimal deformation of $\mathfrak{X}_0$ in the direction of $v$.

20.3. **TGT3: Riemann-Hilbert correspondence and period maps.**

20.4. **TGT4: Local Torelli Theorem.**

20.5. **TGT4a: Small deformations of elliptic K3 surfaces.** In this section we discuss small deformations of an elliptic K3 surface which "preserves" a given elliptic fibration. In the course of this discussion, we'll see many interesting ideas at play. We first set our problem up so that it is well-posed.

Let $\mathfrak{X}_0$ be a complex K3 surface. Consider the universal deformation space $Def(\mathfrak{X}_0)$ of $\mathfrak{X}_0$. More concretely, $Def(\mathfrak{X}_0)$ can be described as follows: there is a small open ball $\Delta \subset H^{0,1}(\mathfrak{X}_0, T^{1,0}_{\mathfrak{X}_0})$ containing the origin 0 and a particular nonzero element $\phi(t) \in (\pi^*_{\mathfrak{X}_0}\mathcal{A}^{0,1}(\mathfrak{X}_0, T^{1,0}_{\mathfrak{X}_0}))$ whose data endows the product of $C^\infty$ manifolds

$$\mathfrak{X}_0 \times \Delta$$

with an integrable complex structure, so that the Kodaira-Spencer map

$$T^{1,0}_0\Delta \to H^{0,1}(\mathfrak{X}_0, T^{1,0}_{\mathfrak{X}_0}),$$

for the family $\pi : \mathfrak{X}_0 \times \Delta \to \Delta$, is exactly the identity map. I will let $\mathfrak{X}_t$ denote the K3 surface which is the fiber over $t \in \Delta$ – or, in other words, the $C^\infty$ manifold $\mathfrak{X}_0$ equipped with the integrable complex structure given by $\phi(t)$ evaluated at $t$. Finally, let

$$P : \Delta \to D \subset \mathbb{P}(H^2(\mathfrak{X}_0; \mathbb{Z}))$$

denote the period map, where $t \mapsto [H^{2,0}(\mathfrak{X}_t)]$.

Now let's suppose that $\mathfrak{X}_0$ has a line bundle $L$, and denote its first Chern class as

$$\ell := c_1(L) \in H^2(\mathfrak{X}_0; \mathbb{Z}) \cap H^{1,1}(\mathfrak{X}_0).$$

Note that $\ell \in H^2(\mathfrak{X}_0; \mathbb{Z}) \cap H^{1,1}(\mathfrak{X}_t) \iff P(t) \in D \cap \mathbb{P}(\ell^\perp_{\mathbb{C}})$. Combining this fact with the local Torelli theorem tells us that we can shrink $\Delta$ and intersect it with a hypersurface to obtain a smooth 19-dimensional $H$ such that

$$\mathfrak{X}_0 \times H \to H$$

is a family where the class $\ell \in H^2(\mathfrak{X}_0;\mathbb{Z})$ defines not only a line bundle on $\mathfrak{X}_0$, but also on $\mathfrak{X}_t$ for every $t \in H$. In particular, these line bundles on $\mathfrak{X}_t$ defined by $\ell \in H^2(\mathfrak{X}_0;\mathbb{Z})$ come from a line bundle $L_{global}$ on $\mathfrak{X}_0 \times H$. Now we arrive to our main claim.

**Theorem 20.13.** *Suppose $\ell \in H^2(\mathfrak{X}_0;\mathbb{Z}) \cap H^{1,1}(\mathfrak{X}_0)$ gives an elliptic fibration on $\mathfrak{X}_0$. Then for $t \in H$ near 0, the line bundle on $\mathfrak{X}_t$ defined by $\ell \in H^2(\mathfrak{X}_0;\mathbb{Z}) \cap H^{1,1}(\mathfrak{X}_t)$ also gives an elliptic fibration on $\mathfrak{X}_t$.*

That $\ell$ gives an elliptic fibration on $\mathfrak{X}_t$ is equivalent to the following three conditions:

(1) $(\ell.\ell)_{\mathfrak{X}_t} = 0$.
(2) $\ell \in \mathrm{Pic}(\mathfrak{X}_t)$ is primitive.
(3) $(\ell.D)_{\mathfrak{X}_t} \geq 0$ for all effective curves $D \subset \mathfrak{X}_t$.

Let's start with the easiest case first.

**Lemma 20.14.** *For all $t \in H$, the element $\ell$ remains primitive in* $\mathrm{Pic}(\mathfrak{X}_t)$.

*Proof.* Assume for the sake of contradiction that this were not true for some $T \in H$. Then $\frac{\ell}{n} \in \mathrm{Pic}(\mathfrak{X}_t) = H^2(\mathfrak{X}_0;\mathbb{Z}) \cap H^{1,1}(\mathfrak{X}_t)$ for some integer $n$ such that $|n| > 1$. Then $\frac{\ell}{n} \in H^2(\mathfrak{X}_0;\mathbb{Z})$. Note by assumption, $\ell \in H^2(\mathfrak{X}_0;\mathbb{Z}) \cap H^{1,1}(\mathfrak{X}_0)$. This implies that $\frac{\ell}{n} \in H^{1,1}(\mathfrak{X}_0)$. But this means that $\frac{\ell}{n} \in \mathrm{Pic}(\mathfrak{X}_0)$ for some $n$ where $|n| > 1$, which gives us the contradiction. □

In essence, being primitive in the Picard group means being primitive in second integral cohomology. Now, before we think about (1) and (3), we would like to show the following.

**Lemma 20.15.** *For all $t \in H$ near 0, $h^0(\mathfrak{X}_t, L_{global,t}) \geq 2$.*

*Proof.* Note that $\chi(L_{\mathrm{global},0}) = 2$ by Rieman Roch for surfaces (4.9), so

$$h^0(L_{\mathrm{global},t}) - h^1(L_{\mathrm{global},t}) + h^2(L_{\mathrm{global},t}) = 2$$

for all $t \in H$ near 0. But note

$$h^2(L_{\mathrm{global},0}) = h^0(L^\vee_{\mathrm{global},0}) = 0,$$

so by upper-semicontinuity, we have $h^2(L_{\mathrm{global},t}) = 0$ for all $t \in H$ near 0. □

**Lemma 20.16.** *For all $t \in H$ near 0, $(\ell.\ell)_{\mathfrak{X}_t} = 0$.*

*Proof.* By the previous lemma, we know that all the line bundles $L_{\mathrm{global},t}$ correspond to effective curves. Then for $t \in H$ near 0, from this deformation of line bundles on these K3 surfaces, we can obtain a family of curves, and by flatness, we must have that the arithmetic genus is 0 for $t \in H$ near 0. Then $(\ell.\ell)_{\mathfrak{X}_t} = 0$, since the self intersection equals two times the arithmetic genus minus two.

An alternative proof to this is the following: first note that

$$(\ell.\ell)_{\mathfrak{X}_t} = \int_{\mathfrak{X}_t} \ell \wedge \ell \in \mathbb{Z}$$

this is continuous in terms of $t$, but takes on an integer value for every $t$. Then by continuity, since $(\ell.\ell)_{\mathfrak{X}_0} = 0$, we must have $(\ell.\ell)_{\mathfrak{X}_t}$ for $t \in H$ near 0. □

Finally, we address the last part.

**Lemma 20.17.** *For $t \in H$ near 0, we have $(\ell.D)_{\mathfrak{X}_t}$ for all effective curves $D \subset \mathfrak{X}_t$.*

*Proof.* Note that since for each $t \in H$, $L_{\text{global},t}$ has a global section, it corresponds to an elliptic curve $E_t \subset \mathfrak{X}_t$. So for $t \in H$ near 0, the family $\mathfrak{X}_0 \times H \to H$ deforms the elliptic curve $E_0$ to $E_t$. Now, suppose $D \subset \mathfrak{X}_t$ is an effective curve. If

$$\dim Supp(E_t \cap D) = 0,$$

then we must have $(\ell.D)_{\mathfrak{X}_t} \geq 0$. On the other hand, if $\dim Supp(E_t \cap D) =\geq 1$, then take an irreducible component of $Supp(E_t \cap D)$ of dimension 1. Note for $t \in H$ near 0, we have that $E_t$ is a smooth elliptic curve, and thus has only one irreducible component. Then we would have to have $E_t \subset D$. Then we can write $D = nE_t + C$ where $C$ is some effective curve so that $\dim Supp(E_t \cap C) = 0$. Then we see that

$$(\ell.D)_{\mathfrak{X}_t} = (E_t.nE_t + C)_{\mathfrak{X}_t} = 0 + (E_t.C)_{\mathfrak{X}_t} \geq 0.$$

□

20.6. **TGT4b: Sometimes many small deformations of a K3 surface are isomorphic.** Suppose that $\mathfrak{X}_0$ is a complex K3 surface such that its automorphism group $Aut(\mathfrak{X}_0)$ is nonzero. Consider the universal deformation space

$$(\pi : \mathfrak{X}_0 \times \Delta \to \Delta, \phi(t)) \text{ where } \Delta \subseteq H^{0,1}(\mathfrak{X}_0, T^{1,0}_{\mathfrak{X}_0}) \text{ and } \xi(t) \in \pi^*_{\mathfrak{X}_0}\mathcal{A}^{0,1}(T^{1,0}_{\mathfrak{X}_0}),$$

where $\xi(t)$ satisfies theorem 20.11. Then we claim that for $\phi \in Aut(\mathfrak{X}_0)$, if the induced linear isomorphism $\phi^* : H^{0,1}(\mathfrak{X}_0, T^{1,0}_{\mathfrak{X}_0}) \to H^{0,1}(\mathfrak{X}_0, T^{1,0}_{\mathfrak{X}_0})$ is nontrivial, then there exist many $t$ and $t'$ such that $\mathfrak{X}_t$ and $\mathfrak{X}_{t'}$ are isomorphic as complex manifolds. In particular, consider the map

$$\begin{tikzcd}
\mathfrak{X}_0 \times \Delta \arrow[r, "\Phi:=(\phi,\phi*)"] \arrow[d] & \mathfrak{X}_0 \times \Delta \arrow[d] \\
\Delta \arrow[r, "\phi^*"'] & \Delta
\end{tikzcd}$$

where we can shrink $\Delta$ if necessary so that $\phi^*$ is well-defined and is an isomorphism. A priori we know that $(\phi, \phi^*)$ is a diffeomorphism of smooth manifolds $\mathfrak{X}_0 \times \Delta$. To show that is defines an actual isomorphism of complex manifolds, it suffices to show that pulling back the integrable structure on $\mathfrak{X}_0 \times \Delta$ defined by $\xi(t)$ along $(\phi, \phi^*)$ defines another integrable complex structure on $\mathfrak{X}_0 \times \Delta$. But indeed it does, because

$$\partial_{\mathfrak{X}_0}\Phi^*(\xi(t)) + \frac{1}{2}[\Phi^*(\xi(t)), \Phi^*(\xi(t))] = \Phi^*[\frac{\partial}{\partial \mathfrak{X}_0}\xi(t) + \frac{1}{2}[\xi(t), \xi(t)]] = 0, \text{ and}$$

$$\frac{\partial}{\partial \bar{t}}\Phi^*(\xi(t)) = \Phi^* \frac{\partial}{\partial \bar{t}}\xi(t) = 0.$$

Thus, if $\phi^*(t) = t'$ where $t' \neq t$, then we find that $\mathfrak{X}_t \cong \mathfrak{X}_{t'}$ as complex manifolds.

### 20.7. **TGT5: Moduli space of marked K3 surfaces (not Hausdorff!)** COME BACK AND FINISH WRITING

**Proposition 20.18.** *The moduli space $\mathcal{M}^{'}$ of marked K3 surfaces is not Hausdorff.*

One way to see that $\mathcal{M}'$ is not Hausdorff is through the Atiyah flop [Ati58]. Consider the family

$$\mathcal{X} \subseteq \mathbb{P}^3_{\mathbb{C}} \times \Delta_\epsilon \to \Delta_\epsilon.$$

defined by the equation

$$(x^4 - 2x^2w^2) + (y^4 - 2y^2w^2) + (z^4 - 2z^2w^2) = 2t^2w^4,$$

where $w, x, y, z$ are the homogeneous coordinates and $\Delta_\epsilon := \{z \in \mathbb{C} \mid |z| < \epsilon\}$, where $\epsilon$ is sufficiently small. How small? Small enough so that $\mathcal{X}_t$ is a smooth K3 surface for every nonzero $t \in \Delta_\epsilon$. Localizing to affine coordinates by setting $w = 1$, our family is described by

$$x^2(x^2 - 2) + y^2(y^2 - 2) + z^2(z^2 - 2) = 2t^2.$$

The Jacobian of this is

$$\begin{pmatrix} 4x(x^2-1) & 4y(y^2-1) & 4z(z^2-1) & 4t \end{pmatrix}.$$

If we just restrict to $\mathcal{X}_t$, we see there is a unique singularity here on the total space, namely at $x = y = z = t = 0$. If we restrict to each fiber $\mathcal{X}_t$, the Jacobian is

$$\begin{pmatrix} 4x(x^2-1) & 4y(y^2-1) & 4z(z^2-1) \end{pmatrix}.$$

We see that for nonzero $t$ sufficiently small, each $\mathcal{X}_t$ is smooth (this can be checked on the other affine patches as well). However, the special fiber $\mathcal{X}_0$ has a singularity at the same unique singularity of the total space $\mathcal{X}$. This singularity happens to be a double point.[17]

Let us briefly mention something about tangent cones, which are a way of extending the notion of a tangent space at a singular point of a variety.

**Example 20.19.** *Consider the nodal cubic $C$, defined by the equation $y^2 - x^2 - x^3 = 0$ in $\mathbb{A}^2_{\mathbb{C}}$. Note $C$ has a singularity at the origin and, in some sense, there are two tangent directions to its singularity. As a closed subscheme, these two tangent directions are exactly given by $y^2 - x^2 = (y - x)(y + x)$.*

[17] At some point, I should learn more about surface singularities. Maybe a good reference is Lectures on Resolution of Singularities by Kollar. Allegedly it contains many concrete and hands on techniques on studying singularities for both curves and surfaces.

In general, suppose you have an affine variety $X \subset \mathbb{A}^k_{\mathbb{C}}$ cut out by the ideal $I \subset \mathbb{C}[X_0, \cdots, X_k]$. For $f \in I$, let $f_{\min}$ be the homogeneous part of minimal degree of $f$. Then let $I_{\min}$ be the ideal generated by $f_{\min, \forall f \in I}$. Then the tangent cone of $X$ at the origin is the closed subscheme given by $I_{\min}$.

In the case of our family $\mathcal{X}$, observe that in the affine patch given by $\operatorname{Spec} \mathbb{C}[x, y, z, t]$, the tangent cone of $\mathcal{X}$ at the singularity $(0,0,0,0)$ is given by

$$x^2 + y^2 + z^2 + t^2 = 0.$$

In particular, this hypersurface has something called a double point. Recall that a hypersurface in $\mathbb{C}^{n+1}$ defined by $f$ (written as a power series centered around the origin) has a singularity if and only if $f_1 = 0$.

**Definition 20.20.** *We say that a hypersurface in $\mathbb{C}^{n+1}$ has a double point at the origin if $f$ has a non-degenerate critical point at the origin. In other words, $f_1 = 0$, and the Hessian of $f$ evaluated at 0 (equivalently, the Hessian of $f_2$ at 0) has nonzero determinant.*

Double points are nice to work with because when the critical point is non-degenerate, the qualitative behavior of the hypersurface at the origin is dictated by $f_1$ and $f_2$, and the positive and negative eigenvalues of the Hessian. If the critical point is degenerate, then higher order terms of $f$ can change the behavior qualitatively around the origin.

Another reason why double points are nice to work with is because of the following.

**Proposition 20.21.** [Ati58] *Let $Z(f) \subset \mathbb{C}^{n+1}$ be a hypersurface with a double point singularity at the origin. Then around the origin, $Z(f)$ is analytically isomorphic to a neighborhood of the vertex of the quadric cone*

$$z_0^2 + \cdots + z_n^2.$$

### 20.8. **TGT6: Moduli space of primitively polarized K3 surfaces.**

### 20.9. **TGT7: Global Torelli Theorem.**

## 21. Positivity: $\mathbb{Q}$-divisors, $\mathbb{R}$-divisors, and ampleness

Let $X$ be a complete $\mathbb{C}$-variety. We want to study certain Cartier divisors on $X$ called ample and nef divisors. More specifically, we want to linearize the locus of these divisors, by $\mathbb{Q}$-linear or $\mathbb{R}$-linear combinations, and study the induced numerical intersection theory. In turn, this will give us many useful tools for studying the geometry of $X$. Before we do this, we should first setup our situation in the integral case.

Let $\operatorname{Div}(X)$ denote the $\mathbb{Z}$-module generated by the Cartier divisors of $X$. Any such element of $\operatorname{Div}(X)$ is called an integral Cartier divisor. We say two integral Cartier divisors $D_1, D_2$ are numerically equivalent, or $D_1 \equiv D_2$, if

$$(D_1.C) = \int_{C_1} c_1(\mathcal{O}_X(D_1)) = \int_{C_2} c_1(\mathcal{O}_X(D_2)) = (D_2.C),$$

for every irreducible curve $C \subseteq X$. We say $D$ is numerically trivial if $(D.C) = 0$ holds for every such $C$. Equivalently, $D$ is numerically trivial if $(D.\gamma) = 0$ for every 1-cycle $\gamma \in H_2(X;\mathbb{Z})$.

**Proposition 21.1.** *Let $X$ be a complete $\mathbb{C}$-variety. Suppose $\{D_i\}_{i=1}^k$ and $\{D_i'\}_{i=1}^k$ are integral Cartier divisors, such that $D_i \equiv D_i'$. Then*

$$(D_1.\cdots.D_k.[V]) = (D_1'.\cdots.D_k'.[V]).$$

*for any subscheme $V \subseteq X$ of pure dimension $k$.*

*Proof.* First, we claim that if $H$ is numerically trivial, then

$$(H.D_2.\cdots.D_k.[V]) = 0.$$

This is true, because $(D_2.\cdots.D_k.[V])$ provides a 2-cycle (not necessarily effective!),

$$\gamma = c_1(\mathcal{O}_X(D_2)).\cdots.c_1(\mathcal{O}_X(D_k)).[V] \in H_2(X;\mathbb{Z})$$

so that if $H$ is numerical trivial, then

$$(H.D_2.\cdots.D_k.[V]) = (H.\gamma) = 0.$$

With this, we see that

$$(D_1.\cdots.D_k.[V]) = (D_1'.\cdots.D_k'.[V]).$$

□

**Definition 21.2.** *Let $X$ be a complete $\mathbb{C}$-variety. Then the Neron-Severi group is defined as*

$$NS(X) := Div(X)/\equiv .$$

Taking sheaf cohomology of the exponential exact sequence, we get

$$\cdots \to H^1(X,\mathcal{O}_X) \to^{\beta} \mathrm{Pic}(X) \to^{\alpha} H^2(X;\mathbb{Z}) \to \cdots$$

Note $\mathrm{Pic}(X)/H^1(X,\mathcal{O}_X)$ injects into $H^2(X;\mathbb{Z})$. Any Cartier divisor with Chern class 0 will be automatically numerically trivial. Then we see that $\mathrm{NS}(X)$ is the quotient of a subgroup of $H^2(X;\mathbb{Z})$. Thus, it is finitely generated. By construction, it also has no torsion. So $\mathrm{NS}(X)$ is a finite free $\mathbb{Z}$-module.

Now we want to consider not only integral Carter divisors, but $\mathbb{Q}$-divisors and $\mathbb{R}$-divisors. In particular, we want to show that turning $\mathrm{NS}(X)$ into a $\mathbb{Q}$ or $\mathbb{R}$-vector space makes sense.

**Definition 21.3.** *Let $X$ be a complete $\mathbb{C}$-variety. A $\mathbb{Q}$-divisor on $X$ is an element of the $\mathbb{Q}$-vector space*

$$Div_{\mathbb{Q}}(X) := Div(X) \otimes_{\mathbb{Z}} \mathbb{Q}.$$

*So $\mathbb{Q}$-divisors are $\mathbb{Q}$-linear combinations of integral divisors. Similarly, a $\mathbb{R}$-divisor is an element of the $\mathbb{R}$-vector space*

$$Div_{\mathbb{R}}(X) := Div(X) \otimes_{\mathbb{Z}} \mathbb{R}.$$

We say a $\mathbb{Q}$-divisor or a $\mathbb{R}$-divisor is effective if it can be written as $\sum c_i D_i$, where $c_i > 0$ and $D_i$ are integral effective divisors. Note there are often multiple ways to write a $\mathbb{Q}$-divisor or $\mathbb{R}$-divisor as a linear combination of integral divisors, so the expressions are not unique. This also means the notion of support of a $\mathbb{Q}$-divisor or a $\mathbb{R}$-divisor is usually not well-defined, but this is fine.

**Definition 21.4.** *Two $\mathbb{Q}$-divisors $D_1, D_2 \in Div_{\mathbb{Q}}(X)$ are numerically equivalent if*

$$(D_1.C) = (D_2.C),$$

*for every irreducible 1-dimensional subvariety $C \subseteq X$. Similarly, two $\mathbb{R}$-divisors $D_1, D_2 \in Div_{\mathbb{R}}(X)$ are numerically equivalent if*

$$(D_1.C) = (D_2.C),$$

*for every irreducible 1-dimensional subvariety $C \subseteq X$.*

We let $D_1 \equiv_{\mathbb{Q}} D_2$ to denote rational numerical equivalence. We let $D_1 \equiv_{\mathbb{R}} D_2$ denote real numerical equivalence

**Definition 21.5.** *Let $X$ be a complete $\mathbb{C}$-variety. The rational Neron-Severi group is defined as*

$$NS_{\mathbb{Q}}(X) := Div_{\mathbb{Q}}(X)/\equiv_{\mathbb{Q}} .$$

*The real Neron-Severi group is defined as*

$$NS_{\mathbb{R}}(X) := Div_{\mathbb{R}}(X)/\equiv_{\mathbb{R}} .$$

**Proposition 21.6.** *Let $X$ be a complete $\mathbb{C}$-variety. The natural map*

$$NS \otimes_{\mathbb{Z}} \mathbb{Q} \to NS_{\mathbb{Q}}(X)$$

*is an isomorphism of $\mathbb{Q}$-vector spaces.*

*Proof.* First let's check that the map is well-defined. If we have a rational linear combination $\sum c_i D_i$, where $c_i \in \mathbb{Q}$ and $D_i \in \mathrm{NS}(X)$ is numerically trivial, then $\sum c_i D_i$ will be rationally numerically trivial. The map is injective because if $D \in \mathrm{NS}_{\mathbb{Q}}(X)$ is rationally numerically trivial, then $(D.C) = 0$ for every irreducible curve $C \subset X$. But $rD \in Div(X)$ for some rational $r$, so $D = rD \otimes \frac{1}{r}$, and $rD$ is numerically trival, so $rD = 0$ in $\mathrm{NS}(X)$. Finally, the map is clearly surjective. □

**Proposition 21.7.** *Let $X$ be a complete $\mathbb{C}$-variety. The natural map*

$$NS \otimes_{\mathbb{Z}} \mathbb{R} \to NS_{\mathbb{R}}(X)$$

*is an isomorphism of $\mathbb{R}$-vector spaces.*

*Proof.* First we check the map is well defined. Suppose $\sum a_i D_i$ is a $\mathbb{R}$-linear combination of numerically trivial divisors. Then clearly $(\sum a_i D_i.C) = 0$ for every irreducible curve $C \subseteq X$, so the image of $\sum a_i D_i$ vanishes. Next, we veirfy injectivity. Suppose $D = \sum c_i D_i \in \mathrm{NS} \otimes_{\mathbb{Z}} \mathbb{R}$ is maps to 0. Then

$$(\sum c_i D_i.C) = 0$$

for every irreducible curve $C \subseteq X$. We claim that $D = \sum c_i D_i$ can be written as $\sum \lambda_i T_i$ where $T_i$ are integral numerically trivial divisors. This is possible because if we let $C_1, \cdots, C_n$ denote generators of $H_2(X; \mathbb{Z})$, then

$$(\sum c_i D_i.C) = \sum (D_i.C_k) c_i = 0$$

forces a system of $n$ integral linear equations in terms of the variables $c_i$. Since they have a solution, their solution must be a real linear combination of integral solutions. Thus, we can write $D = \sum \lambda_i T_i$ as desired. This implies that $D$ is actually the 0 element in $\mathrm{NS}(X) \otimes_{\mathbb{Z}} \mathbb{R}$. Finally, surjectivity is obvious. □

Thus, given a complete $\mathbb{C}$-variety $X$, we have finite dimensional vector spaces $\mathrm{NS}_{\mathbb{Q}}(X)$ and $\mathrm{NS}_{\mathbb{R}}(X)$, and on these vector spaces there is an induced intersection theory, inherited from integral Cartier divisors, which is well-defined in the sense of proposition 21.1 (since the intersections are extended $\mathbb{Q}$ and $\mathbb{R}$-multi-linearly).

We would now like to consider a certain locus of divisors in $\mathrm{NS}_{\mathbb{Q}}(X)$ and $\mathrm{NS}_{\mathbb{R}}(X)$. That is, the locus of ample divisors. It turns out these loci will have the nice structure of open cones. Let's first review the notion of an integral ample divisor.

Let $X$ be a complete $\mathbb{C}$-variety. Since we are dealing with ample divisors, we'll assume that $X$ is projective. Again, anytime we say divisor here, assume we mean Cartier divisor[18] Recall that there are roughly four perspectives on ample line bundles. The first is that if a high enough multiple of the corresponding effective divisor is a hyperplane section of $X$ under some projective embedding.

**Definition 21.8.** *A line bundle $\mathcal{L} \in \mathrm{Pic}(X)$ is ample if $\mathcal{L}^{\otimes n}$ is very ample for some $n$.*

There is also the cohomological perspective:

**Definition 21.9.** *The following are equivalent:*

- *$L$ is ample*
- *Given any coherent sheaf $\mathcal{F}$, there exist integer $m_1(\mathcal{F})$ such that for all $m \geq m_1(\mathcal{F})$, we have*

$$H^i(X, \mathcal{F} \otimes \mathcal{L}^{\otimes m}) = 0, \forall i > 0.$$

[18]The distinction between Weil and Cartier is important when $X$ is not smooth. For example, when $X$ is normal, Cartier divisors inject into Weil divisors, and there are examples of Weil divisors which are not Cartier but are actually $\mathbb{Q}$-Cartier. One example of this phenomenon is a ruling on a quadric cone in $\mathbb{P}^3_{\mathbb{C}}$.

– *Given any coherent sheaf* $\mathcal{F}$*, there exist integer* $n_1(\mathcal{F})$ *such that for all* $n \geq n_1(\mathcal{F})$*, we have*

$$H^0(X, \mathcal{F} \otimes \mathcal{L}^{\otimes n})$$

*is generated by global sections.*

There is also the numerical perspective.

**Theorem 21.10** (Nakai-Moishezon-Kleiman Criterion)**.** [19] $\mathcal{L} \in \mathrm{Pic}(X)$ *is ample if and only if*

$$(\mathcal{L}^{\dim V}.V) = \int_V c_1(\mathcal{L})^{\wedge \dim V} > 0$$

*for every irreducible variety* $V \subseteq X$ *of positive dimension.*

If $X$ is smooth, ampleness of a line bundle can also be detected analytically.

**Definition 21.11.** *A line bundle* $\mathcal{L}$ *on a compact complex manifold is positive if its first Chern class* $c_1(\mathcal{L})$ *is positive.*

There is a very rich and satisfying theory with ample line bundles. One climax of that theory is the Kodaira embedding and Kodaira vanishing theorems. There is another useful aspect to the theory of ample line bundles, namely that of ample cones. It turns out that considering the locus of ample line bundles on a projective $\mathbb{C}$-variety and turning it into a vector space leads to useful insights. One may be tempted to form the ample cone by tensoring the Picard group by $\mathbb{Q}$ (or $\mathbb{R}$), but the not so nice thing about this is that $\mathrm{Pic}(X)$ may have torsion (e.g. Picard group of any curve of positive genus). Also, from the perspective of Theorem 21.10, line bundles $\mathcal{L}$ with intersection 0 with every curve are indistinguishable from the trivial line bundle. So instead, we'll want to study ample cones numerically in the Neron-Severi groups $\mathrm{NS}_{\mathbb{Q}}(X)$ and $\mathrm{NS}_{\mathbb{R}}(X)$. Let's study ampleness over $\mathbb{Q}$ first. The theory of ampleness over $\mathbb{R}$ is more subtle[20], and often will trace back to $\mathbb{Q}$-divisors since $\mathbb{R}$-divisors can be approximated by $\mathbb{Q}$-divisors.

**Definition 21.12.** *Let* $D \in Div_{\mathbb{Q}}(X)$ *be a* $\mathbb{Q}$*-divisor. We say* $D$ *is ample if any of the following equivalent conditions are satisfied:*

1. *$D$ can be written as $\sum c_i D_i$, where $c_i > 0$ is a positive rational number and $D_i$ is integral divisor.*
2. $(D^{\dim V}.V) > 0$ *for every irreducible positive-dimensional subvariety* $V \subseteq X$*.*
3. *There is a positive integer $r$ such that $rD$ is integral and ample.*

Furthermore, the notion of an ample $\mathbb{Q}$-divisor is well defined in $\mathrm{NS}_{\mathbb{Q}}(X)$.

[19] As a consequence of this theorem, ampleness of an integral divisor is well-defined up to numerical equivalence class.

[20] A basic reason for why the theory of $\mathbb{Q}$-divisors is easier is that every such divisor is a rational times an integral divisor. But not every $\mathbb{R}$-divisor is a real number times an integral divisor.

**Proposition 21.13.** *Let $D$ be an ample $\mathbb{Q}$-divisor. Let $T$ be a numerically trivial $\mathbb{Q}$-divisor. Then $D+T$ is also an ample $\mathbb{Q}$-divisor.*

*Proof.* To verify ampleness, it suffices to prove

$$((D+T)^{\dim V}.V) > 0$$

for every irreducible positive dimensional $V \subseteq X$. Note we have

$$((D+T)^{\dim V}.V) = \sum_{j=0}^{\dim V} \binom{\dim V}{j} (D^{\dim V - j}.T^j.V),$$

and we see that when $j > 0$, the summation is 0, and when $j = 0$, we have positivity. □

Next, we show ampleness in $\mathrm{NS}_{\mathbb{Q}}(X)$ is an open condition.

**Proposition 21.14.** *Let $H$ be an ample $\mathbb{Q}$-divisor. Let $E_1, \cdots, E_r$ be arbitrary $\mathbb{Q}$-divisors. Then for sufficiently small $\epsilon_1, \cdots, \epsilon_r$, we have*

$$H + \epsilon_1 E_1 + \cdots + \epsilon_r E_r$$

*remains ample.*

*Proof.* Note that for each $E_i$, we have $m_i H \pm E_i$ is ample for large enough $m_i$. Then $H \pm \lambda_i E_i$ is ample for rational $|\lambda_i| < \frac{1}{m_i}$. Then

$$rH \pm \lambda_1 E_1 \pm \cdots \pm \lambda_r E_r$$

is ample for rational $|\lambda_i| < \frac{1}{m_i}$. So

$$H \pm \frac{\lambda_1}{r} E_1 \pm \cdots \pm \frac{\lambda_r}{r} E_r$$

is ample. This shows the statement. □

We can also define ample $\mathbb{R}$-divisors. We will also prove that they are well-defined in $\mathrm{NS}_{\mathbb{R}}(X)$, and that ampleness is an open condition in this $\mathbb{R}$-vector space – both of these will be proven by approximating $\mathbb{R}$-divisors by $\mathbb{Q}$-divisors.

**Definition 21.15.** *A $\mathbb{R}$-divisor $D$ is ample if we can write*

$$D = \sum a_i A_i,$$

*where $a_i \in \mathbb{R}^+$ and $A_i$ are integral and ample.*

**Theorem 21.16.** [Laz04, Theorem 2.3.18] *A $\mathbb{R}$-divisor $D$ is ample if and only if*

$$(D^{\dim V}.V) > 0,$$

*for all irreducible positive-dimensional $V \subseteq X$.*

Theorem 21.16 is subtle. The reason why we had such a statement so easily for $\mathbb{Q}$-divisors is because we can express every $\mathbb{Q}$-divisor as an integral divisor multipled by a rational number. Then this fact becomes equivalent to Nakai-Moishezon-Kleiman ampleness criterion for integral divisors. But in the case of $\mathbb{R}$-divisors, there are some that are never a multiple of an integral divisor. Nevertheless, Theorem 21.16 still holds. However, at least in these short notes on positivity, we try to prove statements without invoking it in order to avoid circular arguments.

**Proposition 21.17.** *The notion of an ample $\mathbb{R}$-divisor is well-defined up to numerical equivalence class.*

*Proof.* It suffices to prove the following. Let $H$ be an ample $\mathbb{R}$-divisor, and let $B$ be numerically trivial. Then we show that $H+B$ is also an ample $\mathbb{R}$-divisor. Let $B = \sum c_i B_i$. We can assume that the $B_i$ are integral and numerically trivial, from the proof of proposition 21.7.

It suffices to show that $H+rB$ is ample where $H$ is ample $\mathbb{R}$-divisor, $B$ is intregal divisor and numerically trivial, and $r$ is any real number. Let $r_1 < r < r_2$, where $r_1, r_2 \in \mathbb{Q}$. Then there exist $t \in [0,1]$ such that $t(r_1) + (1-t)(r-2) = r$. Then

$$H + rB = t(H + r_1 B) + (1-t)(H + r_2 B),$$

where we know $H + r_1 B$ and $H + r_2 B$ are ample by proposition 21.13. □

Thus, the ample cone in $\mathrm{NS}_{\mathbb{R}}(X)$ is well-defined. Furthermore, it is open.

**Proposition 21.18.** *Let $H$ be an ample $\mathbb{R}$-divisor. Let $E_1, \cdots, E_r$ be arbitrary $\mathbb{R}$-divisors. Then*

$$H + \epsilon_1 E_1 + \cdots + \epsilon_r E_r$$

*is an ample $\mathbb{R}$-divisor for sufficiently small* $0 < |\epsilon_i| << 1$.

*Proof.* Since $H$ is an ample $\mathbb{R}$-divisor, we have

$$H = \sum_{i=1}^{t} c_i H_i,$$

for $c_i > 0$ and $H_i$ integral and ample. Then picking $0 < c < c_1$, we can write

$$H + \epsilon_1 E_1 + \cdots + \epsilon_r E_r = [cH_1 + \sum_{j=1}^{r} \epsilon_j E_j] + [(c_1 - c)H_1 + \sum_{i=2}^{t} c_i H_i].$$

The latter term is an ample $\mathbb{R}$-divisor, so it suffices to show that

$$cH_1 + \sum_{j=1}^{r} \epsilon_j E_j$$

is an ample $\mathbb{R}$-divisor for sufficiently small $\epsilon_j$. But this is

$$c[H_1 + \sum_{j=1}^{r} \epsilon_j \frac{1}{c} E_j].$$

If we expand the $E_j$ as their respective $\mathbb{R}$-linear combinations of integral divisors, it suffices to show that

$$H_1 + \epsilon E$$

is an ample $\mathbb{R}$-divisor for sufficiently small $\epsilon$, where $\epsilon$ can be a real number. But we know from the theory of $\mathbb{Q}$-divisors that can perturb ample $\mathbb{Q}$-divisors, so that there exist a bound $b$ such that for $\epsilon \in \mathbb{Q}$ such that $|\epsilon| < b$, we have $H_1 + \epsilon E$ is ample. We claim that this holds for real $\epsilon$ as well. Pick rational numbers $r_1, r_2$ such that

$$-b < r_1 < \epsilon < r_2 < b.$$

and set $t \in [0, 1]$ such that $tr_1 + (1 - t)r_2 = \epsilon$. Then

$$t(H_1 + r_1 E) + (1 - t)(H_1 + r_2 E) = H_1 + rE$$

is $\mathbb{R}$-ample. □

**Corollary 21.19.** *When $X$ is projective, Propositions 21.14 and 21.18 imply that $NS_{\mathbb{Q}}(X)$ and $NS_{\mathbb{R}}(X)$, as vector spaces, are spanned by $\mathbb{Q}$-ample and $\mathbb{R}$-ample divisors.*

## 22. Positivity: Nef divisors, Kleiman's Theorem

In the previous section, we defined the finite dimensional vector spaces $\mathrm{NS}_{\mathbb{Q}}(X)$ and $\mathrm{NS}_{\mathbb{R}}(X)$, showed that ampleness for $\mathbb{Q}$-divisors and $\mathbb{R}$-divisors are independent of numerical equivalence class, and showed that ampleness for $\mathbb{Q}$-divisors and $\mathbb{R}$-divisors is an open condition.

What about limits of ample divisors? We expect such limits to be characterized by the inequalities:

$$(D^{\dim V}.V) \geq 0, \forall \text{ irreducible positive-dimensional } V \subseteq X,$$

and we'd like to call such divisors *nef divisors.* Indeed, nef divisors are characterized by the above inequality. But it actually suffices to define nef divisors in terms of inequalities against curves. Kleiman's theorem proves that this is sufficient.

Let $X$ be a complete $\mathbb{C}$-variety. In this section we will define nef divisors, prove Kleiman's theorem, and prove precise statements characterizing nef divisors as limits of ample divisors (when there are ample divisors).

**Definition 22.1.** *Let $X$ be a complete $\mathbb{C}$-variety. A Cartier divisor $D$ on $X$ is nef if*

$$(D.C) = \int_C c_1(\mathcal{O}_X(D)) \geq 0$$

*for every irreducible curve $C \subseteq X$. If $D$ is a $\mathbb{R}$-divisor (or $\mathbb{Q}$-divisor), then $D$ is nef if*

$$(D.C) \geq 0.$$

**Theorem 22.2** (Kleiman's Theorem)**.** *Let $X$ be a complete $\mathbb{C}$-variety. Then $D$ is nef if and only if*

$$(D^{\dim V}.V) \geq 0$$

*for all irreducible positive-dimensional $V \subseteq X$.*

*Proof.* The reverse implication is trivially true. We focus on the forward implication. By Chow's lemma, it suffices to assume that $X$ is projective.

Let $D$ be a nef $\mathbb{R}$-divisor of $X$. We induct on the dimension $n = \dim X$. Thus, assume that

$$(D^{\dim V}.V) \geq 0$$

for all irreducible $V \subset X$ such that $0 < \dim V < n$, and $X$ is irreducible. Then we want to show that

$$(D^n) \geq 0.$$

Assume first that $D$ is numerically equivalent to a $\mathbb{Q}$-divisor. The case for $\mathbb{R}$-divisors will follow at the end. Since $X$ is projective, there exist an ample integral divisor $H$. Let

$$P(t) := ((D + tH)^n) = \sum_{j=0}^{n} \binom{n}{j} t^j (D^{n-j}.H^j)$$

be a polynomial in terms of $t \in \mathbb{R}$. By the inductive hypothesis, note that $j \geq 1$, since $H$ is ample, $H^j$ will correspond to an effective $2(n-j)$-cycle in $H_{2(n-j)}(X;\mathbb{Z})$, and intersecting its irreducible components with $D^{n-j}$, by the inductive hypothesis we have that the coefficients of $t^j$ are nonnegative for $j > 0$. On the other hand, assume for the sake of contradiction that the constant term $P(0) = (D^n) < 0$.

Then since $P(0) < 0$ and $P(t) > 0$ for sufficiently large $t$, there exist $t_0 \in \mathbb{R}^+$ such that $P(t_0) = 0$, by continuity of $P(t)$. Let $q > t_0$ be a rational number. We claim that $D + qH$ is ample. Note $D + qH$ is a $\mathbb{Q}$-divisor, by assumption. To show it is ample, it suffices to show that

$$((D + qH)^{\dim V}.V) > 0$$

for all irreducible positive-dimensional $V \subseteq X$. Indeed, we have

$$((D + qH)^{\dim V}.V) = \sum_{j=0}^{\dim V} \binom{\dim V}{j} q^j (D^{\dim V - j}.H^j.V).$$

When $V \subset X$ is a proper subvariety, note that for $j < \dim V$, since $H$ is ample, the intersection $H^j.V$ will correspond to an effective $2(\dim V - j)$ cycle in $H_{2(\dim V - j)}(X;\mathbb{Z})$, so by the inductive hypothesis, $\binom{\dim V}{j} q^j (D^{\dim V - j}.H^j.V) \geq 0$. When $j = \dim V$, we have positivity by ampleness of $H$. When $V = X$, for $0 < j < n$, we have the terms are nonnegative by inductive hypothesis on the irreducible components of $(H^j.V)$, and when

$j = 0$, we have positivity since $P(q) > 0$, and when $j = n$, we have positivity by ampleness of $H$.

Thus, $D + qH$ is ample for rational $q > t_0$. Now write

$$P(q) = Q(q) + R(q) = (D.(D + qH)^{n-1}) + (qH.(D + qH)^{n-1}).$$

For rational $q > t_0$, we have $D + qH$ is ample. So $(D + qH)^{n-1}$ will correspond to an effective 2-cycle, and intersecting $D$ with its irreducible components and using inductive assumption implies $Q(q) \geq 0$. Then by contininuity, since rational $q$ can get arbitrarily close to $t_0$, we have $Q(t_0) \geq 0$.

On the other hand, we see that

$$R(q) = \sum_{j=0}^{n-1} \binom{n-1}{j} q^{j+1} (D^{n-1-j}.H^{j+1}).$$

By inductive assumptions and the $j = n - 1$ term, we see that $R(q) > 0$ and $R(t_0) > 0$. Then $P(t_0) > 0$. Contradiction. Thus, we have shown that if $D$ is numerically equivalent to a nef $\mathbb{Q}$-divisor, then

$$(D^n) \geq 0.$$

Now suppose $D$ is a nef $\mathbb{R}$-divisor. Let $H_1, \cdots, H_r$ be ample divisors which span $\mathrm{NS}_{\mathbb{R}}(X)$. Then consider

$$D(\epsilon_1, \cdots, \epsilon_r) = D + \epsilon_1 H_1 + \cdots + \epsilon_r H_r$$

for all $\epsilon_i > 0$, such that $D(\epsilon_1, \cdots, \epsilon_r)$ is a $\mathbb{Q}$-divisor. Note then by what we've proven, we have $D(\epsilon_1, \cdots, \epsilon_r)$ is a nef $\mathbb{Q}$-divisor, so

$$((D(\epsilon_1, \cdots, \epsilon_r))^n) \geq 0.$$

Then taking $\epsilon_i \to 0$, we find $(D^n) \geq 0$. □

Both the proof of Theorem 22.2 and the following proposition provide evidence of the essential idea that nef divisors are limits of ample divisors.

**Proposition 22.3.** *Let $X$ be a projective $\mathbb{C}$-variety.*

(1) *Let $D$ be a nef $\mathbb{R}$-divisor. Let $H$ any ample $\mathbb{R}$-divisor. Then $D + H$ is ample.*
(2) *Suppose $D_1, D_2$ are arbitrary $\mathbb{R}$-divisors such that $D_1 + \epsilon D_2$ is ample for all $\epsilon > 0$. Then $D_1$ is nef.*

*Proof.* Note (2) is quite easy, because we have $(D_1 + \epsilon D_2.C) > 0$, so $(D_1.C) > -\epsilon(D_2.C) \implies (D_1.C) \geq 0$.

We prove (1). First assume that $D + H$ is a $\mathbb{Q}$-divisor (or numerically equivalent to one). Then to verify ampleness, it suffices to verify that

$$((D + H)^{\dim V}.V) = \sum_{j=0}^{\dim V} \binom{\dim V}{j} (D^j.H^{\dim V - j}.V) > 0$$

for any irreducible positive-dimensional $V \subseteq X$. Indeed this holds by Theorem 22.2 and because of the $(H^{\dim V}.V)$ term. Thus, $D+H$ is ample.

Now suppose $D+H$ is a $\mathbb{R}$-divisor. Then if we let $H_1, \cdots, H_r$ be ample divisors spanning $\mathrm{NS}_{\mathbb{R}}(X)$, we can perturb $H$ so that $H(\epsilon_1, \cdots, \epsilon_r) = H - \epsilon_1 H_1 - \cdots - \epsilon_r H_r$, for sufficiently small $\epsilon_i > 0$, remains ample and $D + H(\epsilon_1, \cdots, \epsilon_r)$ is a $\mathbb{Q}$-divisor. Then by what we just showed, $D + H(\epsilon_1, \cdots, \epsilon_r)$ is an ample divisor, and thus

$$D + H = D + H(\epsilon_1, \cdots, \epsilon_r) + \epsilon_1 H_1 + \cdots + \epsilon_r H_r$$

is an ample $\mathbb{R}$-divisor. □

A nice consequence of this is that we can characterize an ample divisor as: the degree of any curve with respect to $D$ is uniformly bounded below in terms of the degree of $C$ with respect to a known ample divisor.

**Corollary 22.4.** *Let $X$ be a projective $\mathbb{C}$-variety. Let $D$ be a $\mathbb{R}$-divisor and let $H$ be an ample divisor. Then $D$ is ample if and only if there exist a constant $c > 0$ such that*

$$(D.C) \geq c(H.C).$$

*Proof.* Suppose $D$ is ample. Then for sufficiently small $\epsilon$, we have $D - \epsilon H$ is also ample. Then $(D - \epsilon H.C) > 0$ for every irreducible curve $C \subseteq X$, so

$$(D.C) \geq \epsilon(H.C).$$

On the other hand, suppose we have the inequality. Then $D - \epsilon H$ is a nef divisor. Then $D = D - \epsilon H + \epsilon H$ is ample. □

In particular, this is useful because we are checking ampleness against curves. One example of nef divisors is the following.

**Example 22.5.** *Let $X$ be a complete $\mathbb{C}$-variety. Let $D \subseteq X$ be an effective divisor. If the normal bundle $\mathcal{O}_D(D)$ is a nef line bundle on $D$, then $D$ is a nef divisor on $X$.*

*This is because if you want to test nefness of $D$ against irreducible curves $C \subset X$, you are really only worried about curves which are contained in $D$.*

*In particular, this implies that if $X$ is a complex K3 surface, then any irreducible curve $C \subset X$ with nonnegative self-intersection is nef. In particular, for smooth curves, only rational curves on a complex K3 surface fail to be nef. Smooth higher genus curves are always nef.*

## 23. Positivity: nef and ample cones, Kleiman's criterion for ampleness

In this section, we discuss basic properties of nef and ample cones in $\mathrm{NS}_{\mathbb{R}}(X)$. As a consequence, we prove Kleiman's criterion for ampleness – a more practical criterion than the Nakai-Moishezon criterion.

**Definition 23.1.** *Let $X$ be a complete $\mathbb{C}$-variety. We define the nef cone*

$$Nef(X) \subseteq NS_{\mathbb{R}}(X)$$

*to be the locus of all nef divisors. Equivalently, it is all finite nonnegative $\mathbb{R}$-linear combinations of nef divisors.*

**Definition 23.2.** *Let $X$ be a projective $\mathbb{C}$-variety. We define the ample cone*

$$Amp(X) \subseteq NS_{\mathbb{R}}(X)$$

*to be the locus of all ample $\mathbb{R}$-divisors. Equivalently, it is all finite positive $\mathbb{R}$-linear combinations of ample $\mathbb{R}$-divisors (or of integral divisors).*

Note both $\mathrm{Amp}(X)$ and $\mathrm{Nef}(X)$ are convex. Furthermore, the origin is not contained in $\mathrm{Amp}(X)$, so we are not requiring cones to contain 0. In general, cones in a vector space are those loci closed under positive scaling.

**Proposition 23.3.** *Let $X$ be a projective $\mathbb{C}$-variety.*

1. *The nef cone is the closure of the ample cone.*
2. *The ample cone is the interior of the nef cone.*

*Proof.* (1) Note that the nef cone is closed. One way to see this is that since $X$ is projective $\mathbb{C}$-variety, we can choose integral ample $H_1, \cdots, H_n$ as a basis for $\mathrm{NS}_{\mathbb{R}}(X)$. Then every element $D \in \mathrm{Nef}(X)$ is characterized as $(D.C) = (\sum_{i=1}^n c_i H_i.C) \geq 0$. So the possible coefficients $c_i$ are bounded by these integral linear inequalities, which show that $\mathrm{Nef}(X)$ is closed.

Now, we obviously have $\mathrm{Amp}(X) \subseteq \mathrm{Nef}(X)$. Thus $\overline{\mathrm{Amp}(X)} \subseteq \mathrm{Nef}(X)$. Then suppose $D \in \mathrm{Nef}(X)$. Note if $H$ is an integral ample divisor, then $D + \epsilon H$ is ample for all $\epsilon > 0$. This shows that $D$ is a limit of ample $\mathbb{R}$-divisors, so $D \in \overline{\mathrm{Amp}(X)}$. This proves the claim.

(2) We certainly have the ample cone is contained in the interior of the nef cone. Now suppose $D \in \mathrm{Nef}(X)$ is in the interior, but is not contained in the ample cone. This implies that we can perturb $D$ in any direction, and it remains nef. Then let $H$ be an ample $\mathbb{R}$-divisor, and $\epsilon > 0$ a sufficiently small positive number, so that $D - \epsilon H$ remains nef. But then $D = D - \epsilon H + \epsilon H$ is ample, so $D$ is indeed ample.

□

**Remark 23.4.** *If $X$ is a non-projective $\mathbb{C}$-variety, then there is no meaning to the first item of proposition 23.3. But in the case of the second item, it still technically holds if the nef cone has empty interior. Situations where this happens include when $X$ is a complete $\mathbb{C}$-variety that is smooth or $\mathbb{Q}$-factorial* [Laz04, Remark 1.4.24].

We now work our way towards Kleiman's criterion for ampleness. The motivation is this. The Nakai-Moishezon-Kleiman criterion gives a numerical criterion for ampleness.

On the other hand, by Kleiman's Theorem, we are able to test the nefness of a divisor simply against curves. One might wonder whether we could do this for ampleness. It turns out that testing ampleness against just curves is not sufficient, however if we include "limits of curves," then it works. To discuss limits of curves in our numerical setting, we need a vector space.

**Definition 23.5.** *Let $X$ be a complete $\mathbb{C}$-variety. Let $Z_1(X)$ denote the group of one cycles over $\mathbb{R}$; that is, finite $\mathbb{R}$-linear combinations of irreducible curves $C_i \subseteq X$. We say $C_1, C_2 \in Z_1(X)$ are numerically equivalent if*

$$(D.C_1) = (D.C_2),$$

*for every $D \in NS_{\mathbb{R}}(X)$. We let*

$$\frac{Z_1(X)}{\sim}$$

*denote the vector space of one-cycles, up to numerical equivalence.*

Note that there is a perfect pairing

$$\mathrm{NS}_{\mathbb{R}}(X) \times \frac{\mathrm{Z}_1(X)}{\sim} \to \mathbb{R}.$$

In other words, this pairing identifies $\mathrm{NS}_{\mathbb{R}}(X)$ with $(\frac{\mathrm{Z}_1(X)}{\sim})^\vee$.

**Definition 23.6.** *Let the cone of curves of $X$ be the cone*

$$NE(X) \subseteq \frac{Z_1(X)}{\sim}$$

*consisting of finite $\mathbb{R}$-linear combinations $\sum a_i C_i$ where $a_i \in \mathbb{R}_{\geq 0}$ and $C_i$ is an irreducible curve. We let $\overline{NE}(X)$, the Mori cone, denote the closure of $NE(X)$ in the vector space of one-cycles.*[21]

Under the identification, provided by the perfect pairing, of $\mathrm{NS}_{\mathbb{R}}(X)$ with $(\frac{\mathrm{Z}_1(X)}{\sim})^\vee$, we have that $\mathrm{Nef}(X)$ is identified with the dual cone $\overline{\mathrm{NE}}(X)^\vee$. By definition, the dual cone of $\overline{NE}(X)$ is

$$\overline{NE}(X)^\vee := \{\phi \in (\frac{\mathrm{Z}_1(X)}{\sim})^\vee \mid \phi(C) \geq 0,\ \forall C \in \overline{NE}(X)\}.$$

Indeed, any $C \in \overline{NE}(X)$ is the limit of one-cycles $\{C_t = \sum_{i=1}^{n_t} a_{i,t} C_{i,t} \mid a_{i,t} \in \mathbb{R}^{\geq 0}\}_{t=1}^{\infty}$ where $a_{i,t}$, and for $D \in \mathrm{Nef}(X)$, we have

$$\lim_{t\to\infty} (D.C_t) = \lim_{t\to\infty} \sum_{i=1}^{n_t} a_{i,t}(D.C_{i,t}) = (D.C) \geq 0.$$

**Definition 23.7.** *Let $D$ be an element of $NS_{\mathbb{R}}(X)$.*

– *Let $D^\perp$ denote the hyperplane $\{C \in \frac{Z_1(X)}{\sim} | (D.C) = 0\}$.*

[21] The mori cone is extremely important in studying the birational geometry of smooth algebraic varieties.

– *Let* $D_{\geq 0}$ *denote the positive side of the hyperplane:* $\{C \in \frac{Z_1(X)}{\sim} | (D.C) \geq 00\}$.
*From this, the meaning of* $D_{>0}$ *and* $D_{\leq 0}$ *and* $D_{<0}$ *should be clear.*

**Theorem 23.8.** *[Kleiman's Criterion for Ampleness] Let* $X$ *be a projective* $\mathbb{C}$*-variety. Let* $D \in NS_{\mathbb{R}}(X)$*. Then* $D$ *is ample if and only if*

$$\overline{NE}(X) \setminus \{0\} \subseteq D_{>0} \subseteq \frac{Z_1(X)}{\sim}.$$

*Equivalently, if* $||\cdot||$ *is any norm on the vector space* $\frac{Z_1(X)}{\sim}$*, then if we let* $\mathbb{S}$ *denote the unit sphere with respect to* $||\cdot||$*, then*

$$\mathbb{S} \cap \overline{NE}(X) \subseteq \mathbb{S} \cap D_{>0}.$$

*Proof.* Suppose $D$ is ample. Note that $\mathrm{NE}(X) \setminus \{0\} \subseteq D_{>0}$, since $D$ is ample. Then by definition of closure, we have $\overline{\mathrm{NE}}(X) \subseteq D_{\geq 0}$. Assume for the sake of contradiction that

$$\overline{\mathrm{NE}}(X) \setminus \{0\} \nsubseteq D_{>0}.$$

Then there exists some $C \in \overline{\mathrm{NE}}(X) \setminus \{0\}$ such that $(D.C) = 0$. Let $H_1, \cdots, H_n$ be a basis of ample divisors of $\mathrm{NS}_{\mathbb{R}}(X)$. Then we can perturb $D$ by each $H_i$ so that it remains ample. So $(D - \epsilon_i H_i.C) \geq 0 \implies (H_i.C) \leq 0$. But each $H_i$ is ample so $(H_i.C) \geq 0 \implies (H_i.C) = 0$. This implies that $C$ is numerically trivial, so $C = 0$. Thus, we must have

$$\overline{\mathrm{NE}}(X) \setminus \{0\} \subseteq D_{>0}.$$

Now assume the above containment holds. We define the following "taxicab norm." Let $H_1, \cdots, H_r$ be ample divisors of $X$ which together form a basis of $\mathrm{NS}_{\mathbb{R}}(X)$. Let $H = \sum_{i=1}^r H_i$. Then for $C \in \frac{\mathrm{Z}_1(X)}{\sim}$, define

$$||C|| = \sum_{i=1}^r |(C.H_i)|.$$

This defines a norm on $\frac{\mathrm{Z}_1(X)}{\sim}$. Let $\mathbb{S}$ denote the unit sphere in $\frac{\mathrm{Z}_1(X)}{\sim}$ with respect to this norm. Then our assumed containment imply that

$$\mathbb{S} \cap \overline{NE}(X) \subseteq \mathbb{S} \cap D_{>0}.$$

Note $\mathbb{S} \cap \overline{NE}(X)$ is compact (closed and bounded), so the linear functional defined by $D$ restricted to $\mathbb{S} \cap \overline{NE}(X)$ is bounded. Thus, for all $C \in \overline{NE}(X)$, we have

$$(D.C) \geq \epsilon ||C|| = \epsilon \sum_{i=1}^r |(C.H_i)|.$$

Then for every irreducible curve $C$, we have

$$(D.C) \geq \epsilon \sum_{i=1}^r |(H_i.C)| \geq \epsilon (H.C).$$

This implies that $D$ is ample. ◻

Finally, we say something about cones for projective surfaces.

**Example 23.9.** *Let $X$ be a projective surface. Then $NS_{\mathbb{R}}(X)$ and $\frac{Z_1(X)}{\sim}$ are actually the same vector space, since curves now also correspond to Cartier divisors. Thus, the cones $Amp(X), Nef(X),$ and $\overline{NE}(X)$ all live in the same vector space.*

(1) *We have*
$$Nef(X) \subseteq \overline{NE}(X).$$
(2) *Suppose $C$ is an irreducible curve such that $(C.C) \leq 0$. Then $\overline{NE}(X)$ is spanned by the sub-cones*
$$\mathbb{R}_{\geq 0} \cdot [C] \text{ and } C_{\geq 0} \cap \overline{NE}(X).$$
(3) *In addition, $\mathbb{R}_{\geq 0} \cdot [C]$ forms a boundary of $\overline{NE}(X)$. Furthermore, suppose $(C.C) < 0$. Then $\mathbb{R}_{\geq 0} \cdot C$ is an extremal ray of $\overline{NE}(X)$.* [22]

*Proof.* (1) Let $D \in \mathrm{Nef}(X)$. Then for ample $H$, we have $D + tH$ is ample for every $t > 0$. Then $D + tH \in \mathrm{NE}(X)$, so $\lim_{t \to 0} D + tH = D \in \overline{\mathrm{NE}}(X)$.

(2) Let $D \in \overline{NE}(X)$. Then there exist $D_t \in \mathrm{NE}(X)$ such that
$$\lim_{t \to \infty} D_t = D.$$
We can write each one-cycle $D_t$ as
$$D_t = \lambda_t C + \sum_i a_{t,i} C_{t,i} = \lambda_t C + D'_t.$$
where $\lambda_t$ and $a_{t,i} \geq 0$ and $C_{t,i}$ are irreducible reduced curves. Note if we pick a basis of ample divisors $H_1, \cdots, H_n$ for the vector space $\mathrm{NS}_{\mathbb{R}}(X)$, then we can write
$$D'_t = c_{t,1} H_1 + \cdots + c_{t,n} H_n$$
Thinking of each $D'_t$ in this way, and recognizing that each $D'_t \in \mathrm{NE}(X)$, we find that
$$D = (\lim_{t \to \infty} \lambda_t) C + \sum_{j=1}^n (\lim_{t \to \infty} c_{t,j}) H_j,$$
and $\sum_{j=1}^n (\lim_{t \to \infty} c_{t,j}) H_j \in \overline{NE}(X)$, and $(\lim_{t \to \infty} \lambda_t) \geq 0$. Finally, note that
$$\sum_{j=1}^n (\lim_{t \to \infty} c_{t,j}) H_j \in C_{\geq 0} \cap \overline{NE}(X)$$
since
$$(\sum_{j=1}^n (\lim_{t \to \infty} c_{t,j}) H_j . C) = \lim_{t \to \infty} (D'_t . C) = \lim_{t \to \infty} (\sum_i a_{t,i} C_{t,i} . C) \geq 0.$$

[22] Some intuition: if you intersect $\overline{\mathrm{NE}}(X)$ by a hyperplane and get a polygon, then the extremal rays should be thought of as those rays determined by the vertices. In these nice situations, it's easy to see that the extremal rays span the boundary of the cone (and thus also span the entire cone).

(3) First, suppose $(C.C) = 0$. Note that $(C.-)$ defines a linear functional on $\mathrm{NS}_{\mathbb{R}}(X)$. Suppose $H$ is an integral ample divisor. Then $(C.H) > 0$. But then $(C.C - \epsilon H) < 0$ for all $\epsilon > 0$. This shows, using

$$\overline{\mathrm{NE}}(X) = \mathbb{R}_{\geq 0} \cdot [C] + C_{\geq 0} \cap \overline{\mathrm{NE}}(X),$$

that $C - \epsilon H$ is not in $\overline{NE}(X)$ for any $\epsilon > 0$. Thus, $C \in \partial\overline{\mathrm{NE}}(X)$.

Now suppose $(C.C) < 0$. We show that $\mathbb{R}_{\geq 0} \cdot C$ is an extremal ray of $\overline{\mathrm{NE}}(X)$.

- Recall that if $V \subseteq \mathbb{R}^n$ is a closed convex cone, then $\mathbb{R}_{\geq 0} \cdot v$ is an extremal ray if for any $x, y \in V$ we have $x + y \in \mathbb{R}_{\geq 0} \cdot v$ implies $x, y \in \mathbb{R}_{\geq 0} \cdot v$.
- Furthermore, extremal rays always lie on the boundary $\partial V$. Assume for the sake of contradiction that $v$ is not in the boundary. Then for any vector $w$ in $\mathbb{R}^n$, for sufficiently small $\epsilon > 0$, we would have $v - \epsilon w \in V$. Suppose we let $w \in V$. Then $v - \epsilon w + \epsilon w = v$, which means $v - \epsilon w \in \mathbb{R}_{\geq 0} \cdot v$ and $\epsilon w \in \mathbb{R}_{\geq 0} \cdot v$. This means $V = \mathbb{R}_{\geq 0} \cdot v$. But then $v$ must be on the boundary of this 1 dimensional cone in $\mathbb{R}$.

If $D_1, D_2 \in \overline{\mathrm{NE}}(X)$, such that $D_1 + D_2 = \lambda C$, note we can write $D_1 = \lambda_1 C + D_1'$ and $D_2 = \lambda_2 C + D_2'$, where $(C.D_1'), (C.D_2') \geq 0$. Then $(D_1 + D_2.C) = (\lambda_1 + \lambda_2)(C.C) + (D_1' + D_2'.C) = (\lambda C.C) < 0$. In particular, we'd have $D_1' = -D_2'$ numerically, so we see that $\mathbb{R}_{\geq 0} \cdot C$ is an extremal ray, and thus also on the boundary.

□

## 24. A NOTE ON STALKS OF ANALYTIC HIGHER DIRECT IMAGE SHEAVES

Let $\pi : X \to Y$ be a continuous map of topological spaces. Let $\mathcal{F}$ be a sheaf of abelian groups on $X$.

**Definition 24.1.** [Dem12, Definition 4.5] *A subspace $S \subseteq X$ is strongly paracompact if it is Hausdorff and the following property is satisfied: for every covering $\{U_\alpha\}$ of $S$ by open sets in $X$, there exists another such covering $\{V_\beta\}$ and a neighborhood $W$ of $S$ such that each set $W \cap \overline{V_\beta}$ is contained in some $U_\alpha$, and such that every point of $S$ has a neighborhood intersecting only finitely many sets $V_\beta$.*

If $X$ is paracompact (each open cover admits a locally finite refinement) and $S$ is closed, then $S$ is strongly paracompact.

**Theorem 24.2.** [Dem12, Theorem 9.10] *Let $\mathcal{F}$ be a sheaf on $X$, and $S$ a strongly paracompact subspace. When $\Omega$ ranges over open neighborhoods of $S$, we have*

$$H^q(S, \mathcal{F}) = \varinjlim_{\Omega \supset S} H^q(\Omega, \mathcal{F}).$$

Consider the stalk of the $k$-th higher direct image sheaf of $\mathcal{F}$ at $y \in Y$:

$$R^k \pi_* \mathcal{F}|_y = \varinjlim_{y \in U} H^k(\pi^{-1}(U), \mathcal{F}|_U).$$

If the fiber $\pi^{-1}(y)$ is strongly paracompact in $X$, and if the family of open sets $\pi^{-1}(U)$ is a fundamental family of neighborhoods of $\pi^{-1}(y)$, then we have [Dem12, Page 232]

$$R^k\pi_*\mathcal{F}|_y = H^q(\pi^{-1}(y), \mathcal{F}|_{\pi^{-1}(y)}).$$

In particular, if $\pi : X \to Y$ is a submersion between compact complex manifolds, then $R^k\pi_*\mathcal{F}|_y = H^k(\pi^{-1}(y), \mathcal{F}|_{\pi^{-1}(y)})$ for every $y \in Y$.

In the algebraic setting, it's a little different [Har, Corollary 12.9, Grauert]. Let $f : X \to Y$ be a projective morphism of noetherian schemes, with $Y$ being integral, and let $\mathcal{F}$ be a coherent sheaf on $X$, flat over $Y$. Suppose $h^i(\pi^{-1}(y), \mathcal{F})$ is locally constant. Then $R^if_*\mathcal{F}$ is locally free on $Y$, and for every $y$, we have

$$R^if_*\mathcal{F} \otimes k(y) \cong H^i(\pi^{-1}(y), \mathcal{F}|_{\pi^{-1}(y)}).$$

Now if $f : X \to Y$ is a surjective map between projective complete non-singular $\mathbb{C}$-varieties, and $\mathcal{F}$ is coherent, then since $R^kf_*\mathcal{F}$ remains coherent, we can apply GAGA to deduce that in the algebraic category we still have the stalk identification

$$R^kf_*\mathcal{F}|_y = H^k(\pi^{-1}(y), \mathcal{F}|_{\pi^{-1}(y)}).$$

## 25. Positivity: Nef and Mori cones of ruled surfaces

Let's recall some basic facts about ruled surfaces.

**Definition 25.1.** *Let $S$ be a surface, i.e. nonsingular irreducible projective variety of dimension 2, over an algebraically closed field $k$. Let $C$ be a nonsingular curve. Then the surjective morphism $\pi : S \to C$ is a ruled surface if the geometric fibers $S_y$ are isomorphic to $\mathbb{P}^1_k$ for every $y \in C$ and if $\pi$ admits a section.*

**Example 25.2.** *Nonsingular quadrics in $\mathbb{P}^3_k$ are isomorphic to $\mathbb{P}^1_k \times \mathbb{P}^1_k$, and thus are examples of ruled surfaces. In particular, they each have two rulings (one for each factor).*

**Proposition 25.3.** *Let $\pi : S \to C$ be a ruled surface. Let $S_x$ and $S_y$ be fibers over $x, y \in C$. Then*

$$[S_x] = [S_y] \in NS_{\mathbb{R}}(S).$$

*In particular, we can speak of* the fiber class $[F] \in NS_{\mathbb{R}}(X)$ *of a ruled surface.*

*Proof.* For $S_x$ and $S_y$ to be numerically equivalent, we must have

$$(S_x.C) = (S_y.C)$$

for every irreducible curve $C \subset S$. But note a curve $C$ defines an element of $\mathrm{NS}_{\mathbb{R}}(X)$, and since $S$ is projective, $\mathrm{NS}_{\mathbb{R}}(S)$ admits a basis in terms of integral ample divisors. So the class of $C \subset S$ will define an element of $\mathrm{NS}_{\mathbb{R}}(S)$ which may be expressed in terms of linear combinations of integral ample divisors. Thus, it suffices to show that

$$(S_x.H) = (S_y.H)$$

for every integral ample $H$. To prove this, it suffices to assume $H$ is very ample. We now verify that $(S_x.H) = (S_y.H)$. Consider the diagram

$$\begin{tikzcd} S \arrow[rr, hook, "|\mathcal{O}_S(H)|\times\pi"] \arrow[dr, "\pi"'] & & \mathbb{P}^n_k \times C \arrow[dl, "\pi_C"] \\ & C & \end{tikzcd}$$

By flatness, the Hilbert polynomials of $S_x$ and $S_y$ are the same with respect to $H$. In particular, the degree of their Hilbert polynomials is exactly equal to both $(S_x.H)$ and $(S_y.H)$, hence $(S_x.H) = (S_y.H)$. □

**Question 25.4.** *Suppose $X$ is a nonsingular projective irreducible $\mathbb{C}$-variety of dimension $n > 2$. Suppose $\pi : X \to C$ is again a surjective morphism over a smooth curve. I would guess that two fibers $X_a, X_b \subset X$ may not necessarily be numerically equivalent, although I don't have a counterexample at the moment.*

*One would need to check that*

$$(X_a.C) = (X_b.C)$$

*for every irreducible 1-dimensional closed subscheme $C \subset X$. For a given $C$, if there exists a very ample $H$ such that $|\mathcal{O}_X(H)|$ embeds $X$ in a projective space $\mathbb{P}^N$ such that $C \subset X$ comes from intersecting $X$ with a linear subspace $\Lambda = H_1 \cap \cdots \cap H_{N-n-1}$. In this case, both $(X_a.C) = (X_b.C)$ equal the degree of $X_a$ and $X_b$ as closed subschemes of $\mathbb{P}^N$, which are equal.*

*So: does there exist an interesting class of projective varieties, where for every curve on it, there exists an embedding so that the curve comes from intersecting the variety with a linear subspace?*

Ruled surfaces over curves are very much related to projectivizing rank 2 vector bundles over curves. Before we discuss this, let us recall some basic facts about projective bundles. Let $\mathcal{E}$ be a rank $n+1$ vector bundle over $X$. Let $\pi : \mathbb{P}(\mathcal{E}) \to X$ be the projection of the projective bundle onto $X$.

If $U \subset X$ locally trivializes $\mathcal{E}$, then locally $\pi^{-1}(U) \cong U \times \mathbb{P}^n$. Note the pullbacks of $\mathcal{O}_{\mathbb{P}^n}(1)$ to $\pi^{-1}(U_i)$ glue together with respect to a trivializing cover $\{U_i\}$ of $X$ to give a line bundle on $\mathbb{P}(\mathcal{E})$ which we call $\mathcal{O}(1)$. If we are working with complex manifolds, note $\mathcal{O}(1)$ is positive, so if $X$ is compact, then $\mathbb{P}(\mathcal{E})$ is also compact, and thus $\mathbb{P}(\mathcal{E})$ is projective.

**Lemma 25.5.** *Let $X$, $\mathcal{E}$, $\pi : \mathbb{P}(\mathcal{E}) \to X$, and $\mathcal{O}(1)$ be as above.*

(1) *There is a surjection of sheaves $\pi^*\mathcal{E} \to \mathcal{O}(1)$.*
(2) *Given a morphism $g : Y \to X$, diagrams*

$$\begin{tikzcd} & \mathbb{P}(\mathcal{E}) \arrow[d, "\pi"] \\ Y \arrow[ur, dashed, "f"] \arrow[r, "g"'] & X \end{tikzcd}$$

*are equivalent to the data of a surjection of sheaves $g^*\mathcal{E} \to \mathcal{L}$, where $\mathcal{L}$ is an invertible sheaf on $Y$.*

*Proof.* (1) This statement is a relative version of surjections like $\mathcal{O}_{\mathbb{P}^n}^{\oplus n+1} \to \mathcal{O}_{\mathbb{P}^n}(1)$. Pick a trivializing cover $\{U_i\}$ of $X$. Note $\pi^{-1}(U_i) \cong U_i \times \mathbb{P}^n$. We have on $\mathbb{P}^n$ the surjection of sheaves $\mathcal{O}_{\mathbb{P}^n}^{\oplus n+1} \to \mathcal{O}_{\mathbb{P}^n}(1)$. Then we can pullback this surjection onto $U_i \times \mathbb{P}^n$, and thus pull it back onto $\pi^{-1}(U_i)$. Then we see that we can glue these surjections together with respect to $U_i$, and the result is the

$$\pi^*\mathcal{E} \to \mathcal{O}(1),$$

which must then be a surjection of sheaves.

(2) First, suppose we have such a diagram with lift $f$. Then pulling back the surjection $\pi^*\mathcal{E} \to \mathcal{O}(1)$ on $\mathbb{P}(\mathcal{E})$ along $f$ yields a surjection

$$g^*\mathcal{E} \to \mathcal{L} \cong f^*\mathcal{O}(1),$$

as desired. Now we claim that given $g^*\mathcal{E} \to \mathcal{L}$, there exists a unique $f : Y \to \mathbb{P}(\mathcal{E})$ which makes the diagram commute and which identifies the pullback of $\pi^*\mathcal{E} \to \mathcal{O}(1)$ along $f$ with $g^*\mathcal{E} \to \mathcal{L}$.

To see this, it suffices to work locally. Suppose we have a trivializing open $U \subset X$, and consider $g^{-1}(U) \subset Y$. How should we define a map

$$g^{-1}(U) \to \pi^{-1}(U) \cong U \times \mathbb{P}^n?$$

Consider the surjection $g^*\mathcal{E} \to \mathcal{L}$. Then locally the surjection $g^*\mathcal{E}|_{g^{-1}(U)} \cong \mathcal{O}_{g^{-1}(U)}^{\oplus n+1} \to \mathcal{L}|_{g^{-1}(U)}$ means that there are sections $s_0, \cdots, s_n$ over $g^{-1}(U)$ which generate $\mathcal{L}|_{g^{-1}(U)}$. Then we define the map to be

$$p \in g^{-1}(U) \mapsto (\pi(p), [s_0(p) : \cdots : s_n(p)]) \in U \times \mathbb{P}^n \cong \pi^{-1}(U).$$

By construction, the pullback of the surjection $\pi^*\mathcal{E} \to \mathcal{O}(1)$ along this map is exactly $g^*\mathcal{E} \to \mathcal{L}$.

□

**Proposition 25.6.** *If $\pi : X \to C$ is a ruled surface, then there exists a locally free sheaf $\mathcal{E}$ of rank 2 on $C$ such that $X \cong \mathbb{P}(\mathcal{E})$ over $C$. Conversely, every such $\mathbb{P}(\mathcal{E})$ is a ruled surface over $C$.*

*If $\mathcal{E}, \mathcal{E}'$ are two locally free sheaves of rank 2 on $C$, then $\mathbb{P}(\mathcal{E}) \cong \mathbb{P}(\mathcal{E}')$ if and only if $\mathcal{E} \cong \mathcal{E}' \otimes \mathcal{L}$ for some invertible sheaf $\mathcal{L}$.*

*Proof.* First we define a map which associates a ruled surface to a locally free sheaf of rank 2 on $C$. Let $\mathcal{E}$ be such a sheaf on $C$. We claim that $\pi : \mathbb{P}(E) \to C$ is a ruled surface. Indeed, each fiber is isomorphic to $\mathbb{P}^1$. Now we need to produce a section. Pick a neighborhood $U \subset C$ such that $\mathcal{E}|_U$ is free. Then $\mathbb{P}(\mathcal{E})|_U \cong U \times \mathbb{P}^1_k$. Picking a point $\lambda \in \mathbb{P}^1_k$, we obtain a local section on $U$ by sending $p \in U \mapsto (p, \lambda)$. By [Har, Chapter 1, Proposition 6.8], this extends to a section.

Now suppose $\pi : X \to C$ is a ruled surface. Let $D \subset X$ be the divisor which corresponds to a section of the ruled surface. Then $(D.F) = 1$, where $F$ denotes the fiber class of the ruled surface. Note we proved before that the fiber class is a well-defined numerically. Then the line bundle $\mathcal{O}_X(D)$ is such that its restriction to each fiber is isomorphic to $\mathcal{O}_{\mathbb{P}^1}(1)$. Thus, $h^0(\mathcal{O}_X(D)|_{\pi^{-1}(p)})$ is constant as a function of $p \in C$, which implies $\pi_*\mathcal{O}_X(D)$ is a locally free sheaf of rank 2 on $C$. Our goal is to show that $\mathbb{P}(\pi_*\mathcal{O}_X(D)) \cong X$. In particular, we would like to show that there exists $f$ which fits into the diagram

$$\begin{tikzcd} & \mathbb{P}(\pi_*\mathcal{O}_X(D)) \arrow[d] \\ X \arrow[ur, dashed, "f"] \arrow[r, "\pi"'] & C \end{tikzcd}$$

such that $f$ is an isomorphism. First, in order to obtain such a $f$, we need the data of a surjection of sheaves

$$\pi^*\pi_*\mathcal{O}_X(D) \to \mathcal{L},$$

where $\mathcal{L}$ is an invertible sheaf on $X$. Note $\pi_*\mathcal{O}_X(D)$ should be thought of as the rank 2 vector bundle over $C$, where over $p \in C$, the fiber $(\pi_*\mathcal{O}_X(D)) \otimes k(p)$ is identified with $H^0(\mathcal{O}_X(D)|_{\pi^{-1}(p)})$. So $\pi^*\pi_*\mathcal{O}_X(D)$ is a rank 2 vector bundle so that over a point $\lambda \in \pi^{-1}(p)$, its fiber is identified with $H^0(\mathcal{O}_X(D)|_{\pi^{-1}(p)})$. Then we see that there is a surjection of sheaves

$$\pi^*\pi_*\mathcal{O}_X(D) \to \mathcal{O}_X(D),$$

which gives a map $f : X \to \mathbb{P}(\pi_*\mathcal{O}_X(D))$. Finally, we want $f$ to be an isomorphism. But this follows because $X$ surjects onto $C$, and each fiber $\pi^{-1}(p)$ maps isomorphically onto the corresponding fiber of $\mathbb{P}(\pi_*\mathcal{O}_X(D))$ over $C$.

Finally, the correspondence "up to equivalence" falls out of comparing the transition functions for $\mathcal{E}, \mathcal{E}'$, and what isomorphisms of their projectivizations imply. In particular, their transition functions must differ only by invertible scalar functions, which is exactly the data of a line bundle on $C$. □

Let us clarify a subtle point. We just saw in the proof of proposition 25.6, that if $C_0 \subset X = \mathbb{P}(\mathcal{E})$ is the effective divisor associated to a section of a ruled surface $\pi : X \to C$, then

$$X \cong \mathbb{P}(\pi_*\mathcal{O}_X(C_0)),$$

over $C$. But we always have $\pi_*\mathcal{O}(1) \cong \mathcal{E}$. So $\pi_*\mathcal{O}(1)$ and $\pi_*\mathcal{O}_X(C_0)$ will always be isomorphic up to tensoring by a line bundle over $C$.

**Proposition 25.7.** *Let $\pi : X \to C$ be a ruled surface. Let $C_0 \subset X$ be a section, and let $F$ denote the fiber class. Then*

$$\operatorname{Pic}(X) \cong \mathbb{Z}\langle C_0 \rangle \oplus \pi^* \operatorname{Pic} C,$$

*and*

$$NS(X) \cong \mathbb{Z}\langle C_0 \rangle \oplus \mathbb{Z}\langle F \rangle,$$

*satisfying the numerical relations $(C_0.F) = 1$ and $(F.F) = 0$.*

*Proof.* Note that $\pi^* : \operatorname{Pic}(C) \to \operatorname{Pic}(X)$ is injective (just think about transition functions). Now suppose $D \in \operatorname{Pic}(X)$, and suppose $(D.F) = n$. Then consider $D - nC_0$. We have $(D - nC_0.F) = 0$, so $\pi_* \mathcal{O}_X(D - nC_0)$ is a locally free sheaf of rank 1 on $C$, since $h^0(\mathcal{O}_X(D - nC_0)|_{\pi^{-1}(p)})$ is constant as a function of $p \in C$.

Then note $\pi^* \pi_* \mathcal{O}_X(D - nC_0) \cong \mathcal{O}_X(D - nC_0)$. So we see that

$$\operatorname{Pic}(X) \cong \mathbb{Z}\langle C_0 \rangle \oplus \pi^* \operatorname{Pic}(C).$$

For the Neron-Severi group, note that it will be a quotient of $\operatorname{Pic}(X)$. Specifically, note that $\pi^* \operatorname{Pic}(C)$ modulo numerical equivalence is simply $\mathbb{Z}\langle F \rangle$, because the divisors on $C$ are generated by $\mathbb{Z}$-linear combination of points on $C$, and the pullbacks of these will be fiber classes, so

$$\mathrm{NS}(X) \cong \mathbb{Z}\langle C_0 \rangle \oplus \mathbb{Z}\langle F \rangle.$$

Note $(C_0.F) = 1$ since $C_0$ is a section, and $(F.F) = 0$ since distinct fibers do not meet anywhere.

□

To fully describe $\mathrm{NS}(X)$, it remains to determine $(C_0.C_0)$, where $C_0$ is the divisor associated to a section of a ruled surface $\pi : X \to C$. Recall before that $\pi_* \mathcal{O}(1)$ is always isomorphic to $\mathcal{E}$. And for any section $C_0$, we have $\mathbb{P}(\pi_* \mathcal{O}_X(C_0)) \cong \mathbb{P}(\mathcal{E})$. This suggests that perhaps $\mathcal{O}(1)$ can be associated to some section.

**Lemma 25.8.** *Let $\pi : X = \mathbb{P}(\mathcal{E}) \to C$ be a ruled surface, and let $\sigma : C \to X$ be a section associated to the data of a surjection of sheaves*

$$\mathcal{E} \to \mathcal{L} \to 0.$$

*Let $C_0$ denote the effective divisor associated to section $\sigma$. Then*

$$\ker(\mathcal{E} \to \mathcal{L}) \cong \pi_*(\mathcal{O}(1) \otimes \mathcal{O}_X(-C_0)).$$

*Proof.* Consider the short exact sequence

$$0 \to \mathcal{O}_X(-C_0) \to \mathcal{O}_X \to \iota_* \mathcal{O}_{C_0} \to 0.$$

Tensoring, we have the short exact sequence

$$0 \to \mathcal{O}(1) \otimes \mathcal{O}_X(-C_0) \to \mathcal{O}(1) \to \mathcal{O}(1)|_{C_0} \to 0.$$

Pushing forward this short exact sequence, and noting that $\pi_*\mathcal{O}(1) \cong \mathcal{E}$ and $\pi_*\mathcal{O}(1)|_{C_0} \cong \mathcal{L}$,

$$0 \to \pi_*(\mathcal{O}(1) \otimes \mathcal{O}_X(-C_0)) \to \pi_*\mathcal{O}(1) \to \mathcal{L} \to R^1\pi_*(\mathcal{O}(1) \otimes \mathcal{O}_X(-C_0)) \to R^1\pi_*\mathcal{O}(1) \to \cdots$$

Note the higher direct images vanish (they are parameterizing positive degree cohomologies of nonnegative degree line bundles on $\mathbb{P}^1$). So we are left with

$$0 \to \pi_*(\mathcal{O}(1) \otimes \mathcal{O}_X(-C_0)) \to \pi_*\mathcal{O}(1) \to \mathcal{L} \to 0,$$

as desired. □

We will now express any ruled surface in a "normalized form," which is not necessarily unique. Along the way, we'll see that there is a section of the ruled surface associated to the $\mathcal{O}(1)$ obtained from this preferred expression of the ruled surface. Note the $\mathcal{O}(1)$ line bundle on a ruled surface is dependent on the rank 2 vector bundle used – different rank 2 vector bundles lead to different $\mathcal{O}(1)$'s on the same ruled surface.

**Proposition 25.9.** *Let $\pi : X \to C$ be a ruled surface. Then there exists a rank 2 locally free sheaf $\mathcal{E}$ on $C$ such that $\mathbb{P}(\mathcal{E}) \cong X$, $H^0(C, \mathcal{E}) \neq 0$, and $H^0(C, \mathcal{E} \otimes \mathcal{L}) = 0$ for all $\mathcal{L}$ such that* $\deg \mathcal{L} < 0$.

*Furthermore, there exists a section with associated effective divisor $C_0$ such that $\mathcal{O}_X(C_0) \cong \mathcal{O}(1)$, where $\mathcal{O}(1)$ is obtained with respect to the aforementioned $\mathcal{E}$.*

*Proof.* By proposition 25.6, we can write $X = \mathbb{P}(\mathcal{E}')$ for some rank 2 locally free sheaf $\mathcal{E}'$ on $C$. By lemma 25.5, there is a one to one correspondence between sections $\sigma : C \to X$ and surjections of sheaves

$$\mathcal{E}' \to \mathcal{L} \to 0,$$

where $\mathcal{L}$ is a locally free sheaf of rank 1 on $C$. Note the kernel is also a locally free sheaf of rank 1:

$$0 \to K \to \mathcal{E}' \to \mathcal{L} \to 0.$$

If we tensor by some arbitrary line bundle $\mathcal{L}'$, we obtain

$$0 \to K \otimes \mathcal{L}' \to \mathcal{E} \otimes \mathcal{L}' \to \mathcal{L} \otimes \mathcal{L}' \to 0.$$

Taking sheaf cohomology, note we can control whether $\mathcal{E} \otimes \mathcal{L}'$ has global sections because if we twist by an ample enough $\mathcal{L}'$, then $K \otimes \mathcal{L}'$ will have global sections. If we tensor by a line bundle of sufficiently negative degree, we can guarantee that $\mathcal{E} \otimes \mathcal{L}'$ will not have global sections because $K \otimes \mathcal{L}'$ and $\mathcal{L} \otimes \mathcal{L}'$ won't have global sections. Thus, the $\mathcal{E}$ we desire can be obtained. Note there is not necessarily a unique choice of $\mathcal{E}$, but $\deg \mathcal{E} = \deg \bigwedge^2 \mathcal{E}$ is and is an invariant of $X$.

Now, since $\mathcal{E}$ has a global section, there an injection of sheaves

$$0 \to \mathcal{O}_C \to \mathcal{E}.$$

We claim the quotient sheaf $\mathcal{E}/\mathcal{O}_C$ is locally free of rank 1 on $C$. Note any torsion-free sheaf is locally free outside of a codimension at least 2 closed subset. Since $C$, is a curve, it suffices to show that $\mathcal{E}/\mathcal{O}_C$ is torsion-free. Assume for the sake of contradictino that $\mathcal{E}/\mathcal{O}_C$ does have torsion. Then it has a nonzero torsion subsheaf. Let $F \subset \mathcal{E}$ denote the preimage of the torsion subsheaf under the quotient map $\mathcal{E} \to \mathcal{E}/\mathcal{O}_C$. Note we have

$$0 \to \mathcal{F} \to \mathcal{E} \to \mathcal{E}/\mathcal{F} \to 0,$$

and $\mathcal{E}/\mathcal{F}$ is torsion-free, so it is a locally free sheaf of rank 1, so $\mathcal{F}$ is also a locally free sheaf of rank 1. In addition, $\mathcal{O}_C \hookrightarrow \mathcal{F}$, but $\mathcal{O}_C \neq \mathcal{F}$. Since $\mathcal{F}$ has a global section, $\deg \mathcal{F} > 0$, but by tensoring the above short exact sequence by $\mathcal{F}^\vee$, we obtain

$$0 \to \mathcal{O}_C \to \mathcal{E} \otimes \mathcal{F}^\vee \to \mathcal{E}/\mathcal{F} \otimes \mathcal{F}^\vee \to 0.$$

Then $H^0(C, \mathcal{E} \otimes \mathcal{F}^\vee) \neq 0$, violating the defining property of $\mathcal{E}$. Thus, $\mathcal{E}/\mathcal{O}_C$ is torsion-free and thus locally free of rank 1 on $C$.

Altogether, this implies that there is a section, corresponding to the data of

$$0 \to \mathcal{O}_C \to \mathcal{E} \to \mathcal{E}/\mathcal{O}_C \to 0.$$

If $C_0$ denote the effective divisor of the ruled surface associated to that section, then

$$\pi_*(\mathcal{O}(1) \otimes \mathcal{O}_X(-C_0)) \cong \mathcal{O}_C \implies \mathcal{O}_X(C_0) \cong \mathcal{O}(1),$$

as desired. □

Remember: this section on the ruled surface always exists, and so does the $\mathcal{O}(1)$, but the $\mathcal{O}(1)$ depends on $\mathcal{E}$. There could be other $\mathcal{O}(1)$'s, such that $\mathcal{O}_X(C_0) \not\cong \mathcal{O}(1)$.

**Proposition 25.10.** *Let $\pi : X = \mathbb{P}(\mathcal{E}) \to C$ be a ruled surface in normalized form, and let $C_0$ be a section class such that $\mathcal{O}_X(C_0) \cong \mathcal{O}(1)$.*

*Let $D$ be the effective divisor associated to a section $\sigma$, corresponding to the surjection $\mathcal{E} \to \mathcal{L} \to 0$.*

(1) $\deg \mathcal{L} = (C_0.D)$
(2) *We have (abusing notation)*

$$D \sim C_0 + \pi^*\mathcal{L} - \pi^* \bigwedge^2 \mathcal{E}.$$

(3) $(C_0.C_0) = \deg \mathcal{E} := \deg \bigwedge^2 \mathcal{E}$.

*Proof.* (1) This follows because $(C_0.D) = \deg \mathcal{O}(1)|_D = \deg \sigma^*\mathcal{O}(1)|_D = \deg \mathcal{L}$.

(2) Using the short exact sequence

$$0 \to \pi_*(\mathcal{O}(1) \otimes \mathcal{O}_X(-D)) \to \mathcal{E} \to \mathcal{L} \to 0,$$

note $\pi_*(\mathcal{O}(1) \otimes \mathcal{O}_X(-D)) \in \mathrm{Pic}(C)$, and $\pi^*\pi_*(\mathcal{O}(1) \otimes \mathcal{O}_X(-D)) \cong \mathcal{O}(1) \otimes \mathcal{O}_X(-D)$. Then

$$\bigwedge^2 \mathcal{E} \cong \mathcal{L} \otimes \pi_*(\mathcal{O}(1) \otimes \mathcal{O}_X(-D)),$$

and since $\mathcal{O}_X(C_0) \cong \mathcal{O}(1)$, we have

$$C_0 - D \sim (\pi^* \bigwedge^2 \mathcal{E} - \pi^*\mathcal{L}) \implies D \sim C_0 + \pi^*\mathcal{L} - \pi^* \bigwedge^2 \mathcal{E}.$$

(3) Note we have

$$\deg \pi_*(\mathcal{O}(1) \otimes \mathcal{O}_X(-D)) + \deg \mathcal{L} = \deg \mathcal{E}.$$

Then note that

$$\deg \pi_*(\mathcal{O}(1) \otimes \mathcal{O}_X(-D)) = (C_0.\pi^*\pi_*(\mathcal{O}(1) \otimes \mathcal{O}_X(-D))) = (C_0.C_0 - D),$$

so we see that

$$(C_0.C_0 - D) + (C_0.D) = (C_0.C_0) = \deg \mathcal{E}.$$

□

Let $\pi : X = \mathbb{P}(\mathcal{E}) \to C$ be a ruled surface in normalized form. Let $C_0$ denote the section divisor so that $\mathcal{O}_X(C_0) \cong \mathcal{O}(1)$. Let $F$ be the fiber class. Our work thus far has shown that

$$\mathrm{NS}(X) \cong \mathbb{Z}\langle C_0 \rangle \oplus \mathbb{Z}\langle F \rangle,$$

where $(C_0.C_0) = \deg \mathcal{E}, (C_0.F) = 1$, and $(F.F) = 0$. Let's analyze the nef and Mori cones in various cases. Note we must have in this normalized form that $\mathcal{E}$ is semistable. Assume for the sake of contradiction it were not, then:

$$0 \to \mathcal{F} \to \mathcal{E} \to \mathcal{L} \to 0,$$

where $\frac{\deg \mathcal{E}}{2} > \deg \mathcal{L}$. Furthermore, $\deg \mathcal{F} = \deg \mathcal{E} - \deg \mathcal{L} > 0$. Then we would have a section $0 \to \mathcal{O}_C \to \mathcal{E} \otimes \mathcal{F}^\vee$, contradicting the normalized form of $\mathcal{E}$. Thus, we must have $\mathcal{E}$ is semistable.

**Example 25.11** (When $\deg \mathcal{E} = 0$)**.** *Suppose* $\deg \mathcal{E} = 0$*. Any* $\mathbb{R}$*-divisor* $D$ *is of the form*

$$D = aC_0 + bF.$$

*If* $D$ *is a nef* $\mathbb{R}$*-divisor, then at the bare minimum, we must have* $(D.C_0) = b \geq 0$ *and* $(D.F) = a \geq 0$*. Thus, the nef cone must be in the first quadrant of the* $F$*-*$C_0$ *plane. However, note that* $F \cong \mathbb{P}^1$ *and* $C_0 \cong C$ *(abusing notation here) are both irreducible, and* $(F.F) = (C_0.C_0) = 0$*. This implies that* $F$ *and* $C_0$ *are nef divisors. Thus, Nef*$(X)$ *is the entire first quadrant.*

*Now where is the boundary of the Mori cone?*

– *Since $\mathcal{E}$ is semistable. We claim that the Mori cone is simply the first quadrant. Let $C \subset X$ be an effective curve. It suffices to show that $C$ is in the first quadrant. Since* $\operatorname{Pic}(X) \cong \mathbb{Z}\langle C_0\rangle \oplus \pi^* \operatorname{Pic}(C)$, *we know that*

$$\mathcal{O}_X(C) \sim \mathcal{O}(n) \otimes \pi^* A,$$

*for some $n$ and line bundle $A$ on $C$. Note since $F$ is nef, we have that $(F.C) \geq 0$, which implies that $n \geq 0$. Now it remains to show that if we express $C$ numerically: $C \equiv nC_0 + bF$, then $b \geq 0$ as well. In particular, this means we want to show that* $\deg A \geq 0$.

*Note that*

$$H^0(X, \mathcal{O}(n) \otimes \pi^* A) \cong H^0(C, \operatorname{Sym}^n \mathcal{E} \otimes A) \neq 0.$$

*Note since $\mathcal{E}$ is semistable, so is* $\operatorname{Sym}^n \mathcal{E}$. *Then we have an injection of sheaves*

$$0 \to \mathcal{O}_C \to \operatorname{Sym}^n \mathcal{E} \otimes A,$$

*and tensoring, we get an injection*

$$0 \to A^\vee \to \operatorname{Sym}^n \mathcal{E}.$$

*Since* $\operatorname{Sym}^n \mathcal{E}$ *is semistable, this implies that we need $\rho(A^\vee) \leq \rho(\operatorname{Sym}^n \mathcal{E}) = 0 \implies$* $\deg(A^\vee) \leq 0 \implies \deg A \geq 0$. *This completes the proof.*

**Example 25.12** (When $\deg \mathcal{E} < 0$)**.** *In this case,*

$$NS_{\mathbb{R}}(X) \cong \mathbb{Z}\langle C_0\rangle \oplus \mathbb{Z}\langle F\rangle,$$

*where $(C_0.C_0) = \deg \mathcal{E} < 0$, $(C_0.F) = 1$, and $(F.F) = 0$. Let $D = aC_0 + bF$ be some $\mathbb{R}$-divisor. If $D$ is nef, then at the very least, we must have*

$$(D.C_0) = a \deg \mathcal{E} + b \geq 0, \text{ and } (D.F) = a \geq 0.$$

*Then in the $F$-$C_0$ plane, this implies that Nef$(X)$ is contained in the first quadrant, bounded above by the line through the origin with slope $\frac{1}{-\deg \mathcal{E}}$ and bounded below by the horizontal axis. Furthermore, $(C_0.C_0) < 0$, which implies that $\mathbb{R}_{\geq 0} \cdot C_0$ is an extremal ray for $\overline{NE}(X)$.*

**Example 25.13** (When $\deg \mathcal{E} > 0$)**.** *In this case,*

$$NS_{\mathbb{R}}(X) = \mathbb{Z}\langle C_0\rangle \oplus \mathbb{Z}\langle F\rangle,$$

*such that $(C_0.C_0) = \deg \mathcal{E} > 0, (C_0.F) = 1$, and $(F.F) = 0$.*

*Let $D = aF + bC_0$ be some $\mathbb{R}$-divisor. If $D$ is nef, then we must have at least that $(D.C_0) = b \deg \mathcal{E} + a \geq 0$, and $(D.F) = b \geq 0$.*

*Furthermore, note that $F$ and $C_0$ are both nef. In particular, $F$ must form a boundary of both the nef cone and the Mori cone.*

*Altogether, this implies that in the $F$-$C_0$ plane, the nef cone lies in the upper half plane, inclusive of the horizontal axis, and must contain the first quadrant, but must be on the right side of a line with slope $\frac{-1}{\deg \mathcal{E}}$.*

- *Now suppose $\mathcal{E}$ is semistable. What could the cone of effective curves look like? Suppose $C \subset X$ was an effective curve. Then*

$$C \equiv xF + yC_0.$$

*But note $C_0$ and $F$ are both nef. So $(C.F) = y \geq 0, (C.C_0) = x + y \deg \mathcal{E} \geq 0$, so the second boundary of the Mori cone $\overline{NE}(X)$ is bounded by the line $y = \frac{-x}{\deg \mathcal{E}}$.*

*Note if $C \sim nC_0 + \pi^* A$ where $A \in \operatorname{Pic}(C)$, then using the same semistability analysis as example 25.12 only shows that $\deg A^\vee \leq \rho(\operatorname{Sym}^n \mathcal{E})$, so $\deg A$ is bounded below by a negative number. Thus, in this case it's not clear to me that the semistability of $\mathcal{E}$ tells us anything more about the nef and Mori cone.*

## 26. Positivity: Mori's Cone Theorem and a conjecture about cones for blown up $\mathbb{P}^2$'s

Let $X$ be a smooth complex projective variety of dimension $n$. In other words, a compact complex manifold which carries an ample line bundle. Shigefumi Mori's *Cone theorem* tells us that when the canonical divisor $K_X$ is not nef, then (part of) the Mori cone $\overline{\mathrm{NE}}(X)$ admits a particularly nice description with respect to $K_X$ and smooth rational curves on $X$.

**Theorem 26.1** (Mori's Cone Theorem)**.** *Let $X$ be a smooth complex projective variety of dimension $n$. Suppose $K_X$ fails to be nef.*

(1) *There exists countably many smooth rational curves $C_i \subset X$ such that $0 \leq -(K_X.C_i) \leq n+1$ and*

$$\overline{NE}(X) = \overline{NE}(X) \cap (K_X)_{\geq 0} + \sum \mathbb{R}_{\geq 0} \cdot [C_i].$$

(2) *Let $H$ be an ample divisor and fix any $\epsilon > 0$. Then there exist only finitely many $C_i$, say $C_1, \cdots, C_{t_\epsilon}$, such $C_i \in (K_X + \epsilon H)_{\leq 0}$, so that*

$$\overline{NE}(X) = \overline{NE}(X) \cap (K_X + \epsilon H)_{\geq 0} + \sum_{i=1}^{t_\epsilon} \mathbb{R}_{\geq 0} \cdot [C_i].$$

In other words, if $K_X$ fails to be nef, then Mori's cone theorem does not tell us anything about the positive side of the hyperplane $K_X^\perp \subset \frac{Z_1(X)}{\sim}$, but on the negative side of the hyperplane, the Mori cone is generated by extremal rays defined by smooth rational curves, whose intersection number with $K_X$ are bounded by the dimension of $X$.

**Corollary 26.2.** *The Mori cone of smooth complex Fano varieties are polyhedral cones spanned by the classes of smooth rational curves.*

*To see this, note by definition of a Fano variety we have* $-K_X$ *is ample. Then* $(-K_X.C) > 0$ *for any effective curve* $C \subset X$. *Then* $\overline{NE}(X) \cap (K_X \geq 0) = 0$. *Thus, by Mori's Cone theorem, we have*

$$\overline{NE}(X) = \sum \mathbb{R}_{\geq 0} \cdot [C_i],$$

*where* $C_i$ *is a smooth rational curve, with* $0 < -(K_X.C_i) \leq \dim X + 1$.

Let us now illustrate more vividly the phenomena of Mori's Cone theorem, and end with a treacherously hard conjecture.

Suppose $X$ is the blow up of $\mathbb{P}^2$ at $r$ very general points, where $r \geq 10$. Note that for $k >> 0$, we have

$$(\pi^*\mathcal{L})^{\otimes k} \otimes \mathcal{O}_X(-E_1 - \cdots - E_r)$$

is an ample line bundle on $X$, where $E_1, \cdots, E_r$ denote the exceptional divisors obtained by the blowing up processes. Let $e_i := [E_i], \ell := [\pi^*\mathcal{L}] \in \mathrm{NS}(X)$ denote numerical classes. Then there exist sufficiently small $0 < \epsilon << 1$ such that

$$\ell - \epsilon \sum_{i=1}^{r} e_i$$

is an ample $\mathbb{R}$-divisor.

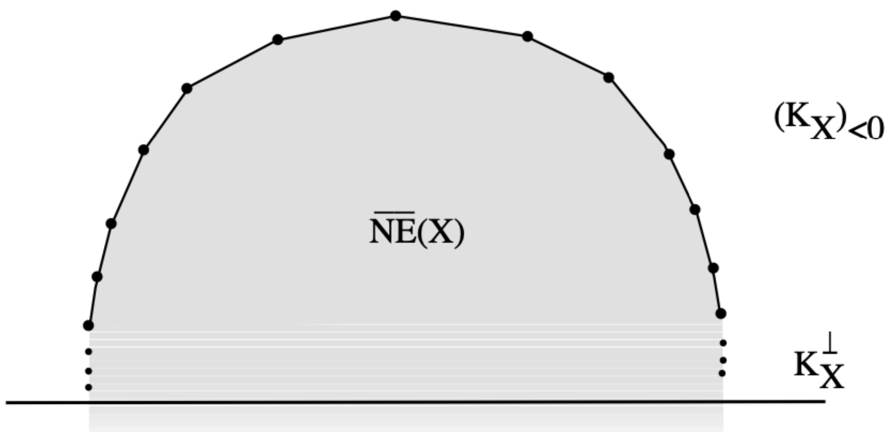
It is known that we can find a sequence of smooth rational curves $C_i \subset X$ such that

$$(C_i.C_i) = -1 \text{ and } (C_i.K_X) = -1,$$

but

$$\lim_{i \to \infty} (C_i.h) = \infty.$$

A good reference for this fact is in [Har, Chapter V, Exercise 4.15]. These $C_i$ define extremal rays for the Mori cone $\overline{\mathrm{NE}}(X)$. In fact, they generate $\overline{\mathrm{NE}}(X) \cap (K_X)_{<0}$. Furthermore, the fact that $(C_i.K_X) = -1$ for every $i$, but $\lim_{i\to\infty}(C_i.h) = \infty$ tell us that the rays $\mathbb{R}_{\geq 0} \cdot [C_i]$ cluster towards the hyperplane $(K_X)^{\perp}$. The picture (which I've robbed from Lazarsfeld's book) looks like so:




The fact that the extremal rays $\mathbb{R}_{\geq 0} \cdot [C_i]$ span $(K_X)_{<0} \cap \overline{\mathrm{NE}}(X)$ is a witness of the first part of Mori's cone theorem. The fact that these rays cluster towards $K_X^{\perp}$ is a witness of the second part of Mori's cone theorem.

**Conjecture 26.3.** *The positive half of the Mori cone, with respect to $K_X$,*

$$\overline{NE}(X) \cap (K_X)_{\geq 0}$$

*consists of classes of nonnegative self-intersection.*

This conjecture is related to Seshadri constants and other conjectures such as the Nagata conjecture. Allegedly Lazarsfeld is quoted as saying that "these are perfect conjectures to ruin a young person's career."

**Remark 26.4.** *Mori's cone theorem was proved using his "bend and break" method, which he developed to prove Hartshorne's conjecture that any smooth complex projective variety which ample tangent bundle is isomorphic to projective space, which earned him a Fields medal, among other achievements.*

**Remark 26.5.** *Mori's cone theorem can be generalized to certain mildly singular complex varieties, but one loses knowledge of whether the $C_i$ are rational. This was carried out by Kawamata and Shokurov, following the ideas of Reid.*

**Remark 26.6.** *Mori's cone theorem paved the way for the minimal model program. Mori showed that when $\dim X = 3$ and if $r$ is an extremal ray in $\overline{NE}(X) \cap (K + \epsilon H)_{\leq 0}$, then there is a mapping $X \to \overline{X}$ which contracts every curve whose class lies in those extremal rays.*

*This was extended to smooth projective varieties of any dimension by Kawamata and Shokurov following ideas of Reid. This contraction theorem opened the door to possibly* *constructing minimal models for varieties of arbitrary dimension. Difficulty arises, however, when after contraction the variety has singularities – this has spurred many important developments since the 80s, 90s and 2000s.*

## 27. Cones of K3 surfaces: testing ampleness against rational curves

Let $X$ be a complex algebraic K3 surface. Recall that we have the cones

$$\mathrm{Amp(X)} \subset \mathrm{Nef}(X).$$

There is another cone which we should mention, namely the positive cone $\mathcal{C}_X \subset \mathrm{NS}_{\mathbb{R}}(X)$. It is defined in the following way: consider the subset $\{\alpha \in \mathrm{NS}_{\mathbb{R}}(X) | (\alpha.\alpha) > 0\}$. Note this subset is a cone. We define $\mathcal{C}_X$ to be that connected component which also contains an ample divisor.

**Proposition 27.1.** *Let $X$ be a complex algebraic K3 surface. The positive cone $\mathcal{C}_X$ is convex. Furthermore,*

$$Amp(X) \subset Nef(X) \subset \overline{\mathcal{C}_X}.$$

*Proof.* First we verify that $\mathcal{C}_X$ is convex. Suppose $x \in \mathcal{C}_X$ and $y \in \mathcal{C}_X$. Then $(x.x) > 0$ and $(y.y) > 0$. By the Hodge index theorem 5.4, since $(y.y) > 0$, we must have $(x.y) \neq 0$. First, we claim that $(x.y) > 0$. Why must this be the case? Because the $(x.-)$ defines a linear functional on $\mathrm{NS}_{\mathbb{R}}(X)$, and in particular, if $(x.y) < 0$, then this would split $\mathcal{C}_X$ into two connected components – but this can't happen since by definition $\mathcal{C}_X$ is connected. Thus, we must have $(x.y) > 0$.

Now note by the Hodge index theorem 5.4 we can write $y = tx+D$. Note $(x.y) = t(x.x) > 0 \implies t > 0$. Then

$$(x + y.x + y) = (1 + t)^2(x.x) + (D.D) > t^2(x.x) + (D.D) = (y.y) > 0,$$

as desired.

Next, we verify the chain of inclusions. We already know the first inclusion by our work in the previous positivity sections. It remains to verify that

$$\mathrm{Nef}(X) \subset \overline{\mathcal{C}_X}.$$

But note that if $D \in \mathrm{Nef}(X)$, and $H$ is an ample divisor, then $D+\epsilon H$ is ample for all $\epsilon > 0$, so $(D + \epsilon H.D + \epsilon H) > 0$ for all $\epsilon > 0$. Then $D + \epsilon H \in \mathcal{C}_X$, so letting $\epsilon \to 0$, we see that $D \in \overline{\mathcal{C}_X}$, as desired. □

Note that if $D$ is a $\mathbb{R}$-divisor such that $(D.D) > 0$, then to show that $D \in \mathcal{C}_X$, it suffices to show that $(D.H) > 0$ for just one ample divisor $H$.

We end this preliminary section with a powerful ampleness criterion that is special to projective K3 surfaces. Recall that, in general, if one has a smooth projective $\mathbb{C}$-variety

$Y$, then one has the Nakai-Moishezon criterion: $D$ is ample if and only if

$$(D^{\dim V}.V) > 0$$

for all irreducible positive-dimensional $V \subset Y$. If $Y$ is a surface then we have another interesting criterion, namely that $D$ is ample if and only if

$$\overline{\mathrm{NE}}(Y) \setminus \{0\} \subset D_{>0},$$

i.e. the Mori cone lies on the positive side of $D^{\perp}$. For projective K3 surfaces, we can test ampleness against rational curves.

**Proposition 27.2.** *Let $X$ be a projective K3 surface. Then*

$$Amp(X) = \{\alpha \in \mathcal{C}_X \mid (\alpha.C) > 0 \textit{ for all smooth rational curves } C \subset X\}.$$

*Proof.* We have

$$\mathrm{Amp}(X) \subseteq \{\alpha \in \mathcal{C}_X \mid (\alpha.C) > 0 \text{ for all } \mathbb{P}^1 \cong C \subset X\}$$

by the Nakai-Moishezon criterion. Now suppose $\alpha \in \mathcal{C}_X$ and $(\alpha.C) > 0$ for all smooth rational curves $C \subset X$. Suppose $C$ is an integral curve on $X$ which is not rational. Then $p_a(C) > 0$, so $(C.C) = 2p_a(C) - 2 \geq 0$, where the equation comes from the fact that $X$ is a K3 surface.

We claim that if $C$ is an integral curve and $(C.C) \geq 0$, then $(D.C) > 0$ for any $D \in \mathcal{C}_X$. Since $(C.C) \not< 0$, we have $(D.C) \neq 0$ by Hodge index theorem. So either $(D.C) > 0$ or $(D.C) < 0$. But note $(H.C) > 0$ for ample $H$. Note $(-.C)$ defines a linear functional on $\mathrm{NS}_{\mathbb{R}}(X)$, so by connectedness of $\mathcal{C}_X$, we must have $(D.C) > 0$ for all $D \in \mathcal{C}_X$.

Altogether, this implies that $(\alpha.C) > 0$ for $p_a(C) > 0$. Since we also have $(\alpha.C)$ for $p_a(C) = 0$, (note $p_a(C) \geq g(C) \geq 0$ and equality forces $p_a(C) = g(C)$ and smoothness of $C$), this means $(\alpha.C) > 0$ for all irreducible 1 dimensional closed subschemes of $X$. Since $(\alpha.\alpha) > 0$ as well, we have by the Nakai-Moishezon criterion that $\alpha$ is an ample $\mathbb{R}$-divisor. □

Note if an algebraic K3 surface $X$ has no rational curves, then Proposition 27.2 implies that to check ampleness, one must simply check that $(D.D) > 0$ and $(D.H) > 0$ for some ample $H$.

In light of testing ampleness against Mori cones for surfaces, Proposition 27.2 hints at a result of Sandor Kovacs: if an algebraic K3 surface $X$ has a rational curve, then its Mori cone $\overline{\mathrm{NE}}(X)$ is a polyhedral cone spanned by extremal rays given by rational curves.

## 28. Cones of K3 surfaces: nef cone as fundamental domain of big cone

Let $X$ be a a complex projective K3 surface. In the last section, we saw that

$$\mathrm{Amp}(X) \subset \mathrm{Nef}(X) \subset \overline{\mathcal{C}_X}.$$

In this section, we will clarify the relationship between the ample/nef cones and the big cone $\overline{\mathcal{C}_X}$ (sometimes also called the positive cone). In particular, we will show the following.

**Theorem 28.1.** *The nef cone of a complex projective K3 surface is a fundamental domain for the action of the Weyl group $W \subset O^+(\mathcal{C}_X)$ on the big cone.*

In other words, there's a group action on the big cone $\mathcal{C}_X$ which moves around the nef cone, in a way similar to how the translation group moves around a fundamental parallelogram in the plane.

Let us first recall the linear algebra of quadratic forms. Let $V$ be a $k$-vector space, where $\mathrm{char}(k) \neq 2$. If $B : V \times V \to k$ is a symmetric bilinear form, then

$$Q : V \to k, \text{ where } Q(v) = B(v, v),$$

is the associated quadratic form. Similarly, given a quadratic form $Q : V \to k$, we can obtain the associated symmetric blinear form

$$B(v_1, v_2) = \frac{1}{2}[Q(v_1 + v_2) - Q(v_1) - Q(v_2)].$$

We say the symmetric bilinear form/quadratic form is non-degenerate if the map

$$V \to V^*, \text{ where } [v \mapsto B(v, -)]$$

is injective. Now let $V$ be a finite-dimensional $k$-vector space with basis $B = (v_1, \cdots, v_n)$. Then the symmetric matrix

$$M(V, B)_{ij} = B(v_i, v_j)$$

is called the matrix of the form.

**Theorem 28.2** (Sylvester)**.** *For every quadratic form over $\mathbb{R}$, there exists a canonical basis with respect to which the matrix form is 0 everywhere, except there are possibly some 1s, -1s, and 0s on the diagonals.*

Given a quadratic form over $\mathbb{R}$, the number of +1s and -1s is called its signature.

Now, fix a lattice $\Lambda$ with signature $(1, n)$ and define $V := \Lambda \otimes_{\mathbb{Z}} \mathbb{R}$. We will now develop a theory of walls and chambers on the big cone/positive cone of $V$. Keep in mind that the $\Lambda$ we are concerned with is $\mathrm{NS}(X)$, equipped with the quadratic form induced by the usual intersection pairing.

First, note that since the the quadratic form on $V$ has signature (1,n), abstractly our data is isomorphic to a $(n + 1)$ dimensional $\mathbb{R}$-vector space $V$ with quadratic form given by $x_0^2 - x_1^2 - \cdots - x_n^2$. Consider the cone $\{x \in V \mid (x.x) > 0\}$. We distinguish one of the components of this cone, and call it the positive cone $\mathcal{C}$. When $V = \mathrm{NS}_{\mathbb{R}}(X)$, the positive cone $\mathcal{C}_X$ is the connected component containing an ample class. Note that two elements

$x, y$, such that $(x.x)$ and $(y.y) > 0$, are in the same connected component only if $(x.y) > 0$, by connectedness of the connected component.

Let $O(1,n)$ denote the group of orthogonal linear isomorphisms of $V$. Thus,

$$O(1,n) = \{\phi \in \mathrm{GL}(V) \mid (x.y) = (\phi(x).\phi(y)), \forall x, y \in V\}.$$

Note $O(1,n)$ preserves the cone $\{x \in V \mid (x.x) > 0\}$, but some orthogonal transformation may interchange the two connected components. We let $O^+(1,n)$ denote the subgroup which maps $\mathcal{C}$ to itself. Note $O^+(1,n)$ is a subgroup of $O(1,n)$ of index two. Next, let

$$\mathcal{C}(1) := \{x \in \mathcal{C} \mid (x.x) = 1\}.$$

Note $\mathcal{C}$ is abstractly isomorphic to

$$\mathcal{C}(1) \times \mathbb{R}_{>0} \cong \{\vec{x} \in \mathbb{R}^{n+1} \mid x_0^2 - x_1^2 - \cdots - x_n^2 = 1, \text{ and } x_0 > 0\} \times \mathbb{R}_{>0}.$$

Since $O^+(1,n)$ acts transitively on $\mathcal{C}(1)$, and the stabilizer of a point $x \in \mathcal{C}(1)$ is isomorphic to the orthogonal group of the negative define space $x^\perp$, we have $\mathcal{C}(1) \cong \mathcal{O}^+(1,n)/\mathcal{O}(n)$.

**Lemma 28.3.** *Any discrete subgroup $H \subset \mathcal{O}^+(1,n)$ acts properly discontinuously on $\mathcal{C}(1)$. In other words, for every $p \in \mathcal{C}(1)$, there exists a neighborhood $U$ such that $U \cap gU = \emptyset$, for every $g \in H$ except those which lie in the stabilizer of $p$.*

Define $O^+(\Lambda) = O^+(1,n) \cap O(\Lambda)$. So these are the orthogonal automorphisms of $V$ which both preserve the positive cone $\mathcal{C}$ and come from a $\mathbb{R}$-linear extension of an automorphism of the lattice preserving the quadratic form.

**Proposition 28.4.** *Let $I \subset O^+(\Lambda)$ be any subset. Then*

$$\bigcup_{\phi \in I} Fix(\phi)$$

*is a closed subset in $\mathcal{C}$, where $Fix(\phi)$ is the locus of points in $\mathcal{C}$ that remain fixed under $\phi$.*

*Proof.* Note it suffices to prove that

$$\bigcup_{\phi \in I} [\mathrm{Fix}(\phi) \cap \mathcal{C}(1)]$$

is a closed subset of $\mathcal{C}(1)$. Let $p \in \mathcal{C}(1) \setminus (\bigcup_{\phi \in I} [\mathrm{Fix}(\phi) \cap \mathcal{C}(1)])$. Since $O^+(\Lambda)$ is a discrete subgroup of $O^+(1,n)$, it acts properly discontinuously on $\mathcal{C}(1)$. Then there exist a neighborhood $U$ of $p$ such that for every $\phi \in I$, we have $U \cap \phi(U) = \emptyset$, except for those $\phi$ which stabilize $p$. But none of the $\phi$ stabilize $p$. And thus we see that $U$ is properly contained in the complement, as desired. □

Now define

$$\Delta := \{x \in \Lambda \mid (x.x) = -2\}.$$

Each $\delta \in \Delta$ defines a reflection automorphism

$$s_\delta : \Lambda \to \Lambda, \text{ where } x \mapsto x + (x.\delta)\delta,$$

which can be extended $\mathbb{R}$-linearly to be defined on $V$. Note that $s_\delta \in O^+(\Lambda)$, since

- for any $x \in V$, we have
$$(s_\delta(x).s_\delta(x)) = (x + (x.\delta)\delta.x + (x.\delta)\delta) = (x.x) + 2(x.\delta)^2 + (x.\delta)^2(\delta.\delta) = (x.x).$$
- and for $x \in \mathcal{C}$,
$$(x.s_\delta(x)) = (x.x + (x.\delta)\delta) = (x.x) + (x.\delta)^2 > 0,$$
so $s_\delta(x) \in \mathcal{C}$.

Note each $s_\delta$ is a reflection so that $s_\delta(\delta) = -\delta$, and is identity on $\delta^\perp_{\mathbb{R}}$. A useful fact about these reflections is that for any $g \in \mathcal{O}(V)$, we have
$$g \circ s_\delta = s_{g(\delta)} \circ g.$$
Indeed, given any $x \in V$, we see that
$$g \circ s_\delta(x) = g(x + (x.\delta)\delta) = g(x) + (x.\delta)g(\delta) = (g(x) + (g(x).g(\delta))g(\delta) = s_{g(\delta)}(x) \circ g.$$

**Definition 28.5.** *We define the Weyl group*
$$W = \langle s_\delta \mid \delta \in \Delta\rangle \subset O^+(\Lambda).$$
*to be the group generated by reflections by roots. We define the walls of the positive cone $\mathcal{C}$ to be*
$$\bigcup_{\delta\in\Delta}[\delta^\perp_{\mathbb{R}} \cap \mathcal{C}].$$

Each wall $\delta^\perp_{\mathbb{R}} \cap \mathcal{C}$ is exactly the points of the positive cone which remain fixed under $s_\delta$. Furthermore, if $\delta, \delta' \in \Delta$, then note $s_{\delta'}(\delta^\perp_{\mathbb{R}}) = (s_{\delta'}(\delta))^\perp_{\mathbb{R}}$, and $s_{\delta'}(\delta) \in \Delta$. Thus, altogether the walls of $\mathcal{C}$ are invariant under the action of the Weyl group.

**Proposition 28.6.** *For every point $p \in \mathcal{C}$, there is a neighborhood $U$ of $p$ such that only finitely many walls of $\mathcal{C}$ intersect $U$.*

*Proof.* Let $p \in \mathcal{C}$. Then take $\frac{p}{\sqrt{(p.p)}}$, and complete it to an orthogonal basis of $\mathbb{R}^{n+1}$, so that we can write every $x \in V$ as $x = x_0 \cdot \frac{p}{\sqrt{(p.p)}} + x'$, where $x' \in (\frac{p}{\sqrt{(p.p)}})^\perp_{\mathbb{R}}$. In particular, we complete the orthogonal basis so that our discriminant form looks like $(x.x) = x_0^2 - x_1^2 - \cdots - x_n^2$. Define the bilinear form $\langle , \rangle := -( , )$ on $(\frac{p}{\sqrt{(p.p)}})^\perp_{\mathbb{R}}$, and $||\cdot||$ the norm obtained from $\langle , \rangle$.

Now suppose we have $x \in \delta^\perp_{\mathbb{R}}$, where $\delta \in \Delta$. Then
$$(\delta.\delta) = -2 = \delta_0^2 - ||\delta'||^2 \implies \delta_0^2 + 2 = ||\delta'||^2 \implies |\delta_0| + \sqrt{2} \geq ||\delta'||.$$
Furthermore,
$$(\delta.x) = 0 \implies \delta_0 x_0 = \langle \delta', x'\rangle \implies |\delta_0| \cdot |x_0| \leq ||\delta'|| \cdot ||x'||,$$

where the last implication holds by Cauchy-Schwarz. Combined with our bound on $||\delta'||$, this means

$$|\delta_0| \cdot |x_0| \le (|\delta_0| + \sqrt{2}) \cdot ||x'|| \implies |\delta_0| \cdot (|x_0| - ||x'||) \le \sqrt{2} \cdot ||x'||.$$

Now if $x$ tends towards $p$, then $||x'|| \to 0$ and $|x_0| \to \sqrt{(p.p)}$, so $|\delta_0| \to 0$. But this means that as $x$ approaches $p$, we must have $||\delta'|| \le \sqrt{2}$.

All of this is to say.. if we pick some arbitrary point $p \in \mathcal{C}$ which lies on the walls, then there is a certain numerical bound on which roots could be cutting out those walls. In particular, since roots are integral lattice points, the numerical bound implies there are only finitely many possible roots that could cut out the walls which $p$ lies on. Thus, the walls are locally finite. This implies the claim. □

We are now developing a picture of the walls and chambers of the positive cone $\mathcal{C}$.

**Definition 28.7.** *Let $\Lambda$ be a lattice of signature $(1, n)$, and let $V := \Lambda \otimes_{\mathbb{Z}} \mathbb{R}$. Let $\mathcal{C}$ be the positive cone/big cone. Let $\Delta = \{\delta \in \Lambda \mid (\delta.\delta) = -2\}$. Then we define a chamber of $\mathcal{C}$ to be a connected component of*

$$\mathcal{C} \setminus \bigcup_{\delta \in \Delta} (\delta_{\mathbb{R}}^{\perp} \cap \mathcal{C}).$$

*Since locally there are finitely many walls in $\mathcal{C}$, for any chamber $\mathcal{C}_i \subset \mathcal{C}$, we have that the cone*

$$\overline{\mathcal{C}_i}$$

*is locally polyhedral in the interior of $\mathcal{C}$.*

The Weyl group $W = \{s_\delta \mid \delta \in \Delta\}$ acts on $\mathcal{C}$, and preserves the walls, since $s_{\delta'}(\delta_{\mathbb{R}}^{\perp}) = (s_{\delta'}(\delta))_{\mathbb{R}}^{\perp}$, and $s_{\delta'}(\delta) \in W$. Thus, the action of the Weyl group on $\mathcal{C}$ must permute the chambers $\mathcal{C}_i$! Let's say a bit more about this action.

First, note any chamber $C_i$ is associated with a set of signs given by

$$(C_i.\delta), \forall \delta \in \Delta,$$

and these signs uniquely determine the chamber $_i$. A positive sign $(\mathcal{C}_i.\delta) > 0$ means $\mathcal{C}_i$ lies on the side of the wall $\delta_{\mathbb{R}}^{\perp}$ that $-\delta$ lies on. Then $\mathcal{C}_i$ is uniquely determined by a valid choice of positive signs $(\mathcal{C}_i.\delta)$ among some of the $\delta \in \Delta$. Note there must be a valid choice, as $(\mathcal{C}_i.\delta)$ and $(\mathcal{C}_i.-\delta)$ cannot be simultaneously positive. Then with respect to the chamber $\mathcal{C}_i$, define

$$\Delta_{+,i} = \{\delta \in \Delta \mid (\mathcal{C}_i.\delta) > 0\},$$

and define

$$\Delta_{+,i} \supseteq \Delta_{\mathcal{C}_i} := \{\delta \in \Delta_{+,i} \mid \delta_{\mathbb{R}}^{\perp} \cap \overline{\mathcal{C}_i} \text{ has codimension 1 }\}.$$

The walls defined by elements of $\Delta_{\mathcal{C}_i}$ should be thought of as those walls which actually cut out $\overline{\mathcal{C}_i}$ as a locally polyhedral cone. Here is a somewhat surprising fact.

**Proposition 28.8.** *Suppose $\mathcal{C}_0 \subset \mathcal{C}$ is a chamber. Let*

$$W_{\mathcal{C}_0} := \langle s_\delta \mid \delta \in \Delta_{\mathcal{C}_0} \rangle$$

*denote the subgroup of the Weyl group generated by reflections by elements of $\Delta_{\mathcal{C}_0}$. Then $W_{\mathcal{C}_0}$ acts transitively on the chambers of $\mathcal{C}$.*

*Proof.* Let $\mathcal{C}_i$ be some chamber. Starting from a point in the chamber $\mathcal{C}_0$, we can draw a path $\gamma : [0,1] \to \mathcal{C}$ from the chamber $\mathcal{C}_0$ to $\mathcal{C}_1$, such that the path intersects only one wall at a time. By compactness of $[0,1]$, it will cross finitely many walls, given by:

$$\delta_0^\perp \mathbb{R}, \cdots, \delta_n^\perp \mathbb{R},$$

where $\delta_i = \Delta_{\mathcal{C}_i} \cap (-\Delta_{\mathcal{C}_{i+1}})$. This tells us that we can go from chamber $\mathcal{C}_0$ to chamber $\mathcal{C}_n$ by the series of reflections

$$s_{\delta_{n-1}} \circ \cdots \circ s_{\delta_0}.$$

We claim that this composition is actually an element of $W_{\mathcal{C}_0}$. We will do the case where $n = 2$. The case of higher $n$ is analogous.

First, we claim that $s_{\delta_0}(\Delta_{\mathcal{C}_0}) = \Delta_{\mathcal{C}_1}$. Indeed, consider $\delta \in \Delta_{\mathcal{C}_1}$. Then we claim $s_{\delta_0}(\delta) \in \Delta_{\mathcal{C}_0}$. First, we see that

$$(C_0.s_{\delta_0}(\delta)) = (s_{\delta_0}(C_1).s_{\delta_0}(\delta)) > 0.$$

Next, we see that $\overline{\mathcal{C}_1} \cap \delta_{\mathbb{R}}^\perp$ is codimension 1 in $\overline{\mathcal{C}_1}$. Then

$$\overline{\mathcal{C}_0} \cap (s_{\delta_0}(\delta))_{\mathbb{R}}^\perp = s_{\delta_0}(\overline{\mathcal{C}_1} \cap \delta_{\mathbb{R}}^\perp),$$

is also codimension 1. Thus, $s_{\delta_0}(\delta) \in \Delta_{\mathcal{C}_0}$. Then the composition

$$s_{\delta_1} \circ s_{\delta_0} = s_{s_{\delta_0}(\delta)} \circ s_{\delta_0} = s_{\delta_0} \circ s_\delta \in W_{\mathcal{C}_0},$$

where $\delta \in \Delta_{\mathcal{C}_0}$, as desired. □

**Proposition 28.9.** *We have $W_{\mathcal{C}_0} = W$, for any chamber $\mathcal{C}_0$ of $\mathcal{C}$.*

*Proof.* Since $W$ is generated by $s_\delta$ for all $\delta \in \Delta$, it suffices to show that $s_\delta \in W_{\mathcal{C}_0}$ for any $\delta \in \Delta$. Since $\delta$ is a root, we have $\delta \in \Delta_{\mathcal{C}_n}$ for some chamber $\mathcal{C}_n$. In other words, any root cuts out a codimension 1 subspace of $\overline{\mathcal{C}_n}$ for some chamber $\mathcal{C}_n$.

Note we can go from chamber $\mathcal{C}_0$ to $\mathcal{C}_n$ by a sequence of reflections

$$s_{\delta_{n-1}} \circ \cdots \circ s_{\delta_0} \in W_{\mathcal{C}_0},$$

where $\delta_i \in \Delta_{\mathcal{C}_i} \cap (-\Delta_{\mathcal{C}_{i+1}})$. But we also showed that $s_{\delta_i}(\Delta_{\mathcal{C}_i}) = \Delta_{\mathcal{C}_{i+1}}$. Thus,

$$s_{\delta_{n-1}} \circ \cdots \circ s_{\delta_0}(\Delta_{\mathcal{C}_0}) = \Delta_{\mathcal{C}_n}.$$

Then

$$\delta = s_{\delta_{n-1}} \circ \cdots \circ s_{\delta_0}(\delta') =: g(\delta')$$

for some $\delta' \in \Delta_{\mathcal{C}_0}$, so $s_\delta = s_{g(\delta')}$. Note $g \circ s_{\delta'} = s_{g(\delta')} \circ g \implies$

$$s_\delta = s_{g(\delta')} = g \circ s_{\delta'} \circ g^{-1} \in W_{\mathcal{C}_0}.$$

□

Finally, we claim that the Weyl group acts not only transitively, but also freely.

**Proposition 28.10.** *The action of the Weyl group on the chambers of $\mathcal{C}$ is transitive and free.*

*Proof.* It remains to verify freeness. Assume for the sake of contradiction that $g \in W$ such that $g$ does not permute the chambers at all, but $g$ is not the identity map on $V$. Now, since $W_{\mathcal{C}_0} = W$, we can write

$$g = s_{\delta_n} \circ \cdots \circ s_{\delta_1} =$$

where $n$ is minimal and $\delta_i \in \Delta_{\mathcal{C}_0}$. If we associate to these reflections a path which crosses one wall at a time, then the walls crossed are

$$(s_{\delta_n} \circ \cdots \circ s_{\delta_j})(\delta^\perp_{j-1\mathbb{R}}).$$

Each wall is crossed twice, so we must have

$$\delta^\perp_n = (s_{\delta_n} \circ \cdots \circ s_{\delta_j})(\delta^\perp_{j-1\mathbb{R}}) = g'(\delta^\perp_{j-1\mathbb{R}}).$$

This implies $g' \circ s_{\delta_{j-1}} = s_{g'(\delta_{j-1})} \circ g' = s_{\delta_n} \circ g' = s_{\delta_{n-1}} \circ \cdots \circ s_{\delta_j}$, contradicting minimality of $g$. □

Finally, let us apply the theory that we have developed to study the relationship between the ample and big cone of an algebraic K3 surfaces.

Let $X$ be a complex projective K3 surface. Note in the previous section we showed that

$$\mathrm{Amp}(X) = \{\alpha \in \mathcal{C}_X \mid (\alpha.C) > 0 \text{ for all } \mathbb{P}^1 \cong C \subset X\}.$$

Furthermore, the ample cone is an open convex cone and thus connected. Thus, the ample cone itself must be a chamber. Furthermore, every rational curve $C \cong \mathbb{P}^1$ is an element of $\Delta_{\mathrm{Amp}(X)}$, i.e. every curve cuts out a codimension 1 wall in $\mathrm{Nef}(X) = \overline{\mathrm{Amp}(X)}$. This is because for any rational curve $C$, the element $x := H + \frac{1}{2}(H.C)C$, where $H$ is some ample divisor, is such that $(x.C) = 0$ but $(x.C') > 0$ for all other rational $C'$.

So $\mathrm{Amp}(X)$ is a chamber of $\mathcal{C}_X$, and $\Delta_{\mathrm{Amp}(X)} = \{$ smooth rational curves $C \subset X\}$.

**Corollary 28.11.** *The Weyl group of $NS_{\mathbb{R}}(X)$ is generated by reflections by the numerical class of smooth rational curves.*

Furthermore, the Weyl group permutes the ample cone transitively and freely to all other chambers in the big cone $\mathcal{C}_X$.

**Proposition 28.12.** *Let $W$ be the group generated by reflections by smooth rational curves. Then the nef cone*

$$Nef(X) = \overline{Amp(X)} \subset \overline{\mathcal{C}_X} \subset NS_{\mathbb{R}}(X)$$

*is a locally polyhedral closed convex cone, which serves as a fundamental domain of $\overline{\mathcal{C}_X}$ with respect to the action of $W$.*

**Corollary 28.13.** *Suppose $\alpha \in NS_{\mathbb{R}}(X)$ is a class such that*

$$(\alpha.\alpha) > 0 \text{ and } (\alpha.\delta) \neq 0 \text{ for all (-2) classes } \delta.$$

*Then there exist smooth rational curves $C_1, \cdots, C_n$ such that*

$$\pm s_{[C_n]} \circ \cdots \circ s_{[C_1]}(\alpha)$$

*is ample. If we drop the assumption that $(\alpha.\delta) \neq 0$ for all (-2) classes, then our resultant is guaranteed to be at least nef.*

## 29. Cones of K3 surfaces: Nef/Mori cones of K3 surfaces

Let $X$ be a complex projective K3 surface. A result of Sandor Kovacs lists all the possibilities for what the Nef and Mori cones of algebraic K3 surfaces can look like and actually occur. In this section, we'll sketch the main ideas. More details can be found in [Huy, Section 8.3], which also provides references for even more details than those details.

First, we have the following relationship between the Mori cone, big cone, and rational curves.

**Proposition 29.1.** *Let $X$ be a complex algebraic K3 surface. Then*

(1) $NE(X) \subset \overline{\mathcal{C}_X} + \sum \mathbb{R}_{\geq 0} \cdot [C_i]$
(2) $\overline{NE(X)} = \overline{\mathcal{C}_X} + \overline{\sum \mathbb{R}_{\geq 0} \cdot [C_i]}$,

*where the $C_i \subset X$ are all the smooth rational curves on $X$.*

*Proof.* (1) This follows directly from the fact that $(C.C) = 2p_a(C) - 2$.

(2) This is a consequence of the Nef cone being a fundamnetal domain for the action of the Weyl group on $\mathcal{C}_X$. We have by the previous item that

$$\overline{\mathrm{NE}(X)} \subseteq \overline{\mathcal{C}_X} + \overline{\sum \mathbb{R}_{\geq 0} \cdot [C_i]}.$$

So suppose that $\alpha \in \overline{\mathcal{C}_X} + \overline{\sum \mathbb{R}_{\geq 0} \cdot [C_i]}$. Then to show that $\alpha \in \overline{\mathrm{NE}(X)}$, it suffices to show that

$$\overline{\mathrm{NE}(X)} \supseteq \overline{\mathcal{C}_X} \supseteq \mathcal{C}_X.$$

To show $\mathcal{C}_X \subset \overline{\mathrm{NE}(X)}$, note it suffices to show that the big cone minus the walls is contained by the Mori cone – in other words:

$$\mathcal{C}_X \setminus (\bigcup_{C_i \subset X} [C_i]^{\perp}_{\mathbb{R}} \cap \mathcal{C}_X) \subseteq \overline{\mathrm{NE}}(X),$$

for all smooth rational curves $C_i \subset X$. But note that if $x \in \mathcal{C}_X \setminus (\bigcup_{C_i \subset X} [C_i]^{\perp}_{\mathbb{R}} \cap \mathcal{C}_X)$, then there exist rational curves $C_1, \cdots, C_n$ such that

$$s_{[C_n]} \circ \cdots \circ s_{[C_1]}(H) = x,$$

for some $H \in \mathrm{Amp}(X)$. Note that

$$s_{[C_1]}(H) = H + (H.C_1)C_1,$$

so $s_{[C_1]}(H) \in \mathrm{NE}(X)$. Furthermore,

$$s_{[C_2]} \circ s_{[C_1]}(H) = H + (H.C_1)C_1 + (H.C_2)C_2 + (H.C_1)(C_1.C_2)C_2 \in \mathrm{NE}(X).$$

Considering

$$s_{[C_3]} \circ s_{[C_2]} \circ s_{[C_1]}(H) = H + (H.C_1)C_1 + (H.C_2)C_2 + (H.C_1)(C_1.C_2)C_2$$

$$+(H.C_3)C_3 + (H.C_1)(C_1.C_3)C_3 + (H.C_2)(C_2.C_3)C_3 + (H.C_1)(C_1.C_2)(C_2.C_3)C_3,$$

we see if $C_1, C_2, C_3$ are all distinct then $s_{[C_3]} \circ s_{[C_2]} \circ s_{[C_1]}(H) \in \mathrm{NE}(X)$. We may worry however whether $C_1$ and $C_3$ being the same whether that would cause problems, but letting $C_1 = C_3$ we see that we get cancellation so that we still have that the element is in the cone of curves.

Thus in general, we have that

$$s_{[C_n]} \circ \cdots \circ s_{[C_1]}(H) = x \in \mathrm{NE}(X),$$

as desired.

□

## 30. Automorphisms of K3 anticanonical divisors of Fano 3 Folds

Let $F$ be a complex Fano 3-fold: a compact complex manifold of dimension 3 with ample $-K_F$. Shokurov [Sho80] proved that the general divisor $X \in |-K_F|$ is smooth, and thus a smooth algebraic K3 surface.

**Question 30.1.** *What do the automorphisms look like of those K3 surfaces which arise as anticanonical divisors of Fano 3-folds?*

Before one approaches Question 30.1, one might ask: which K3 surfaces atually arise as an antincaonical divisor of a Fano 3-fold? First, note such a K3 surface would need to be algebraic. The full answer is provided by Beauville in [Bea02]:

**Theorem 30.2.** *Let $X$ be a complex algebraic K3 surface. Then, in general, $X$ arises as an anticanonical divisor of a Fano 3-Fold if and only if there is an isometry of lattices*

$$(\mathrm{Pic}(X), H) \cong (\mathrm{Pic}(V), -K_V)$$

*sending ample $H$ to $-K_V$, for some Fano 3-fold $V$.*[23] *Note the pairing on* $\mathrm{Pic}(V)$ *is given by* $\langle D, D' \rangle = (D.D'. - K_V)$.

So Theorem 30.2 characterizes the general algebraic K3 surface which arise as a antincanonical divisor of a Fano 3 fold based on its Picard lattice. Note this doesn't mean it holds all the time. For if it did, then we'd have to have $1 \leq \rho(X) \leq 10$ for all K3 surfaces $X$ arising as the anti-canonical divisor of a Fano 3-fold. But consider the fact that the Fermat quartic has Picard rank 20 and it is an anticanonical divisor of $\mathbb{P}^3$.

Note that there is a full classification of Fano 3-folds [Bel25], and their Picard rank is between 1 and 10, inclusive.

**Question 30.3.** *Given a Fano 3-fold $F$, what is the automorphism group* $\mathrm{Aut}(X)$ *of general $X \in |-K_F|$?*

Could it be that for general $X \in |-K_F|$, we have that $\mathrm{Aut}(X)$ is always finite? Always infinite? $\mathrm{Aut}(X)$ can be studied from $\mathrm{Pic}(X)$, so let's first worry about the Picard lattice. By the Lefschetz hyperplane theorem, we know that

$$\rho(F) \leq \rho(X).$$

But can we have equality?

**Theorem 30.4.** *Let $F$ be a Fano 3-fold with very ample $-K_F$. Then for very general $X \in |-K_F|$, we have*

$$\mathrm{Pic}(F) = \mathrm{Pic}(X).$$

*Proof.* This question is very much in the spirit of the well-known Noether thoerem, which states that on a general surface of degree greater than 3 in $\mathbb{P}^3$, any algebraic curve on that surface is cut by another surface in $\mathbb{P}^3$. This theorem was generalized by Lefschetz, using his theory of vanishing cycles, to 3-dimensional complete intersections.

Moishezon uses the ideas of Noether and Lefschetz to prove the following: let $V$ be a nonsingular irreducible algebraic variety, $\dim V = 3$, embedded in $\mathbb{P}^N$. Let $E \subset V$ be a very general hyperplane section. Then in order for every linear equivalence class of divisors on $E$ to be cut out by some linear equivalence class of divisors on $V$, i.e. $\mathrm{Pic}(V)$ surjects on to $\mathrm{Pic}(E)$, it is necessary and sufficient that one of the following two conditions is satisfied: $b_2(V) = b_2(E)$ or $h^{2,0}(E) > h^{2,0}(V)$.

Indeed, in this case since general $X \in |-K_F|$ is a K3 surface, we have $h^{2,0}(X) = 1$, but

$$h^{2,0}(F) = h^{0,2}(F) = \dim H^2(F, \mathcal{O}_F) = \dim H^2(F, \omega_F \otimes \omega_F^{-1}) = 0,$$

by Kodaira-Akizuki-Nakano vanishing. □

[23] Warning: I haven't completely read this paper, but I do not think $X$ must arise as an antincanonical divisor specifically from $V$.

Unfortunately, $-K_F$ is not always very ample. Even among the Fano 3-folds with $\rho(F) = 1$, we may or may not have very ample $-K_F$. But let's consider those Fano 3-folds with very ample $-K_F$. In these cases, we know that

$$(\mathrm{Pic}(F), -K_F) \cong (\mathrm{Pic}(X), -K_F|_X).$$

30.1. **When** $\rho(F) = 1$**.** What happens in the case when $\rho(F) = 1$? These Fano 3-folds are further studied based on their genus $g := \frac{-K_F^3}{2} + 1$. note by HRR we have $h^0(\omega_F^{-1}) = g + 2$. The genus is positive and bounded.

**Proposition 30.5.** *Suppose $\rho(F) = 1$ and $-K_F$ is very ample. Then $\rho(X) = 1$ for general $X \in |-K_F|$ and either*

$$\mathrm{Aut}(X) = \ Id \ \ \text{if } (-K_F)^3 > 2 \text{ and } \mathbb{Z}/2\mathbb{Z} \text{ if } (-K_F)^3 = 2.$$

*Proof.* Since $\rho(F) = 1$ and $-K_F$ is very ample, we have $\rho(X) = 1$. By [Huy, Chapter 15 Corollary 2.12], if $\mathrm{Pic}(X) = \mathbb{Z} \cdot H$, then either $\mathrm{Aut}(X)$ is identity if $(H.H) > 2$ or $\mathbb{Z}/2\mathbb{Z}$ if $(H.H)^2 = 2$. This translates exactly to the numerical condition above. □

The majority of the isomorphism types of Fano 3-Folds with $\rho(F) = 1$ have very ample $-K_F$ and $-K_F^3 > 2$. So for these, we have for general $X \in |-K_F|$ that $\mathrm{Aut}(X)$ is trivial. The few cases where $-K_F$ is not very ample remain unclear to me. But we'll make the following conjecture:

**Conjecture 30.6.** *For all isomorphism type of Fano 3-folds with $\rho(F) = 1$, except the one isomorphism type with $(-K_F^3) = 2$, we have the very general $X \in |-K_F|$ is a smooth K3 surface with automorphism group $\mathrm{Aut}(X) = \{Id\}$.*

**Conjecture 30.7.** *For the remaining Fano 3-fold with $\rho(F) = 1$, $F$ is a double cover of $\mathbb{P}^3$ ramified over a degree 6 hypersurface and $(-K_F^3) = 2$. We conjecture that the very general $X \in |-K_F|$ still has $\rho(X) = 1$, and thus $\mathrm{Aut}(X) = \mathbb{Z}/2\mathbb{Z}$.*

*The reason why is because $|-K_F| : F \to \mathbb{P}^3$ provides the 2 to 1 cover, and so a general $X \in |-K_F|$ will be the preimage of a $\mathbb{P}^2$ intersecting $F$. So the general $X \in |-K_F|$ will admit a 2 to 1 covering of a $\mathbb{P}^2$, and the general such one has automorphism $\mathbb{Z}/2\mathbb{Z}$, given by involution.*

30.2. **When** $\rho(F) \geq 6$**.** Fano 3-folds with $\rho(F) \geq 6$ are a little easier to study, because they are isomorphic to

$$S_{11-\rho(F)} \times \mathbb{P}^1,$$

where $S_{11-\rho(F)}$ is a del Pezzo surface of degree $11 - \rho(F)$. These can be identified with the blow up of $\mathbb{P}^2$ at $\rho(F) - 2$ points, no three of which lie on a line and no six of which lie on a conic [Man86]. We're going to calculate the intersection matrix for a finite sublattice of $(\mathrm{Pic}(F), -K_F)$ in these cases of $\rho(F) \geq 6$, but afterwards we'll restrict our attention to $\rho(F) = 6, 7, 8$ because then we have $-K_F$ is very ample.

I want to first calculate the intersection matrix of the pairing we mentioned before on $\mathrm{Pic}(F)$. Actually, because it is easier and it suffices for our purpose, we will only calculate the intersection matrix for a sublattice which will be isomorphic rationally to $\mathrm{Pic}(F)$. We'll do the case of $\rho(F) = 9$ first; the other cases follow analogously: Consider the projections

$$\begin{tikzcd} & S_2 \times \mathbb{P}^1 \arrow[dl, "\pi_{S_2}"'] \arrow[dr, "\pi_{\mathbb{P}^1}"] & \\ S_2 & & \mathbb{P}^1 \end{tikzcd}$$

Note

$$K_{S_2 \times \mathbb{P}^1} \cong \pi_{S_2}^*(K_{S_2}) \otimes \pi_{\mathbb{P}^1}^*(\mathcal{O}_{\mathbb{P}^1}(-2)) \implies -K_{S_2 \times \mathbb{P}^1} \cong \pi_{S_2}^*(-K_{S_2}) \otimes \pi_{\mathbb{P}^1}^*(\mathcal{O}_{\mathbb{P}^1}(2)).$$

Also note that $\dim | - K_{S_{11-\rho(F)} \times \mathbb{P}^1} = \dim | - K_F| > 0$ for $\rho(F) \geq 6$, as can be checked on Fanography. Also, note that $\dim | - K_{S_{11-\rho(F)}}| > 0$ for $6 \leq \rho(F) \leq 10$. In the cases of $\rho(F) = 6, 7, 8$ this follows from [Man86, First paragraph of Theorem 24.5 proof] (where Manin also determines the dimension of the linear series). Also note in this case we have the anticanonical of the del Pezzo is very ample. In the case of $\rho(F) = 9, 10$ the anticanonical is ample but not very ample, and the fact that there are nontrivial global sections follows from [Dem77, III.2. Théorème 1 ]. I would like to determine an intersection matrix with respect to the intersection form

$$\langle \mathcal{L}_1.\mathcal{L}_2 \rangle_{S_2 \times \mathbb{P}^1} := \langle \mathcal{L}_1.\mathcal{L}_2. - K_{S_2 \times \mathbb{P}^1} \rangle$$

and the following divisors of $\mathrm{Pic}(S_2 \times \mathbb{P}^1)$:

- First we determine a basis for $\mathrm{Pic}(S_2)$... note we can let $S_2$ be the blow up of $\mathbb{P}^2$ at 7 closed points, no three of which lie on one line, and no six of which lie on one conic [Man86, Theorem 24.4]. If we let $E_1, \cdots, E_7 \subset S_2 \times \mathbb{P}^1$ denote the exceptional curves, then the following is a basis for $\mathrm{Pic}(S_2) \cong \mathrm{NS}(S_2)$:

  $$\{E_0, E_1, \cdots, E_7\},$$

  where $E_0$ is defined so that

  $$K_{S_2} \sim -3E_0 + \sum_{i=1}^{7} E_i,$$

  and $E_0^2 = 1, (E_i.E_j) = 0,$ and $(E_i^2) = -1$ for $1 \leq i \leq 7$. Note $\{-K_{S_2}, E_1, \cdots, E_7\}$ will be an index 3 sublattice of $\mathrm{Pic}(S_2)$.
- Then a sublattice for $\mathrm{Pic}(S_2 \times \mathbb{P}^1)$ is:

$$\pi_{S_2}^*(-K_{S_2}) = -K_{S_2} \times \mathbb{P}^1 \text{ and } \pi_{S_2}^*(E_j) = E_j \times \mathbb{P}^1 \text{ for } 1 \leq j \leq 7, \text{ and } \pi_{\mathbb{P}^1}^*(\mathrm{pt}) = S_2 \times \mathrm{pt}.$$

  Note we are abusing notation here when we write $\pi_{S_2}^*(-K_{S_2}) = -K_{S_2} \times \mathbb{P}^1$, where we are treating $-K_{S_2}$ as if it were an effective divisor $D \in | - K_{S_2}|$. $-K_{S_2}$ is really only supposed to indicate the divisor class, but we'll continue to abuse this notation. We'll do this in cases where we look like we are treating $-K_{S_2}$ as a smooth submanifold of $S_2$ – this is justified because there does exist smooth $D \in |-K_{S_2}|$ [Dem77, III.2. Théorème 1].

**Proposition 30.8.** *For $1 \leq j \leq 7$, we have*

$$\langle \pi_{S_2}^*(E_j).\pi_{S_2}^*(E_j)\rangle = -2.$$

*Furthermore,*

$$\langle \pi_{S_2}^*(E_i).\pi_{S_2}^*(E_j)\rangle = 0$$

*for $1 \leq i \neq j \leq 7$.*

*Proof.* The Hirzebruch-Riemann-Roch formula gives us that

$$\chi(\mathcal{O}_{S_2\times\mathbb{P}^1}(D)) = \frac{1}{6}(D^3) - \frac{1}{4}(D^2.K_{S_2\times\mathbb{P}^1}) + \frac{1}{12}D.(K_{S_2\times\mathbb{P}^1})^2 + \frac{1}{12}D.c_2(T_{S_2\times\mathbb{P}^1}) + 1$$

$$\implies \chi(\mathcal{O}_{S_2\times\mathbb{P}^1}(D)) + \chi(\mathcal{O}_{S_2\times\mathbb{P}^1}(-D)) = -\frac{1}{2}(D^2.K_{S_2\times\mathbb{P}^1}) + 2.$$

Let's apply this to the effective divisors $E_j \times \mathbb{P}^1$, for $1 \leq j \leq 7$. Then

$$\implies \chi(\mathcal{O}_{S_2\times\mathbb{P}^1}(E_j\times\mathbb{P}^1)) + \chi(\mathcal{O}_{S_2\times\mathbb{P}^1}(-(E_j\times\mathbb{P}^1))) = -\frac{1}{2}((E_j\times\mathbb{P}^1)^2.K_{S_2\times\mathbb{P}^1}) + 2.$$

We claim that

$$\chi(\mathcal{O}_{S_2\times\mathbb{P}^1}(-(E_j\times\mathbb{P}^1))) = 0.$$

To see this, using additivity of Euler characteristic on the ideal sheaf short exact sequence for $E_j \times \mathbb{P}^1$, we get

$$\chi(\mathcal{O}_{S_2\times\mathbb{P}^1}(-(E_j\times\mathbb{P}^1))) = \chi(\mathcal{O}_{S_2\times\mathbb{P}^1}) - \chi(\mathcal{O}_{E_j\times\mathbb{P}^1}) = \chi(\mathcal{O}_{S_2})\chi(\mathcal{O}_{\mathbb{P}^1}) - \chi(\mathcal{O}_{E_j})\chi(\mathcal{O}_{\mathbb{P}^1}) = 1\cdot 1 - 1\cdot 1 = 0.$$

This leaves us with

$$\chi(\mathcal{O}_{S_2\times\mathbb{P}^1}(E_j\times\mathbb{P}^1)) = -\frac{1}{2}((E_j\times\mathbb{P}^1)^2.K_{S_2\times\mathbb{P}^1}) + 2.$$

Note $\chi(\mathcal{O}_{S_2\times\mathbb{P}^1}(E_j\times\mathbb{P}^1)) = \chi(\mathcal{O}_{S_2}\times\mathbb{P}^1) + \chi(\mathcal{O}_{E_j\times\mathbb{P}^1}(E_j\times\mathbb{P}^1)) = 1 + \chi(\mathcal{O}_{E_j\times\mathbb{P}^1}(E_j\times\mathbb{P}^1))$. Note that

$$\mathcal{O}_{E_j\times\mathbb{P}^1}(E_j\times\mathbb{P}^1) \cong \omega_{E_j\times\mathbb{P}^1} \otimes \omega_{S_2\times\mathbb{P}^1}^{-1}|_{E_j\times\mathbb{P}^1},$$

so by Kodaira vanishing, we have all the higher cohomologies of $\mathcal{O}_{E_j\times\mathbb{P}^1}(E_j\times\mathbb{P}^1)$ vanish. Then

$$\chi(\mathcal{O}_{E_j\times\mathbb{P}^1}(E_j\times\mathbb{P}^1)) = h^0(\mathcal{O}_{E_j\times\mathbb{P}^1}(E_j\times\mathbb{P}^1)).$$

Finally, note that $\mathcal{O}_{E_j\times\mathbb{P}^1}(E_j\times\mathbb{P}^1)$ is the line bundle $\pi_{S_2}^*(\mathcal{O}_{S_2}(E_j))$ restricted to $E_j\times\mathbb{P}^1$. Then it isomorphic to the pullback of $\mathcal{O}_{E_j}(E_j)$ along the projection $E_j\times\mathbb{P}^1 \to E_j$. Note that $E_j\times\mathbb{P}^1 \cong \mathbb{P}^1\times\mathbb{P}^1$, and the line bundles on $\mathbb{P}^1\times\mathbb{P}^1$ are all isomorphic to one of $\mathcal{O}(a,b)$. It suffices to determine the degree of $\mathcal{O}_{E_j}(E_j)$. But note $(E_j.E_j) = -1$ on $S_2$, and this gives the degree of its normal bundle. So $\mathcal{O}_{E_j}(E_j) \cong \mathcal{O}_{\mathbb{P}^1}(-1)$. This implies that $\mathcal{O}_{E_j\times\mathbb{P}^1}(E_j \times \mathbb{P}^1) \cong \mathcal{O}(-1,0)$, thus $h^0(\mathcal{O}_{E_1\times\mathbb{P}^1}(E_1 \times \mathbb{P}^1)) = 0$. Altogether, this implies that

$$1 = -\frac{1}{2}((E_j \times \mathbb{P}^1)^2.K_{S_2\times\mathbb{P}^1}) + 2$$
$$\implies \langle \pi_{S_2}^*(E_j).\pi_{S_2}^*(E_j)\rangle = -2, \text{ for } 1 \le j \le 7.$$

Finally, we have that

$$\langle \pi_{S_2}^*(E_i).\pi_{S_2}^*(E_j)\rangle = (E_i \times \mathbb{P}^1.E_j \times \mathbb{P}^1. - K_{S_2\times\mathbb{P}^1}) = 0,$$

since $(E_i.E_j) = 0$ on $S_2$. □

**Proposition 30.9.** *We have*

$$\langle \pi_{\mathbb{P}^1}^*(pt), \pi_{\mathbb{P}^1}^*(pt)\rangle = 0.$$

*Furthermore,*

$$\langle \pi_{\mathbb{P}^1}^*(pt), \pi_{S_2}^*(E_j)\rangle = 1.$$

*for* $1 \le j \le 7$.

*Proof.* Just as we applied Hirzebruch-Riemann-Roch formula in Proposition 30.8, we have

$$\chi(\mathcal{O}_{S_2\times\mathbb{P}^1}(S_2 \times \text{pt})) + \chi(\mathcal{O}_{S_2\times\mathbb{P}^1}(-(S_2 \times \text{pt}))) = \frac{1}{2}\langle \pi_{\mathbb{P}^1}^*(\text{pt}), \pi_{\mathbb{P}^1}^*(\text{pt})\rangle + 2.$$

Applying additivity of Euler characteristic to the ideal sheaf short exact sequence for $S_2 \times \text{pt}$, we find

$$\chi(\mathcal{O}_{S_2\times\mathbb{P}^1}(-(S_2 \times \text{pt}))) = \chi(\mathcal{O}_{S_2\times\mathbb{P}^1}) - \chi(\mathcal{O}_{S_2\times\text{pt}}) = \chi(\mathcal{O}_{S_2})\chi(\mathcal{O}_{\mathbb{P}^1}) - \chi(\mathcal{O}_{S_2}) = 1 \cdot 1 - 1 = 0.$$

Then we are left with

$$\chi(\mathcal{O}_{S_2\times\mathbb{P}^1}(S_2 \times \text{pt})) = \frac{1}{2}\langle \pi_{\mathbb{P}^1}^*(\text{pt}), \pi_{\mathbb{P}^1}^*(\text{pt})\rangle + 2.$$

We have

$$\chi(\mathcal{O}_{S_2\times\mathbb{P}^1}(S_2 \times \text{pt})) = \chi(\mathcal{O}_{S_2\times\mathbb{P}^1}) + \chi(\mathcal{O}_{S_2\times\text{pt}}(S_2 \times \text{pt})) = 1 + \chi(\mathcal{O}_{S_2\times\text{pt}}(S_2 \times \text{pt}))$$

Once again, by the adjunction formula and Kodaira-Akizuki-Nakano vanishing we have

$$\chi(\mathcal{O}_{S_2\times\text{pt}}(S_2 \times \text{pt})) = h^0(\mathcal{O}_{S_2\times\text{pt}}(S_2 \times \text{pt})).$$

Note $\mathcal{O}_{S_2\times\text{pt}}(S_2 \times \text{pt})$ is the restriction of $\pi_{\mathbb{P}^1}^*(\mathcal{O}_{\mathbb{P}^1}(1))$ to $S_2 \times \text{pt}$, which we then see is a trivial line bundle on $S_2 \times \text{pt}$, so $h^0(\mathcal{O}_{S_2\times\text{pt}}(S_2 \times \text{pt})) = 1$. This implies that

$$\langle \pi_{\mathbb{P}^1}^*(\text{pt}), \pi_{\mathbb{P}^1}^*(\text{pt})\rangle = 0.$$

On the other hand, we have

$$\langle \pi_{\mathbb{P}^1}^*(\text{pt}), \pi_{S_2}^*(E_j)\rangle = (S_2 \times \text{pt}.E_j \times \mathbb{P}^1. - K_{S_2\times\mathbb{P}^1})$$

Note $-K_{S_2\times\mathbb{P}^1} \cong \pi_{S_2}^*(-K_{S_2}) \otimes \pi_{\mathbb{P}^1}^*(-K_{\mathbb{P}^1})$

$$\implies -K_{S_2\times\mathbb{P}^1} \sim [-K_{S_2} \times \mathbb{P}^1] + 2[S_2 \times \text{pt}].$$

Then

$$\langle \pi_{\mathbb{P}^1}^*(\mathrm{pt}), \pi_{S_2}^*(E_j)\rangle = ([S_2\times \mathrm{pt}].[E_j\times\mathbb{P}^1].[-K_{S_2}\times\mathbb{P}^1]) + 2([S_2\times\mathrm{pt}].[E_j\times\mathbb{P}^1].[S_2\times\mathrm{pt}])$$
$$= 1 + 2(\mathcal{O}_{S_2\times\mathbb{P}^1}(S_2\times\mathrm{pt})|_{E_j\times\mathbb{P}^1}.\mathcal{O}_{S_2\times\mathbb{P}^1}(S_2\times\mathrm{pt})|_{E_j\times\mathbb{P}^1}) = 1 + 2(\mathcal{O}(0,1).\mathcal{O}(0,1)) = 1.$$

□

**Proposition 30.10.** *We have*

$$\langle \pi_{S_2}^*(-K_{S_2}), \pi_{S_2}^*(-K_{S_2})\rangle = 4,$$
$$\langle \pi_{S_2}^*(-K_{S_2}), \pi_{S_2}^*(E_j)\rangle = 2, \text{ for } 1\le j\le 7,$$
$$\text{and } \langle \pi_{S_2}^*(-K_{S_2}), \pi_{\mathbb{P}^1}^*(pt)\rangle = 2.$$

*Proof.* We have

$$\chi(\pi_{S_2}^*(-K_{S_2})) + \chi(\pi_{S_2}^*(-K_{S_2})^{-1}) = \frac{1}{2}\langle \pi_{S_2}^*(-K_{S_2}), \pi_{S_2}^*(-K_{S_2})\rangle + 2.$$

Note

$$\chi(\pi_{S_2}^*(-K_{S_2})^{-1}) = \chi(\mathcal{O}_{S_2\times\mathbb{P}^1}) - \chi(\mathcal{O}_{-K_{S_2}\times\mathbb{P}^1}) = 1-\chi(\mathcal{O}_{-K_{S_2}}) = 1-(1-g(-K_{S_2})) = g(-K_{S_2}) = h^1(\mathcal{O}_{-K_{S_2}}) = 1.$$

Then

$$\chi(\pi_{S_2}^*(-K_{S_2})) - 1 = \frac{1}{2}\langle \pi_{S_2}^*(-K_{S_2}), \pi_{S_2}^*(-K_{S_2})\rangle.$$

Using additivity of Euler characteristic and tensoring the ideal sheaf short exact sequence of $\pi_{S_2}^*(-K_{S_2})$ by $\pi_{S_2}^*(-K_{S_2})$, we find

$$\chi(\pi_{S_2}^*(-K_{S_2})) = \chi(\mathcal{O}_{S_2\times\mathbb{P}^1}) + \chi(\mathcal{O}_{-K_{S_2}\times\mathbb{P}^1}(-K_{S_2}\times\mathbb{P}^1)) = 1 + \chi(\mathcal{O}_{-K_{S_2}\times\mathbb{P}^1}(-K_{S_2}\times\mathbb{P}^1))$$
$$= 1 + h^0(\mathcal{O}_{-K_{S_2}\times\mathbb{P}^1}(-K_{S_2}\times\mathbb{P}^1)) = 1 + h^0(\mathcal{O}_{-K_{S_2}}(-K_{S_2})).$$

The fact that $h^0(\mathcal{O}_{-K_{S_2}\times\mathbb{P}^1}(-K_{S_2}\times\mathbb{P}^1)) = h^0(\mathcal{O}_{-K_{S_2}}(-K_{S_2}))$ can be observed by thinking about transition functions and the fact that functions on $\mathbb{P}^1$ must be constant. Now $h^0(\mathcal{O}_{S_2}(-K_{S_2})) = 3$, so we see that $h^0(\mathcal{O}_{-K_{S_2}}(-K_{S_2})) = 2$. So $\chi(\pi_{S_2}^*(-K_{S_2})) = 3$. Thus,

$$\langle \pi_{S_2}^*(-K_{S_2}), \pi_{S_2}^*(-K_{S_2})\rangle = 4.$$

On the other hand, we have

$$\langle \pi_{S_2}^*(-K_{S_2}), \pi_{S_2}^*(E_j)\rangle = ([-K_{S_2}\times\mathbb{P}^1].[E_j\times\mathbb{P}^1].[-K_{S_2\times\mathbb{P}^1}])$$
$$= ([-K_{S_2}\times\mathbb{P}^1].[E_j\times\mathbb{P}^1].[-K_{S_2}\times\mathbb{P}^1]) + 2([-K_{S_2}\times\mathbb{P}^1].[E_j\times\mathbb{P}^1].[S_2\times\mathrm{pt}]).$$
$$= ([-K_{S_2}\times\mathbb{P}^1].[E_j\times\mathbb{P}^1].[-K_{S_2}\times\mathbb{P}^1]) + 2 = (\mathcal{O}(1,0).\mathcal{O}(1,0)) + 2 = 2.$$

Finally, we have

$$\langle \pi_{S_2}^*(-K_{S_2}), \pi_{\mathbb{P}^1}^*(\mathrm{pt})\rangle = ([-K_{S_2}\times\mathbb{P}^1].[S_2\times\mathrm{pt}].[-K_{S_2}\times\mathbb{P}^1] + 2[S_2\times\mathrm{pt}])$$
$$= 2 + 2([-K_{S_2\times\mathbb{P}^1}].[S_2\times\mathrm{pt}].[S_2\times\mathrm{pt}]) = 2.$$

□

**Theorem 30.11.** *The intersection matrix of the pairing*

$$\langle \mathcal{L}_1.\mathcal{L}_2\rangle_{S_2\times\mathbb{P}^1} := \langle \mathcal{L}_1.\mathcal{L}_2.-K_{S_2\times\mathbb{P}^1}\rangle$$

*with respect to the divisors*

$$\{\pi_{S_2}^*(E_1),\cdots,\pi_{S_2}^*(E_7),\pi_{\mathbb{P}^1}^*(pt),\pi_{S_2}^*(-K_{S_2})\}$$

*is*

$$\begin{pmatrix} -2 & 0 & \cdots & 0 & 1 & 2\\ 0 & \ddots & \cdots & \vdots & \vdots & \vdots\\ \vdots & \ddots & \ddots & 0 & \vdots & \vdots\\ 0 & \ddots & 0 & -2 & 1 & \vdots\\ 1 & \cdots & 1 & 1 & 0 & 2\\ 2 & \cdots & \cdots & \cdots & 2 & 4 \end{pmatrix}$$

*The signature of this matrix is $(1,8)$. Note the divisors we have chosen give a $\mathbb{Q}$-basis for $NS_{\mathbb{Q}}(S_2\times\mathbb{P}^1)$, but $\mathbb{Z}$-linearly it is a finite index sublattice of $NS(S_2\times\mathbb{P}^1)$. This is because of the $\pi_{S_2}^*(-K_{S_2})$ term.*

In general, for $\rho(F)\geq 6$, we have the following:

**Theorem 30.12.** *Let $F$ be a Fano 3-fold of Picard rank $\rho(F)\geq 6$. The intersection matrix of the finite index sublattice of* $\mathrm{Pic}(F)\cong\mathrm{Pic}(S_{11-\rho(F)}\times\mathbb{P}^1)$ *given by the basis*

$$\{\pi_{S_{11-\rho(F)}}^*(E_1),\cdots,\pi_{S_{11-\rho(F)}}^*(E_{\rho(F)-2}),\pi_{\mathbb{P}^1}^*(pt),\pi_{S_{11-\rho(F)}}^*(-K_{S_{11-\rho(F)}})\}$$

*is*

$$\begin{pmatrix} -2 & 0 & \cdots & 0 & 1 & 2\\ 0 & \ddots & \cdots & \vdots & \vdots & \vdots\\ \vdots & \ddots & \ddots & 0 & \vdots & \vdots\\ 0 & \ddots & 0 & -2 & 1 & 2\\ 1 & \cdots & 1 & 1 & 0 & 11-\rho(F)\\ 2 & \cdots & \cdots & 2 & 11-\rho(F) & 22-2\rho(F) \end{pmatrix}$$

**Theorem 30.13.** *Let $F$ be a Fano 3-fold of Picard rank $\rho(F)\geq 6$. The intersection matrix of the full lattice* $\mathrm{Pic}(F)\cong\mathrm{Pic}(S_{11-\rho(F)}\times\mathbb{P}^1)$ *with respect to the basis*

$$\{\pi_{S_{11-\rho(F)}}^*(E_1),\cdots,\pi_{S_{11-\rho(F)}}^*(E_{\rho(F)-2}),\pi_{\mathbb{P}^1}^*(pt),\pi_{S_{11-\rho(F)}}^*(E_0)\}$$

*and the pairing* $\langle L_1, L_2\rangle := (L_1.L_2.-K_F)$ *is*

$$\begin{pmatrix} -2 & 0 & \cdots & 0 & 1 & 0 \\ 0 & \ddots & \cdots & \vdots & \vdots & \vdots \\ \vdots & \ddots & \ddots & 0 & \vdots & \vdots \\ 0 & \ddots & 0 & -2 & 1 & 0 \\ 1 & \cdots & 1 & 1 & 0 & 3 \\ 0 & \cdots & \cdots & 0 & 3 & 2 \end{pmatrix}$$

*Proof.* Make the apprioriate change of basis to the matrix in Theorem 30.12, using the fact that $-K_{S_{11-\rho(F)}} + \sum_{j=1}^{\rho(F)-2} E_j \sim 3E_0$ in $\mathrm{Pic}(S_{11-\rho(F)})$. $\square$

When $\rho(F) = 6, 7, 8$ we know that $-K_F$ is very ample, so we can use the result of Shokurov to conclude that for general $X \in |-K_F|$ we have $\mathrm{Pic}(F) \cong \mathrm{Pic}(X)$. If we determined the Picard lattice of $\mathrm{Pic}(X)$, we could compare it to the list of Picard lattices of algebraic K3 surfaces with finite automorphisms [Rou] by computing determinants which measure the size of a lattice's discriminant group. Probably we could do this using the intersection matrix in Propositoin 30.12 and $K_{S_{11-\rho(F)}} \sim -3E_0 + \sum_{i=1}^{\rho(F)-2} E_i$. But actually, we'll do something more geometrically meaningfully. We'll try to find elliptic fibrations on these $X \in |-K_F|$ plus sections, and show that these provide infinite automorphisms.

The case of $\rho(F) = 9, 10$ are trickier; I'm not sure how to go about them since we have $-K_F$ is not very ample, so it's unclear whether $\rho(X)$ can jump for $X \in |-K_F|$. If it does jump, how about the behavior of $\mathrm{Aut}(X)$?

**Conjecture 30.14.** $\mathrm{Aut}(X)$ *is infinite for general* $X \in |-K_F|$ *for* $\rho(X) = 9, 10$. *This might be because of a degeneration argument? One would need to show that for general* $X \in |-K_F|$ *we have* $\rho(X) = \rho(F)$. *Then by degeneration argument, automorphisms can only jump? Huybrechts said something like: take general K3, take graph of general automorphisms, look at graph specialize K3 and take closure of graph. A priori this graph might not be a graph anymore and might have many components, but one component will always be graph of an automorphism.. this argument might be in K3 surface book.*

**Question 30.15.** *When* $\rho(F) = 6, 7$ *or* 8*, we have* $-K_F$ *is base point free. Can general* $X \in |-K_F|$ *contain a smooth rational curve?*

**Question 30.16.** *I conjecture that for* $\rho(F) = 6, 7, 8$ *we have general* $X \in |-K_F|$ *has infinite automorphisms. Can we geometrically realize these automorphisms?*

*Here's a potential way: K3 surfaces (maybe only general? not sure, need to check) with Picard rank* $\geq 5$ *is an elliptic K3 surface. If such a K3 surface admits an elliptic fibration (plus section) with a non-torsion section (non-torsion sectino gives a non-torsion point of the elliptic curve generic fiber), then we can produce infinite automorphisms of the K3* *surface by adding that non-torsion point to itself over and over. This produces infinite automorphisms of the K3 surface.*

Let's see if we can tackle Question 30.16. Let $F$ be a Fano 3-fold with $\rho(F) = 6, 7, 8$. I believe that for general $X \in |-K_F|$ has a smooth rational curve, and thus infinitely many of them. To existence of smooth rational curves, we claim that general $X \in |-K_F|$ admits an elliptic fibration with a section. For simplicity, let $\rho(F) = 6$. The construction is analogous when $\rho(F) = 7, 8$.

Since $\rho(F) = 6$, we have $F \cong S_5 \times \mathbb{P}^1$, where $S_5$ is a del Pezzo surface of degree 5. Let $X \in |-K_F|$ be general. Consider the composition

$$\begin{tikzcd} X \arrow[rr, hook] \arrow[dr, "\pi"'] & & S_5 \times \mathbb{P}^1 \arrow[dl, "\pi_{\mathbb{P}^1}"] \\ & \mathbb{P}^1 & \end{tikzcd}$$

I believe that $\pi$ is an elliptic fibration. To see this, first we need to show $\pi$ is surjective. Suppose $t \in \mathbb{P}^1$, and let $X_t$ denote the fiber over $t$. Note $X_t = X \cap S_5 \times \{t\}$. Note $X_t$ cannot be empty, since

$$-(E_j \times \mathbb{P}^1.S_5 \times \{t\}.X) = 1 \neq 0,$$

by Theorem 30.12. So $\pi$ is surjective.

Now we want to show that the general fiber is an elliptic curve. Note $X_t$ is forced to be 1-dimensional, since $S_5 \times \{t\}$ and $X$ are surfaces in 3-dimensional $F$. Since

$$(S_5 \times \{t\}.S_5 \times \{t\}.X) = 0$$

also by Theorem 30.12, we have the arithmetic genus of $X_t$ is 1. The general fiber is smooth, thus $X_t$ is an elliptic curve for general $t \in \mathbb{P}^1$.

Next, we claim that $\pi : X \to \mathbb{P}^1$ admits a section. Note as a divisor, $X \sim \pi_{S_5}^*(-K_{S_5}) + 2\pi_{\mathbb{P}^1}^*(\mathrm{pt})$, and there is always a meromorphic function on $S_5 \times \mathbb{P}^1$ such that "pt" can be adjusted. Consider that

$$1 = (E_j \times \mathbb{P}^1.S_5 \times \{t\}.X) = (E_j \times \mathbb{P}^1.S_5 \times \{t\}. - K_{S_5} \times \mathbb{P}^1) = (E_j \times \{t\}. - K_{S_5} \times \{t\})_{S_5 \times \{t\}}.$$

I believe this implies that $E_j \times \mathbb{P}^1 \cap X$ defines a section of $\pi : X \to \mathbb{P}^1$. Actually, I think this whole argument holds true when $\rho(F) = 9, 10$.

**Proposition 30.17.** *Let $F \cong S_{11-\rho(F)} \times \mathbb{P}^1$ be a Fano 3-fold of Picard rank $6 \leq \rho(F) \leq 10$. Let $X \in |-K_F|$ be a general anticanonical divisor, and thus a smooth algebraic K3 surface, with $\rho(F) \leq \rho(X)$. In the case of $\rho(F) = 6, 7, 8$ we are certain that $\rho(F) = \rho(X)$.*

*Then $\pi : X \hookrightarrow S_{11-\rho(F)} \times \mathbb{P}^1 \to \mathbb{P}^1$ is an elliptic fibration. Each of the divisors*

$$\pi_{S_{11-\rho(F)}}^*(E_j) = E_j \times \mathbb{P}^1 \subset S_{11-\rho(F)} \times \mathbb{P}^1$$

*for $1 \leq j \leq \rho(F) - 2$ provides a section of $\pi : X \to \mathbb{P}^1$.*

**Remark 30.18.** *Warning: although elliptic curves over $\mathbb{C}$ have only 3 nontrivial torsion points, elliptic curves defined over function fields like $\mathbb{C}(t)$ can have points of order 2, or say, order 5. One cannot naively apply the theory of complex tori to elliptic curves over function fields. So while it may be tempting to say that because we have at least 5 sections, picking one to be a base point, at least one of the other at least 4 sections must be infinite... this is not true.*

Let us now narrow down to the case of $\rho(F) = 6, 7, 8$. Note these are the cases of $\rho(F) \geq 6$ such that $-K_F$ is very ample. Then by Moishezon, the general $X \in |-K_F|$ is a smooth K3 surface such that restriction gives us an isometry of Picard lattices

$$(\mathrm{Pic}_F, -K_F) \cong (\mathrm{Pic}(X), -K_F|_X).$$

Recall that an elliptic fibration splits if it is birationally isomorphic to the base times some elliptic curvce.

**Proposition 30.19.** *Let $X \in |-K_F|$ be general smooth K3 surface, where $\rho(F) = 6, 7,$ or $8$. Consider the elliptic fibration*

$$\pi : X \longrightarrow \mathbb{P}^1$$

*as previously defined. This elliptic fibration does not split over $\mathbb{C}$. In other words, there does not exist some elliptic curve $E_0$ such that $X \to \mathbb{P}^1$ is birationally isomorphic to the projection $E_0 \times \mathbb{P}^1 \to \mathbb{P}^1$.*

*Proof.* In [Saw14], Sawon classifies all isotrivial elliptic K3 surfaces. In particular, if $\pi : X \to \mathbb{P}^1$ is isotrivial, then the $j$-invariant on smooth fibers will be constant as a function of $t \in \mathbb{P}^1$. In particular, if $j \neq 0, 1728$, then $\mathrm{Pic}(X) \geq 17$ [Saw14, Corollary 2] [MS24, Remark 4.17].

Furthermore, if $\pi : X \to \mathbb{P}^1$ was isotrivial with $j$-invariant equal to 1728. Then by [MS24, Remark 4.17], the Picard rank of $X$ must be at least 10, so this cannot happen under our assumptions.

Finally, suppose our $\pi : X \to \mathbb{P}^1$ was isotrivial with $j$-invariant 0. Note elliptic curves with j-invariant 0 are isomorphic to $\mathbb{C}/\langle 1, \xi \rangle$, where $\xi$ is a primitive sixth root of unity, which comes with a $\mathbb{Z}/3$ action. Then there would be a birational fiber-wise automorphism of order 3 on $X$, which extends to a $\mathbb{Z}/3$ action defined everywhere on $X$ by minimality of $X$. Such a $\mathbb{Z}/3$ action is non-symplectic in the sense of [AS08], which computes the possible Picard lattices of $X$. The possible Picard lattices are listed in Table 2, in the '$T(n, k)$' column.

- The only possible Neron-Severi lattices of rank 6 in [AS08, Table 9] are $U \oplus U \oplus A_2$ and $U \oplus U(3) \oplus A_2$. The determinant of these lattices are 3 and -27, respectively.
- There are no picard lattices in [AS08, Table 9] of rank 7.
- The only possible Neron-Severi lattices of rank 8 in [AS08, Table 9] are $U \oplus U \oplus A_2^2$ and $U \oplus U(3) \oplus A_2^2$. The determinants of these lattices are 9 and -81, respectively.

We immediately see that for $\rho(X) = 7$ we cannot have $\pi : X \to \mathbb{P}^1$ isotrivial with j-invariant 0. For $\rho(X) = 6$ or 8, the Picard lattice is given in Theorem 30.13, and the determinant of these two lattices are -80 and -192, respectively. The determinants not matching up with the above cases implies $\pi : X \to \mathbb{P}^1$ cannot be iso-trivial with $j$-invariant 0 for $\rho(X) = 6$ or 8. Altogether, this shows that the elliptic fibrations $\pi : X \to \mathbb{P}^1$ cannot split. □

The reason why we want that $X \to \mathbb{P}^1$ does not split is because knowing that it does not split means that the canonical height of a rational point $p \in X_\eta$ in the generic fiber has finite order if and only if its canonical height $\widehat{h}(p) = 0$ [Sil94, Chapter 3, Theorem 4.3]. This is great because we can compute $\widehat{h}(p)$ purely based on intersection theory on K3 surface $X$ (combined with casework provided by the Shioda-Tate formula of non-split minimal elliptic surfaces admitting a section). In particular,

$$\widehat{h}(p) = \frac{1}{2}\langle P, P\rangle := -\frac{1}{2}((P) - (O) + \Phi_P.(P) - (O) + \Phi_P),$$

where $O \in X_\eta$ is the chosen base point and $(P)$ and $(O)$ are the divisors corresponding to the sections corresponding to the rational points [Sil94, Chapter 3 Section 9]. Furthermore, $\Phi_P$ is defined as the fibral divisor of $X$ such that

$$((P) - (O) + \Phi_P.F_i) = 0$$

for all irreducible components $F_i$ of any singular fiber of $\pi : X \to \mathbb{P}^1$ [Sil94, Chapter 3 Section 9].

**Proposition 30.20.** *Let $X \in |-K_F|$ be a general smooth K3 surface, where $\rho(F) = 6$ or 7. Let $\pi : X \to \mathbb{P}^1$ be the elliptic fibration we have defined and recall that each $(E_j \times \mathbb{P}^1) \cap X$ defines a section of this elliptic fibration, where $1 \leq j \leq \rho(F) - 2$. Let $S_0$ denote one of these sections as a divisor of $X$, and let $S_1$ denote a different section as a divisor of $X$. Let $P_0$ be the point on the generic fiber $X_\eta$ corresponding to $S_0$, and $P_1$ be the point on the generic fiber $X_\eta$ corresponding to $S_1$. If we choose $P_0$ to be the origin on $X_\eta$, then the point $P_1$ has infinite order.*

*Proof.* Since $\pi : X \to \mathbb{P}^1$ does not split, it is equivalent to show

$$(S_1 - S_0 + \Phi_1.S_1 - S_0 + \Phi_1) \neq 0$$

where $\Phi_1$ is as described in [Sil94, Chapter 3, Proposition 8.3]. Note the above expression simplifies to

$$= -4 + 2(S_1.\Phi_1) + 2(-S_0.\Phi_1) + \Phi_1^2 = -4 - \Phi_1^2,$$

since $(S_1 - S_0 + \Phi_1.\Phi_1) = 0$ since $\Phi_1$ is a fibral divisor and by definition of $\Phi_1$. Then it suffices to show that

$$4 + \Phi_1^2 \neq 0 \iff \Phi_1^2 \neq -4.$$

Note $\Phi_1$ is defined as linear combinations of irreducible components of those singular fibers with at least 2 irreducible components of our elliptic fibration $\pi : X \to \mathbb{P}^1$. To show

$\Phi_1^2 \neq -4$, it turns out that it is sufficient to analyze the possible values of $\Phi_1^2$ over just one singular fiber with at least 2 irreducible components.

Note that

$$\rho(X) = 2 + \sum_t (n_t - 1) + \operatorname{rk} \mathrm{MW}(X).$$

Let us do the case of $\rho(X) = 6$ first. We have $\sum_t (n_t - 1) \leq 4$. This means that the only possible singular fibers with at least two irreducible components are those of type $III, I_2, IV, I_3, I_4, I_5$ and $I_0^*$. We analyze each of these cases, with subcases being which irreducible components $S_1$ and $S_0$ might intersect. Note if $S_1$ and $S_0$ intersect the same irreducible component of the singular fiber, then $\Phi_1$ will automatically be zero over this singular fiber, so we only analyze those subcases where they intersect different irreducible components.

- Type $III$ : There are only two irreducible components , so we pair off $S_1$ and $S_0$ to intersect one of each, say $C_1$ and $C_0$, respectively. Note sections always intersect irreducible components at smooth points. If we let $\Phi_1 = \lambda C_1$, then

  $$(S_1 - S_0 + \lambda C_1.C_1) = 0 \implies 1 - 2\lambda = 0 \implies \lambda = \frac{1}{2}.$$

  Then $\Phi_1^2$ over this singular fiber is $\boxed{-\frac{1}{2}}$.
- Type $I_2$ : Again, there are only two irreducible components so let's pair off $S_1$ and $S_0$ to intersect one of each, say $C_1$ and $C_0$ respectively. If we let $\Phi_1 = \lambda C_1$, then

  $$(S_1 - S_0 + \lambda C_1.C_1) = 0 \implies 1 - 2\lambda = 0 \implies \lambda = \frac{1}{2}.$$

  Then $\Phi_1^2$ over this singular fiber is $\boxed{-\frac{1}{2}}$.
- Type $IV$ : There are three irreducible components now, say $C_0, C_1, C_2$, and all three intersect at a common point at 3 distinct tangent directions. Suppose $S_1$ and $S_0$ intersect $C_1$ and $C_0$. By symmetry, this is the only configuration of section intersections, up to isomorphism. Let $\Phi_1 = \lambda_1 C_1 + \lambda_2 C_2$. Then

  $$(S_1 - S_0 + \lambda_1 C_1 + \lambda_2 C_2.C_0) = 0 \implies -1 + \lambda_1 + \lambda_2 = 0,$$

  $$(S_1 - S_0 + \lambda_1 C_1 + \lambda_2 C_2.C_1) = 0 \implies 1 - 2\lambda_1 + \lambda_2 = 0.$$

  Solving this system yields $\Phi_1 = \frac{2}{3}C_1 + \frac{1}{3}C_2$. So $\Phi_1^2$ over this singular fiber contributes $\boxed{\frac{-2}{3}}$.
- Type $I_3$ : There are now three irreducible components, which intersect pairwise transversely. Call them $C_0, C_1, C_2$. WLOG we can let $S_i$ intersect $C_i$ at a non-intersection point. By symmetry, this is the only configuration up to isomorphism. Let $\Phi_1 = \lambda_1 C_1 + \lambda_2 C_2$. Then

  $$(S_1 - S_0 + \lambda_1 C_1 + \lambda_2 C_2.C_0) = 0 \implies -1 + \lambda_1 + \lambda_2 = 0,$$

$$(S_1 - S_0 + \lambda_1 C_1 + \lambda_2 C_2.C_1) = 0 \implies 1 - 2\lambda_1 + \lambda_2 = 0$$
$$\implies \Phi_1 = \frac{2}{3}C_1 + \frac{1}{3}C_2$$

so $\Phi_1^2$ over this fiber contributes $\boxed{\frac{-2}{3}}$.

- Type $I_4$: there are now four irreducible components $C_0, C_1, C_2, C_3$, where $C_i$ intersects $C_j$ transversely once if $i \neq j \mod 2$. Let us fix $S_0$ to intersect $C_0$. There are two configurations here of intersections by $S_1$ and $S_0$, up to symmetry:
  - Subcase 1 of Type $I_4$: Suppose $S_1$ intersects $C_1$. Then let $\Phi_1 = \lambda_1 C_1 + \lambda_2 C_2 + \lambda_3 C_3$. Then
$$(S_1 - S_0 + \Phi_1.C_0) = 0 \implies -1 + \lambda_1 + \lambda_3 = 0,$$
$$(S_1 - S_0 + \Phi_1.C_1) = 0 \implies 1 - 2\lambda_1 + \lambda_2 = 0$$
$$(S_1 - S_0 + \Phi_1.C_2) = 0 \implies \lambda_1 - 2\lambda_2 + \lambda_3 = 0.$$
    Then
$$\Phi_1 = \frac{3}{4}C_1 + \frac{1}{2}C_2 + \frac{1}{4}C_3.$$
    So over this singular fiber and configuration $\Phi_1^2$ contributes $\boxed{\frac{-3}{8}}$.
  - Subcase 2 of Type $I_4$: Suppose $S_1$ intersects $C_2$. Let $\Phi_1 = \lambda_1 C_1 + \lambda_2 C_2 + \lambda_3 C_3$. Then
$$(S_1 - S_0 + \Phi_1.C_0) = 0 \implies -1 + \lambda_1 + \lambda_3 = 0,$$
$$(S_1 - S_0 + \Phi_1.C_1) = 0 \implies -2\lambda_1 + \lambda_2 = 0,$$
$$(S_1 - S_0 + \Phi_1.C_2) = 1 + \lambda_1 - 2\lambda_2 + \lambda_3 = 0.$$
    Then
$$\Phi_1 = \frac{1}{2}C_1 + C_2 + \frac{1}{2}C_3.$$
    Then over this singular fiber and configuration $\Phi_1^2$ contributes $\boxed{-1}$.
- Type $I_5$: In this case there are five irreducible components, $C_0, \cdots, C_4$. Fix $S_0$ to intersect $C_0$. Up to symmetry, there are three configurations for where $S_1$ intersects.
  - Subcase 1 of Type $I_5$: WLOG suppose $S_1$ intersects $C_1$. Let $\Phi_1 = \sum_{i=1}^4 \lambda_i C_i$. Then
$$(S_1 - S_0 + \Phi_1.C_0) = 0 \implies -1 + \lambda_1 + \lambda_4 = 0,$$
$$(S_1 - S_0 + \Phi_1.C_1) = 0 \implies 1 - 2\lambda_1 + \lambda_2 = 0,$$
$$(S_1 - S_0 + \Phi_1.C_2) = 0 \implies \lambda_1 - 2\lambda_2 + \lambda_3 = 0,$$
$$(S_1 - S_0 + \Phi_1.C_3) = 0 \implies \lambda_2 - 2\lambda_3 + \lambda_4 = 0$$
    This implies
$$\Phi_1 = \frac{4}{5}C_1 + \frac{3}{5}C_2 + \frac{2}{5}C_3 + \frac{1}{5}C_4.$$

So over this fiber and configuration $\Phi_1^2$ contributes $\boxed{\dfrac{-4}{5}}$.

– Subcase 2 of Type $I_5$: Now suppose $S_1$ intersects $C_2$. Then

$$(S_1 - S_0 + \Phi_1.C_0) = 0 \implies -1 + \lambda_1 + \lambda_4 = 0,$$

$$(S_1 - S_0 + \Phi_1.C_1) = 0 \implies -2\lambda_1 + \lambda_2 = 0,$$

$$(S_1 - S_0 + \Phi_1.C_2) = 0 \implies 1 + \lambda_1 - 2\lambda_2 + \lambda_3 = 0,$$

$$(S_1 - S_0 + \Phi_1.C_3) = 0 \implies \lambda_2 - 2\lambda_3 + \lambda_4 = 0$$

So

$$\Phi_1 = \frac{3}{5}C_1 + \frac{6}{5}C_2 + \frac{4}{5}C_3 + \frac{2}{5}C_4.$$

Thus, over this singular fiber and configuration $\Phi_1^2$ contributes $\boxed{\dfrac{-6}{5}}$.

– Subcase 3 of Type $I_5$:

$$(S_1 - S_0 + \Phi_1.C_0) = 0 \implies -1 + \lambda_1 + \lambda_4 = 0,$$

$$(S_1 - S_0 + \Phi_1.C_1) = 0 \implies -2\lambda_1 + \lambda_2 = 0,$$

$$(S_1 - S_0 + \Phi_1.C_2) = 0 \implies \lambda_1 - 2\lambda_2 + \lambda_3 = 0,$$

$$(S_1 - S_0 + \Phi_1.C_3) = 0 \implies 1 + \lambda_2 - 2\lambda_3 + \lambda_4 = 0.$$

Then

$$\Phi_1 = \frac{2}{5}C_1 + \frac{4}{5}C_2 + \frac{6}{5}C_3 + \frac{3}{5}C_4.$$

Then over this fiber and configuration, $\Phi_1^2$ contributes $\boxed{\dfrac{-6}{5}}$.

– Type $I_0^*$: Write the $I_0^*$ fiber as $C_0 + C_1 + C_2 + C_3 + 2C_4$, where for $0 \le i \le 3$, we have $C_i$ intersects $C_4$ transversely. There is only one relevant configurartion of section intersections here, where WLOG we let $S_i$ intersect $C_i$. Let $\Phi_1 = \sum_{i=0}^3 \lambda_i C_i$. Then

$$(S_1 - S_0 + \Phi_1.C_0) = 0 \implies -1 - 2\lambda_0,$$

$$(S_1 - S_0 + \Phi_1.C_1) = 0 \implies 1 - 2\lambda_1 = 0$$

$$(S_1 - S_0 + \Phi_1.C_2) = 0 \implies -2\lambda_2 = 0$$

$$(S_1 - S_0 + \Phi_1.C_3) = 0 \implies -2\lambda_3 = 0$$

So $\Phi_1 = -\frac{1}{2}C_0 + \frac{1}{2}C_1$. So over this fiber and configuration $\Phi_1^2$ contributes $\boxed{-1}$.

Combining the above analysis of singular fibers with at least 2 irreducible components, with the equation

$$4 = \sum_t (n_t - 1) + \operatorname{rk} \mathrm{MW}(X),$$

we see that it is impossible for $4 + \Phi_1^2 = 0$. Thus, $S_1$ is a section of infinite order.

Now suppose $\rho(X) = 7$. The possible singular fibers with at least two irreducible components include those which were possible for $\rho(X) = 6$, and the following:

– Type $I_6$: FINISH CASEWORK
– Type $I_1^*$ : FINISH CASEWORK

Combining the above analysis of singular fibers with at least 2 irreducible components, with the equation

$$5 = \sum_t (n_t - 1) + \operatorname{rk} \mathrm{MW}(X),$$

we see that it is impossible for $4 + \Phi_1^2 = 0$. Thus, $S_1$ is a section of infinite order. □

**Corollary 30.21.** *Let $F$ be a Fano 3-fold of Picard rank $\rho(F) = 6$ or $7$. Note $F \cong S_{11-\rho(F)} \times \mathbb{P}^1$, where $S_{11-\rho(F)}$ is a del Pezzo surface of degree $11 - \rho(F)$.*

*Let $X \in |-K_F|$ be a general anticanonical divisor. Then the composition $\pi$:*

$$X \hookrightarrow S_{11-\rho(F)} \times \mathbb{P}^1 \to \mathbb{P}^1$$

*is an elliptic fibration. It admits sections $E_j \times \mathbb{P}^1 \cap X$, which have infinite order. In particular, $X$ has infinitely many smooth rational curves.*

We could also do this for the case of $\rho(F) = 8$. Need to check that $\pi : X \to \mathbb{P}^1$ again does not split over $\mathbb{C}$ here. Then more casework involved to verify $4 + \Phi_1^2 \neq 0$.

**Question 30.22.** *Are the sections $E_j \times \mathbb{P}^1 \cap X$ independent? This would mean $\operatorname{rk} MW(X) \geq \rho(F) - 3$. One can verify pairwise independency by testing whether $4 + (\Phi_1.\Phi_2) \neq 0$. In general, given n points, one can test independency by seeing whether the determinant of the height pairing matrix is nonzero.* *https://mathoverflow.net/questions/359442/what-conditions-are-sufficient-for-two-points-to-be-independent-in-the-mordell-w?noredirect=1#comment1270264_359442*

30.3. **July 14, 2025 Update.** Let $F$ be a Fano 3 fold of Picard rank $\geq 6$. After talking to Giacomo, it seems possible that $\mathrm{Aut}(X)$ is infinite for any $X \in |-K_F|$.

For smooth $X \in |-K_F|$, we know that the lattice in Theorem 30.13 is a sublattice of $\mathrm{Pic}(X)$. As a result of this intersection theory, we know that

$$\pi : X \hookrightarrow F \cong S_{11-\rho(F)} \times \mathbb{P}^1 \to \mathbb{P}^1$$

is an elliptic fibration with at least $E_j \times \mathbb{P}^1$ being sections, for $1 \leq j \leq \rho(F) - 2$. Now, before I was worried about whether or not $\pi : X \to \mathbb{P}^1$ was split. But Giacomo told me that since $X$ is a K3 surface it's immediate that $\pi$ cannot be split. The kodaira dimension is a birational invariant, and for ruled surfaces it is $-\infty$ and for K3 surfaces it is 0. This simplifies what we tried to do before, and doesn't require $\rho(F) = \rho(X)$. In particular, my proof of non-splitting was fundamentally confused – I used very much the work of Sawon, but he is classifying isotrivial elliptic K3 surfaces. These are elliptic fibrations such that each elliptic fiber is isomorphic to each other. But this is not the same as being split. The whole point of showing non-splitness, which I originally confused with isotriviality, is so we can identify height of a section with intersection theory and invoke the Shioda-Tate formula. In summary, the Kodaira dimension ensures our elliptic fibration is not split.

Now up to this point, before we tried to show $\mathrm{Aut}(X)$ is infinite by using the elliptic fibration, choosing a section as a basepoint on the generic fiber, and computing whether the order of another section was infinite or finite using intersection theory. The casework we did also assumed that $\rho(F) = \rho(X)$, and we really only verified the casework for $\rho(F) = \rho(X) = 6$. But Giacomo claims that we can do the following: take smooth $X \in |-K_F|$. Take the elliptic fibration $\pi : X \to \mathbb{P}^1$. Pick a section, say $E_1 \times \mathbb{P}^1$ of the elliptic fibration, and any fiber (smooth or not smooth), say a smooth $E_t$ for some $t \in \mathbb{P}^1$. Then this section plus smooth $E_t$ will define the sublattice

$$\begin{pmatrix} 0 & 1 \\ 1 & -2 \end{pmatrix} \subset \mathrm{Pic}(X).$$

Then let $W = \begin{pmatrix} 0 & 1 \\ 1 & -2 \end{pmatrix}^{\perp} \subset \mathrm{Pic}(X)$. Note $\mathrm{rk}\,\mathrm{Pic}(X) - 2 = \mathrm{rk}\, W$. Now let $W_{\mathrm{root}}$ denote the lattice spanned by all the irreducible components of fibers which the chosen section, in this case $E_1 \times$ , does not intersect. In particular, it will be spanned by the irreducible components of the singular fibers which do not meet $E_1 \times \mathbb{P}^1$. Then

$$|\mathrm{Aut}(X)| = \infty \iff \mathrm{rk}(W) > \mathrm{rk}(W_{\mathrm{root}}).$$

Indeed, by Shiota Tate we have

$$\mathrm{rk}\,\mathrm{Pic}(X) = 2 + \sum_t (n_t - 1) + \mathrm{rk}\, MW_X$$

and $\mathrm{rk}\, W = \sum_t (n_t - 1) + \mathrm{rk}\, MW_X$. Furthermore, $\mathrm{rk}(W_{\mathrm{root}}) \leq \sum (n_t - 1)$. So we have $\mathrm{rk}\, W > \mathrm{rk}(W_{\mathrm{root}}) \iff \mathrm{rk}\, MW_X > 0$. In particular, recall $MW_X$ is defined to be the quotient group

$$0 \to A \to \mathrm{NS}_X \to MW_X \to 0$$

where $A$ is the subgroup generated by all vertical divisors and the section chosen as an origin, in this case $E_1 \times \mathbb{P}^1$. In particular, note $\mathrm{NS}_X \cong \mathrm{Pic}_X$ for a K3 surface. Since we have at least $\rho(F) - 2$ sections for $\pi : X \to \mathbb{P}^1$, we see that $\mathrm{rk}\, MW_X > 0$. Thus, $\mathrm{Aut}(X)$ is infinite for all smooth $X \in | - K_F|$.

**Theorem 30.23.** *Let $F$ be a Fano 3-fold where $\rho(F) \geq 6$. Then general $X \in | - K_F|$ is a smooth algebraic K3 surface. In particular, for each such $X$, we have* $\mathrm{Aut}(X)$ *is infinite.*

Now, at least in the case of $\rho(F) = 6$, we showed that when

## 31. Interesting articles

(1) In http://arxiv.org/abs/1111.4117, Francois Charles, building on the work of Van Lujik, gives an algorithm which conjecturally terminates for finding the Picard number of any K3 surface defined over a number field. If the Hodge conjecture holds for $X \times X$, then the algorithm terminates. More discussion: https://mathoverflow.net/questions/107960/how-to-compute-the-picard-rank-of-a-k3-surface

(2) General introduction: https://gauss.math.yale.edu/~il282/Sveta_S16.pdf

(3) The deeper reason for why Kodaira vanishing holds in characteristic p for algebraic K3 surface is revealed in this paper: P.DeligneandL.Illusie.Relvementsmodulop2etdcompositiond Invent.Math.,89(2):247270,1987.

(4) Picard number of Kummer surfaces: https://dblim.github.io/files/pic_kummer_surface.pdf

(5) Lines on Fermat surface: https://www.researchgate.net/publication/23646972_R_Lines_on_Fermat_surfaces

(6) Gives some techniques/idea for classifying elliptic fibrations on K3 surfaces (when picard number is large): https://arxiv.org/pdf/1312.4421

(7) Which K3 surfaces arise as the smooth anticanonical divisors of a threefold? Described here by Beauville: https://math.univ-cotedazur.fr/~beauvill/pubs/Fano.pdf

(8) Shioda: Period map for 2-tori file:///Users/ray/Downloads/jfs250104.pdf, shows in particular that $H^2(A;\mathbb{Z})$ along with intersection pairing does not distinguish between $A$ or its dual torus $A^\vee$.

(9) The fact that $\pi_* H^2(\tilde{A}, \mathbb{Z})$ is orthogonal to $\bigoplus_{i=1}^{16} \mathbb{Z}[\overline{E_i}]$ in $H^2(X;\mathbb{Z})$ in Kummer surface construction is done in detail by this paper by Beauville: http://www.numdam.org/item/AST_1985__126__99_0.pdf.

(10) Paper by Schutt and Shioda: lines on Fermat quartic surface: https://arxiv.org/pdf/0812.2377

(11) Kuga-Satake Hodge Conjecture, papers with examples in which it holds:
   – https://arxiv.org/pdf/2210.02948
   – https://hal.science/hal-03403477/document

(12) The Kuga-Satake variety of a kummer surface: https://www.mathnet.ru/links/7c539caa8e960ea6aab3fc61bde2f45a/rm2165_eng.pdf.

(13) Griffths on Periods: https://publications.ias.edu/sites/default/files/transcendentalmethod.pdf

(14) A good summative reference for Riemann-Hilbert correspondences, Gauss Manin connection, and variations of Hodge structure:.

(15) bridgeland stability course website by alex perry: https://websites.umich.edu/~arper/stability/, has links to good resources

(16) https://www.cmi.ac.in/~krishna/negative-curves-special-rational-surfaces.pdf negative curves on surfaces. Nagata conjecture. Weak SHGH Conjecture. Bounded negativity conjecture. short document that spells out some major open problems in the theory of negative curves on algebraic surfaces. Huybrechts told me: " The paper also mentions that there is a link to the Nagata Conjecture, which is another one of those that look treacherously easy but are just too hard. Lazarsfeld once said these are perfect conjectures to ruin a young person's career."

(17) Richard Thomas, notes on GIT for bundles? https://arxiv.org/pdf/math/0512411 Also Richard THomas' note on derived categories for working mathematicians is good.

(18) Under certain conditions, recovering a K3 surface from a curve on it: https://www.degruyter.com/document/doi/10.1515/crelle-2019-0025/pdf?srsltid=AfmBOop5SPvuqKKRrHosRngooKxg2A3YTnGCowhdwNNk6UCmLMJRBbAf. uses stability conditions, donaldson-thomas theory, wall-crossing, among other techniques.

(19) Stability conditions on K3 surfaces, Tom Bridgeland: https://arxiv.org/pdf/math/0307164.

(20) List of topics in applications of derived categories: https://ywfan-math.github.io/DerivedCat_Topics.pdf

(21) Determining whether the Cox ring of a variety is finitely generated is a problem of interest from the point of view of birational geometry. Some example papers: https://www.mn.uio.no/math/personer/vit/johnco/papers/pointsonaline.pdf and https://www.mn.uio.no/math/personer/vit/johnco/papers/k3.pdf... look around for other cases?

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